



Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [ ]: !pip install cec2017 --quiet
```

ERROR: Could not find a version that satisfies the requirement cec2017 (from versions: none)
ERROR: No matching distribution found for cec2017

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [3]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

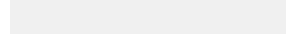
cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [1]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde_mutation_rule.py' in uploaded:
    if 'lblde_mutation_rule' in sys.modules:
        del sys.modules['lblde']
    import lblde_mutation_rule
    from lblde_mutation_rule import LBLDE_MutationRule
    print('lblde_mutation_rule.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

 Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving `lblde_mutation_rule.py` to `lblde_mutation_rule.py`
`lblde_mutation_rule.py` uploaded and imported successfully!

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [4]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [5]: D = 10 # Change this when testing different dimensions
```

```
In [6]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [7]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [8]: def make_wrapper(fn):
        """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
        def wrapper(x):
            val = fn(x.reshape(1, -1))
            if hasattr(val, '__len__'):
                val = float(val[0])
            else:
                val = float(val)
            return val if np.isfinite(val) else 1e18
        return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [10]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE_m on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'LBLDE_m on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE_MutationRule(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'LBLDE_m – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```



```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [11]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

=====

=

LBLDE_m on CEC 2017 F1 – Single Run

=====

=

Function : Shifted and Rotated Bent Cigar (10D)
Bounds : [-100, 100]
Global optimum : 100.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 1.871674e+11 | Diversity = 5.7928e+01

Generation	10:	Best = 1.243388e+10	FES = 1100	Diversity = 1.4148e+01
Generation	20:	Best = 1.569090e+09	FES = 2100	Diversity = 3.5379e+00
Generation	30:	Best = 2.807006e+08	FES = 3100	Diversity = 1.0957e+00
Generation	40:	Best = 7.892445e+07	FES = 4100	Diversity = 8.1056e-01
Generation	50:	Best = 5.618036e+07	FES = 5100	Diversity = 7.9271e-01
Generation	60:	Best = 4.610153e+07	FES = 6100	Diversity = 1.0687e+00
Generation	70:	Best = 4.187752e+07	FES = 7100	Diversity = 2.0873e+00
Generation	80:	Best = 3.719753e+07	FES = 8100	Diversity = 3.7621e+00
Generation	90:	Best = 3.048252e+07	FES = 9100	Diversity = 4.5518e+00
Generation	100:	Best = 2.498019e+07	FES = 10100	Diversity = 3.9337e+00
Generation	110:	Best = 2.198598e+07	FES = 11100	Diversity = 5.3119e+00
Generation	120:	Best = 2.039946e+07	FES = 12100	Diversity = 5.2453e+00
Generation	130:	Best = 1.915564e+07	FES = 13100	Diversity = 6.1589e+00
Generation	140:	Best = 1.827527e+07	FES = 14100	Diversity = 6.7344e+00
Generation	150:	Best = 1.731757e+07	FES = 15100	Diversity = 6.8509e+00
Generation	160:	Best = 1.697799e+07	FES = 16100	Diversity = 7.3877e+00
Generation	170:	Best = 1.658357e+07	FES = 17100	Diversity = 7.8890e+00
Generation	180:	Best = 1.631889e+07	FES = 18100	Diversity = 7.1486e+00
Generation	190:	Best = 1.609733e+07	FES = 19100	Diversity = 6.9873e+00
Generation	200:	Best = 1.579807e+07	FES = 20100	Diversity = 7.9545e+00
Generation	210:	Best = 1.558468e+07	FES = 21100	Diversity = 6.7838e+00
Generation	220:	Best = 1.537639e+07	FES = 22100	Diversity = 7.3773e+00
Generation	230:	Best = 1.530974e+07	FES = 23100	Diversity = 6.0093e+00
Generation	240:	Best = 1.526073e+07	FES = 24100	Diversity = 5.9501e+00
Generation	250:	Best = 1.523441e+07	FES = 25100	Diversity = 5.5600e+00
Generation	260:	Best = 1.521376e+07	FES = 26100	Diversity = 5.3375e+00
Generation	270:	Best = 1.520078e+07	FES = 27100	Diversity = 5.6688e+00
Generation	280:	Best = 1.518960e+07	FES = 28100	Diversity = 4.4701e+00
Generation	290:	Best = 1.517664e+07	FES = 29100	Diversity = 3.9729e+00
Generation	300:	Best = 1.517312e+07	FES = 30100	Diversity = 4.2923e+00
Generation	310:	Best = 1.516879e+07	FES = 31100	Diversity = 3.4807e+00
Generation	320:	Best = 1.516650e+07	FES = 32100	Diversity = 2.4883e+00
Generation	330:	Best = 1.516581e+07	FES = 33100	Diversity = 2.2338e+00
Generation	340:	Best = 1.516486e+07	FES = 34100	Diversity = 1.4557e+00
Generation	350:	Best = 1.516379e+07	FES = 35100	Diversity = 1.7424e+00
Generation	360:	Best = 1.516236e+07	FES = 36100	Diversity = 1.7870e+00
Generation	370:	Best = 1.516177e+07	FES = 37100	Diversity = 1.7861e+00
Generation	380:	Best = 1.516128e+07	FES = 38100	Diversity = 1.3311e+00
Generation	390:	Best = 1.516104e+07	FES = 39100	Diversity = 9.5373e-01
Generation	400:	Best = 1.516083e+07	FES = 40100	Diversity = 8.1781e-01
Generation	410:	Best = 1.516063e+07	FES = 41100	Diversity = 7.8532e-01
Generation	420:	Best = 1.516048e+07	FES = 42100	Diversity = 8.5095e-01
Generation	430:	Best = 1.516044e+07	FES = 43100	Diversity = 7.1724e-01
Generation	440:	Best = 1.516036e+07	FES = 44100	Diversity = 7.1398e-01

Generation	450:	Best = 1.516032e+07	FES = 45100	Diversity = 4.9136e-01
Generation	460:	Best = 1.516031e+07	FES = 46100	Diversity = 3.9321e-01
Generation	470:	Best = 1.516031e+07	FES = 47100	Diversity = 2.2125e-01
Generation	480:	Best = 1.516031e+07	FES = 48100	Diversity = 1.6969e-01
Generation	490:	Best = 1.516031e+07	FES = 49100	Diversity = 9.1203e-02
Generation	500:	Best = 1.516031e+07	FES = 50100	Diversity = 7.7725e-02
Generation	510:	Best = 1.516031e+07	FES = 51100	Diversity = 6.8491e-02
Generation	520:	Best = 1.516031e+07	FES = 52100	Diversity = 6.1486e-02
Generation	530:	Best = 1.516031e+07	FES = 53100	Diversity = 4.3403e-02
Generation	540:	Best = 1.516031e+07	FES = 54100	Diversity = 2.9752e-02
Generation	550:	Best = 1.516031e+07	FES = 55100	Diversity = 1.9343e-02
Generation	560:	Best = 1.516031e+07	FES = 56100	Diversity = 1.6162e-02
Generation	570:	Best = 1.516031e+07	FES = 57100	Diversity = 1.3823e-02
Generation	580:	Best = 1.516031e+07	FES = 58100	Diversity = 1.1812e-02
Generation	590:	Best = 1.516031e+07	FES = 59100	Diversity = 9.1702e-03
Generation	600:	Best = 1.516031e+07	FES = 60100	Diversity = 7.7834e-03
Generation	610:	Best = 1.516031e+07	FES = 61100	Diversity = 5.8190e-03
Generation	620:	Best = 1.516031e+07	FES = 62100	Diversity = 4.9233e-03
Generation	630:	Best = 1.516031e+07	FES = 63100	Diversity = 4.3443e-03
Generation	640:	Best = 1.516031e+07	FES = 64100	Diversity = 3.8698e-03
Generation	650:	Best = 1.516031e+07	FES = 65100	Diversity = 3.2970e-03
Generation	660:	Best = 1.516031e+07	FES = 66100	Diversity = 3.0631e-03
Generation	670:	Best = 1.516031e+07	FES = 67100	Diversity = 2.5921e-03
Generation	680:	Best = 1.516031e+07	FES = 68100	Diversity = 2.5580e-03
Generation	690:	Best = 1.516031e+07	FES = 69100	Diversity = 1.9881e-03
Generation	700:	Best = 1.516031e+07	FES = 70100	Diversity = 1.6256e-03
Generation	710:	Best = 1.516031e+07	FES = 71100	Diversity = 1.4816e-03
Generation	720:	Best = 1.516031e+07	FES = 72100	Diversity = 1.1911e-03
Generation	730:	Best = 1.516031e+07	FES = 73100	Diversity = 1.1203e-03
Generation	740:	Best = 1.516031e+07	FES = 74100	Diversity = 1.1160e-03
Generation	750:	Best = 1.516031e+07	FES = 75100	Diversity = 9.5111e-04
Generation	760:	Best = 1.516031e+07	FES = 76100	Diversity = 8.1776e-04
Generation	770:	Best = 1.516031e+07	FES = 77100	Diversity = 6.8113e-04
Generation	780:	Best = 1.516031e+07	FES = 78100	Diversity = 7.2424e-04
Generation	790:	Best = 1.516031e+07	FES = 79100	Diversity = 5.7993e-04
Generation	800:	Best = 1.516031e+07	FES = 80100	Diversity = 4.4290e-04
Generation	810:	Best = 1.516031e+07	FES = 81100	Diversity = 4.1439e-04
Generation	820:	Best = 1.516031e+07	FES = 82100	Diversity = 5.1452e-04
Generation	830:	Best = 1.516031e+07	FES = 83100	Diversity = 4.3447e-04
Generation	840:	Best = 1.516031e+07	FES = 84100	Diversity = 4.1277e-04
Generation	850:	Best = 1.516031e+07	FES = 85100	Diversity = 3.4670e-04
Generation	860:	Best = 1.516031e+07	FES = 86100	Diversity = 2.5047e-04
Generation	870:	Best = 1.516031e+07	FES = 87100	Diversity = 2.4124e-04
Generation	880:	Best = 1.516031e+07	FES = 88100	Diversity = 2.9397e-04
Generation	890:	Best = 1.516031e+07	FES = 89100	Diversity = 2.9784e-04
Generation	900:	Best = 1.516031e+07	FES = 90100	Diversity = 2.3857e-04
Generation	910:	Best = 1.516031e+07	FES = 91100	Diversity = 1.8816e-04
Generation	920:	Best = 1.516031e+07	FES = 92100	Diversity = 2.1090e-04
Generation	930:	Best = 1.516031e+07	FES = 93100	Diversity = 1.9211e-04
Generation	940:	Best = 1.516031e+07	FES = 94100	Diversity = 1.9196e-04
Generation	950:	Best = 1.516031e+07	FES = 95100	Diversity = 2.0861e-04
Generation	960:	Best = 1.516031e+07	FES = 96100	Diversity = 1.8374e-04
Generation	970:	Best = 1.516031e+07	FES = 97100	Diversity = 1.2276e-04
Generation	980:	Best = 1.516031e+07	FES = 98100	Diversity = 1.0444e-04

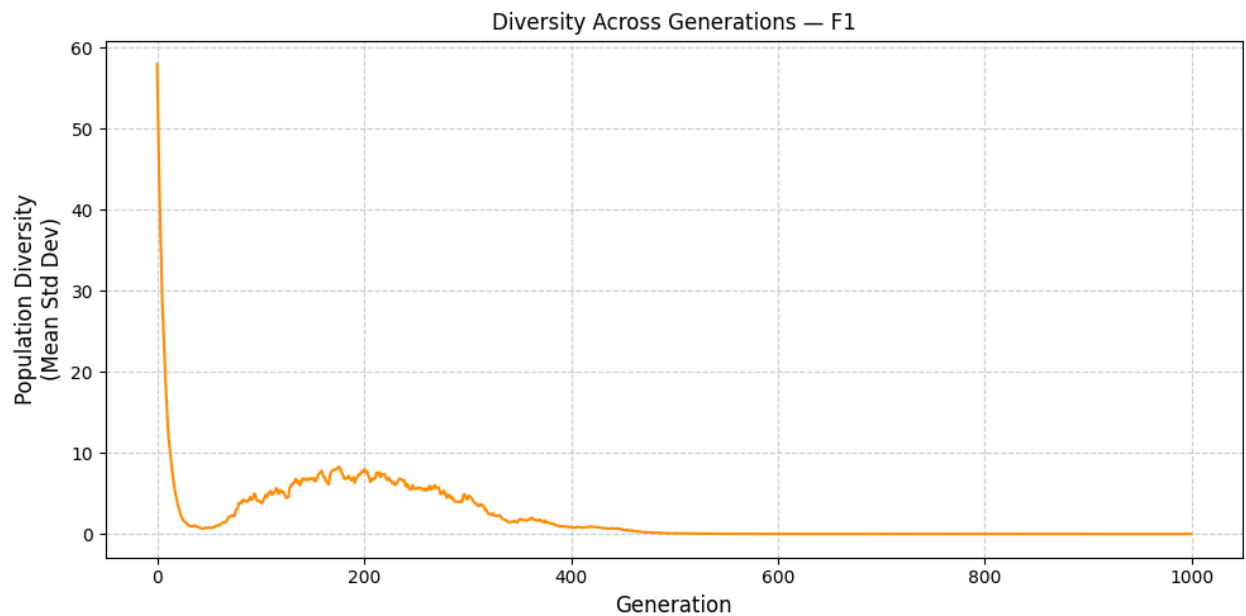
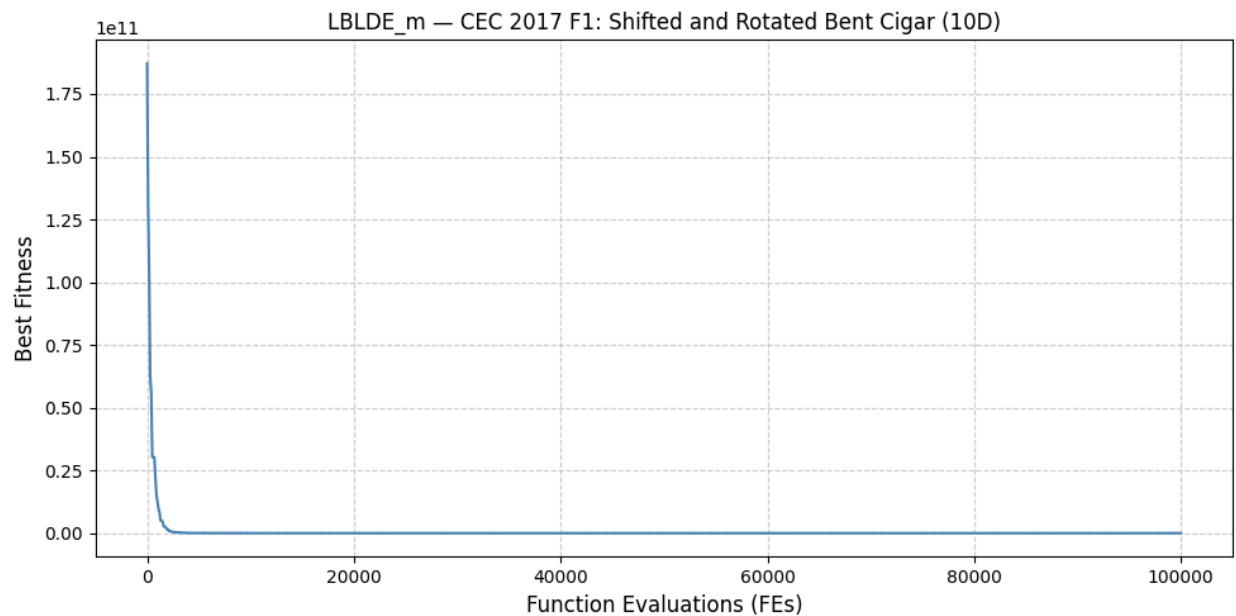
Generation 990: Best = 1.516031e+07 FES = 99100 Diversity = 1.1443e-04

Optimization completed!

Final Best Fitness = 1.516031e+07

Total Function Evaluations = 100000

```
=====
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Results:
  Best Fitness           : 1.516031e+07
  Global Optimum         : 1.000000e+02
  Error (|best-optimum|) : 1.516021e+07
  Best Solution (5 dim)  : [-55.276632   -70.4295284  -29.61013788  -58.32669814
22.08960189]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F1.png

```
In [12]: # =====  
# F2 – Shifted and Rotated Sum of Different Power  
# =====  
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

=====

=

LBLDE_m on CEC 2017 F2 – Single Run

=====

=

Function : Shifted and Rotated Sum of Different Power (Deprecated) (10D)
Bounds : [-100, 100]
Global optimum : 200.0

WARNING: f2 has been deprecated from the CEC 2017 benchmark suite

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.603565e+14 | Diversity = 5.7928e+01

Generation	10:	Best = 3.676803e+07	FES = 1100	Diversity = 2.4539e+01
Generation	20:	Best = 5.171569e+05	FES = 2100	Diversity = 1.0504e+01
Generation	30:	Best = 1.737358e+04	FES = 3100	Diversity = 6.2989e+00
Generation	40:	Best = 3.652031e+03	FES = 4100	Diversity = 2.5152e+00
Generation	50:	Best = 2.668524e+03	FES = 5100	Diversity = 1.2508e+00
Generation	60:	Best = 2.488883e+03	FES = 6100	Diversity = 1.0452e+00
Generation	70:	Best = 2.405264e+03	FES = 7100	Diversity = 6.9493e-01
Generation	80:	Best = 2.383745e+03	FES = 8100	Diversity = 3.6916e-01
Generation	90:	Best = 2.376560e+03	FES = 9100	Diversity = 2.2835e-01
Generation	100:	Best = 2.374242e+03	FES = 10100	Diversity = 1.0476e-01
Generation	110:	Best = 2.373504e+03	FES = 11100	Diversity = 6.9774e-02
Generation	120:	Best = 2.373091e+03	FES = 12100	Diversity = 4.7457e-02
Generation	130:	Best = 2.372823e+03	FES = 13100	Diversity = 3.6413e-02
Generation	140:	Best = 2.372711e+03	FES = 14100	Diversity = 2.5587e-02
Generation	150:	Best = 2.372651e+03	FES = 15100	Diversity = 1.9822e-02
Generation	160:	Best = 2.372616e+03	FES = 16100	Diversity = 1.5757e-02
Generation	170:	Best = 2.372589e+03	FES = 17100	Diversity = 1.2447e-02
Generation	180:	Best = 2.372580e+03	FES = 18100	Diversity = 8.9965e-03
Generation	190:	Best = 2.372573e+03	FES = 19100	Diversity = 9.9027e-03
Generation	200:	Best = 2.372568e+03	FES = 20100	Diversity = 9.8891e-03
Generation	210:	Best = 2.372565e+03	FES = 21100	Diversity = 8.6083e-03
Generation	220:	Best = 2.372563e+03	FES = 22100	Diversity = 8.0436e-03
Generation	230:	Best = 2.372561e+03	FES = 23100	Diversity = 6.9820e-03
Generation	240:	Best = 2.372560e+03	FES = 24100	Diversity = 5.8468e-03
Generation	250:	Best = 2.372559e+03	FES = 25100	Diversity = 4.0036e-03
Generation	260:	Best = 2.372559e+03	FES = 26100	Diversity = 3.9071e-03
Generation	270:	Best = 2.372558e+03	FES = 27100	Diversity = 4.3869e-03
Generation	280:	Best = 2.372558e+03	FES = 28100	Diversity = 3.4387e-03
Generation	290:	Best = 2.372558e+03	FES = 29100	Diversity = 1.8447e-03
Generation	300:	Best = 2.372558e+03	FES = 30100	Diversity = 1.1015e-03
Generation	310:	Best = 2.372558e+03	FES = 31100	Diversity = 7.2900e-04
Generation	320:	Best = 2.372558e+03	FES = 32100	Diversity = 5.0441e-04
Generation	330:	Best = 2.372558e+03	FES = 33100	Diversity = 4.4004e-04
Generation	340:	Best = 2.372558e+03	FES = 34100	Diversity = 2.3813e-04
Generation	350:	Best = 2.372558e+03	FES = 35100	Diversity = 1.5348e-04
Generation	360:	Best = 2.372558e+03	FES = 36100	Diversity = 1.4328e-04
Generation	370:	Best = 2.372558e+03	FES = 37100	Diversity = 6.9524e-05
Generation	380:	Best = 2.372558e+03	FES = 38100	Diversity = 3.9246e-05
Generation	390:	Best = 2.372558e+03	FES = 39100	Diversity = 2.8821e-05
Generation	400:	Best = 2.372558e+03	FES = 40100	Diversity = 1.6493e-05
Generation	410:	Best = 2.372558e+03	FES = 41100	Diversity = 1.3374e-05
Generation	420:	Best = 2.372558e+03	FES = 42100	Diversity = 9.9151e-06
Generation	430:	Best = 2.372558e+03	FES = 43100	Diversity = 6.4423e-06

Generation	440:	Best = 2.372558e+03	FES = 44100	Diversity = 4.8951e-06
Generation	450:	Best = 2.372558e+03	FES = 45100	Diversity = 2.7760e-06
Generation	460:	Best = 2.372558e+03	FES = 46100	Diversity = 2.2199e-06
Generation	470:	Best = 2.372558e+03	FES = 47100	Diversity = 1.7291e-06
Generation	480:	Best = 2.372558e+03	FES = 48100	Diversity = 1.2113e-06
Generation	490:	Best = 2.372558e+03	FES = 49100	Diversity = 9.8912e-07
Generation	500:	Best = 2.372558e+03	FES = 50100	Diversity = 9.0588e-07
Generation	510:	Best = 2.372558e+03	FES = 51100	Diversity = 6.7489e-07
Generation	520:	Best = 2.372558e+03	FES = 52100	Diversity = 4.4201e-07
Generation	530:	Best = 2.372558e+03	FES = 53100	Diversity = 2.7656e-07
Generation	540:	Best = 2.372558e+03	FES = 54100	Diversity = 1.9500e-07
Generation	550:	Best = 2.372558e+03	FES = 55100	Diversity = 2.0284e-07
Generation	560:	Best = 2.372558e+03	FES = 56100	Diversity = 2.3053e-07
Generation	570:	Best = 2.372558e+03	FES = 57100	Diversity = 1.4696e-07
Generation	580:	Best = 2.372558e+03	FES = 58100	Diversity = 1.6041e-07
Generation	590:	Best = 2.372558e+03	FES = 59100	Diversity = 1.5485e-07
Generation	600:	Best = 2.372558e+03	FES = 60100	Diversity = 8.4218e-08
Generation	610:	Best = 2.372558e+03	FES = 61100	Diversity = 7.6243e-08
Generation	620:	Best = 2.372558e+03	FES = 62100	Diversity = 6.7534e-08
Generation	630:	Best = 2.372558e+03	FES = 63100	Diversity = 6.8566e-08
Generation	640:	Best = 2.372558e+03	FES = 64100	Diversity = 6.7449e-08
Generation	650:	Best = 2.372558e+03	FES = 65100	Diversity = 6.0204e-08
Generation	660:	Best = 2.372558e+03	FES = 66100	Diversity = 6.2902e-08
Generation	670:	Best = 2.372558e+03	FES = 67100	Diversity = 6.0705e-08
Generation	680:	Best = 2.372558e+03	FES = 68100	Diversity = 5.7921e-08
Generation	690:	Best = 2.372558e+03	FES = 69100	Diversity = 5.9622e-08
Generation	700:	Best = 2.372558e+03	FES = 70100	Diversity = 2.7786e-08
Generation	710:	Best = 2.372558e+03	FES = 71100	Diversity = 2.1156e-08
Generation	720:	Best = 2.372558e+03	FES = 72100	Diversity = 2.8810e-14
Generation	730:	Best = 2.372558e+03	FES = 73100	Diversity = 2.8810e-14
Generation	740:	Best = 2.372558e+03	FES = 74100	Diversity = 2.8810e-14
Generation	750:	Best = 2.372558e+03	FES = 75100	Diversity = 2.8810e-14
Generation	760:	Best = 2.372558e+03	FES = 76100	Diversity = 2.8810e-14
Generation	770:	Best = 2.372558e+03	FES = 77100	Diversity = 2.8810e-14
Generation	780:	Best = 2.372558e+03	FES = 78100	Diversity = 2.8810e-14
Generation	790:	Best = 2.372558e+03	FES = 79100	Diversity = 2.8810e-14
Generation	800:	Best = 2.372558e+03	FES = 80100	Diversity = 2.8810e-14
Generation	810:	Best = 2.372558e+03	FES = 81100	Diversity = 2.8810e-14
Generation	820:	Best = 2.372558e+03	FES = 82100	Diversity = 2.8810e-14
Generation	830:	Best = 2.372558e+03	FES = 83100	Diversity = 2.8810e-14
Generation	840:	Best = 2.372558e+03	FES = 84100	Diversity = 2.8810e-14
Generation	850:	Best = 2.372558e+03	FES = 85100	Diversity = 2.8810e-14
Generation	860:	Best = 2.372558e+03	FES = 86100	Diversity = 2.8810e-14
Generation	870:	Best = 2.372558e+03	FES = 87100	Diversity = 2.8810e-14
Generation	880:	Best = 2.372558e+03	FES = 88100	Diversity = 2.8810e-14
Generation	890:	Best = 2.372558e+03	FES = 89100	Diversity = 2.8810e-14
Generation	900:	Best = 2.372558e+03	FES = 90100	Diversity = 2.8810e-14
Generation	910:	Best = 2.372558e+03	FES = 91100	Diversity = 2.8810e-14
Generation	920:	Best = 2.372558e+03	FES = 92100	Diversity = 2.8810e-14
Generation	930:	Best = 2.372558e+03	FES = 93100	Diversity = 2.8810e-14
Generation	940:	Best = 2.372558e+03	FES = 94100	Diversity = 2.8810e-14
Generation	950:	Best = 2.372558e+03	FES = 95100	Diversity = 2.8810e-14
Generation	960:	Best = 2.372558e+03	FES = 96100	Diversity = 2.8810e-14
Generation	970:	Best = 2.372558e+03	FES = 97100	Diversity = 2.8810e-14

Generation 980: Best = 2.372558e+03 FES = 98100 Diversity = 2.8810e-14
Generation 990: Best = 2.372558e+03 FES = 99100 Diversity = 2.8810e-14

Optimization completed!

Final Best Fitness = 2.372558e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 2.372558e+03

Global Optimum : 2.000000e+02

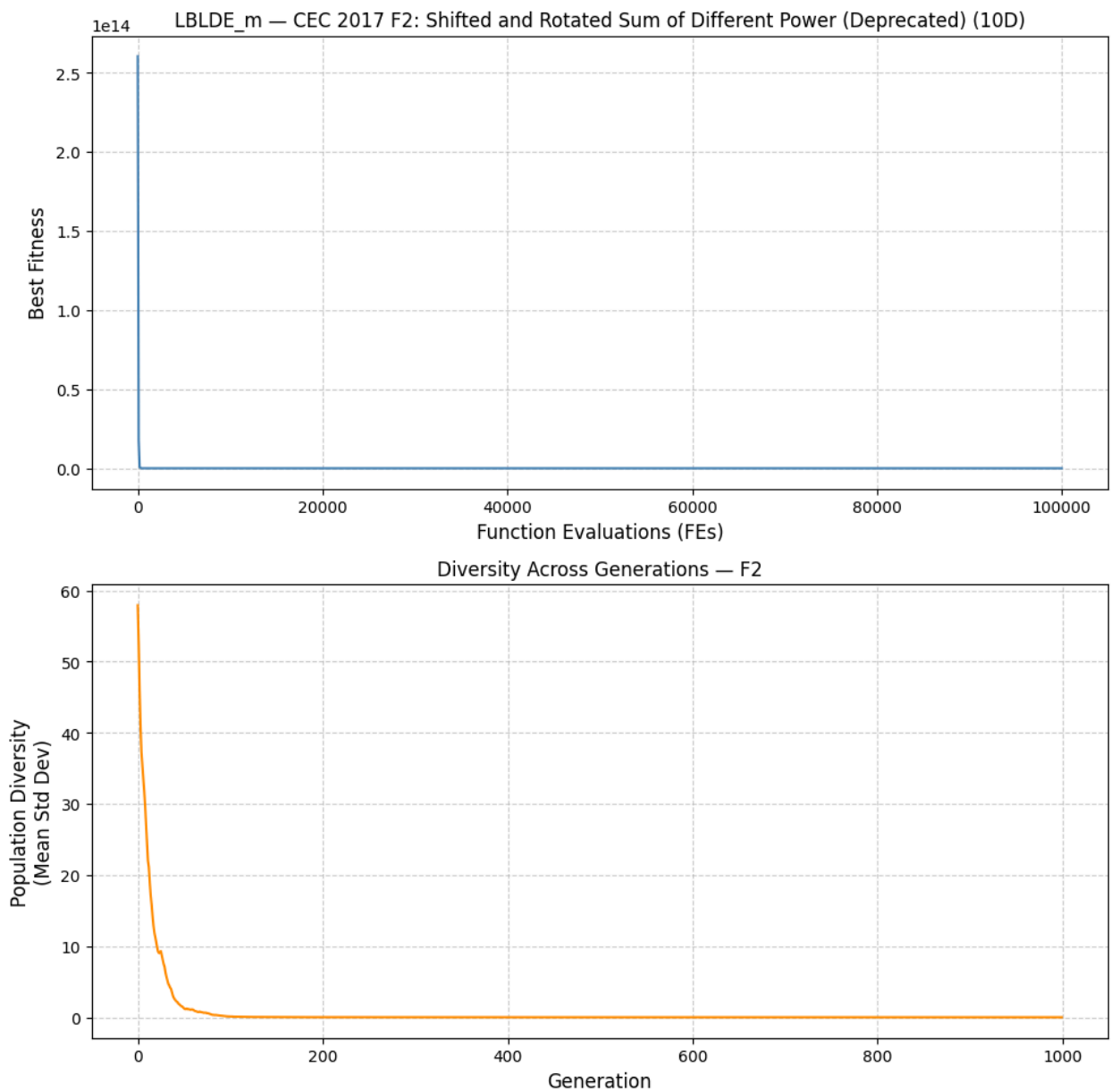
Error (|best-optimum|) : 2.172558e+03

Best Solution (5 dim) : [-28.22252607 -37.94846475 -75.48614396 -0.84266819
4.18271473]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F2.png

```
In [13]: # =====
# F3 – Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

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LBLDE_m on CEC 2017 F3 – Single Run

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Function : Shifted and Rotated Zakharov (10D)

Bounds : [-100, 100]

Global optimum : 300.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.625937e+04 | Diversity = 5.7928e+01

Generation	10:	Best = 1.044934e+04	FES = 1100	Diversity = 4.0412e+01
Generation	20:	Best = 5.114906e+03	FES = 2100	Diversity = 2.4307e+01
Generation	30:	Best = 2.407273e+03	FES = 3100	Diversity = 1.3316e+01
Generation	40:	Best = 1.207255e+03	FES = 4100	Diversity = 7.3778e+00
Generation	50:	Best = 8.612900e+02	FES = 5100	Diversity = 4.1243e+00
Generation	60:	Best = 6.232040e+02	FES = 6100	Diversity = 2.3730e+00
Generation	70:	Best = 5.616293e+02	FES = 7100	Diversity = 1.4500e+00
Generation	80:	Best = 5.085892e+02	FES = 8100	Diversity = 1.0955e+00
Generation	90:	Best = 4.646771e+02	FES = 9100	Diversity = 6.2988e-01
Generation	100:	Best = 4.418431e+02	FES = 10100	Diversity = 5.1798e-01
Generation	110:	Best = 4.273433e+02	FES = 11100	Diversity = 3.9506e-01
Generation	120:	Best = 4.096637e+02	FES = 12100	Diversity = 3.3590e-01
Generation	130:	Best = 4.069689e+02	FES = 13100	Diversity = 1.9625e-01
Generation	140:	Best = 4.056104e+02	FES = 14100	Diversity = 9.7976e-02
Generation	150:	Best = 4.042136e+02	FES = 15100	Diversity = 7.4606e-02
Generation	160:	Best = 4.027027e+02	FES = 16100	Diversity = 6.4613e-02
Generation	170:	Best = 4.020259e+02	FES = 17100	Diversity = 5.0654e-02
Generation	180:	Best = 4.016119e+02	FES = 18100	Diversity = 2.5920e-02
Generation	190:	Best = 4.012126e+02	FES = 19100	Diversity = 2.7466e-02
Generation	200:	Best = 4.009177e+02	FES = 20100	Diversity = 2.6433e-02
Generation	210:	Best = 4.006604e+02	FES = 21100	Diversity = 2.5005e-02
Generation	220:	Best = 4.005632e+02	FES = 22100	Diversity = 1.5847e-02
Generation	230:	Best = 4.004692e+02	FES = 23100	Diversity = 1.2870e-02
Generation	240:	Best = 4.004209e+02	FES = 24100	Diversity = 9.1395e-03
Generation	250:	Best = 4.003752e+02	FES = 25100	Diversity = 1.0061e-02
Generation	260:	Best = 4.003407e+02	FES = 26100	Diversity = 8.6856e-03
Generation	270:	Best = 4.003118e+02	FES = 27100	Diversity = 7.7044e-03
Generation	280:	Best = 4.002997e+02	FES = 28100	Diversity = 5.3588e-03
Generation	290:	Best = 4.002890e+02	FES = 29100	Diversity = 4.4161e-03
Generation	300:	Best = 4.002808e+02	FES = 30100	Diversity = 3.1745e-03
Generation	310:	Best = 4.002756e+02	FES = 31100	Diversity = 2.5976e-03
Generation	320:	Best = 4.002737e+02	FES = 32100	Diversity = 1.3099e-03
Generation	330:	Best = 4.002726e+02	FES = 33100	Diversity = 9.2193e-04
Generation	340:	Best = 4.002720e+02	FES = 34100	Diversity = 6.4755e-04
Generation	350:	Best = 4.002716e+02	FES = 35100	Diversity = 6.1135e-04
Generation	360:	Best = 4.002714e+02	FES = 36100	Diversity = 3.9655e-04
Generation	370:	Best = 4.002713e+02	FES = 37100	Diversity = 2.1895e-04
Generation	380:	Best = 4.002713e+02	FES = 38100	Diversity = 1.3140e-04
Generation	390:	Best = 4.002713e+02	FES = 39100	Diversity = 1.3278e-04
Generation	400:	Best = 4.002713e+02	FES = 40100	Diversity = 1.0520e-04
Generation	410:	Best = 4.002712e+02	FES = 41100	Diversity = 1.0774e-04
Generation	420:	Best = 4.002712e+02	FES = 42100	Diversity = 7.2797e-05
Generation	430:	Best = 4.002712e+02	FES = 43100	Diversity = 6.6608e-05
Generation	440:	Best = 4.002712e+02	FES = 44100	Diversity = 5.4739e-05

Generation	450:	Best = 4.002712e+02	FES = 45100	Diversity = 3.8473e-05
Generation	460:	Best = 4.002712e+02	FES = 46100	Diversity = 3.0910e-05
Generation	470:	Best = 4.002712e+02	FES = 47100	Diversity = 2.1174e-05
Generation	480:	Best = 4.002712e+02	FES = 48100	Diversity = 1.2403e-05
Generation	490:	Best = 4.002712e+02	FES = 49100	Diversity = 1.2437e-05
Generation	500:	Best = 4.002712e+02	FES = 50100	Diversity = 5.3653e-06
Generation	510:	Best = 4.002712e+02	FES = 51100	Diversity = 4.6400e-06
Generation	520:	Best = 4.002712e+02	FES = 52100	Diversity = 3.9982e-06
Generation	530:	Best = 4.002712e+02	FES = 53100	Diversity = 2.3926e-06
Generation	540:	Best = 4.002712e+02	FES = 54100	Diversity = 1.5718e-06
Generation	550:	Best = 4.002712e+02	FES = 55100	Diversity = 1.0104e-06
Generation	560:	Best = 4.002712e+02	FES = 56100	Diversity = 8.2023e-07
Generation	570:	Best = 4.002712e+02	FES = 57100	Diversity = 7.9459e-07
Generation	580:	Best = 4.002712e+02	FES = 58100	Diversity = 5.5352e-07
Generation	590:	Best = 4.002712e+02	FES = 59100	Diversity = 4.3828e-07
Generation	600:	Best = 4.002712e+02	FES = 60100	Diversity = 4.2058e-07
Generation	610:	Best = 4.002712e+02	FES = 61100	Diversity = 3.0396e-07
Generation	620:	Best = 4.002712e+02	FES = 62100	Diversity = 2.1708e-07
Generation	630:	Best = 4.002712e+02	FES = 63100	Diversity = 1.2870e-07
Generation	640:	Best = 4.002712e+02	FES = 64100	Diversity = 5.9414e-08
Generation	650:	Best = 4.002712e+02	FES = 65100	Diversity = 3.1559e-08
Generation	660:	Best = 4.002712e+02	FES = 66100	Diversity = 2.8309e-08
Generation	670:	Best = 4.002712e+02	FES = 67100	Diversity = 1.1739e-08
Generation	680:	Best = 4.002712e+02	FES = 68100	Diversity = 7.4130e-09
Generation	690:	Best = 4.002712e+02	FES = 69100	Diversity = 5.2014e-09
Generation	700:	Best = 4.002712e+02	FES = 70100	Diversity = 1.7336e-09
Generation	710:	Best = 4.002712e+02	FES = 71100	Diversity = 7.2857e-11
Generation	720:	Best = 4.002712e+02	FES = 72100	Diversity = 3.3466e-11
Generation	730:	Best = 4.002712e+02	FES = 73100	Diversity = 8.3031e-13
Generation	740:	Best = 4.002712e+02	FES = 74100	Diversity = 2.6769e-14
Generation	750:	Best = 4.002712e+02	FES = 75100	Diversity = 2.8799e-14
Generation	760:	Best = 4.002712e+02	FES = 76100	Diversity = 2.8799e-14
Generation	770:	Best = 4.002712e+02	FES = 77100	Diversity = 2.8799e-14
Generation	780:	Best = 4.002712e+02	FES = 78100	Diversity = 2.8799e-14
Generation	790:	Best = 4.002712e+02	FES = 79100	Diversity = 2.8799e-14
Generation	800:	Best = 4.002712e+02	FES = 80100	Diversity = 2.8799e-14
Generation	810:	Best = 4.002712e+02	FES = 81100	Diversity = 2.8799e-14
Generation	820:	Best = 4.002712e+02	FES = 82100	Diversity = 2.8799e-14
Generation	830:	Best = 4.002712e+02	FES = 83100	Diversity = 2.8799e-14
Generation	840:	Best = 4.002712e+02	FES = 84100	Diversity = 2.8799e-14
Generation	850:	Best = 4.002712e+02	FES = 85100	Diversity = 2.8799e-14
Generation	860:	Best = 4.002712e+02	FES = 86100	Diversity = 2.8799e-14
Generation	870:	Best = 4.002712e+02	FES = 87100	Diversity = 2.8799e-14
Generation	880:	Best = 4.002712e+02	FES = 88100	Diversity = 2.8799e-14
Generation	890:	Best = 4.002712e+02	FES = 89100	Diversity = 2.8799e-14
Generation	900:	Best = 4.002712e+02	FES = 90100	Diversity = 2.8799e-14
Generation	910:	Best = 4.002712e+02	FES = 91100	Diversity = 2.8799e-14
Generation	920:	Best = 4.002712e+02	FES = 92100	Diversity = 2.8799e-14
Generation	930:	Best = 4.002712e+02	FES = 93100	Diversity = 2.8799e-14
Generation	940:	Best = 4.002712e+02	FES = 94100	Diversity = 2.8799e-14
Generation	950:	Best = 4.002712e+02	FES = 95100	Diversity = 2.8799e-14
Generation	960:	Best = 4.002712e+02	FES = 96100	Diversity = 2.8799e-14
Generation	970:	Best = 4.002712e+02	FES = 97100	Diversity = 2.8799e-14
Generation	980:	Best = 4.002712e+02	FES = 98100	Diversity = 2.8799e-14

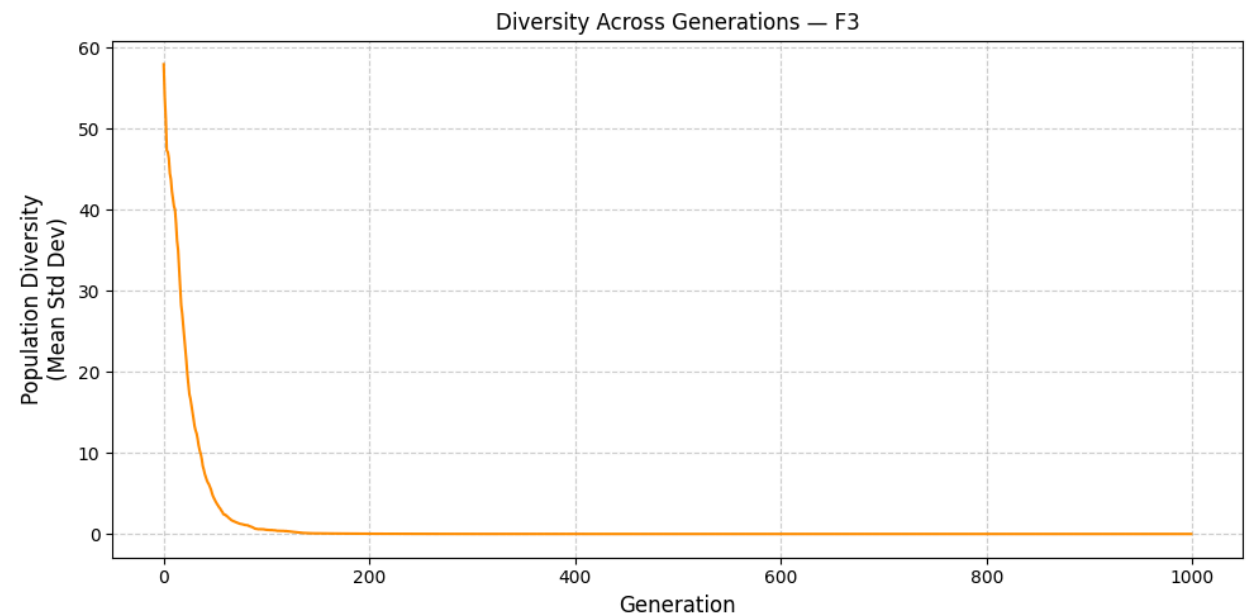
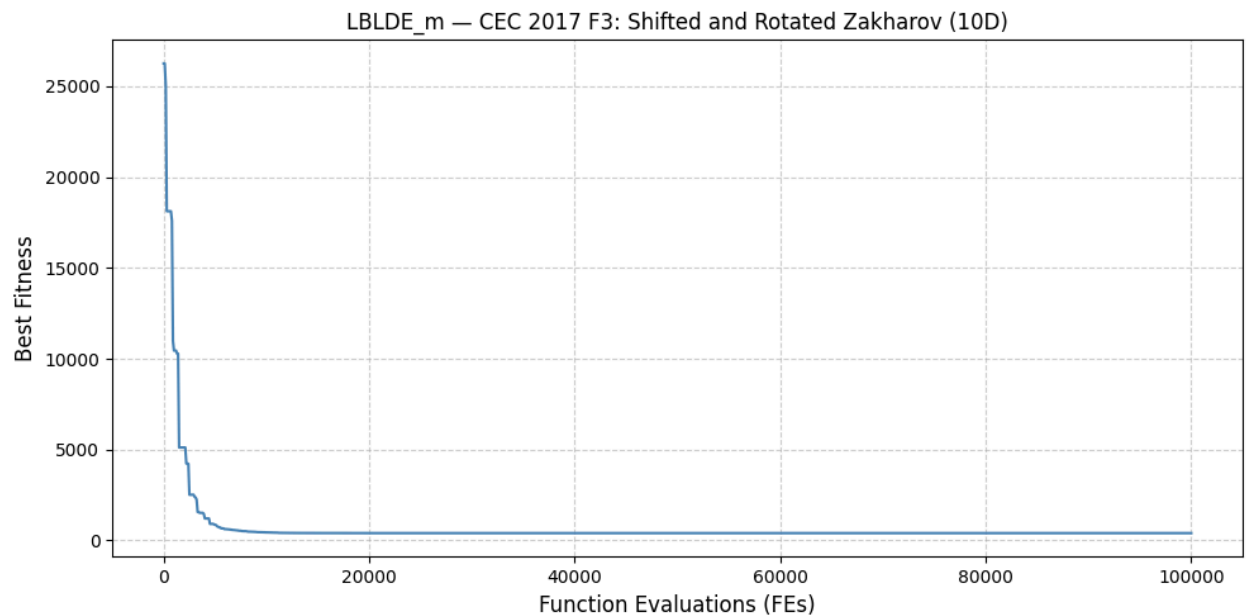
Generation 990: Best = 4.002712e+02 FES = 99100 Diversity = 2.8799e-14

Optimization completed!

Final Best Fitness = 4.002712e+02

Total Function Evaluations = 100000

```
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Results:
  Best Fitness          : 4.002712e+02
  Global Optimum        : 3.000000e+02
  Error (|best-optimum|) : 1.002712e+02
  Best Solution (5 dim)  : [-53.74927029  12.53619655  36.1545046   8.78867921
-46.74896606]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [14]: # =====  
# F4 – Shifted and Rotated Rosenbrock  
# =====  
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosent
```

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LBLDE_m on CEC 2017 F4 – Single Run

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Function : Shifted and Rotated Rosenbrock's (10D)

Bounds : [-100, 100]

Global optimum : 400.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.004346e+03 | Diversity = 5.7928e+01

Generation	10:	Best = 5.038440e+02	FES = 1100	Diversity = 1.7564e+01
Generation	20:	Best = 4.403234e+02	FES = 2100	Diversity = 4.7838e+00
Generation	30:	Best = 4.246175e+02	FES = 3100	Diversity = 1.7035e+00
Generation	40:	Best = 4.188484e+02	FES = 4100	Diversity = 1.5970e+00
Generation	50:	Best = 4.143434e+02	FES = 5100	Diversity = 3.4280e+00
Generation	60:	Best = 4.094306e+02	FES = 6100	Diversity = 5.3827e+00
Generation	70:	Best = 4.061091e+02	FES = 7100	Diversity = 1.1826e+00
Generation	80:	Best = 4.050724e+02	FES = 8100	Diversity = 6.8501e-01
Generation	90:	Best = 4.045000e+02	FES = 9100	Diversity = 4.2875e-01
Generation	100:	Best = 4.041618e+02	FES = 10100	Diversity = 2.7264e-01
Generation	110:	Best = 4.040249e+02	FES = 11100	Diversity = 2.0987e-01
Generation	120:	Best = 4.039475e+02	FES = 12100	Diversity = 9.7044e-02
Generation	130:	Best = 4.039362e+02	FES = 13100	Diversity = 4.8199e-02
Generation	140:	Best = 4.039273e+02	FES = 14100	Diversity = 3.6666e-02
Generation	150:	Best = 4.039230e+02	FES = 15100	Diversity = 2.2802e-02
Generation	160:	Best = 4.039201e+02	FES = 16100	Diversity = 1.7311e-02
Generation	170:	Best = 4.039177e+02	FES = 17100	Diversity = 1.5659e-02
Generation	180:	Best = 4.039167e+02	FES = 18100	Diversity = 1.0726e-02
Generation	190:	Best = 4.039162e+02	FES = 19100	Diversity = 7.7172e-03
Generation	200:	Best = 4.039159e+02	FES = 20100	Diversity = 4.5657e-03
Generation	210:	Best = 4.039158e+02	FES = 21100	Diversity = 2.2235e-03
Generation	220:	Best = 4.039157e+02	FES = 22100	Diversity = 2.4020e-03
Generation	230:	Best = 4.039157e+02	FES = 23100	Diversity = 1.6570e-03
Generation	240:	Best = 4.039157e+02	FES = 24100	Diversity = 1.0992e-03
Generation	250:	Best = 4.039156e+02	FES = 25100	Diversity = 8.7106e-04
Generation	260:	Best = 4.039156e+02	FES = 26100	Diversity = 8.0082e-04
Generation	270:	Best = 4.039156e+02	FES = 27100	Diversity = 3.9478e-04
Generation	280:	Best = 4.039156e+02	FES = 28100	Diversity = 2.8700e-04
Generation	290:	Best = 4.039156e+02	FES = 29100	Diversity = 1.8126e-04
Generation	300:	Best = 4.039156e+02	FES = 30100	Diversity = 2.1835e-04
Generation	310:	Best = 4.039156e+02	FES = 31100	Diversity = 1.4733e-04
Generation	320:	Best = 4.039156e+02	FES = 32100	Diversity = 1.0823e-04
Generation	330:	Best = 4.039156e+02	FES = 33100	Diversity = 7.5928e-05
Generation	340:	Best = 4.039156e+02	FES = 34100	Diversity = 5.5671e-05
Generation	350:	Best = 4.039156e+02	FES = 35100	Diversity = 2.7321e-05
Generation	360:	Best = 4.039156e+02	FES = 36100	Diversity = 1.7449e-05
Generation	370:	Best = 4.039156e+02	FES = 37100	Diversity = 1.2879e-05
Generation	380:	Best = 4.039156e+02	FES = 38100	Diversity = 1.1552e-05
Generation	390:	Best = 4.039156e+02	FES = 39100	Diversity = 7.0477e-06
Generation	400:	Best = 4.039156e+02	FES = 40100	Diversity = 5.9733e-06
Generation	410:	Best = 4.039156e+02	FES = 41100	Diversity = 3.6861e-06
Generation	420:	Best = 4.039156e+02	FES = 42100	Diversity = 3.0874e-06
Generation	430:	Best = 4.039156e+02	FES = 43100	Diversity = 2.0374e-06
Generation	440:	Best = 4.039156e+02	FES = 44100	Diversity = 1.0787e-06

Generation	450:	Best = 4.039156e+02	FES = 45100	Diversity = 1.0991e-06
Generation	460:	Best = 4.039156e+02	FES = 46100	Diversity = 7.0956e-07
Generation	470:	Best = 4.039156e+02	FES = 47100	Diversity = 3.0287e-07
Generation	480:	Best = 4.039156e+02	FES = 48100	Diversity = 1.9643e-07
Generation	490:	Best = 4.039156e+02	FES = 49100	Diversity = 1.3065e-07
Generation	500:	Best = 4.039156e+02	FES = 50100	Diversity = 9.5473e-08
Generation	510:	Best = 4.039156e+02	FES = 51100	Diversity = 5.1290e-08
Generation	520:	Best = 4.039156e+02	FES = 52100	Diversity = 3.1054e-08
Generation	530:	Best = 4.039156e+02	FES = 53100	Diversity = 1.8813e-08
Generation	540:	Best = 4.039156e+02	FES = 54100	Diversity = 9.0120e-09
Generation	550:	Best = 4.039156e+02	FES = 55100	Diversity = 2.4732e-09
Generation	560:	Best = 4.039156e+02	FES = 56100	Diversity = 8.5529e-10
Generation	570:	Best = 4.039156e+02	FES = 57100	Diversity = 6.1638e-11
Generation	580:	Best = 4.039156e+02	FES = 58100	Diversity = 2.9654e-13
Generation	590:	Best = 4.039156e+02	FES = 59100	Diversity = 6.4622e-14
Generation	600:	Best = 4.039156e+02	FES = 60100	Diversity = 4.5943e-14
Generation	610:	Best = 4.039156e+02	FES = 61100	Diversity = 3.4518e-14
Generation	620:	Best = 4.039156e+02	FES = 62100	Diversity = 3.3351e-14
Generation	630:	Best = 4.039156e+02	FES = 63100	Diversity = 3.2241e-14
Generation	640:	Best = 4.039156e+02	FES = 64100	Diversity = 3.2241e-14
Generation	650:	Best = 4.039156e+02	FES = 65100	Diversity = 3.2241e-14
Generation	660:	Best = 4.039156e+02	FES = 66100	Diversity = 3.2241e-14
Generation	670:	Best = 4.039156e+02	FES = 67100	Diversity = 3.2241e-14
Generation	680:	Best = 4.039156e+02	FES = 68100	Diversity = 3.2241e-14
Generation	690:	Best = 4.039156e+02	FES = 69100	Diversity = 3.2241e-14
Generation	700:	Best = 4.039156e+02	FES = 70100	Diversity = 3.2241e-14
Generation	710:	Best = 4.039156e+02	FES = 71100	Diversity = 3.2241e-14
Generation	720:	Best = 4.039156e+02	FES = 72100	Diversity = 3.2241e-14
Generation	730:	Best = 4.039156e+02	FES = 73100	Diversity = 3.2241e-14
Generation	740:	Best = 4.039156e+02	FES = 74100	Diversity = 3.2241e-14
Generation	750:	Best = 4.039156e+02	FES = 75100	Diversity = 3.2241e-14
Generation	760:	Best = 4.039156e+02	FES = 76100	Diversity = 3.2241e-14
Generation	770:	Best = 4.039156e+02	FES = 77100	Diversity = 3.2241e-14
Generation	780:	Best = 4.039156e+02	FES = 78100	Diversity = 3.2241e-14
Generation	790:	Best = 4.039156e+02	FES = 79100	Diversity = 3.2241e-14
Generation	800:	Best = 4.039156e+02	FES = 80100	Diversity = 3.2241e-14
Generation	810:	Best = 4.039156e+02	FES = 81100	Diversity = 3.2241e-14
Generation	820:	Best = 4.039156e+02	FES = 82100	Diversity = 3.2241e-14
Generation	830:	Best = 4.039156e+02	FES = 83100	Diversity = 3.2241e-14
Generation	840:	Best = 4.039156e+02	FES = 84100	Diversity = 3.2241e-14
Generation	850:	Best = 4.039156e+02	FES = 85100	Diversity = 3.2241e-14
Generation	860:	Best = 4.039156e+02	FES = 86100	Diversity = 3.2241e-14
Generation	870:	Best = 4.039156e+02	FES = 87100	Diversity = 3.2241e-14
Generation	880:	Best = 4.039156e+02	FES = 88100	Diversity = 3.2241e-14
Generation	890:	Best = 4.039156e+02	FES = 89100	Diversity = 3.2241e-14
Generation	900:	Best = 4.039156e+02	FES = 90100	Diversity = 3.2241e-14
Generation	910:	Best = 4.039156e+02	FES = 91100	Diversity = 3.2241e-14
Generation	920:	Best = 4.039156e+02	FES = 92100	Diversity = 3.2241e-14
Generation	930:	Best = 4.039156e+02	FES = 93100	Diversity = 3.2241e-14
Generation	940:	Best = 4.039156e+02	FES = 94100	Diversity = 3.2241e-14
Generation	950:	Best = 4.039156e+02	FES = 95100	Diversity = 3.2241e-14
Generation	960:	Best = 4.039156e+02	FES = 96100	Diversity = 3.2241e-14
Generation	970:	Best = 4.039156e+02	FES = 97100	Diversity = 3.2241e-14
Generation	980:	Best = 4.039156e+02	FES = 98100	Diversity = 3.2241e-14

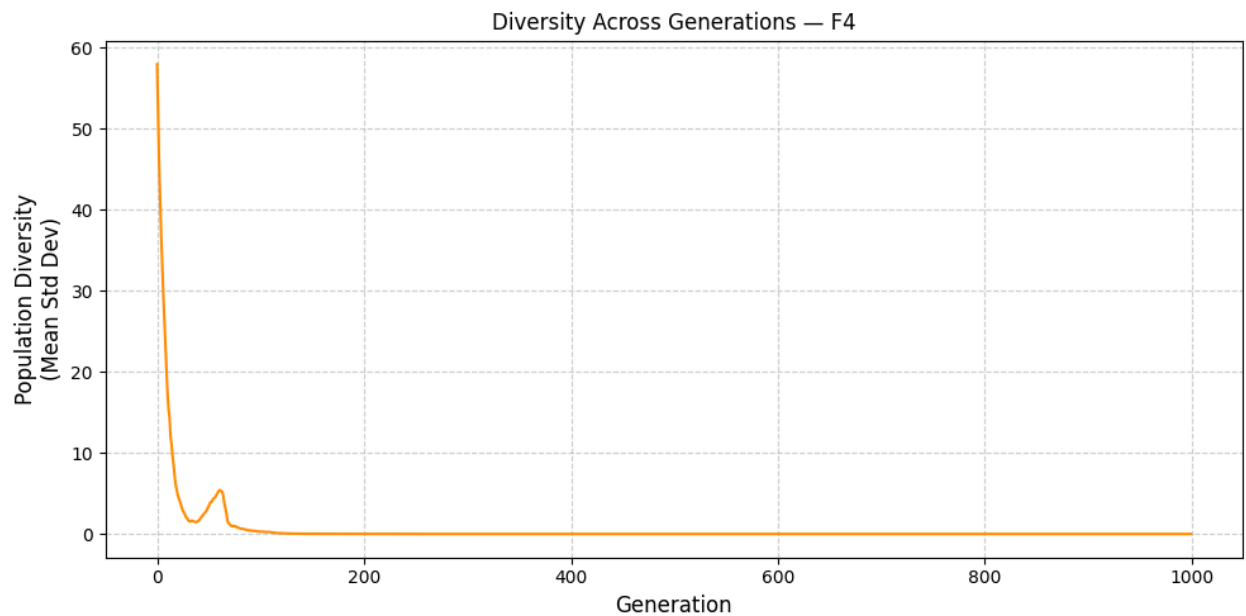
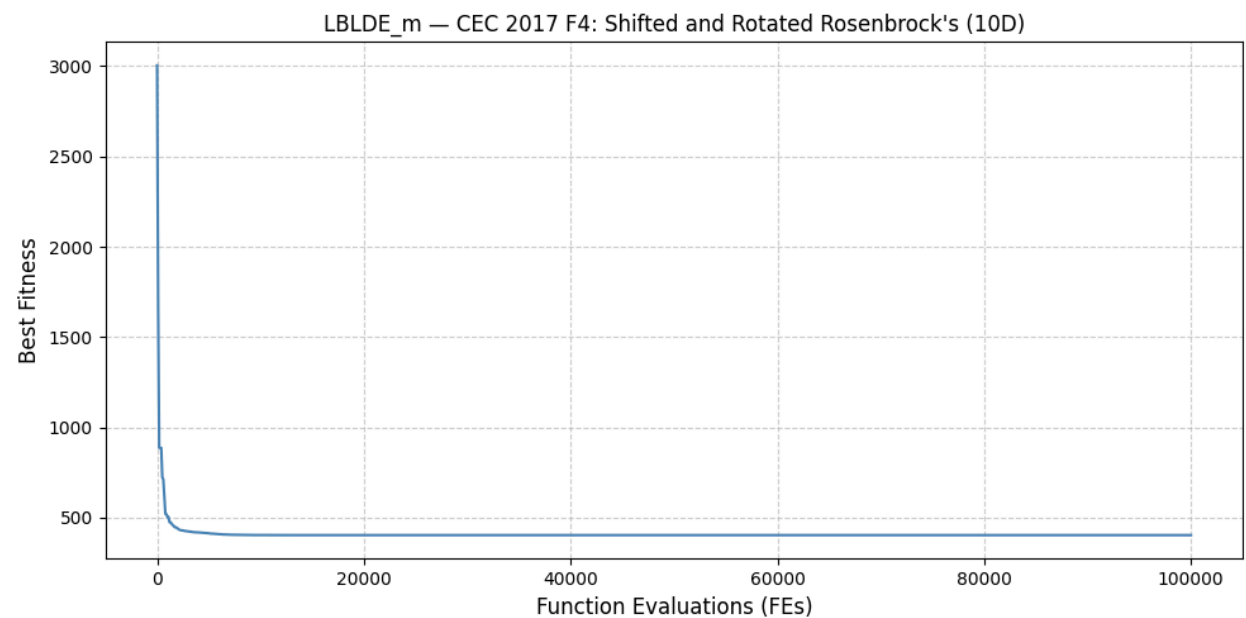
Generation 990: Best = 4.039156e+02 FES = 99100 Diversity = 3.2241e-14

Optimization completed!

Final Best Fitness = 4.039156e+02

Total Function Evaluations = 100000

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Results:
  Best Fitness      : 4.039156e+02
  Global Optimum    : 4.000000e+02
  Error (|best-optimum|) : 3.915600e+00
  Best Solution (5 dim) : [ 56.08048016  47.01672154 -71.50522812  45.07986784
29.50920766]
  D^(0) (init. diversity): 5.7928e+01
=====
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```



Graph saved as convergence_F4.png

```
In [15]: # =====  
# F5 – Shifted and Rotated Rastrigin  
# =====  
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

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LBLDE_m on CEC 2017 F5 – Single Run

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Function : Shifted and Rotated Rastrigin's (10D)
Bounds : [-100, 100]
Global optimum : 500.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 6.377184e+02 | Diversity = 5.7928e+01

Generation	10:	Best = 5.673787e+02	FES = 1100	Diversity = 2.3493e+01
Generation	20:	Best = 5.503506e+02	FES = 2100	Diversity = 1.9775e+01
Generation	30:	Best = 5.374556e+02	FES = 3100	Diversity = 1.7487e+01
Generation	40:	Best = 5.275415e+02	FES = 4100	Diversity = 1.6740e+01
Generation	50:	Best = 5.258471e+02	FES = 5100	Diversity = 1.5586e+01
Generation	60:	Best = 5.258471e+02	FES = 6100	Diversity = 1.5279e+01
Generation	70:	Best = 5.258471e+02	FES = 7100	Diversity = 1.4809e+01
Generation	80:	Best = 5.258423e+02	FES = 8100	Diversity = 1.4491e+01
Generation	90:	Best = 5.220593e+02	FES = 9100	Diversity = 1.4327e+01
Generation	100:	Best = 5.215429e+02	FES = 10100	Diversity = 1.4209e+01
Generation	110:	Best = 5.198554e+02	FES = 11100	Diversity = 1.4015e+01
Generation	120:	Best = 5.185039e+02	FES = 12100	Diversity = 1.4289e+01
Generation	130:	Best = 5.181752e+02	FES = 13100	Diversity = 1.4192e+01
Generation	140:	Best = 5.131305e+02	FES = 14100	Diversity = 1.3608e+01
Generation	150:	Best = 5.087229e+02	FES = 15100	Diversity = 1.3238e+01
Generation	160:	Best = 5.087229e+02	FES = 16100	Diversity = 1.3117e+01
Generation	170:	Best = 5.082951e+02	FES = 17100	Diversity = 1.3030e+01
Generation	180:	Best = 5.082951e+02	FES = 18100	Diversity = 1.3089e+01
Generation	190:	Best = 5.082951e+02	FES = 19100	Diversity = 1.2937e+01
Generation	200:	Best = 5.082951e+02	FES = 20100	Diversity = 1.2824e+01
Generation	210:	Best = 5.075432e+02	FES = 21100	Diversity = 1.2851e+01
Generation	220:	Best = 5.072297e+02	FES = 22100	Diversity = 1.2576e+01
Generation	230:	Best = 5.072297e+02	FES = 23100	Diversity = 1.2463e+01
Generation	240:	Best = 5.072004e+02	FES = 24100	Diversity = 1.2487e+01
Generation	250:	Best = 5.068917e+02	FES = 25100	Diversity = 1.2418e+01
Generation	260:	Best = 5.065306e+02	FES = 26100	Diversity = 1.2408e+01
Generation	270:	Best = 5.064488e+02	FES = 27100	Diversity = 1.2286e+01
Generation	280:	Best = 5.063653e+02	FES = 28100	Diversity = 1.2170e+01
Generation	290:	Best = 5.063486e+02	FES = 29100	Diversity = 1.2125e+01
Generation	300:	Best = 5.062143e+02	FES = 30100	Diversity = 1.1915e+01
Generation	310:	Best = 5.054188e+02	FES = 31100	Diversity = 1.1702e+01
Generation	320:	Best = 5.052975e+02	FES = 32100	Diversity = 1.1652e+01
Generation	330:	Best = 5.052889e+02	FES = 33100	Diversity = 1.1563e+01
Generation	340:	Best = 5.052881e+02	FES = 34100	Diversity = 1.1539e+01
Generation	350:	Best = 5.052880e+02	FES = 35100	Diversity = 1.1370e+01
Generation	360:	Best = 5.052880e+02	FES = 36100	Diversity = 1.1364e+01
Generation	370:	Best = 5.052880e+02	FES = 37100	Diversity = 1.1130e+01
Generation	380:	Best = 5.052880e+02	FES = 38100	Diversity = 1.1028e+01
Generation	390:	Best = 5.052880e+02	FES = 39100	Diversity = 1.0961e+01
Generation	400:	Best = 5.052880e+02	FES = 40100	Diversity = 1.0894e+01
Generation	410:	Best = 5.052880e+02	FES = 41100	Diversity = 1.0714e+01
Generation	420:	Best = 5.052880e+02	FES = 42100	Diversity = 1.0623e+01
Generation	430:	Best = 5.052880e+02	FES = 43100	Diversity = 1.0661e+01
Generation	440:	Best = 5.052141e+02	FES = 44100	Diversity = 1.0651e+01

Generation	450:	Best = 5.048377e+02	FES = 45100	Diversity = 1.0660e+01
Generation	460:	Best = 5.047622e+02	FES = 46100	Diversity = 1.0608e+01
Generation	470:	Best = 5.047515e+02	FES = 47100	Diversity = 1.0437e+01
Generation	480:	Best = 5.047344e+02	FES = 48100	Diversity = 1.0369e+01
Generation	490:	Best = 5.047245e+02	FES = 49100	Diversity = 1.0244e+01
Generation	500:	Best = 5.047245e+02	FES = 50100	Diversity = 1.0259e+01
Generation	510:	Best = 5.046148e+02	FES = 51100	Diversity = 1.0270e+01
Generation	520:	Best = 5.046148e+02	FES = 52100	Diversity = 1.0125e+01
Generation	530:	Best = 5.046148e+02	FES = 53100	Diversity = 9.9264e+00
Generation	540:	Best = 5.045789e+02	FES = 54100	Diversity = 9.6051e+00
Generation	550:	Best = 5.043389e+02	FES = 55100	Diversity = 9.3272e+00
Generation	560:	Best = 5.037628e+02	FES = 56100	Diversity = 8.9614e+00
Generation	570:	Best = 5.035067e+02	FES = 57100	Diversity = 8.7034e+00
Generation	580:	Best = 5.035067e+02	FES = 58100	Diversity = 8.6868e+00
Generation	590:	Best = 5.035067e+02	FES = 59100	Diversity = 8.3981e+00
Generation	600:	Best = 5.035063e+02	FES = 60100	Diversity = 8.3649e+00
Generation	610:	Best = 5.035063e+02	FES = 61100	Diversity = 8.2285e+00
Generation	620:	Best = 5.035063e+02	FES = 62100	Diversity = 8.0878e+00
Generation	630:	Best = 5.031769e+02	FES = 63100	Diversity = 7.7493e+00
Generation	640:	Best = 5.031763e+02	FES = 64100	Diversity = 7.6558e+00
Generation	650:	Best = 5.031752e+02	FES = 65100	Diversity = 7.4464e+00
Generation	660:	Best = 5.030559e+02	FES = 66100	Diversity = 7.3520e+00
Generation	670:	Best = 5.022678e+02	FES = 67100	Diversity = 7.1246e+00
Generation	680:	Best = 5.022678e+02	FES = 68100	Diversity = 6.7797e+00
Generation	690:	Best = 5.022271e+02	FES = 69100	Diversity = 6.4372e+00
Generation	700:	Best = 5.021646e+02	FES = 70100	Diversity = 6.1879e+00
Generation	710:	Best = 5.020521e+02	FES = 71100	Diversity = 6.1743e+00
Generation	720:	Best = 5.020130e+02	FES = 72100	Diversity = 6.1839e+00
Generation	730:	Best = 5.020126e+02	FES = 73100	Diversity = 6.1035e+00
Generation	740:	Best = 5.020124e+02	FES = 74100	Diversity = 5.6897e+00
Generation	750:	Best = 5.020117e+02	FES = 75100	Diversity = 5.6457e+00
Generation	760:	Best = 5.020117e+02	FES = 76100	Diversity = 5.5413e+00
Generation	770:	Best = 5.020117e+02	FES = 77100	Diversity = 5.4584e+00
Generation	780:	Best = 5.020117e+02	FES = 78100	Diversity = 5.4477e+00
Generation	790:	Best = 5.020117e+02	FES = 79100	Diversity = 5.3790e+00
Generation	800:	Best = 5.020117e+02	FES = 80100	Diversity = 5.3663e+00
Generation	810:	Best = 5.020097e+02	FES = 81100	Diversity = 5.4029e+00
Generation	820:	Best = 5.020091e+02	FES = 82100	Diversity = 5.2394e+00
Generation	830:	Best = 5.020091e+02	FES = 83100	Diversity = 5.1785e+00
Generation	840:	Best = 5.020047e+02	FES = 84100	Diversity = 5.1788e+00
Generation	850:	Best = 5.020047e+02	FES = 85100	Diversity = 5.1805e+00
Generation	860:	Best = 5.020047e+02	FES = 86100	Diversity = 5.1152e+00
Generation	870:	Best = 5.020003e+02	FES = 87100	Diversity = 5.0721e+00
Generation	880:	Best = 5.019998e+02	FES = 88100	Diversity = 4.9946e+00
Generation	890:	Best = 5.019994e+02	FES = 89100	Diversity = 4.9795e+00
Generation	900:	Best = 5.019993e+02	FES = 90100	Diversity = 4.9748e+00
Generation	910:	Best = 5.019993e+02	FES = 91100	Diversity = 4.9484e+00
Generation	920:	Best = 5.019993e+02	FES = 92100	Diversity = 4.8089e+00
Generation	930:	Best = 5.019987e+02	FES = 93100	Diversity = 4.8252e+00
Generation	940:	Best = 5.019987e+02	FES = 94100	Diversity = 4.8202e+00
Generation	950:	Best = 5.019987e+02	FES = 95100	Diversity = 4.8038e+00
Generation	960:	Best = 5.019987e+02	FES = 96100	Diversity = 4.8029e+00
Generation	970:	Best = 5.019987e+02	FES = 97100	Diversity = 4.7439e+00
Generation	980:	Best = 5.019920e+02	FES = 98100	Diversity = 4.5671e+00

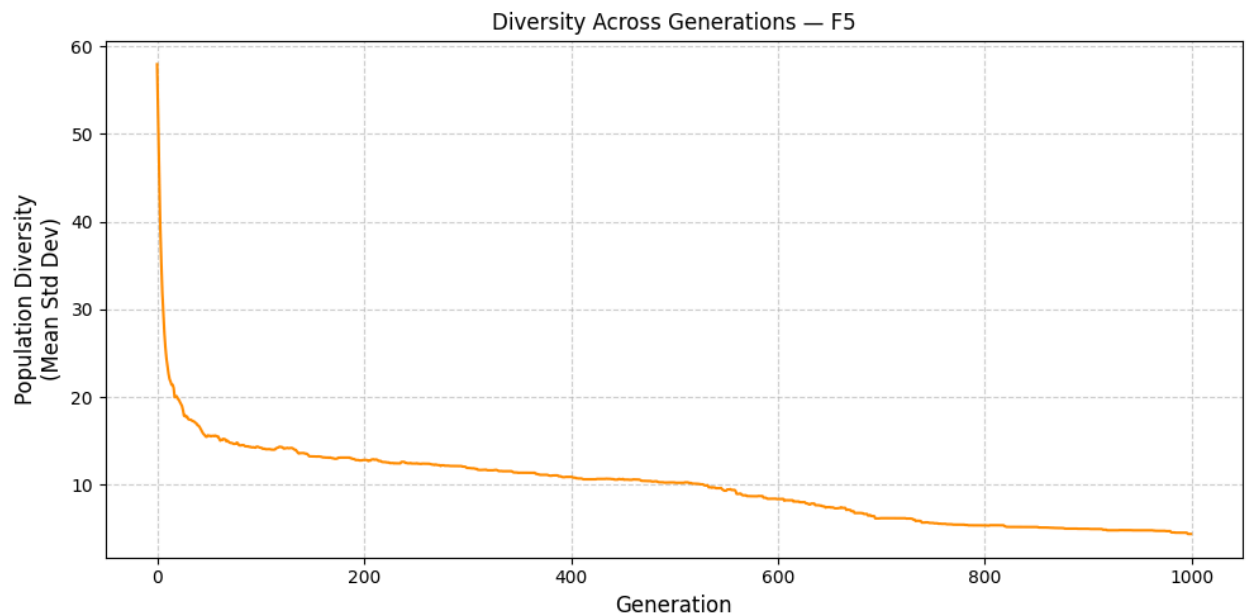
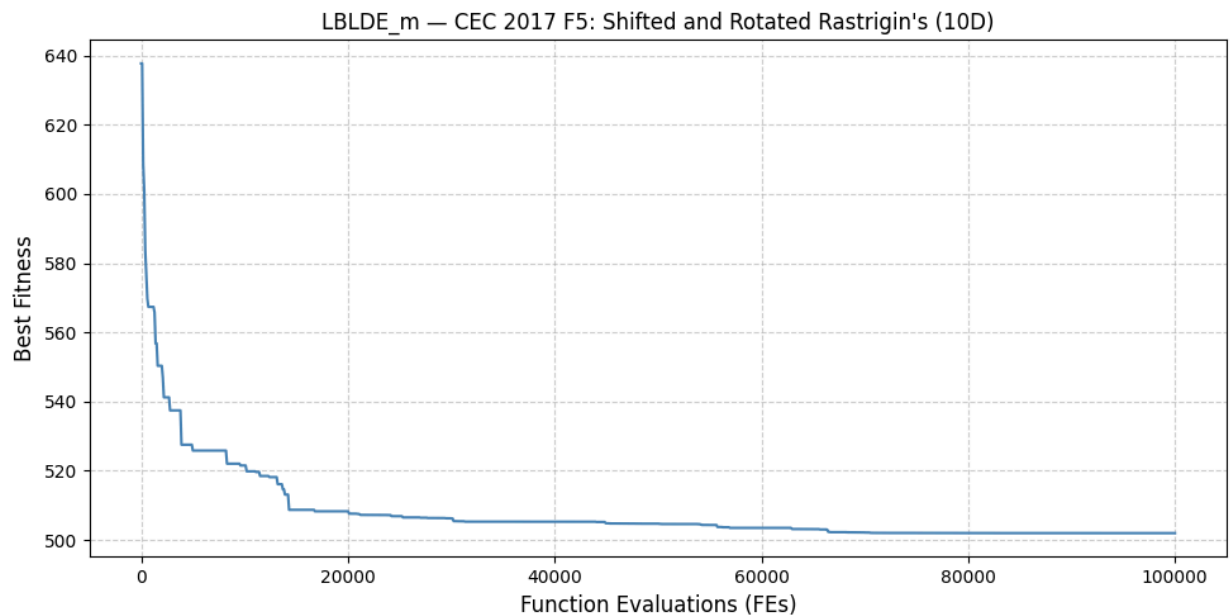
Generation 990: Best = 5.019918e+02 FES = 99100 Diversity = 4.5375e+00

Optimization completed!

Final Best Fitness = 5.019918e+02

Total Function Evaluations = 100000

```
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=
Results:
  Best Fitness      : 5.019918e+02
  Global Optimum    : 5.000000e+02
  Error (|best-optimum|) : 1.991785e+00
  Best Solution (5 dim) : [-19.07795832  44.33265969 -31.70979154 -42.55890826
14.25802185]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F5.png

```
In [16]: # =====  
# F6 – Shifted and Rotated Schaffer's F7  
# =====  
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

=====

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LBLDE_m on CEC 2017 F6 – Single Run

=====

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Function : Shifted and Rotated Schaffer's F7 (10D)

Bounds : [-100, 100]

Global optimum : 600.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 7.140452e+02 | Diversity = 5.7928e+01

Generation	10:	Best = 6.292080e+02	FES = 1100	Diversity = 2.1218e+01
Generation	20:	Best = 6.159205e+02	FES = 2100	Diversity = 8.6178e+00
Generation	30:	Best = 6.068768e+02	FES = 3100	Diversity = 5.4726e+00
Generation	40:	Best = 6.029094e+02	FES = 4100	Diversity = 1.9900e+00
Generation	50:	Best = 6.021060e+02	FES = 5100	Diversity = 1.4481e+00
Generation	60:	Best = 6.016875e+02	FES = 6100	Diversity = 1.7013e+00
Generation	70:	Best = 6.014871e+02	FES = 7100	Diversity = 1.9534e+00
Generation	80:	Best = 6.010905e+02	FES = 8100	Diversity = 2.1915e+00
Generation	90:	Best = 6.006035e+02	FES = 9100	Diversity = 2.0857e+00
Generation	100:	Best = 6.004683e+02	FES = 10100	Diversity = 1.9663e+00
Generation	110:	Best = 6.003346e+02	FES = 11100	Diversity = 1.6914e+00
Generation	120:	Best = 6.002711e+02	FES = 12100	Diversity = 7.8307e-01
Generation	130:	Best = 6.002475e+02	FES = 13100	Diversity = 6.0847e-01
Generation	140:	Best = 6.002342e+02	FES = 14100	Diversity = 2.3613e-01
Generation	150:	Best = 6.002248e+02	FES = 15100	Diversity = 1.8922e-01
Generation	160:	Best = 6.002182e+02	FES = 16100	Diversity = 1.1517e-01
Generation	170:	Best = 6.002147e+02	FES = 17100	Diversity = 1.0129e-01
Generation	180:	Best = 6.001771e+02	FES = 18100	Diversity = 9.5557e-02
Generation	190:	Best = 6.001633e+02	FES = 19100	Diversity = 7.1289e-02
Generation	200:	Best = 6.001633e+02	FES = 20100	Diversity = 6.3484e-02
Generation	210:	Best = 6.001524e+02	FES = 21100	Diversity = 6.2618e-02
Generation	220:	Best = 6.001524e+02	FES = 22100	Diversity = 6.6090e-02
Generation	230:	Best = 6.001506e+02	FES = 23100	Diversity = 6.8334e-02
Generation	240:	Best = 6.001441e+02	FES = 24100	Diversity = 6.3548e-02
Generation	250:	Best = 6.001246e+02	FES = 25100	Diversity = 5.7118e-02
Generation	260:	Best = 6.001153e+02	FES = 26100	Diversity = 4.9567e-02
Generation	270:	Best = 6.001053e+02	FES = 27100	Diversity = 4.5381e-02
Generation	280:	Best = 6.001015e+02	FES = 28100	Diversity = 3.9694e-02
Generation	290:	Best = 6.000910e+02	FES = 29100	Diversity = 2.7788e-02
Generation	300:	Best = 6.000907e+02	FES = 30100	Diversity = 2.4638e-02
Generation	310:	Best = 6.000875e+02	FES = 31100	Diversity = 1.8372e-02
Generation	320:	Best = 6.000792e+02	FES = 32100	Diversity = 1.2077e-02
Generation	330:	Best = 6.000792e+02	FES = 33100	Diversity = 7.5225e-03
Generation	340:	Best = 6.000757e+02	FES = 34100	Diversity = 5.8663e-03
Generation	350:	Best = 6.000739e+02	FES = 35100	Diversity = 4.6042e-03
Generation	360:	Best = 6.000681e+02	FES = 36100	Diversity = 3.3600e-03
Generation	370:	Best = 6.000663e+02	FES = 37100	Diversity = 2.4665e-03
Generation	380:	Best = 6.000649e+02	FES = 38100	Diversity = 1.6538e-03
Generation	390:	Best = 6.000648e+02	FES = 39100	Diversity = 1.1717e-03
Generation	400:	Best = 6.000638e+02	FES = 40100	Diversity = 7.7046e-04
Generation	410:	Best = 6.000633e+02	FES = 41100	Diversity = 5.7141e-04
Generation	420:	Best = 6.000629e+02	FES = 42100	Diversity = 3.2981e-04
Generation	430:	Best = 6.000625e+02	FES = 43100	Diversity = 2.7336e-04
Generation	440:	Best = 6.000620e+02	FES = 44100	Diversity = 2.2875e-04

Generation	450:	Best = 6.000618e+02	FES = 45100	Diversity = 1.5393e-04
Generation	460:	Best = 6.000608e+02	FES = 46100	Diversity = 1.6924e-04
Generation	470:	Best = 6.000604e+02	FES = 47100	Diversity = 1.8033e-04
Generation	480:	Best = 6.000604e+02	FES = 48100	Diversity = 1.4968e-04
Generation	490:	Best = 6.000597e+02	FES = 49100	Diversity = 1.3859e-04
Generation	500:	Best = 6.000595e+02	FES = 50100	Diversity = 1.0553e-04
Generation	510:	Best = 6.000594e+02	FES = 51100	Diversity = 1.0368e-04
Generation	520:	Best = 6.000593e+02	FES = 52100	Diversity = 1.1381e-04
Generation	530:	Best = 6.000590e+02	FES = 53100	Diversity = 1.0807e-04
Generation	540:	Best = 6.000590e+02	FES = 54100	Diversity = 9.7315e-05
Generation	550:	Best = 6.000589e+02	FES = 55100	Diversity = 9.8877e-05
Generation	560:	Best = 6.000588e+02	FES = 56100	Diversity = 9.0689e-05
Generation	570:	Best = 6.000588e+02	FES = 57100	Diversity = 6.3339e-05
Generation	580:	Best = 6.000588e+02	FES = 58100	Diversity = 7.5124e-05
Generation	590:	Best = 6.000586e+02	FES = 59100	Diversity = 1.6271e-04
Generation	600:	Best = 6.000586e+02	FES = 60100	Diversity = 2.9350e-04
Generation	610:	Best = 6.000585e+02	FES = 61100	Diversity = 2.7747e-04
Generation	620:	Best = 6.000584e+02	FES = 62100	Diversity = 4.1426e-04
Generation	630:	Best = 6.000583e+02	FES = 63100	Diversity = 4.9152e-04
Generation	640:	Best = 6.000583e+02	FES = 64100	Diversity = 4.6882e-04
Generation	650:	Best = 6.000582e+02	FES = 65100	Diversity = 5.3299e-04
Generation	660:	Best = 6.000582e+02	FES = 66100	Diversity = 5.5309e-04
Generation	670:	Best = 6.000582e+02	FES = 67100	Diversity = 5.3396e-04
Generation	680:	Best = 6.000582e+02	FES = 68100	Diversity = 7.1139e-04
Generation	690:	Best = 6.000582e+02	FES = 69100	Diversity = 7.9186e-04
Generation	700:	Best = 6.000582e+02	FES = 70100	Diversity = 7.5884e-04
Generation	710:	Best = 6.000581e+02	FES = 71100	Diversity = 7.3421e-04
Generation	720:	Best = 6.000581e+02	FES = 72100	Diversity = 7.3936e-04
Generation	730:	Best = 6.000581e+02	FES = 73100	Diversity = 1.2487e-03
Generation	740:	Best = 6.000581e+02	FES = 74100	Diversity = 1.6325e-03
Generation	750:	Best = 6.000581e+02	FES = 75100	Diversity = 1.7112e-03
Generation	760:	Best = 6.000581e+02	FES = 76100	Diversity = 1.8021e-03
Generation	770:	Best = 6.000580e+02	FES = 77100	Diversity = 1.3998e-03
Generation	780:	Best = 6.000579e+02	FES = 78100	Diversity = 8.6477e-04
Generation	790:	Best = 6.000579e+02	FES = 79100	Diversity = 5.7928e-04
Generation	800:	Best = 6.000579e+02	FES = 80100	Diversity = 4.7174e-04
Generation	810:	Best = 6.000579e+02	FES = 81100	Diversity = 4.7318e-04
Generation	820:	Best = 6.000579e+02	FES = 82100	Diversity = 4.7027e-04
Generation	830:	Best = 6.000579e+02	FES = 83100	Diversity = 4.0729e-04
Generation	840:	Best = 6.000579e+02	FES = 84100	Diversity = 4.3133e-04
Generation	850:	Best = 6.000579e+02	FES = 85100	Diversity = 3.3903e-04
Generation	860:	Best = 6.000579e+02	FES = 86100	Diversity = 3.8986e-04
Generation	870:	Best = 6.000579e+02	FES = 87100	Diversity = 3.5326e-04
Generation	880:	Best = 6.000579e+02	FES = 88100	Diversity = 3.1810e-04
Generation	890:	Best = 6.000579e+02	FES = 89100	Diversity = 3.6832e-04
Generation	900:	Best = 6.000579e+02	FES = 90100	Diversity = 4.1350e-04
Generation	910:	Best = 6.000579e+02	FES = 91100	Diversity = 4.7636e-04
Generation	920:	Best = 6.000579e+02	FES = 92100	Diversity = 7.8428e-04
Generation	930:	Best = 6.000578e+02	FES = 93100	Diversity = 9.9287e-04
Generation	940:	Best = 6.000578e+02	FES = 94100	Diversity = 1.3098e-03
Generation	950:	Best = 6.000578e+02	FES = 95100	Diversity = 1.6465e-03
Generation	960:	Best = 6.000578e+02	FES = 96100	Diversity = 1.3692e-03
Generation	970:	Best = 6.000578e+02	FES = 97100	Diversity = 1.2982e-03
Generation	980:	Best = 6.000578e+02	FES = 98100	Diversity = 1.1463e-03

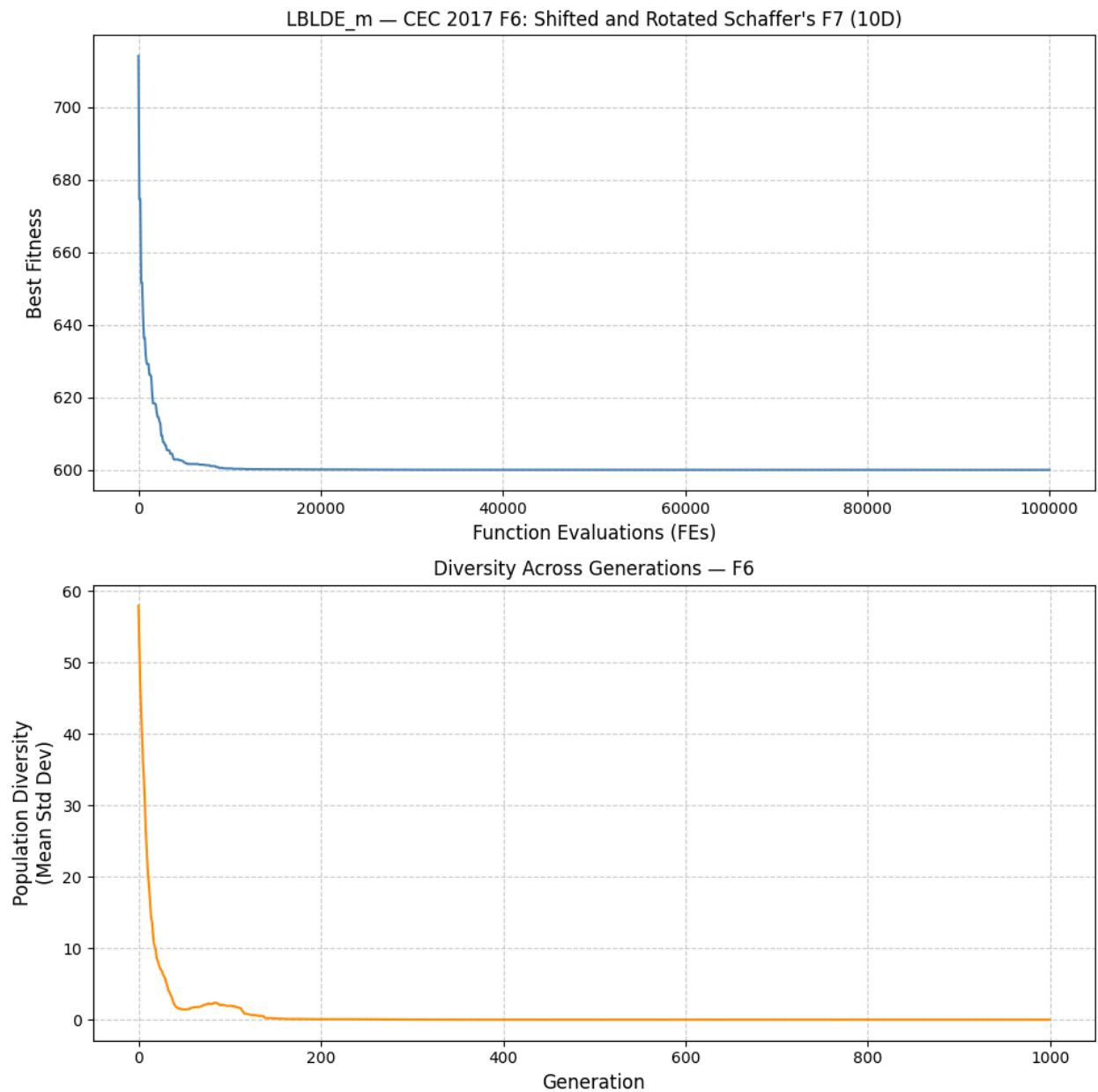
Generation 990: Best = 6.000578e+02 FES = 99100 Diversity = 8.4185e-04

Optimization completed!

Final Best Fitness = 6.000578e+02

Total Function Evaluations = 100000

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Results:
  Best Fitness      : 6.000578e+02
  Global Optimum    : 6.000000e+02
  Error (|best-optimum|) : 5.775676e-02
  Best Solution (5 dim) : [ 79.08939298 -24.49827746 -34.67392638 -44.68162091
75.99787966]
  D^(0) (init. diversity): 5.7928e+01
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Graph saved as convergence_F6.png

```
In [17]: # =====  
# F7 – Shifted and Rotated Lunacek Bi-Rastrigin  
# =====  
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

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LBLDE_m on CEC 2017 F7 – Single Run

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Function : Shifted and Rotated Lunacek Bi-Rastrigin's (10D)

Bounds : [-100, 100]

Global optimum : 700.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 1.202264e+03 | Diversity = 5.7928e+01

Generation 10:	Best = 7.892390e+02	FES = 1100	Diversity = 1.2520e+01
Generation 20:	Best = 7.471439e+02	FES = 2100	Diversity = 8.4922e+00
Generation 30:	Best = 7.436953e+02	FES = 3100	Diversity = 7.0454e+00
Generation 40:	Best = 7.411768e+02	FES = 4100	Diversity = 6.7862e+00
Generation 50:	Best = 7.396145e+02	FES = 5100	Diversity = 6.2869e+00
Generation 60:	Best = 7.373214e+02	FES = 6100	Diversity = 6.0975e+00
Generation 70:	Best = 7.373214e+02	FES = 7100	Diversity = 6.1603e+00
Generation 80:	Best = 7.351530e+02	FES = 8100	Diversity = 5.7986e+00
Generation 90:	Best = 7.351530e+02	FES = 9100	Diversity = 5.7278e+00
Generation 100:	Best = 7.351530e+02	FES = 10100	Diversity = 5.6421e+00
Generation 110:	Best = 7.223262e+02	FES = 11100	Diversity = 5.4755e+00
Generation 120:	Best = 7.223262e+02	FES = 12100	Diversity = 5.3454e+00
Generation 130:	Best = 7.213796e+02	FES = 13100	Diversity = 5.2519e+00
Generation 140:	Best = 7.213796e+02	FES = 14100	Diversity = 5.2693e+00
Generation 150:	Best = 7.213796e+02	FES = 15100	Diversity = 5.1807e+00
Generation 160:	Best = 7.213796e+02	FES = 16100	Diversity = 5.0861e+00
Generation 170:	Best = 7.213796e+02	FES = 17100	Diversity = 5.1343e+00
Generation 180:	Best = 7.213376e+02	FES = 18100	Diversity = 5.0757e+00
Generation 190:	Best = 7.213376e+02	FES = 19100	Diversity = 5.0542e+00
Generation 200:	Best = 7.213376e+02	FES = 20100	Diversity = 4.9964e+00
Generation 210:	Best = 7.207953e+02	FES = 21100	Diversity = 4.9355e+00
Generation 220:	Best = 7.207953e+02	FES = 22100	Diversity = 4.8789e+00
Generation 230:	Best = 7.207953e+02	FES = 23100	Diversity = 4.8618e+00
Generation 240:	Best = 7.204066e+02	FES = 24100	Diversity = 4.8499e+00
Generation 250:	Best = 7.204066e+02	FES = 25100	Diversity = 4.8450e+00
Generation 260:	Best = 7.203815e+02	FES = 26100	Diversity = 4.8629e+00
Generation 270:	Best = 7.203814e+02	FES = 27100	Diversity = 4.8387e+00
Generation 280:	Best = 7.197524e+02	FES = 28100	Diversity = 4.8082e+00
Generation 290:	Best = 7.197524e+02	FES = 29100	Diversity = 4.7883e+00
Generation 300:	Best = 7.197524e+02	FES = 30100	Diversity = 4.7406e+00
Generation 310:	Best = 7.197524e+02	FES = 31100	Diversity = 4.7127e+00
Generation 320:	Best = 7.197524e+02	FES = 32100	Diversity = 4.7459e+00
Generation 330:	Best = 7.197524e+02	FES = 33100	Diversity = 4.7420e+00
Generation 340:	Best = 7.195244e+02	FES = 34100	Diversity = 4.6763e+00
Generation 350:	Best = 7.188078e+02	FES = 35100	Diversity = 4.6830e+00
Generation 360:	Best = 7.187311e+02	FES = 36100	Diversity = 4.6065e+00
Generation 370:	Best = 7.185962e+02	FES = 37100	Diversity = 4.6000e+00
Generation 380:	Best = 7.185962e+02	FES = 38100	Diversity = 4.5829e+00
Generation 390:	Best = 7.185962e+02	FES = 39100	Diversity = 4.5704e+00
Generation 400:	Best = 7.185688e+02	FES = 40100	Diversity = 4.6091e+00
Generation 410:	Best = 7.185688e+02	FES = 41100	Diversity = 4.5923e+00
Generation 420:	Best = 7.185688e+02	FES = 42100	Diversity = 4.5372e+00
Generation 430:	Best = 7.185688e+02	FES = 43100	Diversity = 4.5305e+00
Generation 440:	Best = 7.185688e+02	FES = 44100	Diversity = 4.5029e+00

Generation	450:	Best = 7.185688e+02	FES = 45100	Diversity = 4.4885e+00
Generation	460:	Best = 7.185688e+02	FES = 46100	Diversity = 4.5003e+00
Generation	470:	Best = 7.185599e+02	FES = 47100	Diversity = 4.4958e+00
Generation	480:	Best = 7.185599e+02	FES = 48100	Diversity = 4.4853e+00
Generation	490:	Best = 7.182868e+02	FES = 49100	Diversity = 4.4833e+00
Generation	500:	Best = 7.181018e+02	FES = 50100	Diversity = 4.4656e+00
Generation	510:	Best = 7.167295e+02	FES = 51100	Diversity = 4.4341e+00
Generation	520:	Best = 7.167295e+02	FES = 52100	Diversity = 4.4335e+00
Generation	530:	Best = 7.167295e+02	FES = 53100	Diversity = 4.4527e+00
Generation	540:	Best = 7.167295e+02	FES = 54100	Diversity = 4.4180e+00
Generation	550:	Best = 7.167295e+02	FES = 55100	Diversity = 4.4128e+00
Generation	560:	Best = 7.167295e+02	FES = 56100	Diversity = 4.4127e+00
Generation	570:	Best = 7.167295e+02	FES = 57100	Diversity = 4.4112e+00
Generation	580:	Best = 7.165062e+02	FES = 58100	Diversity = 4.4181e+00
Generation	590:	Best = 7.165062e+02	FES = 59100	Diversity = 4.4207e+00
Generation	600:	Best = 7.162202e+02	FES = 60100	Diversity = 4.3968e+00
Generation	610:	Best = 7.162202e+02	FES = 61100	Diversity = 4.3961e+00
Generation	620:	Best = 7.152487e+02	FES = 62100	Diversity = 4.3628e+00
Generation	630:	Best = 7.152487e+02	FES = 63100	Diversity = 4.3834e+00
Generation	640:	Best = 7.152487e+02	FES = 64100	Diversity = 4.4055e+00
Generation	650:	Best = 7.152487e+02	FES = 65100	Diversity = 4.3789e+00
Generation	660:	Best = 7.152391e+02	FES = 66100	Diversity = 4.3235e+00
Generation	670:	Best = 7.147806e+02	FES = 67100	Diversity = 4.2886e+00
Generation	680:	Best = 7.147613e+02	FES = 68100	Diversity = 4.2788e+00
Generation	690:	Best = 7.143928e+02	FES = 69100	Diversity = 4.2809e+00
Generation	700:	Best = 7.143757e+02	FES = 70100	Diversity = 4.2726e+00
Generation	710:	Best = 7.142642e+02	FES = 71100	Diversity = 4.2400e+00
Generation	720:	Best = 7.142607e+02	FES = 72100	Diversity = 4.2367e+00
Generation	730:	Best = 7.142607e+02	FES = 73100	Diversity = 4.1730e+00
Generation	740:	Best = 7.142598e+02	FES = 74100	Diversity = 4.1641e+00
Generation	750:	Best = 7.142596e+02	FES = 75100	Diversity = 4.1518e+00
Generation	760:	Best = 7.142596e+02	FES = 76100	Diversity = 4.1119e+00
Generation	770:	Best = 7.142586e+02	FES = 77100	Diversity = 4.1116e+00
Generation	780:	Best = 7.142586e+02	FES = 78100	Diversity = 4.1117e+00
Generation	790:	Best = 7.142586e+02	FES = 79100	Diversity = 4.1236e+00
Generation	800:	Best = 7.142586e+02	FES = 80100	Diversity = 4.1237e+00
Generation	810:	Best = 7.142586e+02	FES = 81100	Diversity = 4.1227e+00
Generation	820:	Best = 7.142586e+02	FES = 82100	Diversity = 4.1135e+00
Generation	830:	Best = 7.142569e+02	FES = 83100	Diversity = 4.1375e+00
Generation	840:	Best = 7.142567e+02	FES = 84100	Diversity = 4.1369e+00
Generation	850:	Best = 7.142567e+02	FES = 85100	Diversity = 4.1498e+00
Generation	860:	Best = 7.142567e+02	FES = 86100	Diversity = 4.1513e+00
Generation	870:	Best = 7.137374e+02	FES = 87100	Diversity = 4.1449e+00
Generation	880:	Best = 7.137374e+02	FES = 88100	Diversity = 4.1475e+00
Generation	890:	Best = 7.125088e+02	FES = 89100	Diversity = 4.1465e+00
Generation	900:	Best = 7.125077e+02	FES = 90100	Diversity = 4.1368e+00
Generation	910:	Best = 7.124751e+02	FES = 91100	Diversity = 4.1354e+00
Generation	920:	Best = 7.124751e+02	FES = 92100	Diversity = 4.1334e+00
Generation	930:	Best = 7.124750e+02	FES = 93100	Diversity = 4.1171e+00
Generation	940:	Best = 7.124750e+02	FES = 94100	Diversity = 4.1056e+00
Generation	950:	Best = 7.124593e+02	FES = 95100	Diversity = 4.1025e+00
Generation	960:	Best = 7.124573e+02	FES = 96100	Diversity = 4.0691e+00
Generation	970:	Best = 7.124552e+02	FES = 97100	Diversity = 4.0665e+00
Generation	980:	Best = 7.124539e+02	FES = 98100	Diversity = 4.0607e+00

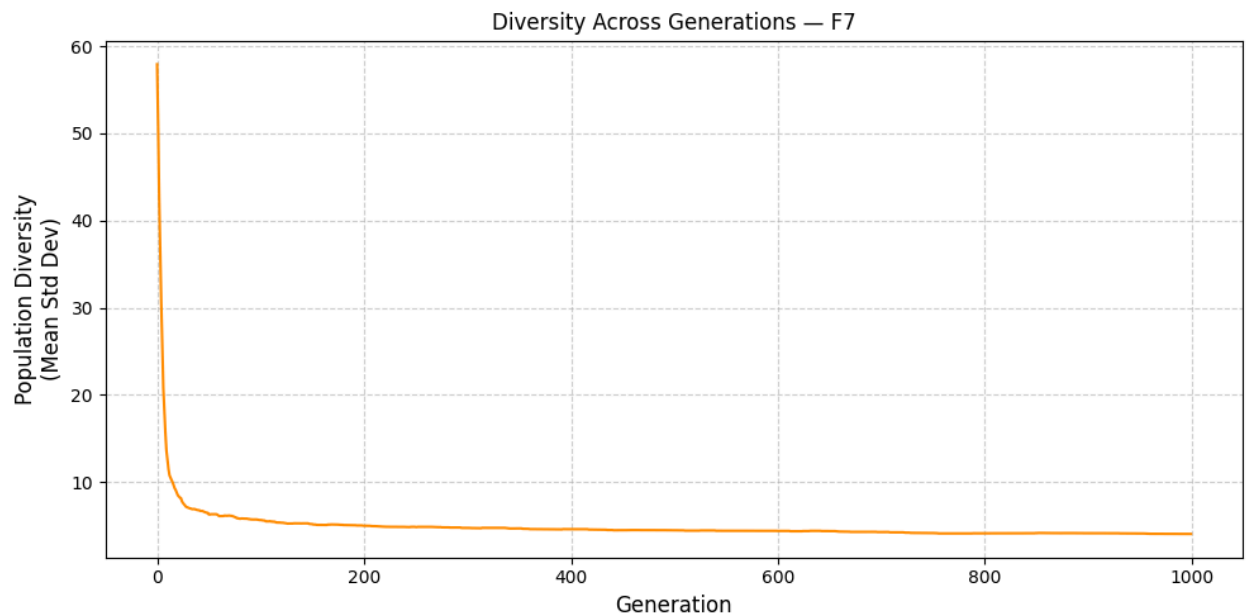
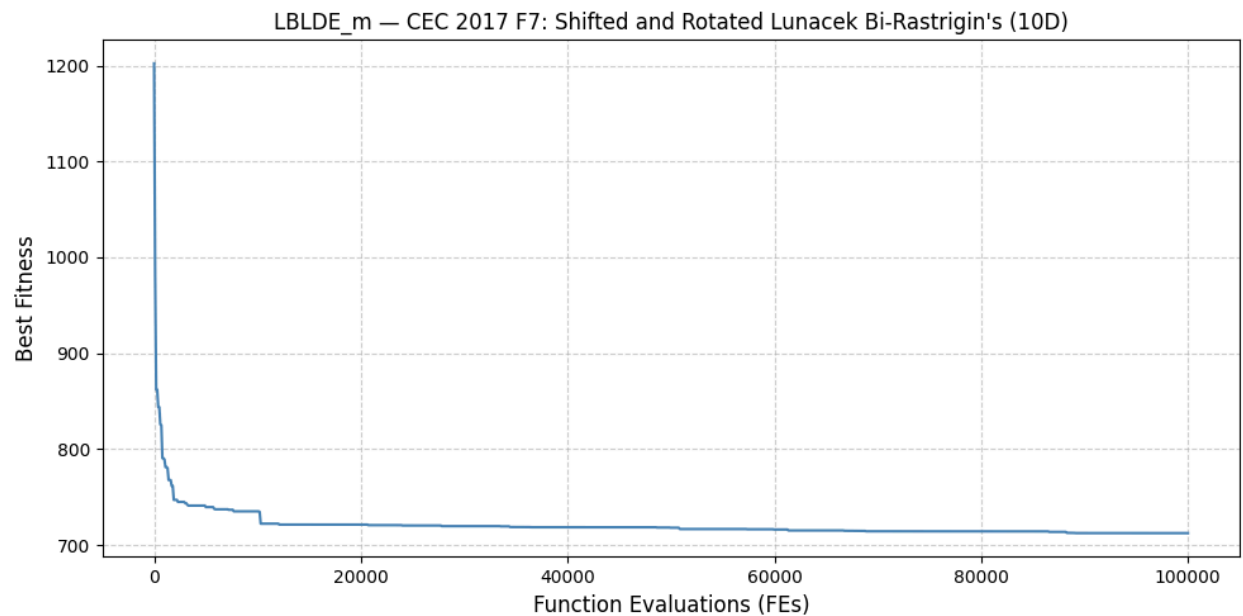
Generation 990: Best = 7.124535e+02 FES = 99100 Diversity = 4.0498e+00

Optimization completed!

Final Best Fitness = 7.124535e+02

Total Function Evaluations = 100000

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Results:
  Best Fitness           : 7.124535e+02
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 1.245351e+01
  Best Solution (5 dim)  : [-19.03751401  15.4760852 -29.68248874 -12.99410738
14.02113466]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F7.png

```
In [18]: # =====  
# F8 – Shifted and Rotated Non-Continuous Rastrigin  
# =====  
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

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LBLDE_m on CEC 2017 F8 – Single Run

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Function : Shifted and Rotated Non-Continuous Rastrigin's (10D)

Bounds : [-100, 100]

Global optimum : 800.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 9.505966e+02 | Diversity = 5.7928e+01

Generation	10:	Best = 8.604851e+02	FES = 1100	Diversity = 2.3781e+01
Generation	20:	Best = 8.419766e+02	FES = 2100	Diversity = 1.8002e+01
Generation	30:	Best = 8.298290e+02	FES = 3100	Diversity = 1.6080e+01
Generation	40:	Best = 8.298290e+02	FES = 4100	Diversity = 1.5595e+01
Generation	50:	Best = 8.239656e+02	FES = 5100	Diversity = 1.4662e+01
Generation	60:	Best = 8.155328e+02	FES = 6100	Diversity = 1.4394e+01
Generation	70:	Best = 8.145272e+02	FES = 7100	Diversity = 1.3987e+01
Generation	80:	Best = 8.103535e+02	FES = 8100	Diversity = 1.3779e+01
Generation	90:	Best = 8.101322e+02	FES = 9100	Diversity = 1.3545e+01
Generation	100:	Best = 8.086399e+02	FES = 10100	Diversity = 1.3088e+01
Generation	110:	Best = 8.075581e+02	FES = 11100	Diversity = 1.2376e+01
Generation	120:	Best = 8.073874e+02	FES = 12100	Diversity = 1.1495e+01
Generation	130:	Best = 8.068912e+02	FES = 13100	Diversity = 1.0512e+01
Generation	140:	Best = 8.062875e+02	FES = 14100	Diversity = 9.5356e+00
Generation	150:	Best = 8.061806e+02	FES = 15100	Diversity = 8.8290e+00
Generation	160:	Best = 8.048103e+02	FES = 16100	Diversity = 8.1224e+00
Generation	170:	Best = 8.046082e+02	FES = 17100	Diversity = 7.5867e+00
Generation	180:	Best = 8.045256e+02	FES = 18100	Diversity = 6.9427e+00
Generation	190:	Best = 8.044842e+02	FES = 19100	Diversity = 6.4906e+00
Generation	200:	Best = 8.043072e+02	FES = 20100	Diversity = 5.9486e+00
Generation	210:	Best = 8.041447e+02	FES = 21100	Diversity = 5.3785e+00
Generation	220:	Best = 8.040896e+02	FES = 22100	Diversity = 5.1226e+00
Generation	230:	Best = 8.040609e+02	FES = 23100	Diversity = 4.9348e+00
Generation	240:	Best = 8.040369e+02	FES = 24100	Diversity = 4.4340e+00
Generation	250:	Best = 8.040261e+02	FES = 25100	Diversity = 4.3775e+00
Generation	260:	Best = 8.040044e+02	FES = 26100	Diversity = 4.2852e+00
Generation	270:	Best = 8.039967e+02	FES = 27100	Diversity = 3.9497e+00
Generation	280:	Best = 8.039854e+02	FES = 28100	Diversity = 3.8969e+00
Generation	290:	Best = 8.039842e+02	FES = 29100	Diversity = 3.7911e+00
Generation	300:	Best = 8.039832e+02	FES = 30100	Diversity = 3.7724e+00
Generation	310:	Best = 8.039829e+02	FES = 31100	Diversity = 3.4840e+00
Generation	320:	Best = 8.039828e+02	FES = 32100	Diversity = 3.4579e+00
Generation	330:	Best = 8.039826e+02	FES = 33100	Diversity = 3.3674e+00
Generation	340:	Best = 8.039826e+02	FES = 34100	Diversity = 3.3630e+00
Generation	350:	Best = 8.039826e+02	FES = 35100	Diversity = 3.3630e+00
Generation	360:	Best = 8.039825e+02	FES = 36100	Diversity = 3.3548e+00
Generation	370:	Best = 8.039825e+02	FES = 37100	Diversity = 2.9343e+00
Generation	380:	Best = 8.039825e+02	FES = 38100	Diversity = 2.8826e+00
Generation	390:	Best = 8.039825e+02	FES = 39100	Diversity = 2.8824e+00
Generation	400:	Best = 8.039825e+02	FES = 40100	Diversity = 2.8065e+00
Generation	410:	Best = 8.039825e+02	FES = 41100	Diversity = 2.6777e+00
Generation	420:	Best = 8.039825e+02	FES = 42100	Diversity = 2.6531e+00
Generation	430:	Best = 8.039825e+02	FES = 43100	Diversity = 3.5794e-04
Generation	440:	Best = 8.039825e+02	FES = 44100	Diversity = 2.3726e-04

Generation	450:	Best = 8.039825e+02	FES = 45100	Diversity = 1.6807e-04
Generation	460:	Best = 8.039825e+02	FES = 46100	Diversity = 1.0056e-04
Generation	470:	Best = 8.039825e+02	FES = 47100	Diversity = 7.4031e-05
Generation	480:	Best = 8.039825e+02	FES = 48100	Diversity = 6.4684e-05
Generation	490:	Best = 8.039825e+02	FES = 49100	Diversity = 4.4172e-05
Generation	500:	Best = 8.039825e+02	FES = 50100	Diversity = 4.2708e-05
Generation	510:	Best = 8.039825e+02	FES = 51100	Diversity = 2.7419e-05
Generation	520:	Best = 8.039825e+02	FES = 52100	Diversity = 1.7802e-05
Generation	530:	Best = 8.039825e+02	FES = 53100	Diversity = 1.4168e-05
Generation	540:	Best = 8.039825e+02	FES = 54100	Diversity = 1.3348e-05
Generation	550:	Best = 8.039825e+02	FES = 55100	Diversity = 8.5488e-06
Generation	560:	Best = 8.039825e+02	FES = 56100	Diversity = 4.9496e-06
Generation	570:	Best = 8.039825e+02	FES = 57100	Diversity = 3.5711e-06
Generation	580:	Best = 8.039825e+02	FES = 58100	Diversity = 2.6885e-06
Generation	590:	Best = 8.039825e+02	FES = 59100	Diversity = 1.6507e-06
Generation	600:	Best = 8.039825e+02	FES = 60100	Diversity = 1.5615e-06
Generation	610:	Best = 8.039825e+02	FES = 61100	Diversity = 1.3016e-06
Generation	620:	Best = 8.039825e+02	FES = 62100	Diversity = 1.3002e-06
Generation	630:	Best = 8.039825e+02	FES = 63100	Diversity = 8.5121e-07
Generation	640:	Best = 8.039825e+02	FES = 64100	Diversity = 4.4772e-07
Generation	650:	Best = 8.039825e+02	FES = 65100	Diversity = 4.2693e-07
Generation	660:	Best = 8.039825e+02	FES = 66100	Diversity = 2.8855e-07
Generation	670:	Best = 8.039825e+02	FES = 67100	Diversity = 2.8954e-07
Generation	680:	Best = 8.039825e+02	FES = 68100	Diversity = 2.5896e-07
Generation	690:	Best = 8.039825e+02	FES = 69100	Diversity = 2.2510e-07
Generation	700:	Best = 8.039825e+02	FES = 70100	Diversity = 1.9278e-07
Generation	710:	Best = 8.039825e+02	FES = 71100	Diversity = 1.3537e-07
Generation	720:	Best = 8.039825e+02	FES = 72100	Diversity = 1.0362e-07
Generation	730:	Best = 8.039825e+02	FES = 73100	Diversity = 9.4657e-08
Generation	740:	Best = 8.039825e+02	FES = 74100	Diversity = 5.8861e-08
Generation	750:	Best = 8.039825e+02	FES = 75100	Diversity = 3.4427e-08
Generation	760:	Best = 8.039825e+02	FES = 76100	Diversity = 2.9152e-08
Generation	770:	Best = 8.039825e+02	FES = 77100	Diversity = 1.8772e-08
Generation	780:	Best = 8.039825e+02	FES = 78100	Diversity = 2.8001e-09
Generation	790:	Best = 8.039825e+02	FES = 79100	Diversity = 5.6657e-10
Generation	800:	Best = 8.039825e+02	FES = 80100	Diversity = 5.7203e-11
Generation	810:	Best = 8.039825e+02	FES = 81100	Diversity = 1.1923e-11
Generation	820:	Best = 8.039825e+02	FES = 82100	Diversity = 4.2042e-12
Generation	830:	Best = 8.039825e+02	FES = 83100	Diversity = 2.0411e-12
Generation	840:	Best = 8.039825e+02	FES = 84100	Diversity = 2.2000e-13
Generation	850:	Best = 8.039825e+02	FES = 85100	Diversity = 3.6294e-14
Generation	860:	Best = 8.039825e+02	FES = 86100	Diversity = 3.5794e-14
Generation	870:	Best = 8.039825e+02	FES = 87100	Diversity = 3.5794e-14
Generation	880:	Best = 8.039825e+02	FES = 88100	Diversity = 3.5794e-14
Generation	890:	Best = 8.039825e+02	FES = 89100	Diversity = 3.5794e-14
Generation	900:	Best = 8.039825e+02	FES = 90100	Diversity = 3.5794e-14
Generation	910:	Best = 8.039825e+02	FES = 91100	Diversity = 3.5794e-14
Generation	920:	Best = 8.039825e+02	FES = 92100	Diversity = 3.5794e-14
Generation	930:	Best = 8.039825e+02	FES = 93100	Diversity = 3.5794e-14
Generation	940:	Best = 8.039825e+02	FES = 94100	Diversity = 3.5794e-14
Generation	950:	Best = 8.039825e+02	FES = 95100	Diversity = 3.5794e-14
Generation	960:	Best = 8.039825e+02	FES = 96100	Diversity = 3.5794e-14
Generation	970:	Best = 8.039825e+02	FES = 97100	Diversity = 3.5794e-14
Generation	980:	Best = 8.039825e+02	FES = 98100	Diversity = 3.5794e-14

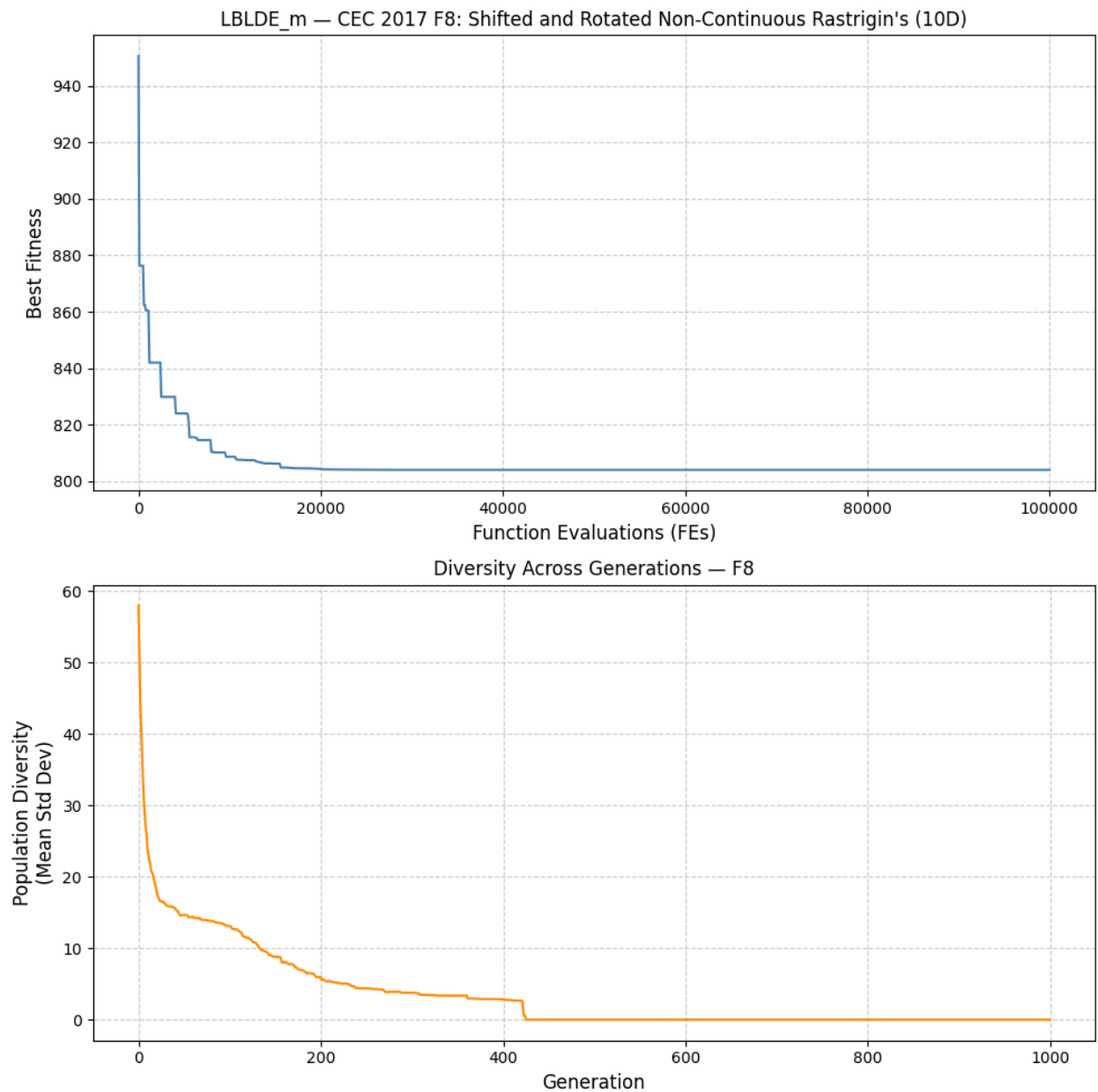
Generation 990: Best = 8.039825e+02 FES = 99100 Diversity = 3.5794e-14

Optimization completed!

Final Best Fitness = 8.039825e+02

Total Function Evaluations = 100000

```
=====
=
Results:
  Best Fitness           : 8.039825e+02
  Global Optimum         : 8.000000e+02
  Error (|best-optimum|) : 3.982451e+00
  Best Solution (5 dim)  : [ 33.56500615 -47.15197238  19.13692495 -34.27646996
-31.99053033]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F8.png

```
In [19]: # =====  
# F9 – Shifted and Rotated Levy  
# =====  
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

```

=====
=
LBLDE_m on CEC 2017 F9 – Single Run
=====
=
Function      : Shifted and Rotated Levy (10D)
Bounds       : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 5.7928e+01 | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 6.112557e+03 | Diversity = 5.7928e+01
Generation 10: Best = 1.265937e+03 FES = 1100 Diversity = 2.1455e+01
Generation 20: Best = 9.823649e+02 FES = 2100 Diversity = 8.3759e+00
Generation 30: Best = 9.141061e+02 FES = 3100 Diversity = 3.3460e+00
Generation 40: Best = 9.027662e+02 FES = 4100 Diversity = 1.4020e+00
Generation 50: Best = 9.014648e+02 FES = 5100 Diversity = 8.7722e-01
Generation 60: Best = 9.006654e+02 FES = 6100 Diversity = 3.6434e-01
Generation 70: Best = 9.002985e+02 FES = 7100 Diversity = 2.5958e-01
Generation 80: Best = 9.002428e+02 FES = 8100 Diversity = 6.8930e-02
Generation 90: Best = 9.002190e+02 FES = 9100 Diversity = 4.9366e-02
Generation 100: Best = 9.001987e+02 FES = 10100 Diversity = 3.3110e-02
Generation 110: Best = 9.001978e+02 FES = 11100 Diversity = 1.1414e-02
Generation 120: Best = 9.001975e+02 FES = 12100 Diversity = 4.9655e-03
Generation 130: Best = 9.001975e+02 FES = 13100 Diversity = 2.2062e-03
Generation 140: Best = 9.001974e+02 FES = 14100 Diversity = 8.8939e-04
Generation 150: Best = 9.001974e+02 FES = 15100 Diversity = 6.9732e-04
Generation 160: Best = 9.001974e+02 FES = 16100 Diversity = 4.7903e-04
Generation 170: Best = 9.001974e+02 FES = 17100 Diversity = 2.2255e-04
Generation 180: Best = 9.001974e+02 FES = 18100 Diversity = 1.7599e-04
Generation 190: Best = 9.001974e+02 FES = 19100 Diversity = 1.2203e-04
Generation 200: Best = 9.001974e+02 FES = 20100 Diversity = 1.4043e-04
Generation 210: Best = 9.001974e+02 FES = 21100 Diversity = 1.0949e-04
Generation 220: Best = 9.001974e+02 FES = 22100 Diversity = 5.8152e-05
Generation 230: Best = 9.001974e+02 FES = 23100 Diversity = 3.9171e-05
Generation 240: Best = 9.001974e+02 FES = 24100 Diversity = 2.2255e-05
Generation 250: Best = 9.001974e+02 FES = 25100 Diversity = 1.8389e-05
Generation 260: Best = 9.001974e+02 FES = 26100 Diversity = 1.6815e-05
Generation 270: Best = 9.001974e+02 FES = 27100 Diversity = 1.1842e-05
Generation 280: Best = 9.001974e+02 FES = 28100 Diversity = 8.3341e-06
Generation 290: Best = 9.001974e+02 FES = 29100 Diversity = 6.3472e-06
Generation 300: Best = 9.001974e+02 FES = 30100 Diversity = 4.6931e-06
Generation 310: Best = 9.001974e+02 FES = 31100 Diversity = 3.1217e-06
Generation 320: Best = 9.001974e+02 FES = 32100 Diversity = 2.6220e-06
Generation 330: Best = 9.001974e+02 FES = 33100 Diversity = 1.5824e-06
Generation 340: Best = 9.001974e+02 FES = 34100 Diversity = 1.9111e-06
Generation 350: Best = 9.001974e+02 FES = 35100 Diversity = 1.2839e-06
Generation 360: Best = 9.001974e+02 FES = 36100 Diversity = 9.6508e-07
Generation 370: Best = 9.001974e+02 FES = 37100 Diversity = 9.4568e-07
Generation 380: Best = 9.001974e+02 FES = 38100 Diversity = 9.4941e-07
Generation 390: Best = 9.001974e+02 FES = 39100 Diversity = 4.7983e-07
Generation 400: Best = 9.001974e+02 FES = 40100 Diversity = 3.9099e-07
Generation 410: Best = 9.001974e+02 FES = 41100 Diversity = 3.1904e-07
Generation 420: Best = 9.001974e+02 FES = 42100 Diversity = 3.5689e-07
Generation 430: Best = 9.001974e+02 FES = 43100 Diversity = 2.0079e-07
Generation 440: Best = 9.001974e+02 FES = 44100 Diversity = 2.5936e-07

```

[illegible]

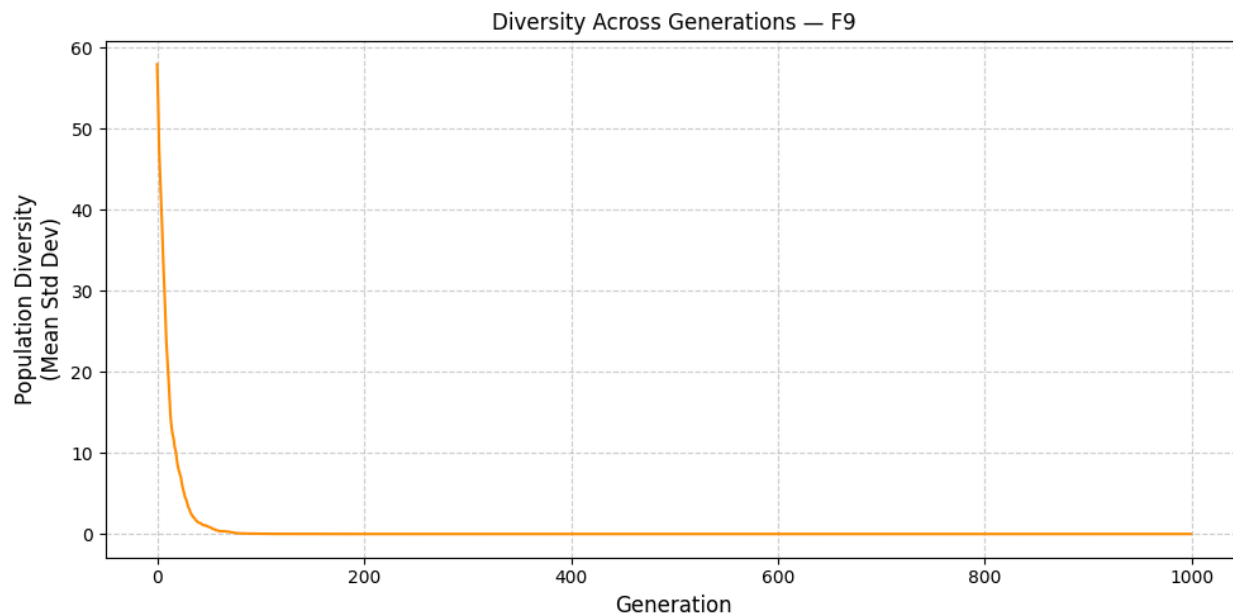
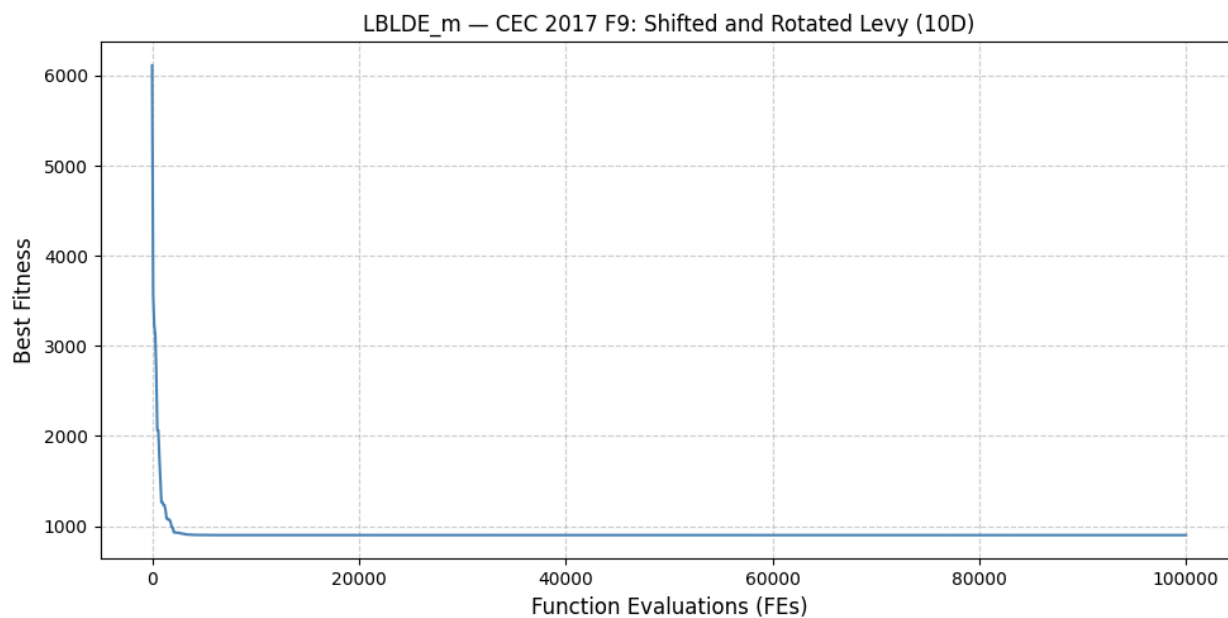
Generation 990: Best = 9.001974e+02 FES = 99100 Diversity = 3.5086e-14

Optimization completed!

Final Best Fitness = 9.001974e+02

Total Function Evaluations = 100000

```
=====
=
Results:
  Best Fitness      : 9.001974e+02
  Global Optimum    : 9.000000e+02
  Error (|best-optimum|) : 1.974188e-01
  Best Solution (5 dim) : [-23.65722059  0.11593598  21.13111609  53.08333042
-18.6432765 ]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F9.png

```
In [20]: # =====  
# F10 – Shifted and Rotated Schwefel  
# =====  
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

=====

=

LBLDE_m on CEC 2017 F10 – Single Run

=====

=

Function : Shifted and Rotated Schwefel's (10D)
Bounds : [-100, 100]
Global optimum : 1000.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 3.749402e+03 | Diversity = 5.7928e+01

Generation	10:	Best = 2.885491e+03	FES = 1100	Diversity = 4.8046e+01
Generation	20:	Best = 2.783544e+03	FES = 2100	Diversity = 4.4235e+01
Generation	30:	Best = 2.306192e+03	FES = 3100	Diversity = 4.3397e+01
Generation	40:	Best = 2.306192e+03	FES = 4100	Diversity = 4.3142e+01
Generation	50:	Best = 2.172570e+03	FES = 5100	Diversity = 4.2486e+01
Generation	60:	Best = 2.172570e+03	FES = 6100	Diversity = 4.2366e+01
Generation	70:	Best = 2.172570e+03	FES = 7100	Diversity = 4.1654e+01
Generation	80:	Best = 1.884825e+03	FES = 8100	Diversity = 4.0635e+01
Generation	90:	Best = 1.884825e+03	FES = 9100	Diversity = 3.8660e+01
Generation	100:	Best = 1.769840e+03	FES = 10100	Diversity = 3.8906e+01
Generation	110:	Best = 1.666056e+03	FES = 11100	Diversity = 3.9248e+01
Generation	120:	Best = 1.666056e+03	FES = 12100	Diversity = 3.8641e+01
Generation	130:	Best = 1.616253e+03	FES = 13100	Diversity = 3.8374e+01
Generation	140:	Best = 1.607053e+03	FES = 14100	Diversity = 3.8073e+01
Generation	150:	Best = 1.587366e+03	FES = 15100	Diversity = 3.7588e+01
Generation	160:	Best = 1.584448e+03	FES = 16100	Diversity = 3.7106e+01
Generation	170:	Best = 1.583108e+03	FES = 17100	Diversity = 3.6341e+01
Generation	180:	Best = 1.583098e+03	FES = 18100	Diversity = 3.6036e+01
Generation	190:	Best = 1.494471e+03	FES = 19100	Diversity = 3.6148e+01
Generation	200:	Best = 1.491985e+03	FES = 20100	Diversity = 3.5912e+01
Generation	210:	Best = 1.425834e+03	FES = 21100	Diversity = 3.4944e+01
Generation	220:	Best = 1.424236e+03	FES = 22100	Diversity = 3.4324e+01
Generation	230:	Best = 1.411722e+03	FES = 23100	Diversity = 3.3834e+01
Generation	240:	Best = 1.359259e+03	FES = 24100	Diversity = 3.3401e+01
Generation	250:	Best = 1.358438e+03	FES = 25100	Diversity = 3.2907e+01
Generation	260:	Best = 1.336522e+03	FES = 26100	Diversity = 3.1305e+01
Generation	270:	Best = 1.336522e+03	FES = 27100	Diversity = 3.1291e+01
Generation	280:	Best = 1.336473e+03	FES = 28100	Diversity = 3.0582e+01
Generation	290:	Best = 1.333178e+03	FES = 29100	Diversity = 3.0579e+01
Generation	300:	Best = 1.333178e+03	FES = 30100	Diversity = 3.0315e+01
Generation	310:	Best = 1.333178e+03	FES = 31100	Diversity = 3.0511e+01
Generation	320:	Best = 1.312809e+03	FES = 32100	Diversity = 3.0745e+01
Generation	330:	Best = 1.307720e+03	FES = 33100	Diversity = 3.0242e+01
Generation	340:	Best = 1.277239e+03	FES = 34100	Diversity = 2.9968e+01
Generation	350:	Best = 1.277239e+03	FES = 35100	Diversity = 2.9719e+01
Generation	360:	Best = 1.276092e+03	FES = 36100	Diversity = 2.9029e+01
Generation	370:	Best = 1.276092e+03	FES = 37100	Diversity = 2.8996e+01
Generation	380:	Best = 1.275919e+03	FES = 38100	Diversity = 2.8023e+01
Generation	390:	Best = 1.257494e+03	FES = 39100	Diversity = 2.7646e+01
Generation	400:	Best = 1.257494e+03	FES = 40100	Diversity = 2.7653e+01
Generation	410:	Best = 1.257494e+03	FES = 41100	Diversity = 2.7147e+01
Generation	420:	Best = 1.190396e+03	FES = 42100	Diversity = 2.6998e+01
Generation	430:	Best = 1.190164e+03	FES = 43100	Diversity = 2.7497e+01
Generation	440:	Best = 1.190164e+03	FES = 44100	Diversity = 2.7327e+01

Generation	450:	Best = 1.190164e+03	FES = 45100	Diversity = 2.7032e+01
Generation	460:	Best = 1.178390e+03	FES = 46100	Diversity = 2.6966e+01
Generation	470:	Best = 1.178329e+03	FES = 47100	Diversity = 2.6851e+01
Generation	480:	Best = 1.128312e+03	FES = 48100	Diversity = 2.6897e+01
Generation	490:	Best = 1.128312e+03	FES = 49100	Diversity = 2.5845e+01
Generation	500:	Best = 1.128312e+03	FES = 50100	Diversity = 2.6564e+01
Generation	510:	Best = 1.066103e+03	FES = 51100	Diversity = 2.6421e+01
Generation	520:	Best = 1.065331e+03	FES = 52100	Diversity = 2.6459e+01
Generation	530:	Best = 1.063917e+03	FES = 53100	Diversity = 2.6442e+01
Generation	540:	Best = 1.058223e+03	FES = 54100	Diversity = 2.6374e+01
Generation	550:	Best = 1.022920e+03	FES = 55100	Diversity = 2.6220e+01
Generation	560:	Best = 1.020039e+03	FES = 56100	Diversity = 2.5954e+01
Generation	570:	Best = 1.019163e+03	FES = 57100	Diversity = 2.6004e+01
Generation	580:	Best = 1.018689e+03	FES = 58100	Diversity = 2.6144e+01
Generation	590:	Best = 1.018463e+03	FES = 59100	Diversity = 2.5465e+01
Generation	600:	Best = 1.018452e+03	FES = 60100	Diversity = 2.5320e+01
Generation	610:	Best = 1.018451e+03	FES = 61100	Diversity = 2.4863e+01
Generation	620:	Best = 1.018451e+03	FES = 62100	Diversity = 2.4847e+01
Generation	630:	Best = 1.018451e+03	FES = 63100	Diversity = 2.4923e+01
Generation	640:	Best = 1.018451e+03	FES = 64100	Diversity = 2.4759e+01
Generation	650:	Best = 1.016427e+03	FES = 65100	Diversity = 2.4927e+01
Generation	660:	Best = 1.016427e+03	FES = 66100	Diversity = 2.4820e+01
Generation	670:	Best = 1.016427e+03	FES = 67100	Diversity = 2.4758e+01
Generation	680:	Best = 1.016427e+03	FES = 68100	Diversity = 2.4858e+01
Generation	690:	Best = 1.016427e+03	FES = 69100	Diversity = 2.4859e+01
Generation	700:	Best = 1.016427e+03	FES = 70100	Diversity = 2.4783e+01
Generation	710:	Best = 1.016427e+03	FES = 71100	Diversity = 2.4607e+01
Generation	720:	Best = 1.015745e+03	FES = 72100	Diversity = 2.4518e+01
Generation	730:	Best = 1.015664e+03	FES = 73100	Diversity = 2.4051e+01
Generation	740:	Best = 1.015663e+03	FES = 74100	Diversity = 2.3610e+01
Generation	750:	Best = 1.015655e+03	FES = 75100	Diversity = 2.3326e+01
Generation	760:	Best = 1.014464e+03	FES = 76100	Diversity = 2.3233e+01
Generation	770:	Best = 1.014445e+03	FES = 77100	Diversity = 2.3115e+01
Generation	780:	Best = 1.014021e+03	FES = 78100	Diversity = 2.3010e+01
Generation	790:	Best = 1.012041e+03	FES = 79100	Diversity = 2.3062e+01
Generation	800:	Best = 1.004645e+03	FES = 80100	Diversity = 2.2997e+01
Generation	810:	Best = 1.004644e+03	FES = 81100	Diversity = 2.2596e+01
Generation	820:	Best = 1.004644e+03	FES = 82100	Diversity = 2.1795e+01
Generation	830:	Best = 1.004644e+03	FES = 83100	Diversity = 2.1643e+01
Generation	840:	Best = 1.004644e+03	FES = 84100	Diversity = 2.1400e+01
Generation	850:	Best = 1.004644e+03	FES = 85100	Diversity = 2.0993e+01
Generation	860:	Best = 1.004430e+03	FES = 86100	Diversity = 2.0681e+01
Generation	870:	Best = 1.004403e+03	FES = 87100	Diversity = 2.0424e+01
Generation	880:	Best = 1.004399e+03	FES = 88100	Diversity = 2.0096e+01
Generation	890:	Best = 1.004399e+03	FES = 89100	Diversity = 1.9226e+01
Generation	900:	Best = 1.004399e+03	FES = 90100	Diversity = 1.8035e+01
Generation	910:	Best = 1.004399e+03	FES = 91100	Diversity = 1.6553e+01
Generation	920:	Best = 1.003745e+03	FES = 92100	Diversity = 1.4594e+01
Generation	930:	Best = 1.003522e+03	FES = 93100	Diversity = 1.3736e+01
Generation	940:	Best = 1.003502e+03	FES = 94100	Diversity = 1.3434e+01
Generation	950:	Best = 1.003502e+03	FES = 95100	Diversity = 1.3066e+01
Generation	960:	Best = 1.003502e+03	FES = 96100	Diversity = 1.2567e+01
Generation	970:	Best = 1.003502e+03	FES = 97100	Diversity = 1.2137e+01
Generation	980:	Best = 1.003502e+03	FES = 98100	Diversity = 1.1111e+01

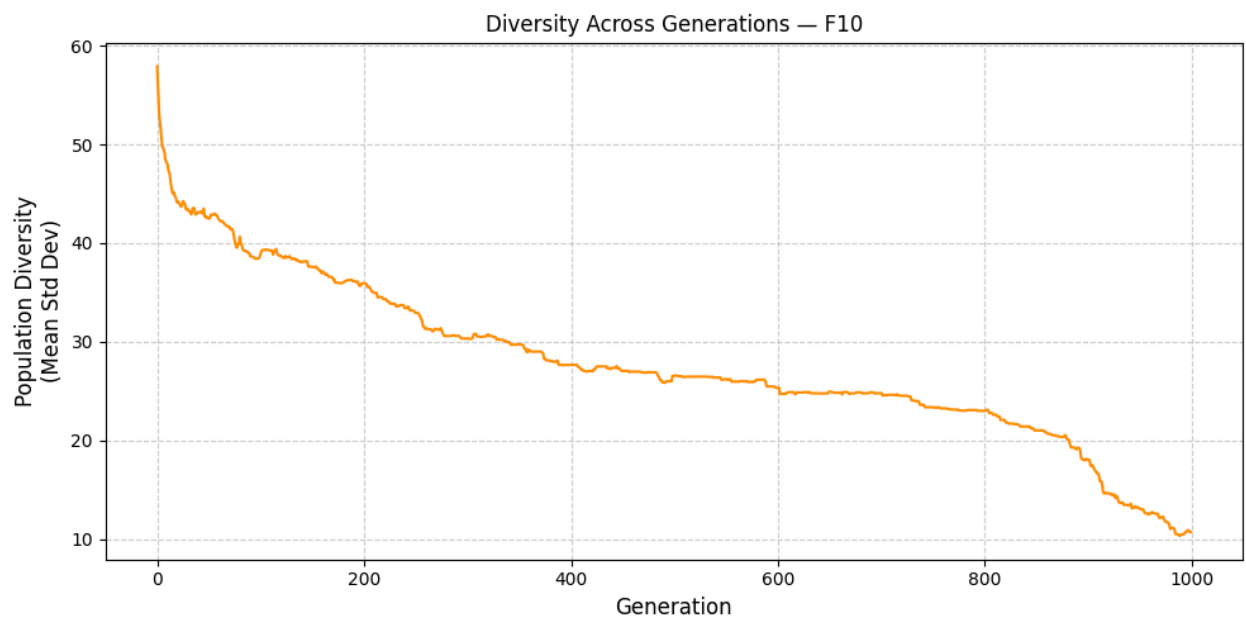
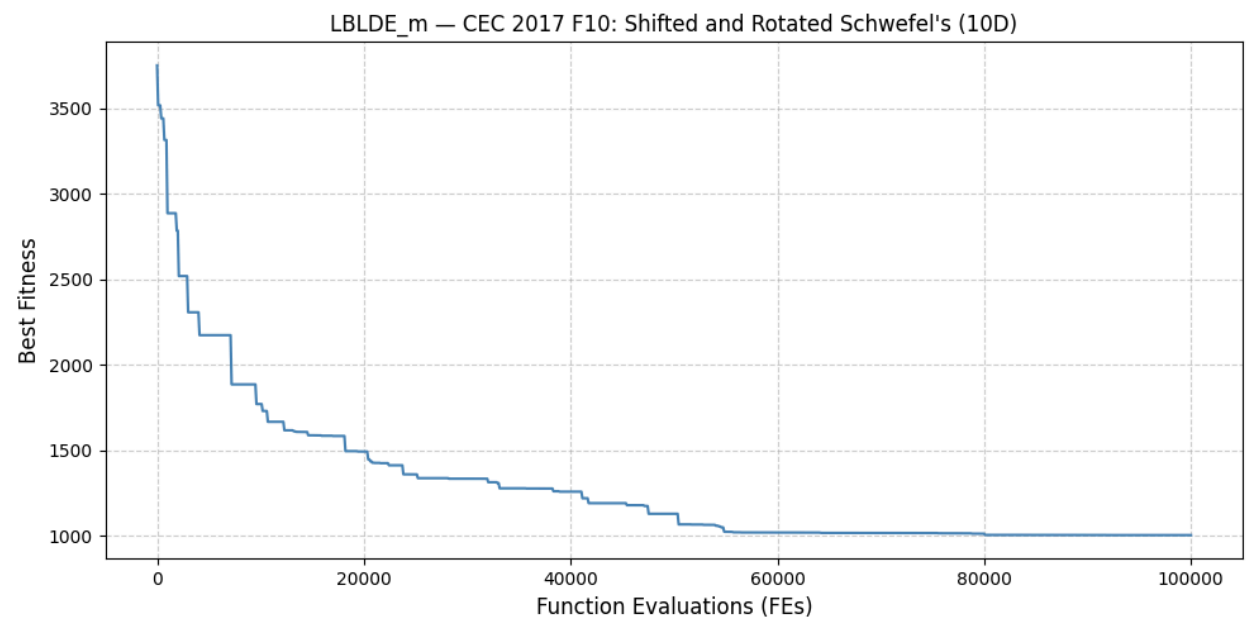
Generation 990: Best = 1.003487e+03 FES = 99100 Diversity = 1.0460e+01

Optimization completed!

Final Best Fitness = 1.003487e+03

Total Function Evaluations = 100000

```
=====
=
Results:
  Best Fitness          : 1.003487e+03
  Global Optimum        : 1.000000e+03
  Error (|best-optimum|) : 3.487353e+00
  Best Solution (5 dim)  : [-36.71341555 -19.70114632  49.87794018  60.49377035
22.64223522]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [21]: # =====  
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)  
# =====  
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Rosenbrock; Rastrigin')
```

=====

=

LBLDE_m on CEC 2017 F11 – Single Run

=====

=

Function : Hybrid: Zakharov; Rosenbrock; Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 1100.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.292232e+04 | Diversity = 5.7928e+01

Generation	10:	Best = 1.278040e+03	FES = 1100	Diversity = 2.5734e+01
Generation	20:	Best = 1.152736e+03	FES = 2100	Diversity = 9.2399e+00
Generation	30:	Best = 1.128431e+03	FES = 3100	Diversity = 5.5725e+00
Generation	40:	Best = 1.121178e+03	FES = 4100	Diversity = 4.2241e+00
Generation	50:	Best = 1.118226e+03	FES = 5100	Diversity = 3.8309e+00
Generation	60:	Best = 1.115263e+03	FES = 6100	Diversity = 3.2852e+00
Generation	70:	Best = 1.114778e+03	FES = 7100	Diversity = 3.0257e+00
Generation	80:	Best = 1.112912e+03	FES = 8100	Diversity = 3.0287e+00
Generation	90:	Best = 1.111805e+03	FES = 9100	Diversity = 2.9600e+00
Generation	100:	Best = 1.111367e+03	FES = 10100	Diversity = 3.2522e+00
Generation	110:	Best = 1.111189e+03	FES = 11100	Diversity = 3.1654e+00
Generation	120:	Best = 1.111174e+03	FES = 12100	Diversity = 3.1850e+00
Generation	130:	Best = 1.110646e+03	FES = 13100	Diversity = 3.1085e+00
Generation	140:	Best = 1.110195e+03	FES = 14100	Diversity = 3.0287e+00
Generation	150:	Best = 1.110148e+03	FES = 15100	Diversity = 2.7622e+00
Generation	160:	Best = 1.110122e+03	FES = 16100	Diversity = 2.4471e+00
Generation	170:	Best = 1.110098e+03	FES = 17100	Diversity = 2.4118e+00
Generation	180:	Best = 1.110096e+03	FES = 18100	Diversity = 2.2542e+00
Generation	190:	Best = 1.110094e+03	FES = 19100	Diversity = 2.2028e+00
Generation	200:	Best = 1.110075e+03	FES = 20100	Diversity = 2.1250e+00
Generation	210:	Best = 1.110071e+03	FES = 21100	Diversity = 1.9475e+00
Generation	220:	Best = 1.110063e+03	FES = 22100	Diversity = 1.8637e+00
Generation	230:	Best = 1.110057e+03	FES = 23100	Diversity = 1.7986e+00
Generation	240:	Best = 1.110049e+03	FES = 24100	Diversity = 1.7469e+00
Generation	250:	Best = 1.110040e+03	FES = 25100	Diversity = 1.6526e+00
Generation	260:	Best = 1.110033e+03	FES = 26100	Diversity = 1.4968e+00
Generation	270:	Best = 1.110029e+03	FES = 27100	Diversity = 1.4395e+00
Generation	280:	Best = 1.110025e+03	FES = 28100	Diversity = 3.9352e-01
Generation	290:	Best = 1.110022e+03	FES = 29100	Diversity = 4.7195e-02
Generation	300:	Best = 1.110022e+03	FES = 30100	Diversity = 4.1459e-02
Generation	310:	Best = 1.110021e+03	FES = 31100	Diversity = 2.9901e-02
Generation	320:	Best = 1.110020e+03	FES = 32100	Diversity = 2.4791e-02
Generation	330:	Best = 1.110020e+03	FES = 33100	Diversity = 2.2368e-02
Generation	340:	Best = 1.110020e+03	FES = 34100	Diversity = 1.6704e-02
Generation	350:	Best = 1.110019e+03	FES = 35100	Diversity = 1.4114e-02
Generation	360:	Best = 1.110019e+03	FES = 36100	Diversity = 1.3399e-02
Generation	370:	Best = 1.110019e+03	FES = 37100	Diversity = 1.1801e-02
Generation	380:	Best = 1.110019e+03	FES = 38100	Diversity = 8.6199e-03
Generation	390:	Best = 1.110019e+03	FES = 39100	Diversity = 6.1330e-03
Generation	400:	Best = 1.110019e+03	FES = 40100	Diversity = 8.4182e-03
Generation	410:	Best = 1.110019e+03	FES = 41100	Diversity = 5.7717e-03
Generation	420:	Best = 1.110019e+03	FES = 42100	Diversity = 4.5537e-03
Generation	430:	Best = 1.110019e+03	FES = 43100	Diversity = 3.6084e-03
Generation	440:	Best = 1.110019e+03	FES = 44100	Diversity = 3.1376e-03

Generation	450:	Best = 1.110019e+03	FES = 45100	Diversity = 2.1989e-03
Generation	460:	Best = 1.110019e+03	FES = 46100	Diversity = 2.0752e-03
Generation	470:	Best = 1.110019e+03	FES = 47100	Diversity = 2.1418e-03
Generation	480:	Best = 1.110019e+03	FES = 48100	Diversity = 1.7170e-03
Generation	490:	Best = 1.110019e+03	FES = 49100	Diversity = 1.1405e-03
Generation	500:	Best = 1.110019e+03	FES = 50100	Diversity = 7.5564e-04
Generation	510:	Best = 1.110019e+03	FES = 51100	Diversity = 4.5181e-04
Generation	520:	Best = 1.110019e+03	FES = 52100	Diversity = 4.3625e-04
Generation	530:	Best = 1.110019e+03	FES = 53100	Diversity = 2.9527e-04
Generation	540:	Best = 1.110019e+03	FES = 54100	Diversity = 3.2214e-04
Generation	550:	Best = 1.110019e+03	FES = 55100	Diversity = 2.7960e-04
Generation	560:	Best = 1.110019e+03	FES = 56100	Diversity = 3.0958e-04
Generation	570:	Best = 1.110019e+03	FES = 57100	Diversity = 2.7931e-04
Generation	580:	Best = 1.110019e+03	FES = 58100	Diversity = 2.7974e-04
Generation	590:	Best = 1.110019e+03	FES = 59100	Diversity = 2.2242e-04
Generation	600:	Best = 1.110019e+03	FES = 60100	Diversity = 1.8516e-04
Generation	610:	Best = 1.110019e+03	FES = 61100	Diversity = 1.3786e-04
Generation	620:	Best = 1.110019e+03	FES = 62100	Diversity = 1.3190e-04
Generation	630:	Best = 1.110019e+03	FES = 63100	Diversity = 1.2176e-04
Generation	640:	Best = 1.110019e+03	FES = 64100	Diversity = 8.9777e-05
Generation	650:	Best = 1.110019e+03	FES = 65100	Diversity = 6.7948e-05
Generation	660:	Best = 1.110019e+03	FES = 66100	Diversity = 3.8839e-05
Generation	670:	Best = 1.110019e+03	FES = 67100	Diversity = 3.9724e-05
Generation	680:	Best = 1.110019e+03	FES = 68100	Diversity = 3.5081e-05
Generation	690:	Best = 1.110019e+03	FES = 69100	Diversity = 2.9387e-05
Generation	700:	Best = 1.110019e+03	FES = 70100	Diversity = 2.4355e-05
Generation	710:	Best = 1.110019e+03	FES = 71100	Diversity = 2.2524e-05
Generation	720:	Best = 1.110019e+03	FES = 72100	Diversity = 1.9955e-05
Generation	730:	Best = 1.110019e+03	FES = 73100	Diversity = 1.7847e-05
Generation	740:	Best = 1.110019e+03	FES = 74100	Diversity = 1.7410e-05
Generation	750:	Best = 1.110019e+03	FES = 75100	Diversity = 1.6694e-05
Generation	760:	Best = 1.110019e+03	FES = 76100	Diversity = 1.4508e-05
Generation	770:	Best = 1.110019e+03	FES = 77100	Diversity = 1.2937e-05
Generation	780:	Best = 1.110019e+03	FES = 78100	Diversity = 1.4384e-05
Generation	790:	Best = 1.110019e+03	FES = 79100	Diversity = 1.2786e-05
Generation	800:	Best = 1.110019e+03	FES = 80100	Diversity = 1.2169e-05
Generation	810:	Best = 1.110019e+03	FES = 81100	Diversity = 1.1924e-05
Generation	820:	Best = 1.110019e+03	FES = 82100	Diversity = 1.0064e-05
Generation	830:	Best = 1.110019e+03	FES = 83100	Diversity = 8.1771e-06
Generation	840:	Best = 1.110019e+03	FES = 84100	Diversity = 6.4551e-06
Generation	850:	Best = 1.110019e+03	FES = 85100	Diversity = 5.2361e-06
Generation	860:	Best = 1.110019e+03	FES = 86100	Diversity = 4.0130e-06
Generation	870:	Best = 1.110019e+03	FES = 87100	Diversity = 4.8442e-06
Generation	880:	Best = 1.110019e+03	FES = 88100	Diversity = 4.6416e-06
Generation	890:	Best = 1.110019e+03	FES = 89100	Diversity = 3.7784e-06
Generation	900:	Best = 1.110019e+03	FES = 90100	Diversity = 3.1128e-06
Generation	910:	Best = 1.110019e+03	FES = 91100	Diversity = 2.7798e-06
Generation	920:	Best = 1.110019e+03	FES = 92100	Diversity = 3.4478e-06
Generation	930:	Best = 1.110019e+03	FES = 93100	Diversity = 2.6910e-06
Generation	940:	Best = 1.110019e+03	FES = 94100	Diversity = 2.2116e-06
Generation	950:	Best = 1.110019e+03	FES = 95100	Diversity = 2.1092e-06
Generation	960:	Best = 1.110019e+03	FES = 96100	Diversity = 2.3452e-06
Generation	970:	Best = 1.110019e+03	FES = 97100	Diversity = 1.7550e-06
Generation	980:	Best = 1.110019e+03	FES = 98100	Diversity = 1.9494e-06

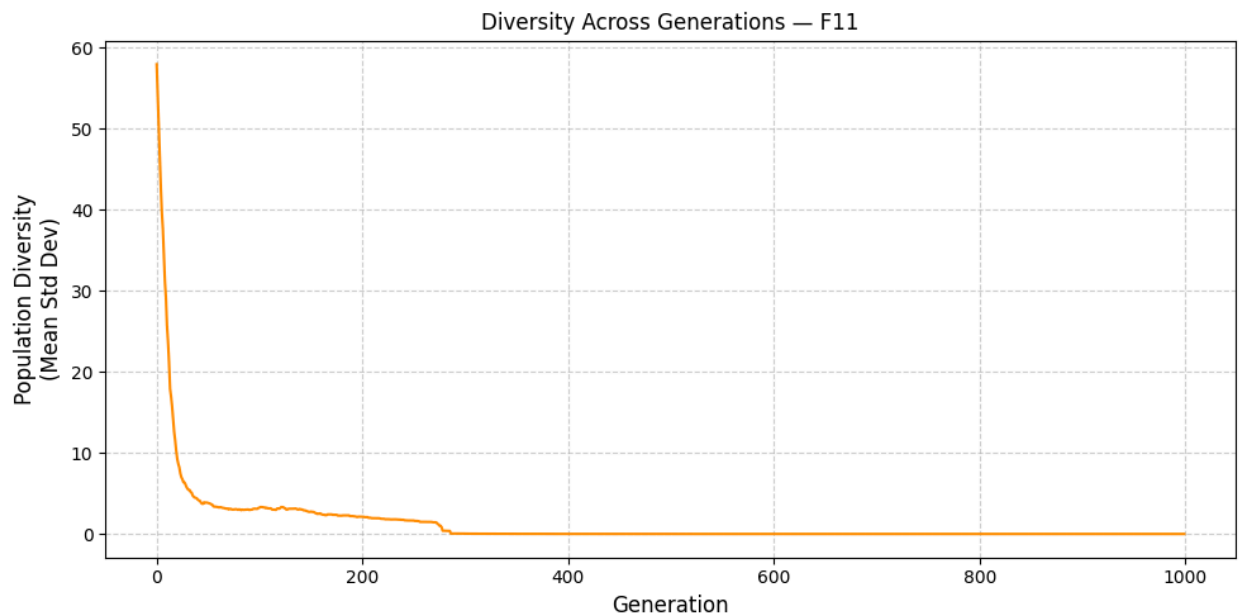
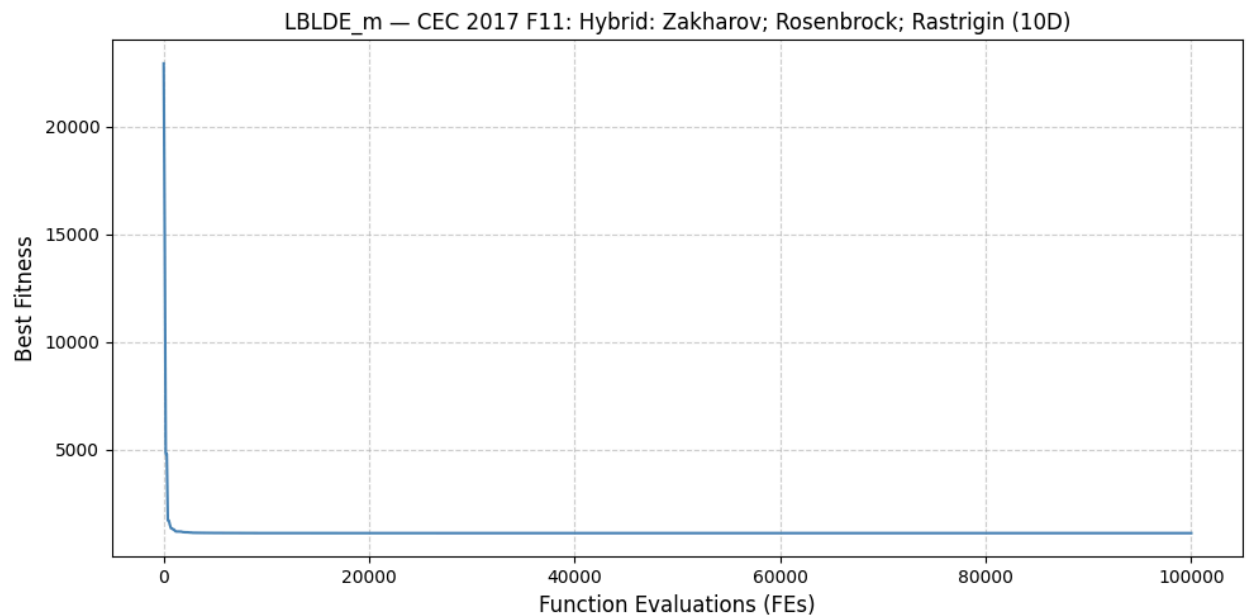
Generation 990: Best = 1.110019e+03 FES = 99100 Diversity = 1.6201e-06

Optimization completed!

Final Best Fitness = 1.110019e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness           : 1.110019e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 1.001882e+01
  Best Solution (5 dim)  : [ -4.58331028 -49.98691725 -38.85991119 18.76160803
19.75000083]
  D^(0) (init. diversity): 5.7928e+01
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```



Graph saved as convergence_F11.png

```
In [22]: # =====  
# F12 – Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)  
# =====  
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```

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LBLDE_m on CEC 2017 F12 – Single Run

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Function : Hybrid: Elliptic; Schwefel; Bent Cigar (10D)
Bounds : [-100, 100]
Global optimum : 1200.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 7.052719e+09 | Diversity = 5.7928e+01

Generation	10:	Best = 6.041338e+07	FES = 1100	Diversity = 2.1874e+01
Generation	20:	Best = 4.913739e+06	FES = 2100	Diversity = 1.1628e+01
Generation	30:	Best = 7.803958e+05	FES = 3100	Diversity = 1.0538e+01
Generation	40:	Best = 5.218510e+04	FES = 4100	Diversity = 9.0477e+00
Generation	50:	Best = 1.195518e+04	FES = 5100	Diversity = 9.5197e+00
Generation	60:	Best = 1.195518e+04	FES = 6100	Diversity = 7.0929e+00
Generation	70:	Best = 6.885572e+03	FES = 7100	Diversity = 7.2510e+00
Generation	80:	Best = 4.315890e+03	FES = 8100	Diversity = 8.2961e+00
Generation	90:	Best = 3.643766e+03	FES = 9100	Diversity = 8.6074e+00
Generation	100:	Best = 2.513406e+03	FES = 10100	Diversity = 7.9678e+00
Generation	110:	Best = 2.216494e+03	FES = 11100	Diversity = 7.8602e+00
Generation	120:	Best = 1.903093e+03	FES = 12100	Diversity = 7.9490e+00
Generation	130:	Best = 1.842486e+03	FES = 13100	Diversity = 7.6289e+00
Generation	140:	Best = 1.674500e+03	FES = 14100	Diversity = 7.9763e+00
Generation	150:	Best = 1.674500e+03	FES = 15100	Diversity = 7.8942e+00
Generation	160:	Best = 1.587056e+03	FES = 16100	Diversity = 7.5183e+00
Generation	170:	Best = 1.559075e+03	FES = 17100	Diversity = 7.3125e+00
Generation	180:	Best = 1.420181e+03	FES = 18100	Diversity = 6.7459e+00
Generation	190:	Best = 1.411766e+03	FES = 19100	Diversity = 6.1651e+00
Generation	200:	Best = 1.361902e+03	FES = 20100	Diversity = 5.8777e+00
Generation	210:	Best = 1.361536e+03	FES = 21100	Diversity = 5.5992e+00
Generation	220:	Best = 1.361536e+03	FES = 22100	Diversity = 5.0398e+00
Generation	230:	Best = 1.344918e+03	FES = 23100	Diversity = 4.3684e+00
Generation	240:	Best = 1.344918e+03	FES = 24100	Diversity = 3.6016e+00
Generation	250:	Best = 1.340770e+03	FES = 25100	Diversity = 2.9688e+00
Generation	260:	Best = 1.332866e+03	FES = 26100	Diversity = 2.6184e+00
Generation	270:	Best = 1.332731e+03	FES = 27100	Diversity = 2.2765e+00
Generation	280:	Best = 1.332731e+03	FES = 28100	Diversity = 1.9223e+00
Generation	290:	Best = 1.326727e+03	FES = 29100	Diversity = 1.7776e+00
Generation	300:	Best = 1.326674e+03	FES = 30100	Diversity = 1.6755e+00
Generation	310:	Best = 1.324975e+03	FES = 31100	Diversity = 1.6036e+00
Generation	320:	Best = 1.324975e+03	FES = 32100	Diversity = 1.5737e+00
Generation	330:	Best = 1.324678e+03	FES = 33100	Diversity = 1.5711e+00
Generation	340:	Best = 1.323759e+03	FES = 34100	Diversity = 7.4920e-01
Generation	350:	Best = 1.320695e+03	FES = 35100	Diversity = 4.2426e-01
Generation	360:	Best = 1.320516e+03	FES = 36100	Diversity = 3.8277e-01
Generation	370:	Best = 1.320281e+03	FES = 37100	Diversity = 3.1863e-01
Generation	380:	Best = 1.319924e+03	FES = 38100	Diversity = 2.5841e-01
Generation	390:	Best = 1.319600e+03	FES = 39100	Diversity = 1.9287e-01
Generation	400:	Best = 1.319496e+03	FES = 40100	Diversity = 1.3810e-01
Generation	410:	Best = 1.319040e+03	FES = 41100	Diversity = 1.1844e-01
Generation	420:	Best = 1.319031e+03	FES = 42100	Diversity = 1.0105e-01
Generation	430:	Best = 1.319012e+03	FES = 43100	Diversity = 1.0095e-01
Generation	440:	Best = 1.319006e+03	FES = 44100	Diversity = 8.6375e-02

Generation	450:	Best = 1.318956e+03	FES = 45100	Diversity = 7.0335e-02
Generation	460:	Best = 1.318956e+03	FES = 46100	Diversity = 6.3275e-02
Generation	470:	Best = 1.318946e+03	FES = 47100	Diversity = 5.3413e-02
Generation	480:	Best = 1.318946e+03	FES = 48100	Diversity = 5.2315e-02
Generation	490:	Best = 1.318934e+03	FES = 49100	Diversity = 4.5645e-02
Generation	500:	Best = 1.318934e+03	FES = 50100	Diversity = 4.3025e-02
Generation	510:	Best = 1.318929e+03	FES = 51100	Diversity = 4.1449e-02
Generation	520:	Best = 1.318754e+03	FES = 52100	Diversity = 4.1405e-02
Generation	530:	Best = 1.318754e+03	FES = 53100	Diversity = 4.0777e-02
Generation	540:	Best = 1.318754e+03	FES = 54100	Diversity = 3.9467e-02
Generation	550:	Best = 1.318754e+03	FES = 55100	Diversity = 3.8607e-02
Generation	560:	Best = 1.318754e+03	FES = 56100	Diversity = 3.6296e-02
Generation	570:	Best = 1.318754e+03	FES = 57100	Diversity = 3.5978e-02
Generation	580:	Best = 1.318754e+03	FES = 58100	Diversity = 3.4708e-02
Generation	590:	Best = 1.318754e+03	FES = 59100	Diversity = 2.9629e-02
Generation	600:	Best = 1.318754e+03	FES = 60100	Diversity = 2.8757e-02
Generation	610:	Best = 1.318699e+03	FES = 61100	Diversity = 2.8950e-02
Generation	620:	Best = 1.318699e+03	FES = 62100	Diversity = 3.1153e-02
Generation	630:	Best = 1.318699e+03	FES = 63100	Diversity = 2.9462e-02
Generation	640:	Best = 1.318669e+03	FES = 64100	Diversity = 2.8468e-02
Generation	650:	Best = 1.318669e+03	FES = 65100	Diversity = 2.5331e-02
Generation	660:	Best = 1.318669e+03	FES = 66100	Diversity = 2.2779e-02
Generation	670:	Best = 1.318669e+03	FES = 67100	Diversity = 2.2319e-02
Generation	680:	Best = 1.318666e+03	FES = 68100	Diversity = 2.0150e-02
Generation	690:	Best = 1.318663e+03	FES = 69100	Diversity = 1.5432e-02
Generation	700:	Best = 1.318662e+03	FES = 70100	Diversity = 1.3172e-02
Generation	710:	Best = 1.318661e+03	FES = 71100	Diversity = 1.2406e-02
Generation	720:	Best = 1.318657e+03	FES = 72100	Diversity = 1.1390e-02
Generation	730:	Best = 1.318657e+03	FES = 73100	Diversity = 1.1017e-02
Generation	740:	Best = 1.318657e+03	FES = 74100	Diversity = 9.8475e-03
Generation	750:	Best = 1.318655e+03	FES = 75100	Diversity = 9.2979e-03
Generation	760:	Best = 1.318655e+03	FES = 76100	Diversity = 8.0597e-03
Generation	770:	Best = 1.318654e+03	FES = 77100	Diversity = 7.9617e-03
Generation	780:	Best = 1.318654e+03	FES = 78100	Diversity = 7.4592e-03
Generation	790:	Best = 1.318654e+03	FES = 79100	Diversity = 6.7119e-03
Generation	800:	Best = 1.318654e+03	FES = 80100	Diversity = 5.3691e-03
Generation	810:	Best = 1.318654e+03	FES = 81100	Diversity = 4.8126e-03
Generation	820:	Best = 1.318654e+03	FES = 82100	Diversity = 4.3809e-03
Generation	830:	Best = 1.318654e+03	FES = 83100	Diversity = 3.5131e-03
Generation	840:	Best = 1.318654e+03	FES = 84100	Diversity = 3.3637e-03
Generation	850:	Best = 1.318653e+03	FES = 85100	Diversity = 3.2562e-03
Generation	860:	Best = 1.318653e+03	FES = 86100	Diversity = 3.3552e-03
Generation	870:	Best = 1.318653e+03	FES = 87100	Diversity = 3.2927e-03
Generation	880:	Best = 1.318653e+03	FES = 88100	Diversity = 3.2636e-03
Generation	890:	Best = 1.318653e+03	FES = 89100	Diversity = 3.0732e-03
Generation	900:	Best = 1.318653e+03	FES = 90100	Diversity = 2.8693e-03
Generation	910:	Best = 1.318652e+03	FES = 91100	Diversity = 2.7906e-03
Generation	920:	Best = 1.318652e+03	FES = 92100	Diversity = 2.4937e-03
Generation	930:	Best = 1.318652e+03	FES = 93100	Diversity = 2.3419e-03
Generation	940:	Best = 1.318652e+03	FES = 94100	Diversity = 2.1409e-03
Generation	950:	Best = 1.318652e+03	FES = 95100	Diversity = 2.1357e-03
Generation	960:	Best = 1.318652e+03	FES = 96100	Diversity = 2.0096e-03
Generation	970:	Best = 1.318652e+03	FES = 97100	Diversity = 1.7797e-03
Generation	980:	Best = 1.318652e+03	FES = 98100	Diversity = 1.7872e-03

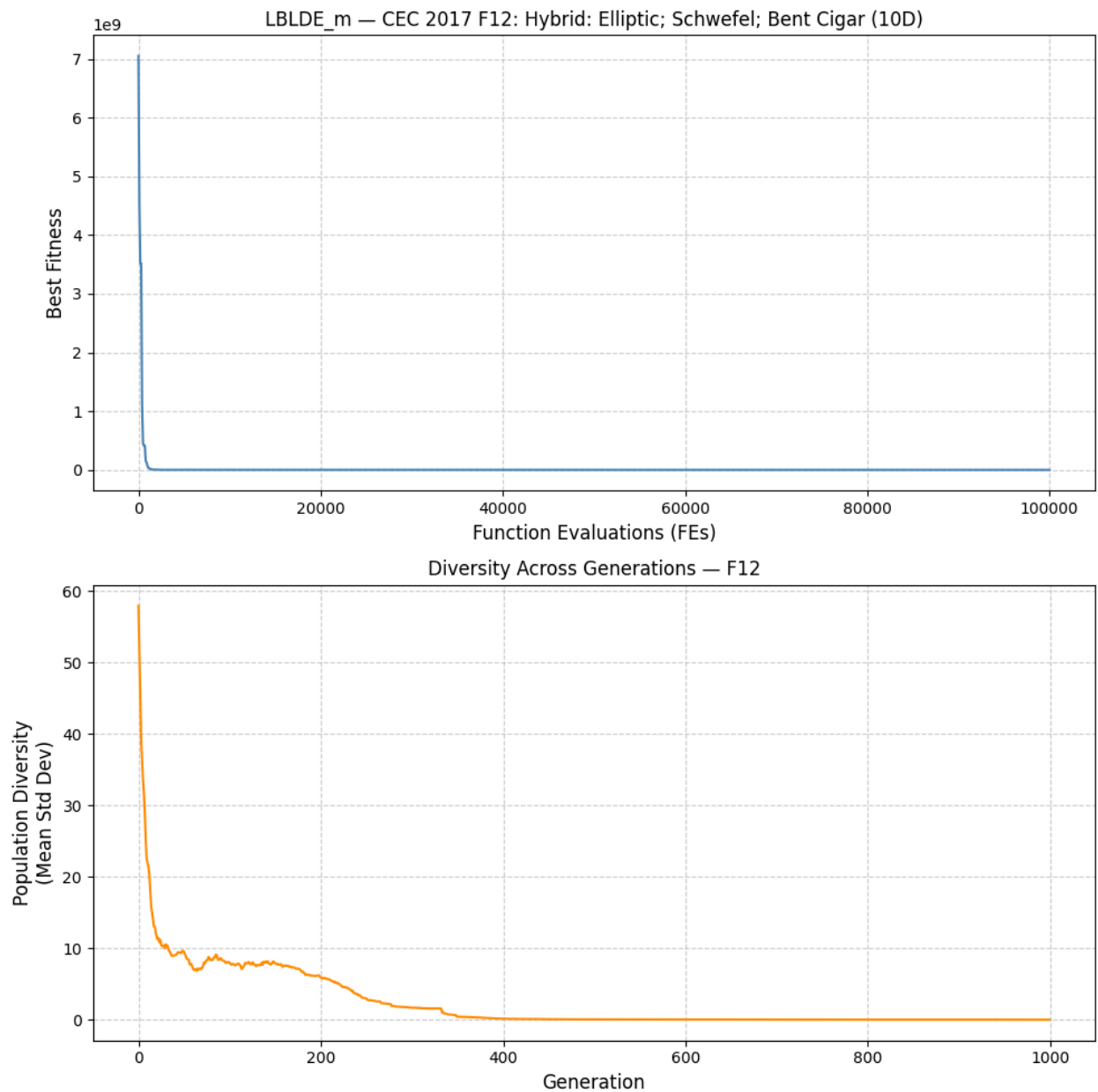
Generation 990: Best = 1.318652e+03 FES = 99100 Diversity = 1.8933e-03

Optimization completed!

Final Best Fitness = 1.318652e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness           : 1.318652e+03
  Global Optimum         : 1.200000e+03
  Error (|best-optimum|) : 1.186518e+02
  Best Solution (5 dim)  : [ 65.24440581  2.58269091 -7.74480717  8.32148606
-34.48574823]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F12.png

```
In [23]: # =====  
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)  
# =====  
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

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LBLDE_m on CEC 2017 F13 – Single Run

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Function : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 1300.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 1.212051e+09 | Diversity = 5.7928e+01

Generation	10:	Best = 3.801345e+05	FES = 1100	Diversity = 2.7386e+01
Generation	20:	Best = 1.609519e+04	FES = 2100	Diversity = 2.5851e+01
Generation	30:	Best = 3.131154e+03	FES = 3100	Diversity = 2.5452e+01
Generation	40:	Best = 1.923229e+03	FES = 4100	Diversity = 1.7464e+01
Generation	50:	Best = 1.617218e+03	FES = 5100	Diversity = 1.4334e+01
Generation	60:	Best = 1.505835e+03	FES = 6100	Diversity = 1.0800e+01
Generation	70:	Best = 1.334744e+03	FES = 7100	Diversity = 8.7064e+00
Generation	80:	Best = 1.334744e+03	FES = 8100	Diversity = 7.5680e+00
Generation	90:	Best = 1.334744e+03	FES = 9100	Diversity = 7.1764e+00
Generation	100:	Best = 1.320385e+03	FES = 10100	Diversity = 6.8602e+00
Generation	110:	Best = 1.315588e+03	FES = 11100	Diversity = 6.5510e+00
Generation	120:	Best = 1.307756e+03	FES = 12100	Diversity = 6.5415e+00
Generation	130:	Best = 1.307756e+03	FES = 13100	Diversity = 6.7070e+00
Generation	140:	Best = 1.307756e+03	FES = 14100	Diversity = 6.2721e+00
Generation	150:	Best = 1.307756e+03	FES = 15100	Diversity = 6.3258e+00
Generation	160:	Best = 1.306438e+03	FES = 16100	Diversity = 6.2807e+00
Generation	170:	Best = 1.306438e+03	FES = 17100	Diversity = 6.0270e+00
Generation	180:	Best = 1.306317e+03	FES = 18100	Diversity = 6.0057e+00
Generation	190:	Best = 1.304689e+03	FES = 19100	Diversity = 5.9386e+00
Generation	200:	Best = 1.304689e+03	FES = 20100	Diversity = 5.6429e+00
Generation	210:	Best = 1.304676e+03	FES = 21100	Diversity = 5.6267e+00
Generation	220:	Best = 1.302599e+03	FES = 22100	Diversity = 5.4631e+00
Generation	230:	Best = 1.302599e+03	FES = 23100	Diversity = 5.1592e+00
Generation	240:	Best = 1.302599e+03	FES = 24100	Diversity = 4.8586e+00
Generation	250:	Best = 1.302203e+03	FES = 25100	Diversity = 4.7770e+00
Generation	260:	Best = 1.302057e+03	FES = 26100	Diversity = 4.4084e+00
Generation	270:	Best = 1.302014e+03	FES = 27100	Diversity = 4.3722e+00
Generation	280:	Best = 1.301976e+03	FES = 28100	Diversity = 4.2530e+00
Generation	290:	Best = 1.301918e+03	FES = 29100	Diversity = 4.1594e+00
Generation	300:	Best = 1.301896e+03	FES = 30100	Diversity = 4.0762e+00
Generation	310:	Best = 1.301890e+03	FES = 31100	Diversity = 3.9897e+00
Generation	320:	Best = 1.301890e+03	FES = 32100	Diversity = 3.9414e+00
Generation	330:	Best = 1.301745e+03	FES = 33100	Diversity = 3.8490e+00
Generation	340:	Best = 1.301519e+03	FES = 34100	Diversity = 3.7836e+00
Generation	350:	Best = 1.301519e+03	FES = 35100	Diversity = 3.7923e+00
Generation	360:	Best = 1.301519e+03	FES = 36100	Diversity = 3.5930e+00
Generation	370:	Best = 1.301518e+03	FES = 37100	Diversity = 3.5832e+00
Generation	380:	Best = 1.301337e+03	FES = 38100	Diversity = 3.5659e+00
Generation	390:	Best = 1.301337e+03	FES = 39100	Diversity = 3.5411e+00
Generation	400:	Best = 1.301337e+03	FES = 40100	Diversity = 3.5120e+00
Generation	410:	Best = 1.301333e+03	FES = 41100	Diversity = 3.3862e+00
Generation	420:	Best = 1.301333e+03	FES = 42100	Diversity = 3.3760e+00
Generation	430:	Best = 1.301333e+03	FES = 43100	Diversity = 3.3485e+00
Generation	440:	Best = 1.301333e+03	FES = 44100	Diversity = 3.2876e+00

Generation	450:	Best = 1.301333e+03	FES = 45100	Diversity = 3.2383e+00
Generation	460:	Best = 1.301333e+03	FES = 46100	Diversity = 3.2045e+00
Generation	470:	Best = 1.301256e+03	FES = 47100	Diversity = 3.1112e+00
Generation	480:	Best = 1.301256e+03	FES = 48100	Diversity = 3.1019e+00
Generation	490:	Best = 1.301182e+03	FES = 49100	Diversity = 2.9483e+00
Generation	500:	Best = 1.301182e+03	FES = 50100	Diversity = 2.7932e+00
Generation	510:	Best = 1.301182e+03	FES = 51100	Diversity = 2.2970e+00
Generation	520:	Best = 1.301125e+03	FES = 52100	Diversity = 1.4432e+00
Generation	530:	Best = 1.301100e+03	FES = 53100	Diversity = 1.1103e+00
Generation	540:	Best = 1.301048e+03	FES = 54100	Diversity = 1.0335e+00
Generation	550:	Best = 1.301022e+03	FES = 55100	Diversity = 1.0353e+00
Generation	560:	Best = 1.301004e+03	FES = 56100	Diversity = 9.1359e-01
Generation	570:	Best = 1.300999e+03	FES = 57100	Diversity = 6.7552e-01
Generation	580:	Best = 1.300997e+03	FES = 58100	Diversity = 5.7328e-01
Generation	590:	Best = 1.300996e+03	FES = 59100	Diversity = 4.0000e-01
Generation	600:	Best = 1.300995e+03	FES = 60100	Diversity = 3.0773e-01
Generation	610:	Best = 1.300995e+03	FES = 61100	Diversity = 2.0744e-01
Generation	620:	Best = 1.300995e+03	FES = 62100	Diversity = 1.3978e-01
Generation	630:	Best = 1.300995e+03	FES = 63100	Diversity = 9.7018e-02
Generation	640:	Best = 1.300995e+03	FES = 64100	Diversity = 7.5420e-02
Generation	650:	Best = 1.300995e+03	FES = 65100	Diversity = 6.3795e-02
Generation	660:	Best = 1.300995e+03	FES = 66100	Diversity = 4.9291e-02
Generation	670:	Best = 1.300995e+03	FES = 67100	Diversity = 3.8852e-02
Generation	680:	Best = 1.300995e+03	FES = 68100	Diversity = 2.8893e-02
Generation	690:	Best = 1.300995e+03	FES = 69100	Diversity = 2.3232e-02
Generation	700:	Best = 1.300995e+03	FES = 70100	Diversity = 1.7711e-02
Generation	710:	Best = 1.300995e+03	FES = 71100	Diversity = 1.3591e-02
Generation	720:	Best = 1.300995e+03	FES = 72100	Diversity = 1.2366e-02
Generation	730:	Best = 1.300995e+03	FES = 73100	Diversity = 1.0316e-02
Generation	740:	Best = 1.300995e+03	FES = 74100	Diversity = 7.9036e-03
Generation	750:	Best = 1.300995e+03	FES = 75100	Diversity = 7.9923e-03
Generation	760:	Best = 1.300995e+03	FES = 76100	Diversity = 8.0904e-03
Generation	770:	Best = 1.300995e+03	FES = 77100	Diversity = 6.7104e-03
Generation	780:	Best = 1.300995e+03	FES = 78100	Diversity = 6.2999e-03
Generation	790:	Best = 1.300995e+03	FES = 79100	Diversity = 5.3331e-03
Generation	800:	Best = 1.300995e+03	FES = 80100	Diversity = 4.5191e-03
Generation	810:	Best = 1.300995e+03	FES = 81100	Diversity = 3.5749e-03
Generation	820:	Best = 1.300995e+03	FES = 82100	Diversity = 2.8839e-03
Generation	830:	Best = 1.300995e+03	FES = 83100	Diversity = 2.8686e-03
Generation	840:	Best = 1.300995e+03	FES = 84100	Diversity = 2.4547e-03
Generation	850:	Best = 1.300995e+03	FES = 85100	Diversity = 1.7827e-03
Generation	860:	Best = 1.300995e+03	FES = 86100	Diversity = 1.5968e-03
Generation	870:	Best = 1.300995e+03	FES = 87100	Diversity = 1.4350e-03
Generation	880:	Best = 1.300995e+03	FES = 88100	Diversity = 1.2849e-03
Generation	890:	Best = 1.300995e+03	FES = 89100	Diversity = 1.0939e-03
Generation	900:	Best = 1.300995e+03	FES = 90100	Diversity = 1.0120e-03
Generation	910:	Best = 1.300995e+03	FES = 91100	Diversity = 8.8842e-04
Generation	920:	Best = 1.300995e+03	FES = 92100	Diversity = 9.4285e-04
Generation	930:	Best = 1.300995e+03	FES = 93100	Diversity = 8.0105e-04
Generation	940:	Best = 1.300995e+03	FES = 94100	Diversity = 7.3109e-04
Generation	950:	Best = 1.300995e+03	FES = 95100	Diversity = 6.8072e-04
Generation	960:	Best = 1.300995e+03	FES = 96100	Diversity = 6.5147e-04
Generation	970:	Best = 1.300995e+03	FES = 97100	Diversity = 5.4451e-04
Generation	980:	Best = 1.300995e+03	FES = 98100	Diversity = 4.0108e-04

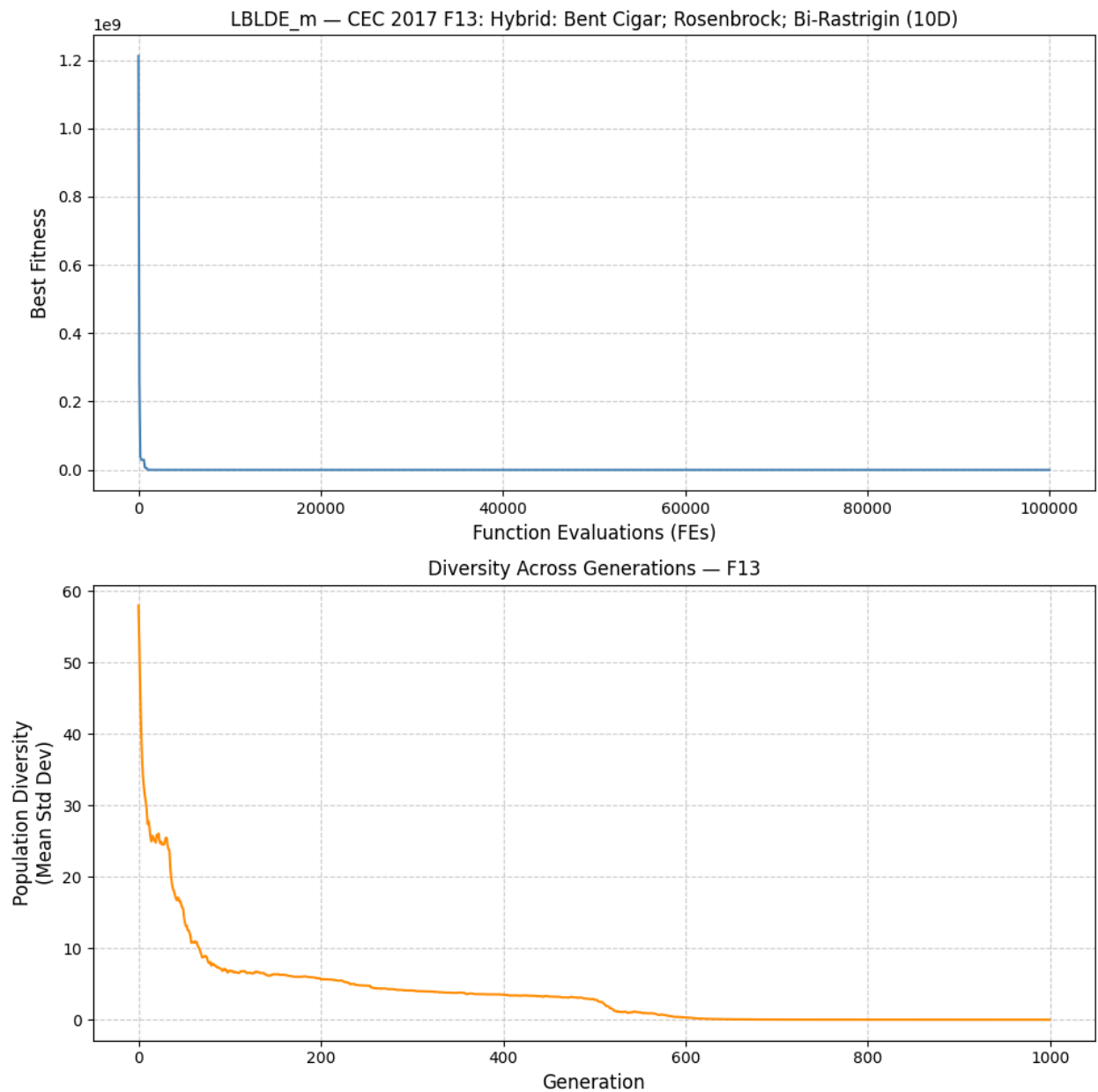
Generation 990: Best = 1.300995e+03 FES = 99100 Diversity = 3.0573e-04

Optimization completed!

Final Best Fitness = 1.300995e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness           : 1.300995e+03
  Global Optimum         : 1.300000e+03
  Error (|best-optimum|) : 9.949601e-01
  Best Solution (5 dim)  : [ 0.55897979 -73.58219046 -6.24588079 -14.50780689
25.61481445]
  D^(0) (init. diversity): 5.7928e+01
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```



Graph saved as convergence_F13.png

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In [24]: # =====  
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)  
# =====  
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

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LBLDE_m on CEC 2017 F14 – Single Run

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Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 1400.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.917787e+04 | Diversity = 5.7928e+01

Generation	10:	Best = 4.604853e+03	FES = 1100	Diversity = 3.2568e+01
Generation	20:	Best = 1.991489e+03	FES = 2100	Diversity = 2.5161e+01
Generation	30:	Best = 1.666800e+03	FES = 3100	Diversity = 2.1105e+01
Generation	40:	Best = 1.556155e+03	FES = 4100	Diversity = 1.9741e+01
Generation	50:	Best = 1.479517e+03	FES = 5100	Diversity = 1.8048e+01
Generation	60:	Best = 1.453979e+03	FES = 6100	Diversity = 1.2972e+01
Generation	70:	Best = 1.439425e+03	FES = 7100	Diversity = 9.6709e+00
Generation	80:	Best = 1.430381e+03	FES = 8100	Diversity = 8.0469e+00
Generation	90:	Best = 1.428208e+03	FES = 9100	Diversity = 6.1708e+00
Generation	100:	Best = 1.426450e+03	FES = 10100	Diversity = 3.3377e+00
Generation	110:	Best = 1.422960e+03	FES = 11100	Diversity = 1.7296e+00
Generation	120:	Best = 1.422396e+03	FES = 12100	Diversity = 1.8710e+00
Generation	130:	Best = 1.422294e+03	FES = 13100	Diversity = 2.5255e+00
Generation	140:	Best = 1.421644e+03	FES = 14100	Diversity = 3.1116e+00
Generation	150:	Best = 1.421123e+03	FES = 15100	Diversity = 4.2269e+00
Generation	160:	Best = 1.416321e+03	FES = 16100	Diversity = 5.2826e+00
Generation	170:	Best = 1.402385e+03	FES = 17100	Diversity = 8.4891e-01
Generation	180:	Best = 1.402383e+03	FES = 18100	Diversity = 2.5951e-01
Generation	190:	Best = 1.402113e+03	FES = 19100	Diversity = 1.1730e-01
Generation	200:	Best = 1.402032e+03	FES = 20100	Diversity = 5.6183e-02
Generation	210:	Best = 1.402006e+03	FES = 21100	Diversity = 3.8821e-02
Generation	220:	Best = 1.401995e+03	FES = 22100	Diversity = 2.2053e-02
Generation	230:	Best = 1.401992e+03	FES = 23100	Diversity = 1.3791e-02
Generation	240:	Best = 1.401991e+03	FES = 24100	Diversity = 8.9947e-03
Generation	250:	Best = 1.401990e+03	FES = 25100	Diversity = 6.1213e-03
Generation	260:	Best = 1.401990e+03	FES = 26100	Diversity = 3.1930e-03
Generation	270:	Best = 1.401990e+03	FES = 27100	Diversity = 2.1759e-03
Generation	280:	Best = 1.401990e+03	FES = 28100	Diversity = 1.2700e-03
Generation	290:	Best = 1.401990e+03	FES = 29100	Diversity = 8.2939e-04
Generation	300:	Best = 1.401990e+03	FES = 30100	Diversity = 4.5719e-04
Generation	310:	Best = 1.401990e+03	FES = 31100	Diversity = 2.7901e-04
Generation	320:	Best = 1.401990e+03	FES = 32100	Diversity = 1.5828e-04
Generation	330:	Best = 1.401990e+03	FES = 33100	Diversity = 9.6544e-05
Generation	340:	Best = 1.401990e+03	FES = 34100	Diversity = 6.2197e-05
Generation	350:	Best = 1.401990e+03	FES = 35100	Diversity = 3.3813e-05
Generation	360:	Best = 1.401990e+03	FES = 36100	Diversity = 1.7790e-05
Generation	370:	Best = 1.401990e+03	FES = 37100	Diversity = 7.4285e-06
Generation	380:	Best = 1.401990e+03	FES = 38100	Diversity = 4.2912e-06
Generation	390:	Best = 1.401990e+03	FES = 39100	Diversity = 2.6697e-06
Generation	400:	Best = 1.401990e+03	FES = 40100	Diversity = 1.5194e-06
Generation	410:	Best = 1.401990e+03	FES = 41100	Diversity = 8.6355e-07
Generation	420:	Best = 1.401990e+03	FES = 42100	Diversity = 4.5663e-07
Generation	430:	Best = 1.401990e+03	FES = 43100	Diversity = 2.5583e-07
Generation	440:	Best = 1.401990e+03	FES = 44100	Diversity = 1.6684e-07

Generation	450:	Best = 1.401990e+03	FES = 45100	Diversity = 1.0962e-07
Generation	460:	Best = 1.401990e+03	FES = 46100	Diversity = 7.9752e-08
Generation	470:	Best = 1.401990e+03	FES = 47100	Diversity = 6.1641e-08
Generation	480:	Best = 1.401990e+03	FES = 48100	Diversity = 4.8033e-08
Generation	490:	Best = 1.401990e+03	FES = 49100	Diversity = 3.7365e-08
Generation	500:	Best = 1.401990e+03	FES = 50100	Diversity = 2.0829e-08
Generation	510:	Best = 1.401990e+03	FES = 51100	Diversity = 1.0089e-08
Generation	520:	Best = 1.401990e+03	FES = 52100	Diversity = 6.3853e-09
Generation	530:	Best = 1.401990e+03	FES = 53100	Diversity = 5.6413e-09
Generation	540:	Best = 1.401990e+03	FES = 54100	Diversity = 4.0015e-09
Generation	550:	Best = 1.401990e+03	FES = 55100	Diversity = 2.3977e-09
Generation	560:	Best = 1.401990e+03	FES = 56100	Diversity = 2.3363e-09
Generation	570:	Best = 1.401990e+03	FES = 57100	Diversity = 8.2200e-10
Generation	580:	Best = 1.401990e+03	FES = 58100	Diversity = 7.9166e-10
Generation	590:	Best = 1.401990e+03	FES = 59100	Diversity = 5.1616e-10
Generation	600:	Best = 1.401990e+03	FES = 60100	Diversity = 3.0055e-10
Generation	610:	Best = 1.401990e+03	FES = 61100	Diversity = 2.6000e-10
Generation	620:	Best = 1.401990e+03	FES = 62100	Diversity = 2.4558e-10
Generation	630:	Best = 1.401990e+03	FES = 63100	Diversity = 2.0326e-10
Generation	640:	Best = 1.401990e+03	FES = 64100	Diversity = 1.9164e-10
Generation	650:	Best = 1.401990e+03	FES = 65100	Diversity = 3.2631e-11
Generation	660:	Best = 1.401990e+03	FES = 66100	Diversity = 6.7998e-13
Generation	670:	Best = 1.401990e+03	FES = 67100	Diversity = 5.8453e-14
Generation	680:	Best = 1.401990e+03	FES = 68100	Diversity = 5.8265e-14
Generation	690:	Best = 1.401990e+03	FES = 69100	Diversity = 5.8265e-14
Generation	700:	Best = 1.401990e+03	FES = 70100	Diversity = 5.8265e-14
Generation	710:	Best = 1.401990e+03	FES = 71100	Diversity = 5.8265e-14
Generation	720:	Best = 1.401990e+03	FES = 72100	Diversity = 5.8265e-14
Generation	730:	Best = 1.401990e+03	FES = 73100	Diversity = 5.8265e-14
Generation	740:	Best = 1.401990e+03	FES = 74100	Diversity = 5.8265e-14
Generation	750:	Best = 1.401990e+03	FES = 75100	Diversity = 5.8265e-14
Generation	760:	Best = 1.401990e+03	FES = 76100	Diversity = 5.8265e-14
Generation	770:	Best = 1.401990e+03	FES = 77100	Diversity = 5.8265e-14
Generation	780:	Best = 1.401990e+03	FES = 78100	Diversity = 5.8265e-14
Generation	790:	Best = 1.401990e+03	FES = 79100	Diversity = 5.8265e-14
Generation	800:	Best = 1.401990e+03	FES = 80100	Diversity = 5.8265e-14
Generation	810:	Best = 1.401990e+03	FES = 81100	Diversity = 5.8265e-14
Generation	820:	Best = 1.401990e+03	FES = 82100	Diversity = 5.8265e-14
Generation	830:	Best = 1.401990e+03	FES = 83100	Diversity = 5.8265e-14
Generation	840:	Best = 1.401990e+03	FES = 84100	Diversity = 5.8265e-14
Generation	850:	Best = 1.401990e+03	FES = 85100	Diversity = 5.8265e-14
Generation	860:	Best = 1.401990e+03	FES = 86100	Diversity = 5.8265e-14
Generation	870:	Best = 1.401990e+03	FES = 87100	Diversity = 5.8265e-14
Generation	880:	Best = 1.401990e+03	FES = 88100	Diversity = 5.8265e-14
Generation	890:	Best = 1.401990e+03	FES = 89100	Diversity = 5.8265e-14
Generation	900:	Best = 1.401990e+03	FES = 90100	Diversity = 5.8265e-14
Generation	910:	Best = 1.401990e+03	FES = 91100	Diversity = 5.8265e-14
Generation	920:	Best = 1.401990e+03	FES = 92100	Diversity = 5.8265e-14
Generation	930:	Best = 1.401990e+03	FES = 93100	Diversity = 5.8265e-14
Generation	940:	Best = 1.401990e+03	FES = 94100	Diversity = 5.8265e-14
Generation	950:	Best = 1.401990e+03	FES = 95100	Diversity = 5.8265e-14
Generation	960:	Best = 1.401990e+03	FES = 96100	Diversity = 5.8265e-14
Generation	970:	Best = 1.401990e+03	FES = 97100	Diversity = 5.8265e-14
Generation	980:	Best = 1.401990e+03	FES = 98100	Diversity = 5.8265e-14

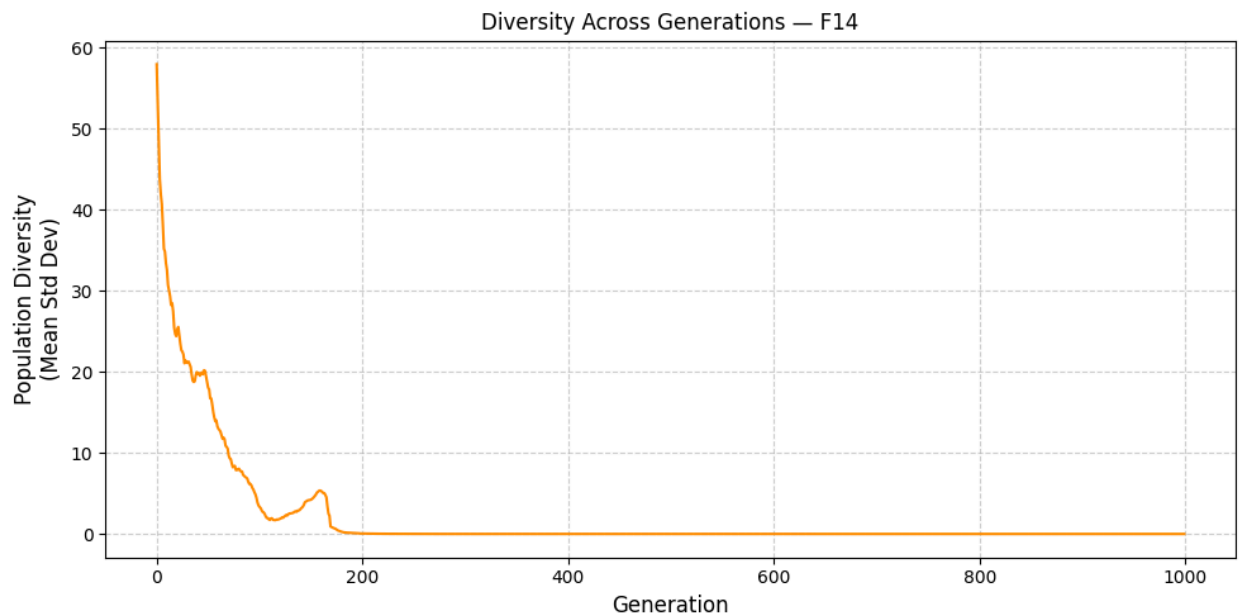
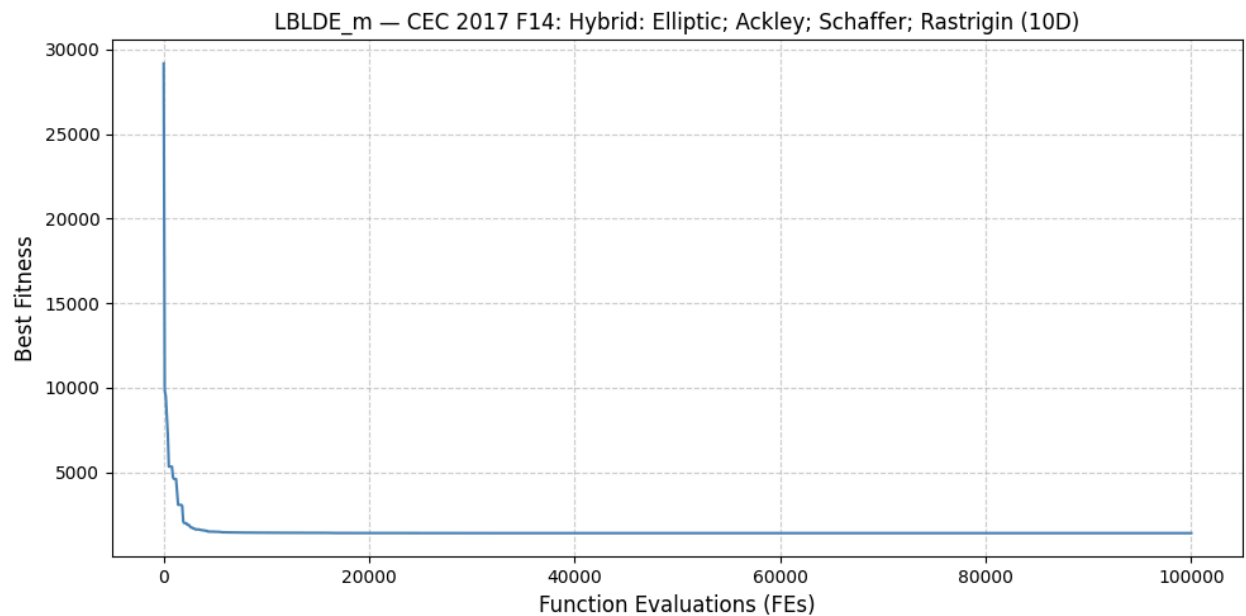
Generation 990: Best = 1.401990e+03 FES = 99100 Diversity = 5.8265e-14

Optimization completed!

Final Best Fitness = 1.401990e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness          : 1.401990e+03
  Global Optimum        : 1.400000e+03
  Error (|best-optimum|) : 1.989918e+00
  Best Solution (5 dim)  : [-46.48782027 -70.77608488 66.34140996 -43.14980833
39.20852308]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F14.png

```
In [25]: # =====  
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)  
# =====  
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock')
```

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LBLDE_m on CEC 2017 F15 – Single Run

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (10D)

Bounds : [-100, 100]

Global optimum : 1500.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.114909e+07 | Diversity = 5.7928e+01

Generation	10:	Best = 6.168391e+03	FES = 1100	Diversity = 3.3983e+01
Generation	20:	Best = 2.219679e+03	FES = 2100	Diversity = 2.5255e+01
Generation	30:	Best = 1.580374e+03	FES = 3100	Diversity = 1.4541e+01
Generation	40:	Best = 1.540833e+03	FES = 4100	Diversity = 7.1437e+00
Generation	50:	Best = 1.533040e+03	FES = 5100	Diversity = 5.5449e+00
Generation	60:	Best = 1.524574e+03	FES = 6100	Diversity = 4.8127e+00
Generation	70:	Best = 1.520500e+03	FES = 7100	Diversity = 4.3565e+00
Generation	80:	Best = 1.517386e+03	FES = 8100	Diversity = 4.3244e+00
Generation	90:	Best = 1.515501e+03	FES = 9100	Diversity = 3.7616e+00
Generation	100:	Best = 1.513310e+03	FES = 10100	Diversity = 2.6807e+00
Generation	110:	Best = 1.513220e+03	FES = 11100	Diversity = 1.7347e+00
Generation	120:	Best = 1.511421e+03	FES = 12100	Diversity = 1.5438e+00
Generation	130:	Best = 1.509779e+03	FES = 13100	Diversity = 1.7149e+00
Generation	140:	Best = 1.508594e+03	FES = 14100	Diversity = 1.8074e+00
Generation	150:	Best = 1.507622e+03	FES = 15100	Diversity = 2.0374e+00
Generation	160:	Best = 1.506087e+03	FES = 16100	Diversity = 2.1809e+00
Generation	170:	Best = 1.505955e+03	FES = 17100	Diversity = 1.9405e+00
Generation	180:	Best = 1.505653e+03	FES = 18100	Diversity = 1.8721e+00
Generation	190:	Best = 1.505511e+03	FES = 19100	Diversity = 1.8857e+00
Generation	200:	Best = 1.505437e+03	FES = 20100	Diversity = 2.0968e+00
Generation	210:	Best = 1.505385e+03	FES = 21100	Diversity = 1.8895e+00
Generation	220:	Best = 1.505364e+03	FES = 22100	Diversity = 1.7859e+00
Generation	230:	Best = 1.505261e+03	FES = 23100	Diversity = 1.6527e+00
Generation	240:	Best = 1.505214e+03	FES = 24100	Diversity = 1.6428e+00
Generation	250:	Best = 1.505211e+03	FES = 25100	Diversity = 1.6398e+00
Generation	260:	Best = 1.505211e+03	FES = 26100	Diversity = 1.5913e+00
Generation	270:	Best = 1.505211e+03	FES = 27100	Diversity = 1.6066e+00
Generation	280:	Best = 1.505205e+03	FES = 28100	Diversity = 1.6335e+00
Generation	290:	Best = 1.505148e+03	FES = 29100	Diversity = 1.6404e+00
Generation	300:	Best = 1.505138e+03	FES = 30100	Diversity = 1.5698e+00
Generation	310:	Best = 1.505083e+03	FES = 31100	Diversity = 1.5164e+00
Generation	320:	Best = 1.505075e+03	FES = 32100	Diversity = 1.5302e+00
Generation	330:	Best = 1.505075e+03	FES = 33100	Diversity = 1.5483e+00
Generation	340:	Best = 1.505059e+03	FES = 34100	Diversity = 1.5239e+00
Generation	350:	Best = 1.505058e+03	FES = 35100	Diversity = 1.5225e+00
Generation	360:	Best = 1.505057e+03	FES = 36100	Diversity = 1.4341e+00
Generation	370:	Best = 1.505057e+03	FES = 37100	Diversity = 1.3498e+00
Generation	380:	Best = 1.505054e+03	FES = 38100	Diversity = 1.2984e+00
Generation	390:	Best = 1.505054e+03	FES = 39100	Diversity = 1.2979e+00
Generation	400:	Best = 1.505054e+03	FES = 40100	Diversity = 1.1952e+00
Generation	410:	Best = 1.505054e+03	FES = 41100	Diversity = 1.1581e+00
Generation	420:	Best = 1.505053e+03	FES = 42100	Diversity = 1.1015e+00
Generation	430:	Best = 1.505053e+03	FES = 43100	Diversity = 1.0657e+00
Generation	440:	Best = 1.505053e+03	FES = 44100	Diversity = 1.0440e+00

Generation	450:	Best = 1.505053e+03	FES = 45100	Diversity = 9.7632e-01
Generation	460:	Best = 1.505009e+03	FES = 46100	Diversity = 5.0423e-01
Generation	470:	Best = 1.504962e+03	FES = 47100	Diversity = 4.2277e-02
Generation	480:	Best = 1.504922e+03	FES = 48100	Diversity = 5.3807e-02
Generation	490:	Best = 1.504863e+03	FES = 49100	Diversity = 5.0924e-02
Generation	500:	Best = 1.504863e+03	FES = 50100	Diversity = 4.9211e-02
Generation	510:	Best = 1.504851e+03	FES = 51100	Diversity = 4.3830e-02
Generation	520:	Best = 1.504833e+03	FES = 52100	Diversity = 4.1930e-02
Generation	530:	Best = 1.504823e+03	FES = 53100	Diversity = 4.0679e-02
Generation	540:	Best = 1.504816e+03	FES = 54100	Diversity = 3.9728e-02
Generation	550:	Best = 1.504816e+03	FES = 55100	Diversity = 3.5852e-02
Generation	560:	Best = 1.504799e+03	FES = 56100	Diversity = 3.4082e-02
Generation	570:	Best = 1.504794e+03	FES = 57100	Diversity = 2.8982e-02
Generation	580:	Best = 1.504787e+03	FES = 58100	Diversity = 2.6086e-02
Generation	590:	Best = 1.504786e+03	FES = 59100	Diversity = 2.1016e-02
Generation	600:	Best = 1.504781e+03	FES = 60100	Diversity = 1.8253e-02
Generation	610:	Best = 1.504769e+03	FES = 61100	Diversity = 1.6767e-02
Generation	620:	Best = 1.504769e+03	FES = 62100	Diversity = 1.7840e-02
Generation	630:	Best = 1.504766e+03	FES = 63100	Diversity = 1.6114e-02
Generation	640:	Best = 1.504765e+03	FES = 64100	Diversity = 1.4408e-02
Generation	650:	Best = 1.504765e+03	FES = 65100	Diversity = 1.4897e-02
Generation	660:	Best = 1.504763e+03	FES = 66100	Diversity = 1.4822e-02
Generation	670:	Best = 1.504751e+03	FES = 67100	Diversity = 1.4583e-02
Generation	680:	Best = 1.504751e+03	FES = 68100	Diversity = 1.4118e-02
Generation	690:	Best = 1.504745e+03	FES = 69100	Diversity = 1.2715e-02
Generation	700:	Best = 1.504745e+03	FES = 70100	Diversity = 1.2901e-02
Generation	710:	Best = 1.504745e+03	FES = 71100	Diversity = 1.2282e-02
Generation	720:	Best = 1.504738e+03	FES = 72100	Diversity = 9.3840e-03
Generation	730:	Best = 1.504738e+03	FES = 73100	Diversity = 8.5996e-03
Generation	740:	Best = 1.504735e+03	FES = 74100	Diversity = 7.9319e-03
Generation	750:	Best = 1.504735e+03	FES = 75100	Diversity = 7.3978e-03
Generation	760:	Best = 1.504734e+03	FES = 76100	Diversity = 6.5034e-03
Generation	770:	Best = 1.504734e+03	FES = 77100	Diversity = 5.6420e-03
Generation	780:	Best = 1.504734e+03	FES = 78100	Diversity = 4.4619e-03
Generation	790:	Best = 1.504734e+03	FES = 79100	Diversity = 4.3447e-03
Generation	800:	Best = 1.504734e+03	FES = 80100	Diversity = 3.8988e-03
Generation	810:	Best = 1.504734e+03	FES = 81100	Diversity = 3.8375e-03
Generation	820:	Best = 1.504734e+03	FES = 82100	Diversity = 3.6848e-03
Generation	830:	Best = 1.504733e+03	FES = 83100	Diversity = 3.4751e-03
Generation	840:	Best = 1.504733e+03	FES = 84100	Diversity = 3.3500e-03
Generation	850:	Best = 1.504733e+03	FES = 85100	Diversity = 3.2942e-03
Generation	860:	Best = 1.504733e+03	FES = 86100	Diversity = 3.0848e-03
Generation	870:	Best = 1.504733e+03	FES = 87100	Diversity = 2.4642e-03
Generation	880:	Best = 1.504733e+03	FES = 88100	Diversity = 1.9144e-03
Generation	890:	Best = 1.504733e+03	FES = 89100	Diversity = 1.7104e-03
Generation	900:	Best = 1.504733e+03	FES = 90100	Diversity = 1.6665e-03
Generation	910:	Best = 1.504733e+03	FES = 91100	Diversity = 1.4840e-03
Generation	920:	Best = 1.504733e+03	FES = 92100	Diversity = 1.3593e-03
Generation	930:	Best = 1.504733e+03	FES = 93100	Diversity = 1.3543e-03
Generation	940:	Best = 1.504733e+03	FES = 94100	Diversity = 1.1805e-03
Generation	950:	Best = 1.504733e+03	FES = 95100	Diversity = 1.0544e-03
Generation	960:	Best = 1.504733e+03	FES = 96100	Diversity = 9.2760e-04
Generation	970:	Best = 1.504733e+03	FES = 97100	Diversity = 9.1511e-04
Generation	980:	Best = 1.504733e+03	FES = 98100	Diversity = 9.2893e-04

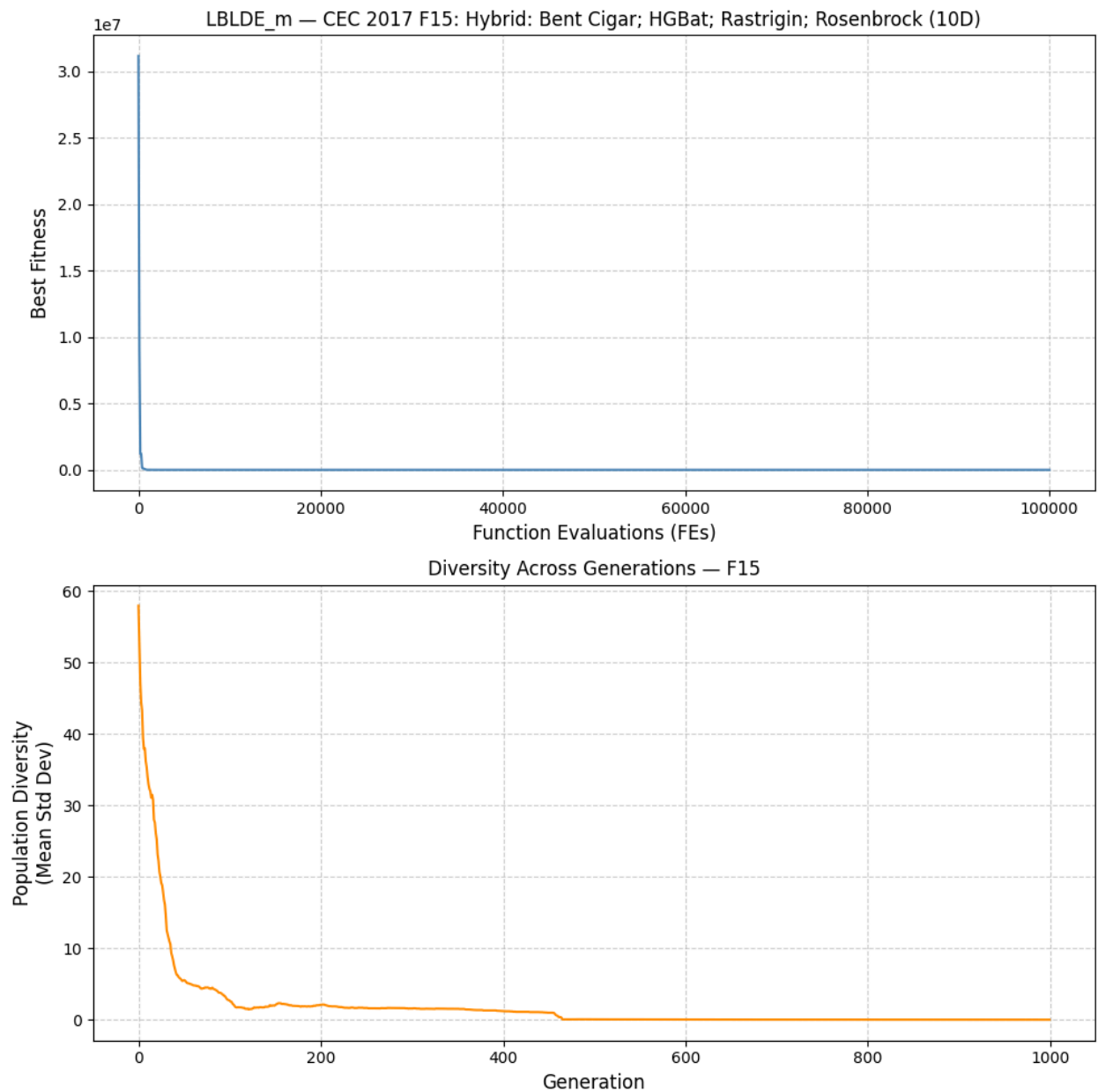
Generation 990: Best = 1.504733e+03 FES = 99100 Diversity = 9.2070e-04

Optimization completed!

Final Best Fitness = 1.504733e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness          : 1.504733e+03
  Global Optimum        : 1.500000e+03
  Error (|best-optimum|) : 4.732737e+00
  Best Solution (5 dim)  : [ 1.20557172  1.38208448  43.98841795 -41.75791148
55.61096107]
  D^(0) (init. diversity): 5.7928e+01
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Graph saved as convergence_F15.png

```
In [26]: # =====  
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)  
# =====  
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

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LBLDE_m on CEC 2017 F16 – Single Run

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Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (10D)

Bounds : [-100, 100]

Global optimum : 1600.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.512498e+03 | Diversity = 5.7928e+01

Generation	10:	Best = 2.113147e+03	FES = 1100	Diversity = 3.3318e+01
Generation	20:	Best = 1.860467e+03	FES = 2100	Diversity = 2.9068e+01
Generation	30:	Best = 1.734070e+03	FES = 3100	Diversity = 2.6609e+01
Generation	40:	Best = 1.668069e+03	FES = 4100	Diversity = 2.3746e+01
Generation	50:	Best = 1.607526e+03	FES = 5100	Diversity = 2.1241e+01
Generation	60:	Best = 1.605668e+03	FES = 6100	Diversity = 1.8504e+01
Generation	70:	Best = 1.605043e+03	FES = 7100	Diversity = 1.5549e+01
Generation	80:	Best = 1.603952e+03	FES = 8100	Diversity = 1.2768e+01
Generation	90:	Best = 1.603543e+03	FES = 9100	Diversity = 8.0003e+00
Generation	100:	Best = 1.603096e+03	FES = 10100	Diversity = 4.2427e+00
Generation	110:	Best = 1.602637e+03	FES = 11100	Diversity = 4.5022e+00
Generation	120:	Best = 1.602463e+03	FES = 12100	Diversity = 2.6593e+00
Generation	130:	Best = 1.602369e+03	FES = 13100	Diversity = 8.4499e-01
Generation	140:	Best = 1.602235e+03	FES = 14100	Diversity = 7.2121e-01
Generation	150:	Best = 1.602197e+03	FES = 15100	Diversity = 6.9636e-01
Generation	160:	Best = 1.602124e+03	FES = 16100	Diversity = 6.8683e-01
Generation	170:	Best = 1.602102e+03	FES = 17100	Diversity = 5.9938e-01
Generation	180:	Best = 1.602096e+03	FES = 18100	Diversity = 4.9598e-01
Generation	190:	Best = 1.602091e+03	FES = 19100	Diversity = 4.6523e-01
Generation	200:	Best = 1.602082e+03	FES = 20100	Diversity = 4.4504e-01
Generation	210:	Best = 1.602080e+03	FES = 21100	Diversity = 3.8971e-01
Generation	220:	Best = 1.602069e+03	FES = 22100	Diversity = 3.7919e-01
Generation	230:	Best = 1.602066e+03	FES = 23100	Diversity = 3.4528e-01
Generation	240:	Best = 1.602064e+03	FES = 24100	Diversity = 3.1720e-01
Generation	250:	Best = 1.602063e+03	FES = 25100	Diversity = 2.9782e-01
Generation	260:	Best = 1.602063e+03	FES = 26100	Diversity = 2.9667e-01
Generation	270:	Best = 1.602062e+03	FES = 27100	Diversity = 2.9054e-01
Generation	280:	Best = 1.602062e+03	FES = 28100	Diversity = 2.7783e-01
Generation	290:	Best = 1.602062e+03	FES = 29100	Diversity = 2.7351e-01
Generation	300:	Best = 1.602062e+03	FES = 30100	Diversity = 2.7197e-01
Generation	310:	Best = 1.602061e+03	FES = 31100	Diversity = 2.6822e-01
Generation	320:	Best = 1.602060e+03	FES = 32100	Diversity = 2.6179e-01
Generation	330:	Best = 1.602059e+03	FES = 33100	Diversity = 2.5582e-01
Generation	340:	Best = 1.602059e+03	FES = 34100	Diversity = 2.5243e-01
Generation	350:	Best = 1.602059e+03	FES = 35100	Diversity = 2.4049e-01
Generation	360:	Best = 1.602059e+03	FES = 36100	Diversity = 2.3638e-01
Generation	370:	Best = 1.602059e+03	FES = 37100	Diversity = 2.2875e-01
Generation	380:	Best = 1.602059e+03	FES = 38100	Diversity = 2.2476e-01
Generation	390:	Best = 1.602058e+03	FES = 39100	Diversity = 2.1969e-01
Generation	400:	Best = 1.602058e+03	FES = 40100	Diversity = 2.1972e-01
Generation	410:	Best = 1.602058e+03	FES = 41100	Diversity = 2.1856e-01
Generation	420:	Best = 1.602058e+03	FES = 42100	Diversity = 2.1699e-01
Generation	430:	Best = 1.602058e+03	FES = 43100	Diversity = 2.1759e-01
Generation	440:	Best = 1.602058e+03	FES = 44100	Diversity = 2.1291e-01

Generation	450:	Best = 1.602058e+03	FES = 45100	Diversity = 2.0958e-01
Generation	460:	Best = 1.602058e+03	FES = 46100	Diversity = 1.9752e-01
Generation	470:	Best = 1.602058e+03	FES = 47100	Diversity = 1.8692e-01
Generation	480:	Best = 1.602058e+03	FES = 48100	Diversity = 1.6419e-01
Generation	490:	Best = 1.602058e+03	FES = 49100	Diversity = 1.5342e-01
Generation	500:	Best = 1.602058e+03	FES = 50100	Diversity = 1.5101e-01
Generation	510:	Best = 1.602058e+03	FES = 51100	Diversity = 1.5102e-01
Generation	520:	Best = 1.602058e+03	FES = 52100	Diversity = 1.4941e-01
Generation	530:	Best = 1.602058e+03	FES = 53100	Diversity = 1.4457e-01
Generation	540:	Best = 1.602058e+03	FES = 54100	Diversity = 6.1393e-02
Generation	550:	Best = 1.602058e+03	FES = 55100	Diversity = 4.8387e-04
Generation	560:	Best = 1.602058e+03	FES = 56100	Diversity = 3.6654e-04
Generation	570:	Best = 1.602058e+03	FES = 57100	Diversity = 2.4896e-04
Generation	580:	Best = 1.602058e+03	FES = 58100	Diversity = 1.7577e-04
Generation	590:	Best = 1.602058e+03	FES = 59100	Diversity = 1.9177e-04
Generation	600:	Best = 1.602058e+03	FES = 60100	Diversity = 2.0742e-04
Generation	610:	Best = 1.602058e+03	FES = 61100	Diversity = 1.2667e-04
Generation	620:	Best = 1.602058e+03	FES = 62100	Diversity = 9.8387e-05
Generation	630:	Best = 1.602058e+03	FES = 63100	Diversity = 8.3393e-05
Generation	640:	Best = 1.602058e+03	FES = 64100	Diversity = 6.8509e-05
Generation	650:	Best = 1.602058e+03	FES = 65100	Diversity = 6.4961e-05
Generation	660:	Best = 1.602058e+03	FES = 66100	Diversity = 4.5992e-05
Generation	670:	Best = 1.602058e+03	FES = 67100	Diversity = 3.6708e-05
Generation	680:	Best = 1.602058e+03	FES = 68100	Diversity = 3.1656e-05
Generation	690:	Best = 1.602058e+03	FES = 69100	Diversity = 2.6052e-05
Generation	700:	Best = 1.602058e+03	FES = 70100	Diversity = 2.2822e-05
Generation	710:	Best = 1.602058e+03	FES = 71100	Diversity = 1.4826e-05
Generation	720:	Best = 1.602058e+03	FES = 72100	Diversity = 1.0012e-05
Generation	730:	Best = 1.602058e+03	FES = 73100	Diversity = 8.7061e-06
Generation	740:	Best = 1.602058e+03	FES = 74100	Diversity = 6.9450e-06
Generation	750:	Best = 1.602058e+03	FES = 75100	Diversity = 5.7892e-06
Generation	760:	Best = 1.602058e+03	FES = 76100	Diversity = 4.7880e-06
Generation	770:	Best = 1.602058e+03	FES = 77100	Diversity = 3.1907e-06
Generation	780:	Best = 1.602058e+03	FES = 78100	Diversity = 1.9339e-06
Generation	790:	Best = 1.602058e+03	FES = 79100	Diversity = 2.2880e-06
Generation	800:	Best = 1.602058e+03	FES = 80100	Diversity = 1.7134e-06
Generation	810:	Best = 1.602058e+03	FES = 81100	Diversity = 1.0548e-06
Generation	820:	Best = 1.602058e+03	FES = 82100	Diversity = 8.7520e-07
Generation	830:	Best = 1.602058e+03	FES = 83100	Diversity = 7.2794e-07
Generation	840:	Best = 1.602058e+03	FES = 84100	Diversity = 7.7371e-07
Generation	850:	Best = 1.602058e+03	FES = 85100	Diversity = 6.9694e-07
Generation	860:	Best = 1.602058e+03	FES = 86100	Diversity = 5.7109e-07
Generation	870:	Best = 1.602058e+03	FES = 87100	Diversity = 3.9614e-07
Generation	880:	Best = 1.602058e+03	FES = 88100	Diversity = 3.6432e-07
Generation	890:	Best = 1.602058e+03	FES = 89100	Diversity = 3.6098e-07
Generation	900:	Best = 1.602058e+03	FES = 90100	Diversity = 2.9707e-07
Generation	910:	Best = 1.602058e+03	FES = 91100	Diversity = 2.6889e-07
Generation	920:	Best = 1.602058e+03	FES = 92100	Diversity = 2.4282e-07
Generation	930:	Best = 1.602058e+03	FES = 93100	Diversity = 1.8479e-07
Generation	940:	Best = 1.602058e+03	FES = 94100	Diversity = 9.1925e-08
Generation	950:	Best = 1.602058e+03	FES = 95100	Diversity = 9.3131e-08
Generation	960:	Best = 1.602058e+03	FES = 96100	Diversity = 8.6172e-08
Generation	970:	Best = 1.602058e+03	FES = 97100	Diversity = 4.9108e-08
Generation	980:	Best = 1.602058e+03	FES = 98100	Diversity = 4.2395e-08

Generation 990: Best = 1.602058e+03 FES = 99100 Diversity = 4.0389e-08

Optimization completed!

Final Best Fitness = 1.602058e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 1.602058e+03

Global Optimum : 1.600000e+03

Error (|best-optimum|) : 2.057710e+00

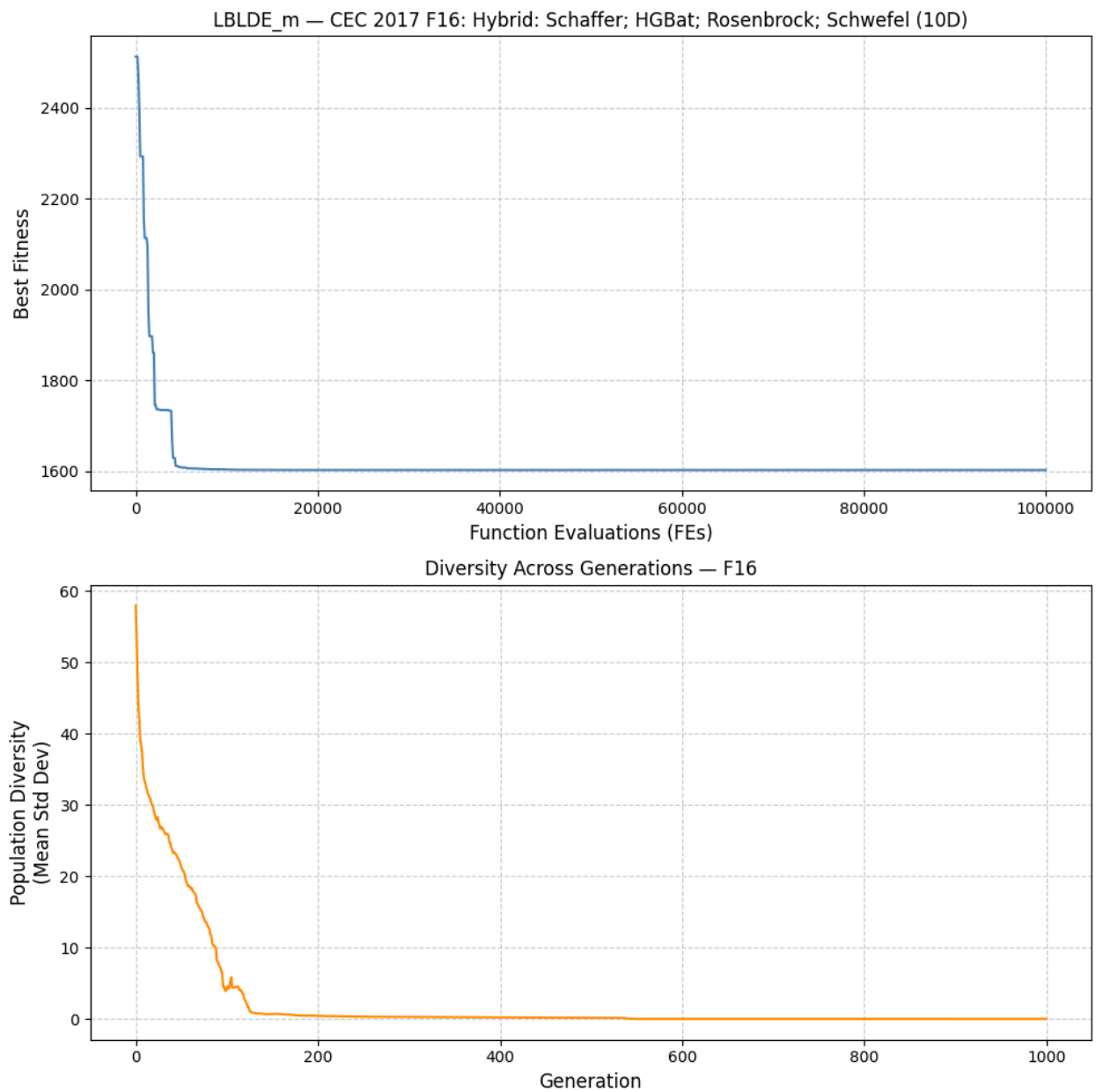
Best Solution (5 dim) : [-4.35296086e-02 6.19829356e+01 4.15366103e+01
-3.23055975e+01

-3.95786266e+01]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F16.png

```
In [27]: # =====
# F17 – Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

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LBLDE_m on CEC 2017 F17 – Single Run

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Function : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 1700.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.471803e+03 | Diversity = 5.7928e+01

Generation 10: Best = 1.864212e+03 FES = 1100 Diversity = 3.3686e+01

Generation 20: Best = 1.783559e+03 FES = 2100 Diversity = 2.7388e+01

Generation 30: Best = 1.757840e+03 FES = 3100 Diversity = 2.3542e+01

Generation 40: Best = 1.731211e+03 FES = 4100 Diversity = 2.0183e+01

Generation 50: Best = 1.728428e+03 FES = 5100 Diversity = 1.7379e+01

Generation 60: Best = 1.725244e+03 FES = 6100 Diversity = 1.4963e+01

Generation 70: Best = 1.719629e+03 FES = 7100 Diversity = 1.3699e+01

Generation 80: Best = 1.706756e+03 FES = 8100 Diversity = 1.1948e+01

Generation 90: Best = 1.703644e+03 FES = 9100 Diversity = 8.7387e+00

Generation 100: Best = 1.703041e+03 FES = 10100 Diversity = 5.0856e+00

Generation 110: Best = 1.702894e+03 FES = 11100 Diversity = 4.3732e+00

Generation 120: Best = 1.702826e+03 FES = 12100 Diversity = 4.2226e+00

Generation 130: Best = 1.702795e+03 FES = 13100 Diversity = 5.9165e-01

Generation 140: Best = 1.702777e+03 FES = 14100 Diversity = 1.3206e-01

Generation 150: Best = 1.702770e+03 FES = 15100 Diversity = 2.8067e-02

Generation 160: Best = 1.702764e+03 FES = 16100 Diversity = 2.0703e-02

Generation 170: Best = 1.702759e+03 FES = 17100 Diversity = 1.4349e-02

Generation 180: Best = 1.702758e+03 FES = 18100 Diversity = 1.2478e-02

Generation 190: Best = 1.702758e+03 FES = 19100 Diversity = 6.4232e-03

Generation 200: Best = 1.702758e+03 FES = 20100 Diversity = 4.0651e-03

Generation 210: Best = 1.702757e+03 FES = 21100 Diversity = 3.1435e-03

Generation 220: Best = 1.702757e+03 FES = 22100 Diversity = 2.7878e-03

Generation 230: Best = 1.702757e+03 FES = 23100 Diversity = 2.2188e-03

Generation 240: Best = 1.702757e+03 FES = 24100 Diversity = 1.9266e-03

Generation 250: Best = 1.702757e+03 FES = 25100 Diversity = 1.2567e-03

Generation 260: Best = 1.702757e+03 FES = 26100 Diversity = 1.1028e-03

Generation 270: Best = 1.702757e+03 FES = 27100 Diversity = 8.4059e-04

Generation 280: Best = 1.702757e+03 FES = 28100 Diversity = 5.8980e-04

Generation 290: Best = 1.702757e+03 FES = 29100 Diversity = 4.1721e-04

Generation 300: Best = 1.702757e+03 FES = 30100 Diversity = 3.2012e-04

Generation 310: Best = 1.702757e+03 FES = 31100 Diversity = 3.1865e-04

Generation 320: Best = 1.702757e+03 FES = 32100 Diversity = 2.3456e-04

Generation 330: Best = 1.702757e+03 FES = 33100 Diversity = 1.6517e-04

Generation 340: Best = 1.702757e+03 FES = 34100 Diversity = 1.9129e-04

Generation 350: Best = 1.702757e+03 FES = 35100 Diversity = 1.4711e-04

Generation 360: Best = 1.702757e+03 FES = 36100 Diversity = 1.1031e-04

Generation 370: Best = 1.702757e+03 FES = 37100 Diversity = 9.2522e-05

Generation 380: Best = 1.702757e+03 FES = 38100 Diversity = 8.1756e-05

Generation 390: Best = 1.702757e+03 FES = 39100 Diversity = 6.9298e-05

Generation 400: Best = 1.702757e+03 FES = 40100 Diversity = 6.3286e-05

Generation 410: Best = 1.702757e+03 FES = 41100 Diversity = 4.6362e-05

Generation 420: Best = 1.702757e+03 FES = 42100 Diversity = 3.3011e-05

Generation 430: Best = 1.702757e+03 FES = 43100 Diversity = 2.9252e-05

Generation	440:	Best = 1.702757e+03	FES = 44100	Diversity = 2.5077e-05
Generation	450:	Best = 1.702757e+03	FES = 45100	Diversity = 2.4078e-05
Generation	460:	Best = 1.702757e+03	FES = 46100	Diversity = 2.1660e-05
Generation	470:	Best = 1.702757e+03	FES = 47100	Diversity = 1.8823e-05
Generation	480:	Best = 1.702757e+03	FES = 48100	Diversity = 1.8277e-05
Generation	490:	Best = 1.702757e+03	FES = 49100	Diversity = 1.4180e-05
Generation	500:	Best = 1.702757e+03	FES = 50100	Diversity = 1.0908e-05
Generation	510:	Best = 1.702757e+03	FES = 51100	Diversity = 9.4224e-06
Generation	520:	Best = 1.702757e+03	FES = 52100	Diversity = 7.5803e-06
Generation	530:	Best = 1.702757e+03	FES = 53100	Diversity = 7.5406e-06
Generation	540:	Best = 1.702757e+03	FES = 54100	Diversity = 7.0150e-06
Generation	550:	Best = 1.702757e+03	FES = 55100	Diversity = 5.8036e-06
Generation	560:	Best = 1.702757e+03	FES = 56100	Diversity = 4.3736e-06
Generation	570:	Best = 1.702757e+03	FES = 57100	Diversity = 4.0217e-06
Generation	580:	Best = 1.702757e+03	FES = 58100	Diversity = 3.4102e-06
Generation	590:	Best = 1.702757e+03	FES = 59100	Diversity = 3.5032e-06
Generation	600:	Best = 1.702757e+03	FES = 60100	Diversity = 2.4952e-06
Generation	610:	Best = 1.702757e+03	FES = 61100	Diversity = 1.5882e-06
Generation	620:	Best = 1.702757e+03	FES = 62100	Diversity = 1.3246e-06
Generation	630:	Best = 1.702757e+03	FES = 63100	Diversity = 9.6407e-07
Generation	640:	Best = 1.702757e+03	FES = 64100	Diversity = 1.2603e-06
Generation	650:	Best = 1.702757e+03	FES = 65100	Diversity = 1.0128e-06
Generation	660:	Best = 1.702757e+03	FES = 66100	Diversity = 7.4259e-07
Generation	670:	Best = 1.702757e+03	FES = 67100	Diversity = 6.3185e-07
Generation	680:	Best = 1.702757e+03	FES = 68100	Diversity = 5.8688e-07
Generation	690:	Best = 1.702757e+03	FES = 69100	Diversity = 5.9140e-07
Generation	700:	Best = 1.702757e+03	FES = 70100	Diversity = 6.5796e-07
Generation	710:	Best = 1.702757e+03	FES = 71100	Diversity = 6.0065e-07
Generation	720:	Best = 1.702757e+03	FES = 72100	Diversity = 5.4520e-07
Generation	730:	Best = 1.702757e+03	FES = 73100	Diversity = 5.6842e-07
Generation	740:	Best = 1.702757e+03	FES = 74100	Diversity = 5.4762e-07
Generation	750:	Best = 1.702757e+03	FES = 75100	Diversity = 6.1644e-07
Generation	760:	Best = 1.702757e+03	FES = 76100	Diversity = 4.6465e-07
Generation	770:	Best = 1.702757e+03	FES = 77100	Diversity = 3.4848e-07
Generation	780:	Best = 1.702757e+03	FES = 78100	Diversity = 2.8195e-07
Generation	790:	Best = 1.702757e+03	FES = 79100	Diversity = 3.2037e-07
Generation	800:	Best = 1.702757e+03	FES = 80100	Diversity = 2.2993e-07
Generation	810:	Best = 1.702757e+03	FES = 81100	Diversity = 2.4622e-07
Generation	820:	Best = 1.702757e+03	FES = 82100	Diversity = 2.1217e-07
Generation	830:	Best = 1.702757e+03	FES = 83100	Diversity = 1.8953e-07
Generation	840:	Best = 1.702757e+03	FES = 84100	Diversity = 1.5241e-07
Generation	850:	Best = 1.702757e+03	FES = 85100	Diversity = 1.4704e-07
Generation	860:	Best = 1.702757e+03	FES = 86100	Diversity = 1.3867e-07
Generation	870:	Best = 1.702757e+03	FES = 87100	Diversity = 1.3220e-07
Generation	880:	Best = 1.702757e+03	FES = 88100	Diversity = 1.2737e-07
Generation	890:	Best = 1.702757e+03	FES = 89100	Diversity = 1.3579e-07
Generation	900:	Best = 1.702757e+03	FES = 90100	Diversity = 1.4502e-07
Generation	910:	Best = 1.702757e+03	FES = 91100	Diversity = 1.3503e-07
Generation	920:	Best = 1.702757e+03	FES = 92100	Diversity = 1.3863e-07
Generation	930:	Best = 1.702757e+03	FES = 93100	Diversity = 1.1736e-07
Generation	940:	Best = 1.702757e+03	FES = 94100	Diversity = 1.0207e-07
Generation	950:	Best = 1.702757e+03	FES = 95100	Diversity = 1.0369e-07
Generation	960:	Best = 1.702757e+03	FES = 96100	Diversity = 9.6085e-08
Generation	970:	Best = 1.702757e+03	FES = 97100	Diversity = 1.0676e-07

Generation 980: Best = 1.702757e+03 FES = 98100 Diversity = 9.6359e-08
Generation 990: Best = 1.702757e+03 FES = 99100 Diversity = 8.7107e-08

Optimization completed!

Final Best Fitness = 1.702757e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 1.702757e+03

Global Optimum : 1.700000e+03

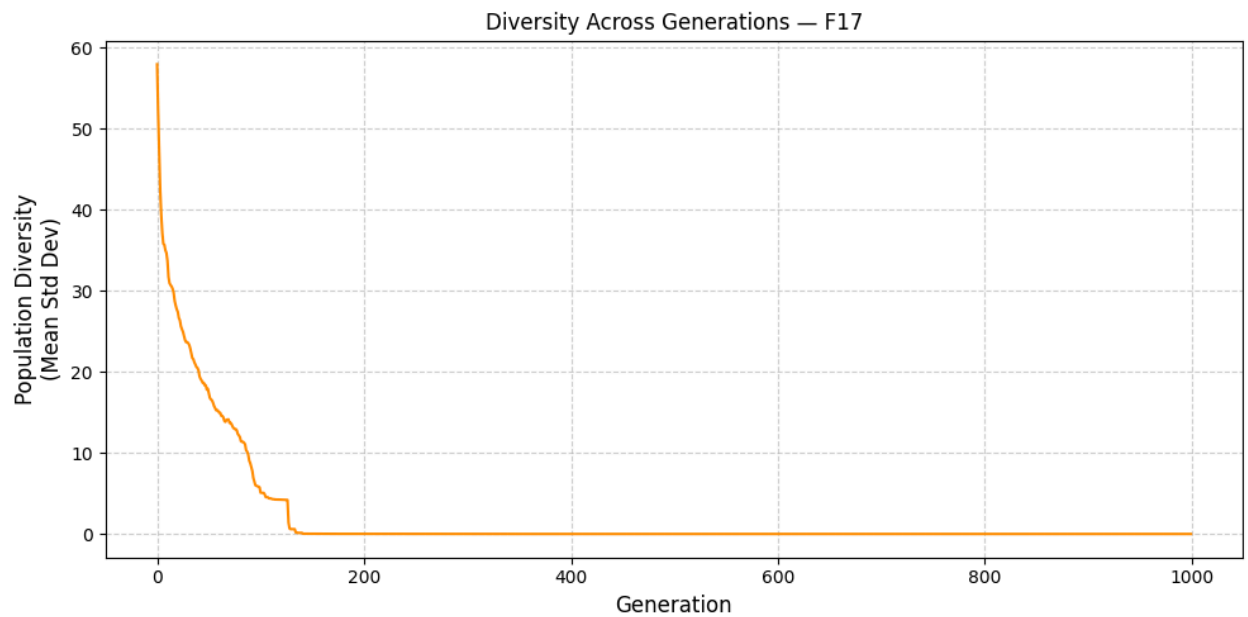
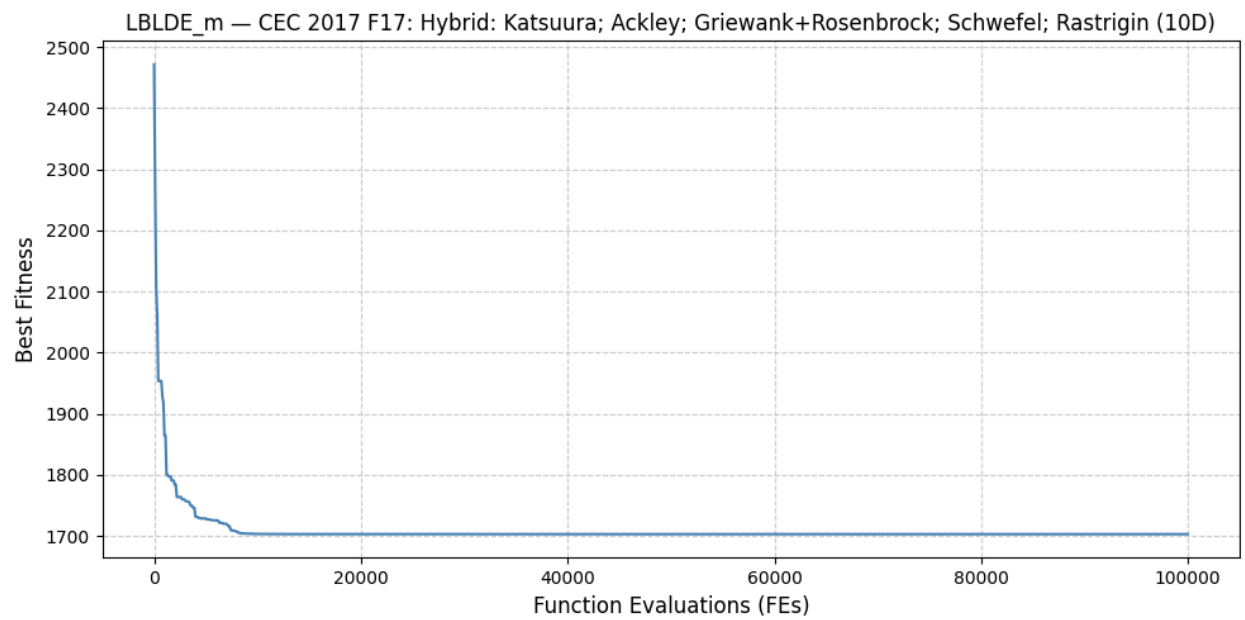
Error (|best-optimum|) : 2.756999e+00

Best Solution (5 dim) : [-46.83959786 -7.16024381 -22.35742795 15.85610283
-53.05441624]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F17.png

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In [28]: # =====
# F18 – Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

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LBLDE_m on CEC 2017 F18 – Single Run

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Function : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (10D)
Bounds : [-100, 100]
Global optimum : 1800.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 6.135676e+08 | Diversity = 5.7928e+01

Generation	10:	Best = 5.240816e+05	FES = 1100	Diversity = 3.1245e+01
Generation	20:	Best = 2.576724e+04	FES = 2100	Diversity = 2.4885e+01
Generation	30:	Best = 4.449429e+03	FES = 3100	Diversity = 1.9727e+01
Generation	40:	Best = 2.662322e+03	FES = 4100	Diversity = 1.9028e+01
Generation	50:	Best = 2.051073e+03	FES = 5100	Diversity = 1.8533e+01
Generation	60:	Best = 1.915077e+03	FES = 6100	Diversity = 1.6462e+01
Generation	70:	Best = 1.862401e+03	FES = 7100	Diversity = 1.6234e+01
Generation	80:	Best = 1.840315e+03	FES = 8100	Diversity = 1.3335e+01
Generation	90:	Best = 1.829597e+03	FES = 9100	Diversity = 1.2831e+01
Generation	100:	Best = 1.819479e+03	FES = 10100	Diversity = 1.1997e+01
Generation	110:	Best = 1.817049e+03	FES = 11100	Diversity = 8.2001e+00
Generation	120:	Best = 1.810196e+03	FES = 12100	Diversity = 5.6680e+00
Generation	130:	Best = 1.807869e+03	FES = 13100	Diversity = 2.3167e+00
Generation	140:	Best = 1.804165e+03	FES = 14100	Diversity = 1.2169e+00
Generation	150:	Best = 1.802234e+03	FES = 15100	Diversity = 1.1561e+00
Generation	160:	Best = 1.801673e+03	FES = 16100	Diversity = 9.9632e-01
Generation	170:	Best = 1.801656e+03	FES = 17100	Diversity = 7.9858e-01
Generation	180:	Best = 1.801572e+03	FES = 18100	Diversity = 6.8313e-01
Generation	190:	Best = 1.801418e+03	FES = 19100	Diversity = 7.7556e-01
Generation	200:	Best = 1.801368e+03	FES = 20100	Diversity = 8.0197e-01
Generation	210:	Best = 1.801362e+03	FES = 21100	Diversity = 8.7768e-01
Generation	220:	Best = 1.801306e+03	FES = 22100	Diversity = 9.1698e-01
Generation	230:	Best = 1.801290e+03	FES = 23100	Diversity = 9.5638e-01
Generation	240:	Best = 1.801282e+03	FES = 24100	Diversity = 9.9189e-01
Generation	250:	Best = 1.801231e+03	FES = 25100	Diversity = 9.9755e-01
Generation	260:	Best = 1.801218e+03	FES = 26100	Diversity = 9.4273e-01
Generation	270:	Best = 1.801213e+03	FES = 27100	Diversity = 8.3015e-01
Generation	280:	Best = 1.801213e+03	FES = 28100	Diversity = 6.0638e-01
Generation	290:	Best = 1.801189e+03	FES = 29100	Diversity = 4.7953e-01
Generation	300:	Best = 1.801186e+03	FES = 30100	Diversity = 4.7862e-01
Generation	310:	Best = 1.801114e+03	FES = 31100	Diversity = 4.7068e-01
Generation	320:	Best = 1.801114e+03	FES = 32100	Diversity = 4.4095e-01
Generation	330:	Best = 1.801114e+03	FES = 33100	Diversity = 4.3356e-01
Generation	340:	Best = 1.801104e+03	FES = 34100	Diversity = 3.6115e-01
Generation	350:	Best = 1.801104e+03	FES = 35100	Diversity = 3.3870e-01
Generation	360:	Best = 1.801103e+03	FES = 36100	Diversity = 3.3104e-01
Generation	370:	Best = 1.801103e+03	FES = 37100	Diversity = 3.2228e-01
Generation	380:	Best = 1.801103e+03	FES = 38100	Diversity = 3.0924e-01
Generation	390:	Best = 1.801103e+03	FES = 39100	Diversity = 3.1681e-01
Generation	400:	Best = 1.801103e+03	FES = 40100	Diversity = 3.1544e-01
Generation	410:	Best = 1.801101e+03	FES = 41100	Diversity = 3.0859e-01
Generation	420:	Best = 1.801101e+03	FES = 42100	Diversity = 3.0140e-01
Generation	430:	Best = 1.801094e+03	FES = 43100	Diversity = 2.9140e-01
Generation	440:	Best = 1.801092e+03	FES = 44100	Diversity = 2.8364e-01

Generation	450:	Best = 1.801086e+03	FES = 45100	Diversity = 2.8212e-01
Generation	460:	Best = 1.801085e+03	FES = 46100	Diversity = 2.7914e-01
Generation	470:	Best = 1.801085e+03	FES = 47100	Diversity = 2.8033e-01
Generation	480:	Best = 1.801085e+03	FES = 48100	Diversity = 2.6829e-01
Generation	490:	Best = 1.801085e+03	FES = 49100	Diversity = 2.6902e-01
Generation	500:	Best = 1.801084e+03	FES = 50100	Diversity = 2.7000e-01
Generation	510:	Best = 1.801084e+03	FES = 51100	Diversity = 2.4336e-01
Generation	520:	Best = 1.801082e+03	FES = 52100	Diversity = 2.3226e-01
Generation	530:	Best = 1.801082e+03	FES = 53100	Diversity = 2.3269e-01
Generation	540:	Best = 1.801082e+03	FES = 54100	Diversity = 2.3122e-01
Generation	550:	Best = 1.801080e+03	FES = 55100	Diversity = 2.3097e-01
Generation	560:	Best = 1.801080e+03	FES = 56100	Diversity = 2.3068e-01
Generation	570:	Best = 1.801079e+03	FES = 57100	Diversity = 2.0047e-01
Generation	580:	Best = 1.801079e+03	FES = 58100	Diversity = 1.9904e-01
Generation	590:	Best = 1.801078e+03	FES = 59100	Diversity = 1.8535e-01
Generation	600:	Best = 1.801078e+03	FES = 60100	Diversity = 1.5635e-01
Generation	610:	Best = 1.801078e+03	FES = 61100	Diversity = 1.4193e-01
Generation	620:	Best = 1.801078e+03	FES = 62100	Diversity = 1.3809e-01
Generation	630:	Best = 1.801077e+03	FES = 63100	Diversity = 1.4001e-01
Generation	640:	Best = 1.801077e+03	FES = 64100	Diversity = 1.3556e-01
Generation	650:	Best = 1.801077e+03	FES = 65100	Diversity = 1.3522e-01
Generation	660:	Best = 1.801077e+03	FES = 66100	Diversity = 1.3491e-01
Generation	670:	Best = 1.801077e+03	FES = 67100	Diversity = 1.3447e-01
Generation	680:	Best = 1.801077e+03	FES = 68100	Diversity = 1.2982e-01
Generation	690:	Best = 1.801077e+03	FES = 69100	Diversity = 1.2653e-01
Generation	700:	Best = 1.801077e+03	FES = 70100	Diversity = 1.1229e-01
Generation	710:	Best = 1.801077e+03	FES = 71100	Diversity = 1.0516e-01
Generation	720:	Best = 1.801077e+03	FES = 72100	Diversity = 9.4554e-02
Generation	730:	Best = 1.801077e+03	FES = 73100	Diversity = 9.0329e-02
Generation	740:	Best = 1.801077e+03	FES = 74100	Diversity = 7.4734e-02
Generation	750:	Best = 1.801077e+03	FES = 75100	Diversity = 7.3108e-02
Generation	760:	Best = 1.801077e+03	FES = 76100	Diversity = 6.4459e-02
Generation	770:	Best = 1.801077e+03	FES = 77100	Diversity = 5.8985e-02
Generation	780:	Best = 1.801077e+03	FES = 78100	Diversity = 4.7925e-02
Generation	790:	Best = 1.801077e+03	FES = 79100	Diversity = 2.5608e-02
Generation	800:	Best = 1.801077e+03	FES = 80100	Diversity = 1.6534e-02
Generation	810:	Best = 1.801077e+03	FES = 81100	Diversity = 1.4385e-02
Generation	820:	Best = 1.801077e+03	FES = 82100	Diversity = 1.3723e-02
Generation	830:	Best = 1.801077e+03	FES = 83100	Diversity = 1.2916e-02
Generation	840:	Best = 1.801077e+03	FES = 84100	Diversity = 1.2642e-02
Generation	850:	Best = 1.801077e+03	FES = 85100	Diversity = 1.1674e-02
Generation	860:	Best = 1.801077e+03	FES = 86100	Diversity = 1.1144e-02
Generation	870:	Best = 1.801077e+03	FES = 87100	Diversity = 1.1075e-02
Generation	880:	Best = 1.801077e+03	FES = 88100	Diversity = 1.0855e-02
Generation	890:	Best = 1.801077e+03	FES = 89100	Diversity = 1.0271e-02
Generation	900:	Best = 1.801077e+03	FES = 90100	Diversity = 9.9101e-03
Generation	910:	Best = 1.801077e+03	FES = 91100	Diversity = 9.8922e-03
Generation	920:	Best = 1.801077e+03	FES = 92100	Diversity = 1.0007e-02
Generation	930:	Best = 1.801077e+03	FES = 93100	Diversity = 9.7072e-03
Generation	940:	Best = 1.801077e+03	FES = 94100	Diversity = 9.6182e-03
Generation	950:	Best = 1.801077e+03	FES = 95100	Diversity = 9.3428e-03
Generation	960:	Best = 1.801077e+03	FES = 96100	Diversity = 9.0192e-03
Generation	970:	Best = 1.801077e+03	FES = 97100	Diversity = 8.8959e-03
Generation	980:	Best = 1.801077e+03	FES = 98100	Diversity = 8.9225e-03

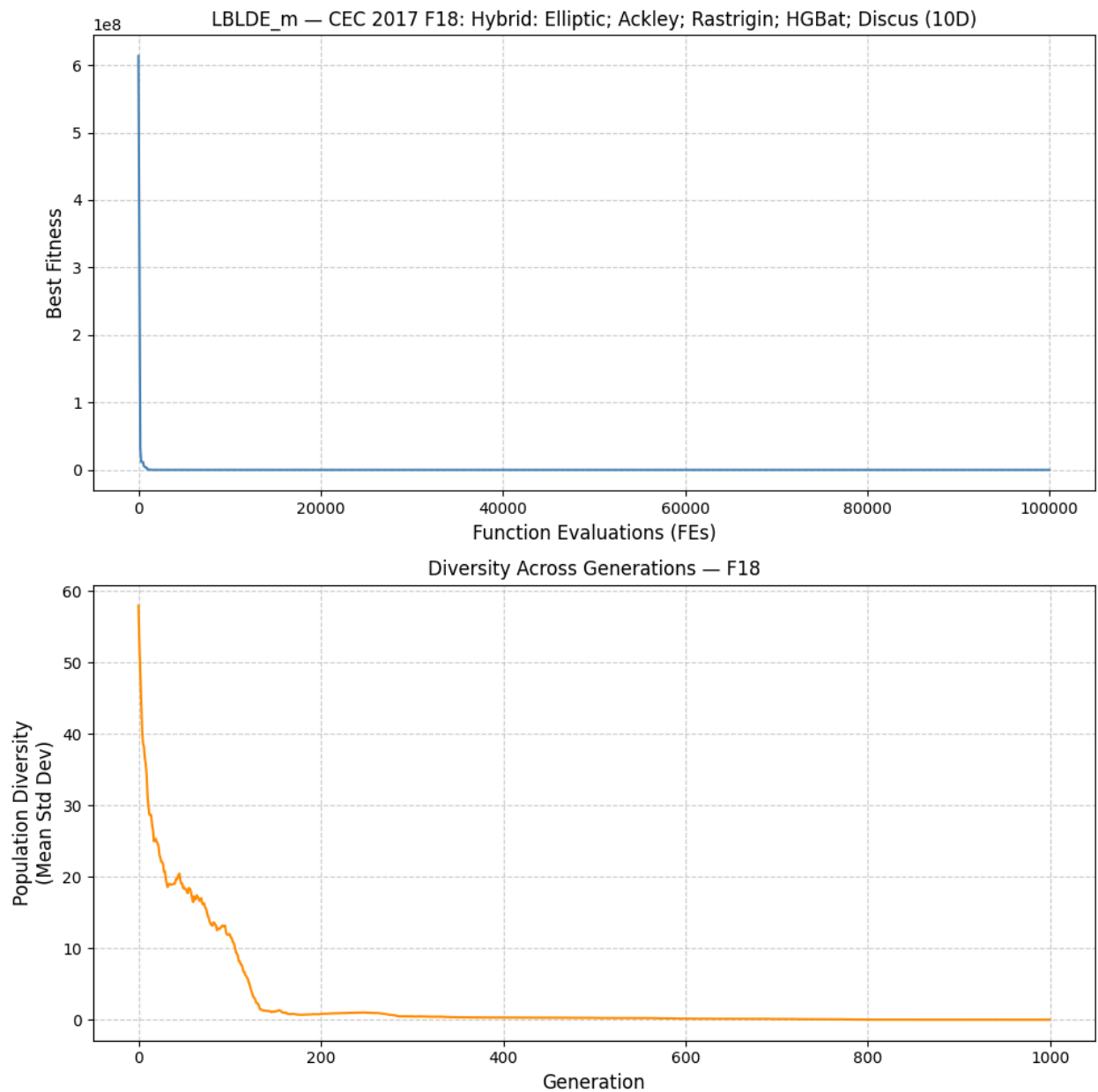
Generation 990: Best = 1.801077e+03 FES = 99100 Diversity = 8.7507e-03

Optimization completed!

Final Best Fitness = 1.801077e+03

Total Function Evaluations = 100000

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Results:
  Best Fitness          : 1.801077e+03
  Global Optimum        : 1.800000e+03
  Error (|best-optimum|) : 1.077054e+00
  Best Solution (5 dim)  : [ 18.15733447  1.84457212  21.4772699  -54.39927714
-11.41493853]
  D^(0) (init. diversity): 5.7928e+01
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```



Graph saved as convergence_F18.png

```
In [29]: # =====  
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst  
# =====  
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

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LBLDE_m on CEC 2017 F19 – Single Run

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Function : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (10D)

Bounds : [-100, 100]

Global optimum : 1900.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.530389e+08 | Diversity = 5.7928e+01

Generation 10: Best = 1.034464e+04 FES = 1100 Diversity = 3.1511e+01

Generation 20: Best = 2.495840e+03 FES = 2100 Diversity = 2.0336e+01

Generation 30: Best = 1.965970e+03 FES = 3100 Diversity = 1.3400e+01

Generation 40: Best = 1.937925e+03 FES = 4100 Diversity = 1.1468e+01

Generation 50: Best = 1.925879e+03 FES = 5100 Diversity = 1.2359e+01

Generation 60: Best = 1.914031e+03 FES = 6100 Diversity = 1.3038e+01

Generation 70: Best = 1.911233e+03 FES = 7100 Diversity = 1.2436e+01

Generation 80: Best = 1.907288e+03 FES = 8100 Diversity = 8.6172e+00

Generation 90: Best = 1.905855e+03 FES = 9100 Diversity = 5.3112e+00

Generation 100: Best = 1.903744e+03 FES = 10100 Diversity = 4.1195e+00

Generation 110: Best = 1.902662e+03 FES = 11100 Diversity = 4.4843e+00

Generation 120: Best = 1.902394e+03 FES = 12100 Diversity = 4.5093e+00

Generation 130: Best = 1.902164e+03 FES = 13100 Diversity = 1.9587e+00

Generation 140: Best = 1.902066e+03 FES = 14100 Diversity = 1.1823e+00

Generation 150: Best = 1.902049e+03 FES = 15100 Diversity = 8.7742e-01

Generation 160: Best = 1.902026e+03 FES = 16100 Diversity = 7.4039e-01

Generation 170: Best = 1.902021e+03 FES = 17100 Diversity = 6.0898e-01

Generation 180: Best = 1.902015e+03 FES = 18100 Diversity = 5.4100e-01

Generation 190: Best = 1.902010e+03 FES = 19100 Diversity = 4.9198e-01

Generation 200: Best = 1.902008e+03 FES = 20100 Diversity = 4.6828e-01

Generation 210: Best = 1.902008e+03 FES = 21100 Diversity = 4.0845e-01

Generation 220: Best = 1.902003e+03 FES = 22100 Diversity = 3.7775e-01

Generation 230: Best = 1.902002e+03 FES = 23100 Diversity = 3.5676e-01

Generation 240: Best = 1.902002e+03 FES = 24100 Diversity = 3.4303e-01

Generation 250: Best = 1.901996e+03 FES = 25100 Diversity = 3.1328e-01

Generation 260: Best = 1.901995e+03 FES = 26100 Diversity = 3.0270e-01

Generation 270: Best = 1.901994e+03 FES = 27100 Diversity = 2.9538e-01

Generation 280: Best = 1.901994e+03 FES = 28100 Diversity = 2.8870e-01

Generation 290: Best = 1.901993e+03 FES = 29100 Diversity = 2.7976e-01

Generation 300: Best = 1.901993e+03 FES = 30100 Diversity = 2.7096e-01

Generation 310: Best = 1.901993e+03 FES = 31100 Diversity = 2.6861e-01

Generation 320: Best = 1.901993e+03 FES = 32100 Diversity = 2.6918e-01

Generation 330: Best = 1.901993e+03 FES = 33100 Diversity = 2.6713e-01

Generation 340: Best = 1.901993e+03 FES = 34100 Diversity = 2.6744e-01

Generation 350: Best = 1.901993e+03 FES = 35100 Diversity = 2.6227e-01

Generation 360: Best = 1.901993e+03 FES = 36100 Diversity = 2.5566e-01

Generation 370: Best = 1.901993e+03 FES = 37100 Diversity = 2.4741e-01

Generation 380: Best = 1.901993e+03 FES = 38100 Diversity = 2.5035e-01

Generation 390: Best = 1.901993e+03 FES = 39100 Diversity = 2.5357e-01

Generation 400: Best = 1.901993e+03 FES = 40100 Diversity = 2.4415e-01

Generation 410: Best = 1.901993e+03 FES = 41100 Diversity = 2.4395e-01

Generation 420: Best = 1.901993e+03 FES = 42100 Diversity = 2.4242e-01

Generation 430: Best = 1.901993e+03 FES = 43100 Diversity = 2.4162e-01

Generation	440:	Best = 1.901993e+03	FES = 44100	Diversity = 2.4104e-01
Generation	450:	Best = 1.901993e+03	FES = 45100	Diversity = 2.3987e-01
Generation	460:	Best = 1.901993e+03	FES = 46100	Diversity = 2.4042e-01
Generation	470:	Best = 1.901993e+03	FES = 47100	Diversity = 2.3966e-01
Generation	480:	Best = 1.901993e+03	FES = 48100	Diversity = 2.3838e-01
Generation	490:	Best = 1.901991e+03	FES = 49100	Diversity = 2.3309e-01
Generation	500:	Best = 1.901991e+03	FES = 50100	Diversity = 2.2892e-01
Generation	510:	Best = 1.901991e+03	FES = 51100	Diversity = 2.1519e-01
Generation	520:	Best = 1.901991e+03	FES = 52100	Diversity = 2.1188e-01
Generation	530:	Best = 1.901991e+03	FES = 53100	Diversity = 2.1621e-01
Generation	540:	Best = 1.901991e+03	FES = 54100	Diversity = 2.1598e-01
Generation	550:	Best = 1.901991e+03	FES = 55100	Diversity = 2.1508e-01
Generation	560:	Best = 1.901991e+03	FES = 56100	Diversity = 2.0588e-01
Generation	570:	Best = 1.901991e+03	FES = 57100	Diversity = 2.0145e-01
Generation	580:	Best = 1.901991e+03	FES = 58100	Diversity = 2.0132e-01
Generation	590:	Best = 1.901991e+03	FES = 59100	Diversity = 2.0169e-01
Generation	600:	Best = 1.901991e+03	FES = 60100	Diversity = 2.0083e-01
Generation	610:	Best = 1.901991e+03	FES = 61100	Diversity = 2.0030e-01
Generation	620:	Best = 1.901991e+03	FES = 62100	Diversity = 1.9346e-01
Generation	630:	Best = 1.901991e+03	FES = 63100	Diversity = 1.9280e-01
Generation	640:	Best = 1.901991e+03	FES = 64100	Diversity = 1.9233e-01
Generation	650:	Best = 1.901991e+03	FES = 65100	Diversity = 1.9051e-01
Generation	660:	Best = 1.901991e+03	FES = 66100	Diversity = 1.9052e-01
Generation	670:	Best = 1.901991e+03	FES = 67100	Diversity = 1.9087e-01
Generation	680:	Best = 1.901991e+03	FES = 68100	Diversity = 1.9087e-01
Generation	690:	Best = 1.901991e+03	FES = 69100	Diversity = 1.8901e-01
Generation	700:	Best = 1.901991e+03	FES = 70100	Diversity = 1.8395e-01
Generation	710:	Best = 1.901991e+03	FES = 71100	Diversity = 1.7815e-01
Generation	720:	Best = 1.901991e+03	FES = 72100	Diversity = 1.7396e-01
Generation	730:	Best = 1.901991e+03	FES = 73100	Diversity = 1.7325e-01
Generation	740:	Best = 1.901991e+03	FES = 74100	Diversity = 1.7042e-01
Generation	750:	Best = 1.901991e+03	FES = 75100	Diversity = 1.6438e-01
Generation	760:	Best = 1.901991e+03	FES = 76100	Diversity = 1.6375e-01
Generation	770:	Best = 1.901991e+03	FES = 77100	Diversity = 1.6303e-01
Generation	780:	Best = 1.901991e+03	FES = 78100	Diversity = 1.5599e-01
Generation	790:	Best = 1.901991e+03	FES = 79100	Diversity = 1.5489e-01
Generation	800:	Best = 1.901991e+03	FES = 80100	Diversity = 1.5334e-01
Generation	810:	Best = 1.901991e+03	FES = 81100	Diversity = 1.4502e-01
Generation	820:	Best = 1.901991e+03	FES = 82100	Diversity = 1.4502e-01
Generation	830:	Best = 1.901991e+03	FES = 83100	Diversity = 1.4362e-01
Generation	840:	Best = 1.901991e+03	FES = 84100	Diversity = 1.4362e-01
Generation	850:	Best = 1.901991e+03	FES = 85100	Diversity = 1.4362e-01
Generation	860:	Best = 1.901991e+03	FES = 86100	Diversity = 1.4170e-01
Generation	870:	Best = 1.901991e+03	FES = 87100	Diversity = 1.4171e-01
Generation	880:	Best = 1.901991e+03	FES = 88100	Diversity = 1.4172e-01
Generation	890:	Best = 1.901991e+03	FES = 89100	Diversity = 2.2412e-02
Generation	900:	Best = 1.901991e+03	FES = 90100	Diversity = 8.2255e-05
Generation	910:	Best = 1.901991e+03	FES = 91100	Diversity = 2.8132e-05
Generation	920:	Best = 1.901991e+03	FES = 92100	Diversity = 1.1748e-05
Generation	930:	Best = 1.901991e+03	FES = 93100	Diversity = 8.1056e-06
Generation	940:	Best = 1.901991e+03	FES = 94100	Diversity = 6.2261e-06
Generation	950:	Best = 1.901991e+03	FES = 95100	Diversity = 6.1867e-06
Generation	960:	Best = 1.901991e+03	FES = 96100	Diversity = 4.8663e-06
Generation	970:	Best = 1.901991e+03	FES = 97100	Diversity = 4.3359e-06

Generation 980: Best = 1.901991e+03 FES = 98100 Diversity = 4.1016e-06
Generation 990: Best = 1.901991e+03 FES = 99100 Diversity = 3.0357e-06

Optimization completed!

Final Best Fitness = 1.901991e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 1.901991e+03

Global Optimum : 1.900000e+03

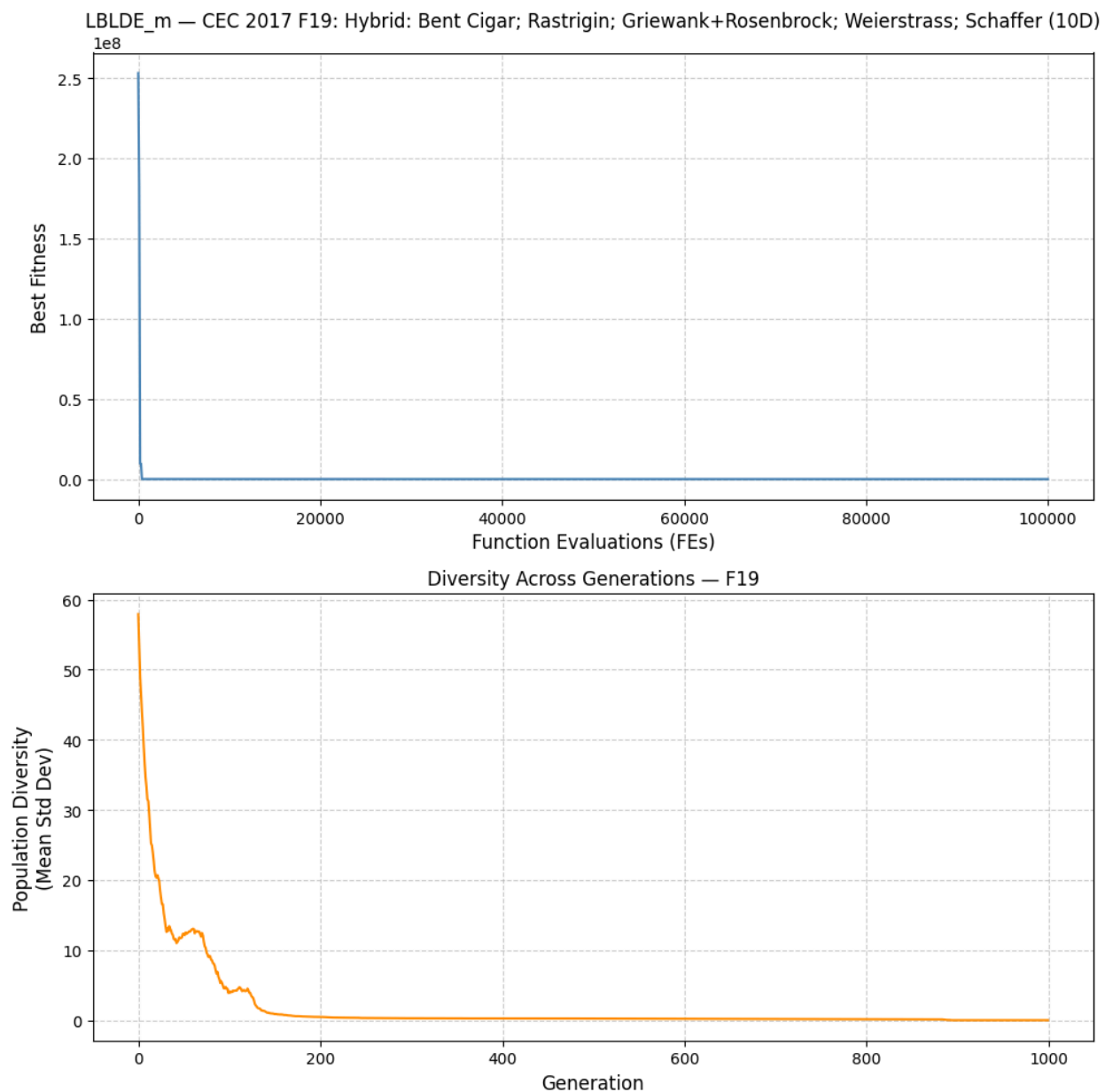
Error (|best-optimum|) : 1.991105e+00

Best Solution (5 dim) : [79.28581829 4.01094764 -35.04044539 35.23712245
25.83828843]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F19.png

```
In [30]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

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LBLDE_m on CEC 2017 F20 – Single Run

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Function : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer (10D)

Bounds : [-100, 100]

Global optimum : 2000.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.391561e+03 | Diversity = 5.7928e+01

Generation 10: Best = 2.216701e+03 FES = 1100 Diversity = 4.1789e+01

Generation 20: Best = 2.076062e+03 FES = 2100 Diversity = 2.9336e+01

Generation 30: Best = 2.074336e+03 FES = 3100 Diversity = 1.9178e+01

Generation 40: Best = 2.052052e+03 FES = 4100 Diversity = 1.2211e+01

Generation 50: Best = 2.044088e+03 FES = 5100 Diversity = 9.2983e+00

Generation 60: Best = 2.040549e+03 FES = 6100 Diversity = 5.8560e+00

Generation 70: Best = 2.034269e+03 FES = 7100 Diversity = 4.8810e+00

Generation 80: Best = 2.030512e+03 FES = 8100 Diversity = 5.9967e+00

Generation 90: Best = 2.027621e+03 FES = 9100 Diversity = 6.3430e+00

Generation 100: Best = 2.026788e+03 FES = 10100 Diversity = 2.4131e+00

Generation 110: Best = 2.007727e+03 FES = 11100 Diversity = 9.7128e-01

Generation 120: Best = 2.006342e+03 FES = 12100 Diversity = 7.5019e-01

Generation 130: Best = 2.005847e+03 FES = 13100 Diversity = 8.2660e-02

Generation 140: Best = 2.005667e+03 FES = 14100 Diversity = 2.6058e-02

Generation 150: Best = 2.005620e+03 FES = 15100 Diversity = 1.5572e-02

Generation 160: Best = 2.005607e+03 FES = 16100 Diversity = 9.0965e-03

Generation 170: Best = 2.005602e+03 FES = 17100 Diversity = 4.4369e-03

Generation 180: Best = 2.005600e+03 FES = 18100 Diversity = 3.0553e-03

Generation 190: Best = 2.005599e+03 FES = 19100 Diversity = 1.6210e-03

Generation 200: Best = 2.005599e+03 FES = 20100 Diversity = 5.8037e-04

Generation 210: Best = 2.005599e+03 FES = 21100 Diversity = 2.5481e-04

Generation 220: Best = 2.005599e+03 FES = 22100 Diversity = 1.0885e-04

Generation 230: Best = 2.005599e+03 FES = 23100 Diversity = 5.9721e-05

Generation 240: Best = 2.005599e+03 FES = 24100 Diversity = 2.5786e-05

Generation 250: Best = 2.005599e+03 FES = 25100 Diversity = 1.4637e-05

Generation 260: Best = 2.005599e+03 FES = 26100 Diversity = 7.0912e-06

Generation 270: Best = 2.005599e+03 FES = 27100 Diversity = 2.7426e-06

Generation 280: Best = 2.005599e+03 FES = 28100 Diversity = 1.5237e-06

Generation 290: Best = 2.005599e+03 FES = 29100 Diversity = 4.5695e-07

Generation 300: Best = 2.005599e+03 FES = 30100 Diversity = 2.2718e-07

Generation 310: Best = 2.005599e+03 FES = 31100 Diversity = 1.2529e-07

Generation 320: Best = 2.005599e+03 FES = 32100 Diversity = 7.6238e-08

Generation 330: Best = 2.005599e+03 FES = 33100 Diversity = 6.4351e-08

Generation 340: Best = 2.005599e+03 FES = 34100 Diversity = 4.4512e-08

Generation 350: Best = 2.005599e+03 FES = 35100 Diversity = 2.8220e-08

Generation 360: Best = 2.005599e+03 FES = 36100 Diversity = 1.6572e-08

Generation 370: Best = 2.005599e+03 FES = 37100 Diversity = 1.1411e-08

Generation 380: Best = 2.005599e+03 FES = 38100 Diversity = 8.0836e-09

Generation 390: Best = 2.005599e+03 FES = 39100 Diversity = 6.9622e-09

Generation 400: Best = 2.005599e+03 FES = 40100 Diversity = 3.9832e-09

Generation 410: Best = 2.005599e+03 FES = 41100 Diversity = 5.2470e-09

Generation 420: Best = 2.005599e+03 FES = 42100 Diversity = 2.7294e-09

Generation 430: Best = 2.005599e+03 FES = 43100 Diversity = 2.4862e-09

Generation	440:	Best = 2.005599e+03	FES = 44100	Diversity = 2.0851e-09
Generation	450:	Best = 2.005599e+03	FES = 45100	Diversity = 8.1266e-10
Generation	460:	Best = 2.005599e+03	FES = 46100	Diversity = 2.1614e-10
Generation	470:	Best = 2.005599e+03	FES = 47100	Diversity = 7.4892e-11
Generation	480:	Best = 2.005599e+03	FES = 48100	Diversity = 7.9129e-11
Generation	490:	Best = 2.005599e+03	FES = 49100	Diversity = 7.5054e-11
Generation	500:	Best = 2.005599e+03	FES = 50100	Diversity = 5.8255e-11
Generation	510:	Best = 2.005599e+03	FES = 51100	Diversity = 6.6006e-11
Generation	520:	Best = 2.005599e+03	FES = 52100	Diversity = 6.8314e-11
Generation	530:	Best = 2.005599e+03	FES = 53100	Diversity = 6.3517e-11
Generation	540:	Best = 2.005599e+03	FES = 54100	Diversity = 5.4526e-11
Generation	550:	Best = 2.005599e+03	FES = 55100	Diversity = 5.1695e-11
Generation	560:	Best = 2.005599e+03	FES = 56100	Diversity = 5.3236e-11
Generation	570:	Best = 2.005599e+03	FES = 57100	Diversity = 5.0379e-11
Generation	580:	Best = 2.005599e+03	FES = 58100	Diversity = 4.8587e-11
Generation	590:	Best = 2.005599e+03	FES = 59100	Diversity = 4.9221e-11
Generation	600:	Best = 2.005599e+03	FES = 60100	Diversity = 4.9118e-11
Generation	610:	Best = 2.005599e+03	FES = 61100	Diversity = 4.9221e-11
Generation	620:	Best = 2.005599e+03	FES = 62100	Diversity = 4.4092e-14
Generation	630:	Best = 2.005599e+03	FES = 63100	Diversity = 4.3832e-14
Generation	640:	Best = 2.005599e+03	FES = 64100	Diversity = 4.3832e-14
Generation	650:	Best = 2.005599e+03	FES = 65100	Diversity = 4.3832e-14
Generation	660:	Best = 2.005599e+03	FES = 66100	Diversity = 4.3832e-14
Generation	670:	Best = 2.005599e+03	FES = 67100	Diversity = 4.3832e-14
Generation	680:	Best = 2.005599e+03	FES = 68100	Diversity = 4.3832e-14
Generation	690:	Best = 2.005599e+03	FES = 69100	Diversity = 4.3832e-14
Generation	700:	Best = 2.005599e+03	FES = 70100	Diversity = 4.3832e-14
Generation	710:	Best = 2.005599e+03	FES = 71100	Diversity = 4.3832e-14
Generation	720:	Best = 2.005599e+03	FES = 72100	Diversity = 4.3832e-14
Generation	730:	Best = 2.005599e+03	FES = 73100	Diversity = 4.3832e-14
Generation	740:	Best = 2.005599e+03	FES = 74100	Diversity = 4.3832e-14
Generation	750:	Best = 2.005599e+03	FES = 75100	Diversity = 4.3832e-14
Generation	760:	Best = 2.005599e+03	FES = 76100	Diversity = 4.3832e-14
Generation	770:	Best = 2.005599e+03	FES = 77100	Diversity = 4.3832e-14
Generation	780:	Best = 2.005599e+03	FES = 78100	Diversity = 4.3832e-14
Generation	790:	Best = 2.005599e+03	FES = 79100	Diversity = 4.3832e-14
Generation	800:	Best = 2.005599e+03	FES = 80100	Diversity = 4.3832e-14
Generation	810:	Best = 2.005599e+03	FES = 81100	Diversity = 4.3832e-14
Generation	820:	Best = 2.005599e+03	FES = 82100	Diversity = 4.3832e-14
Generation	830:	Best = 2.005599e+03	FES = 83100	Diversity = 4.3832e-14
Generation	840:	Best = 2.005599e+03	FES = 84100	Diversity = 4.3832e-14
Generation	850:	Best = 2.005599e+03	FES = 85100	Diversity = 4.3832e-14
Generation	860:	Best = 2.005599e+03	FES = 86100	Diversity = 4.3832e-14
Generation	870:	Best = 2.005599e+03	FES = 87100	Diversity = 4.3832e-14
Generation	880:	Best = 2.005599e+03	FES = 88100	Diversity = 4.3832e-14
Generation	890:	Best = 2.005599e+03	FES = 89100	Diversity = 4.3832e-14
Generation	900:	Best = 2.005599e+03	FES = 90100	Diversity = 4.3832e-14
Generation	910:	Best = 2.005599e+03	FES = 91100	Diversity = 4.3832e-14
Generation	920:	Best = 2.005599e+03	FES = 92100	Diversity = 4.3832e-14
Generation	930:	Best = 2.005599e+03	FES = 93100	Diversity = 4.3832e-14
Generation	940:	Best = 2.005599e+03	FES = 94100	Diversity = 4.3832e-14
Generation	950:	Best = 2.005599e+03	FES = 95100	Diversity = 4.3832e-14
Generation	960:	Best = 2.005599e+03	FES = 96100	Diversity = 4.3832e-14
Generation	970:	Best = 2.005599e+03	FES = 97100	Diversity = 4.3832e-14

Generation 980: Best = 2.005599e+03 FES = 98100 Diversity = 4.3832e-14
Generation 990: Best = 2.005599e+03 FES = 99100 Diversity = 4.3832e-14

Optimization completed!

Final Best Fitness = 2.005599e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 2.005599e+03

Global Optimum : 2.000000e+03

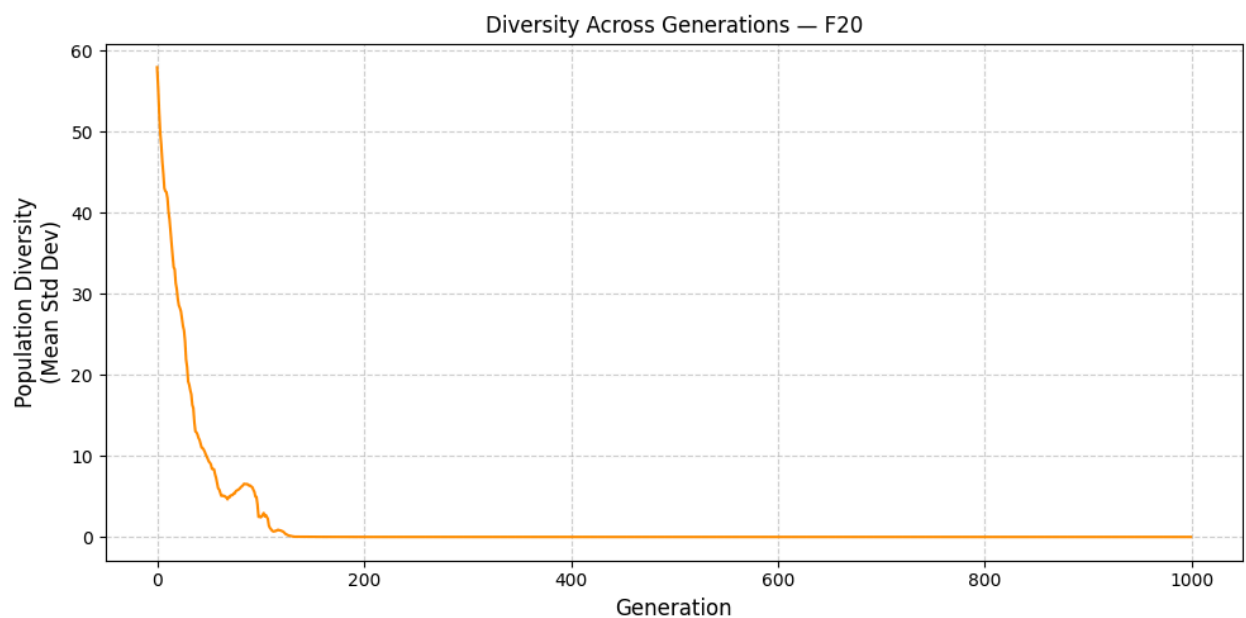
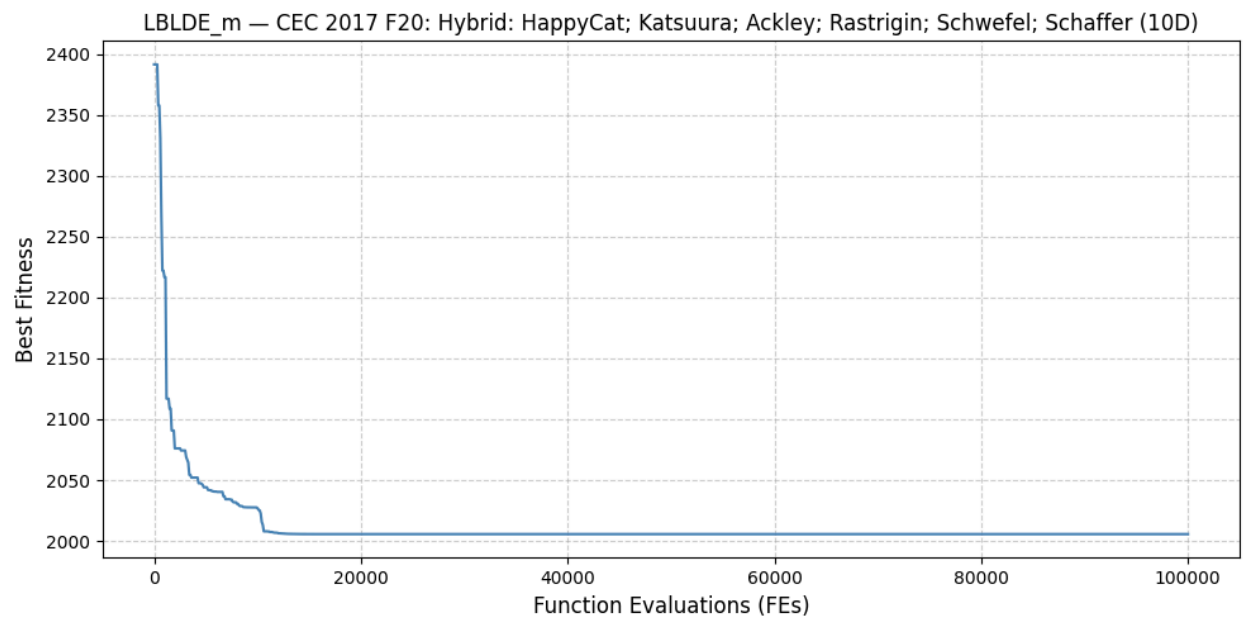
Error (|best-optimum|) : 5.599194e+00

Best Solution (5 dim) : [-30.59057282 -21.23048973 -3.23188763 61.04536957
85.63534254]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [31]: # =====
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

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LBLDE_m on CEC 2017 F21 – Single Run

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Function : Composition: Rosenbrock; Elliptic; Rastrigin (10D)

Bounds : [-100, 100]

Global optimum : 2100.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 2.446690e+03 | Diversity = 5.7928e+01

Generation 10: Best = 2.346661e+03 FES = 1100 Diversity = 2.9860e+01

Generation 20: Best = 2.346661e+03 FES = 2100 Diversity = 2.1916e+01

Generation 30: Best = 2.330556e+03 FES = 3100 Diversity = 2.0424e+01

Generation 40: Best = 2.293271e+03 FES = 4100 Diversity = 1.9067e+01

Generation 50: Best = 2.268566e+03 FES = 5100 Diversity = 1.8246e+01

Generation 60: Best = 2.268566e+03 FES = 6100 Diversity = 2.1490e+01

Generation 70: Best = 2.238438e+03 FES = 7100 Diversity = 2.2489e+01

Generation 80: Best = 2.237637e+03 FES = 8100 Diversity = 2.5641e+01

Generation 90: Best = 2.229023e+03 FES = 9100 Diversity = 3.0437e+01

Generation 100: Best = 2.220101e+03 FES = 10100 Diversity = 3.0112e+01

Generation 110: Best = 2.215737e+03 FES = 11100 Diversity = 5.2176e+00

Generation 120: Best = 2.210736e+03 FES = 12100 Diversity = 5.8254e+00

Generation 130: Best = 2.208991e+03 FES = 13100 Diversity = 3.6856e+00

Generation 140: Best = 2.207689e+03 FES = 14100 Diversity = 3.6144e+00

Generation 150: Best = 2.206315e+03 FES = 15100 Diversity = 3.5689e+00

Generation 160: Best = 2.205073e+03 FES = 16100 Diversity = 4.1733e+00

Generation 170: Best = 2.204330e+03 FES = 17100 Diversity = 2.8674e+00

Generation 180: Best = 2.203855e+03 FES = 18100 Diversity = 2.8499e+00

Generation 190: Best = 2.203445e+03 FES = 19100 Diversity = 2.7040e+00

Generation 200: Best = 2.202899e+03 FES = 20100 Diversity = 2.4254e+00

Generation 210: Best = 2.202457e+03 FES = 21100 Diversity = 2.3078e+00

Generation 220: Best = 2.201974e+03 FES = 22100 Diversity = 1.8598e+00

Generation 230: Best = 2.201860e+03 FES = 23100 Diversity = 1.5354e+00

Generation 240: Best = 2.201511e+03 FES = 24100 Diversity = 1.3011e+00

Generation 250: Best = 2.201404e+03 FES = 25100 Diversity = 1.0008e+00

Generation 260: Best = 2.201308e+03 FES = 26100 Diversity = 8.9590e-01

Generation 270: Best = 2.201277e+03 FES = 27100 Diversity = 7.5162e-01

Generation 280: Best = 2.201175e+03 FES = 28100 Diversity = 5.4744e-01

Generation 290: Best = 2.201135e+03 FES = 29100 Diversity = 3.7883e-01

Generation 300: Best = 2.201107e+03 FES = 30100 Diversity = 2.8742e-01

Generation 310: Best = 2.201076e+03 FES = 31100 Diversity = 2.7253e-01

Generation 320: Best = 2.201044e+03 FES = 32100 Diversity = 2.4259e-01

Generation 330: Best = 2.201029e+03 FES = 33100 Diversity = 1.7988e-01

Generation 340: Best = 2.201011e+03 FES = 34100 Diversity = 1.6388e-01

Generation 350: Best = 2.200996e+03 FES = 35100 Diversity = 1.3737e-01

Generation 360: Best = 2.200978e+03 FES = 36100 Diversity = 1.5142e-01

Generation 370: Best = 2.200961e+03 FES = 37100 Diversity = 1.5195e-01

Generation 380: Best = 2.200954e+03 FES = 38100 Diversity = 1.2566e-01

Generation 390: Best = 2.200946e+03 FES = 39100 Diversity = 9.8266e-02

Generation 400: Best = 2.200940e+03 FES = 40100 Diversity = 7.7222e-02

Generation 410: Best = 2.200935e+03 FES = 41100 Diversity = 6.0996e-02

Generation 420: Best = 2.200934e+03 FES = 42100 Diversity = 5.1687e-02

Generation 430: Best = 2.200932e+03 FES = 43100 Diversity = 4.1459e-02

Generation 440: Best = 2.200931e+03 FES = 44100 Diversity = 3.6032e-02

Generation	450:	Best = 2.200930e+03	FES = 45100	Diversity = 2.4761e-02
Generation	460:	Best = 2.200929e+03	FES = 46100	Diversity = 2.4148e-02
Generation	470:	Best = 2.200929e+03	FES = 47100	Diversity = 1.5815e-02
Generation	480:	Best = 2.200929e+03	FES = 48100	Diversity = 1.2709e-02
Generation	490:	Best = 2.200929e+03	FES = 49100	Diversity = 6.6110e-03
Generation	500:	Best = 2.200929e+03	FES = 50100	Diversity = 6.8496e-03
Generation	510:	Best = 2.200929e+03	FES = 51100	Diversity = 4.2114e-03
Generation	520:	Best = 2.200929e+03	FES = 52100	Diversity = 3.9358e-03
Generation	530:	Best = 2.200928e+03	FES = 53100	Diversity = 3.2441e-03
Generation	540:	Best = 2.200928e+03	FES = 54100	Diversity = 3.6833e-03
Generation	550:	Best = 2.200928e+03	FES = 55100	Diversity = 2.5532e-03
Generation	560:	Best = 2.200928e+03	FES = 56100	Diversity = 2.2800e-03
Generation	570:	Best = 2.200928e+03	FES = 57100	Diversity = 2.1846e-03
Generation	580:	Best = 2.200928e+03	FES = 58100	Diversity = 2.2801e-03
Generation	590:	Best = 2.200928e+03	FES = 59100	Diversity = 1.4138e-03
Generation	600:	Best = 2.200928e+03	FES = 60100	Diversity = 8.4101e-04
Generation	610:	Best = 2.200928e+03	FES = 61100	Diversity = 9.1014e-04
Generation	620:	Best = 2.200928e+03	FES = 62100	Diversity = 7.7824e-04
Generation	630:	Best = 2.200928e+03	FES = 63100	Diversity = 4.6290e-04
Generation	640:	Best = 2.200928e+03	FES = 64100	Diversity = 3.2556e-04
Generation	650:	Best = 2.200928e+03	FES = 65100	Diversity = 2.6078e-04
Generation	660:	Best = 2.200928e+03	FES = 66100	Diversity = 2.0337e-04
Generation	670:	Best = 2.200928e+03	FES = 67100	Diversity = 1.9440e-04
Generation	680:	Best = 2.200928e+03	FES = 68100	Diversity = 1.9666e-04
Generation	690:	Best = 2.200928e+03	FES = 69100	Diversity = 1.7331e-04
Generation	700:	Best = 2.200928e+03	FES = 70100	Diversity = 1.3305e-04
Generation	710:	Best = 2.200928e+03	FES = 71100	Diversity = 1.4660e-04
Generation	720:	Best = 2.200928e+03	FES = 72100	Diversity = 1.6018e-04
Generation	730:	Best = 2.200928e+03	FES = 73100	Diversity = 8.9067e-05
Generation	740:	Best = 2.200928e+03	FES = 74100	Diversity = 1.0944e-04
Generation	750:	Best = 2.200928e+03	FES = 75100	Diversity = 9.4278e-05
Generation	760:	Best = 2.200928e+03	FES = 76100	Diversity = 9.0558e-05
Generation	770:	Best = 2.200928e+03	FES = 77100	Diversity = 8.6122e-05
Generation	780:	Best = 2.200928e+03	FES = 78100	Diversity = 4.0365e-05
Generation	790:	Best = 2.200928e+03	FES = 79100	Diversity = 2.5673e-05
Generation	800:	Best = 2.200928e+03	FES = 80100	Diversity = 1.6680e-05
Generation	810:	Best = 2.200928e+03	FES = 81100	Diversity = 1.5890e-05
Generation	820:	Best = 2.200928e+03	FES = 82100	Diversity = 1.3594e-05
Generation	830:	Best = 2.200928e+03	FES = 83100	Diversity = 1.1075e-05
Generation	840:	Best = 2.200928e+03	FES = 84100	Diversity = 1.1287e-05
Generation	850:	Best = 2.200928e+03	FES = 85100	Diversity = 7.4489e-06
Generation	860:	Best = 2.200928e+03	FES = 86100	Diversity = 7.5126e-06
Generation	870:	Best = 2.200928e+03	FES = 87100	Diversity = 8.3839e-06
Generation	880:	Best = 2.200928e+03	FES = 88100	Diversity = 7.6810e-06
Generation	890:	Best = 2.200928e+03	FES = 89100	Diversity = 5.4962e-06
Generation	900:	Best = 2.200928e+03	FES = 90100	Diversity = 5.2714e-06
Generation	910:	Best = 2.200928e+03	FES = 91100	Diversity = 3.4063e-06
Generation	920:	Best = 2.200928e+03	FES = 92100	Diversity = 1.8878e-06
Generation	930:	Best = 2.200928e+03	FES = 93100	Diversity = 1.9146e-06
Generation	940:	Best = 2.200928e+03	FES = 94100	Diversity = 1.1465e-06
Generation	950:	Best = 2.200928e+03	FES = 95100	Diversity = 1.3502e-06
Generation	960:	Best = 2.200928e+03	FES = 96100	Diversity = 6.6467e-07
Generation	970:	Best = 2.200928e+03	FES = 97100	Diversity = 2.6564e-07
Generation	980:	Best = 2.200928e+03	FES = 98100	Diversity = 1.4215e-07

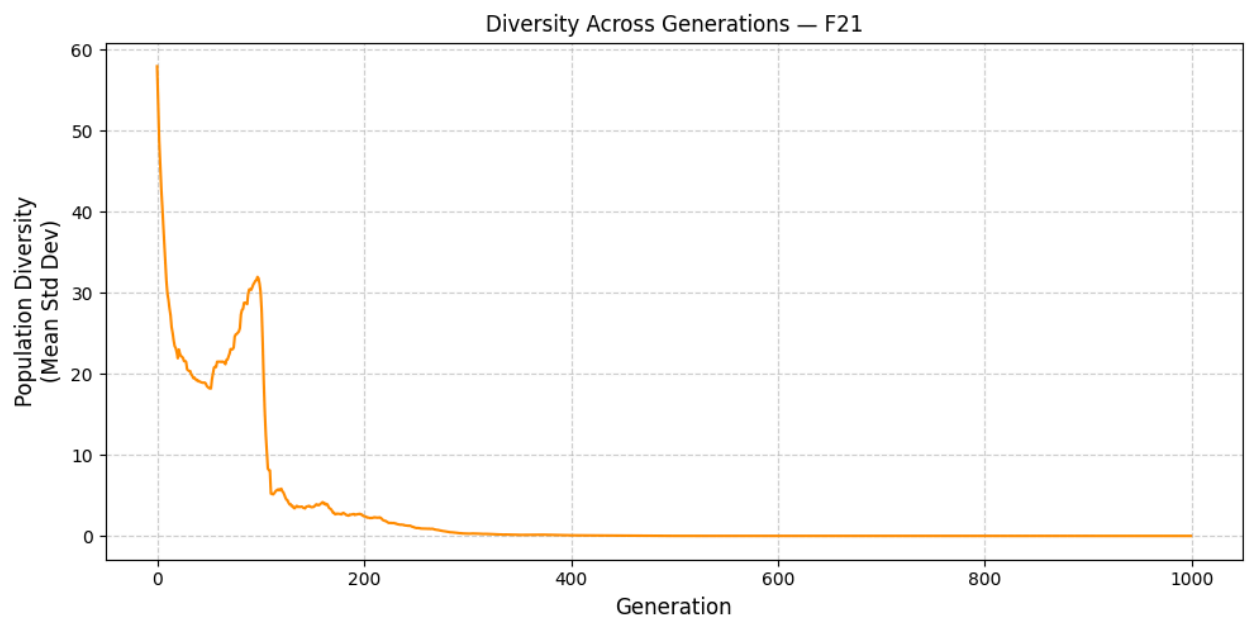
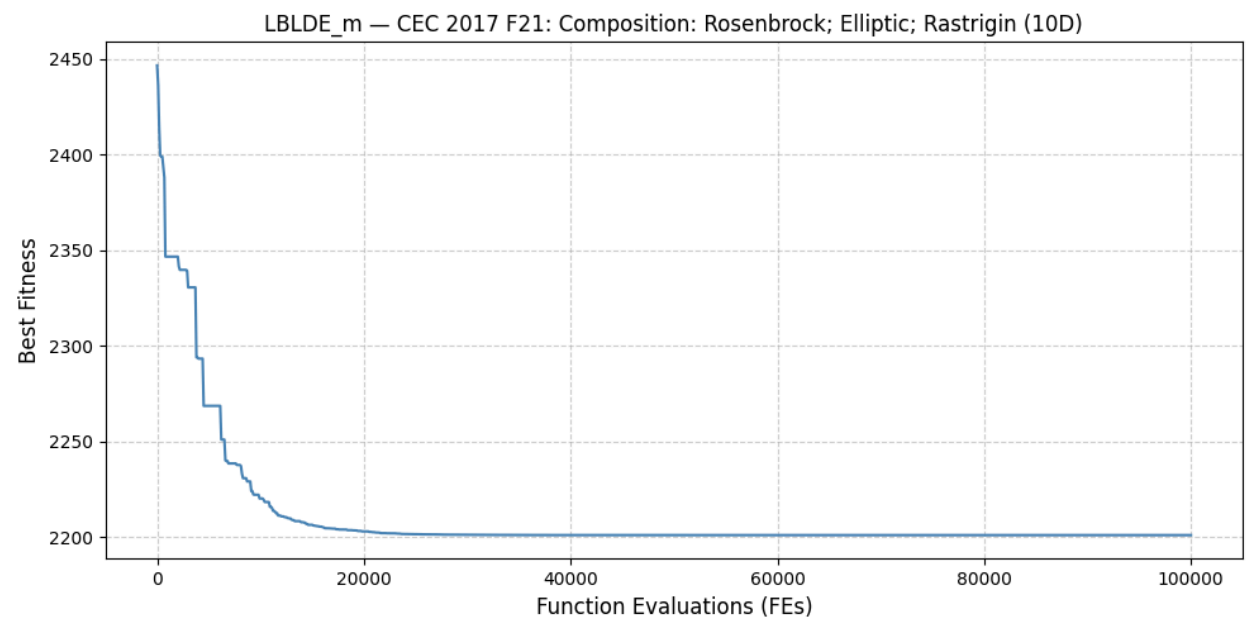
Generation 990: Best = 2.200928e+03 FES = 99100 Diversity = 1.2847e-08

Optimization completed!

Final Best Fitness = 2.200928e+03

Total Function Evaluations = 100000

```
=====
=
Results:
  Best Fitness           : 2.200928e+03
  Global Optimum         : 2.100000e+03
  Error (|best-optimum|) : 1.009283e+02
  Best Solution (5 dim)  : [ 32.52812707 -61.19158807 -44.91460464 -71.32584962
-25.01733072]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F21.png

```
In [32]: # =====  
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)  
# =====  
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

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LBLDE_m on CEC 2017 F22 – Single Run

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Function : Composition: Rastrigin; Griewank; Schwefel (10D)

Bounds : [-100, 100]

Global optimum : 2200.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.528003e+03 | Diversity = 5.7928e+01

Generation 10:	Best = 2.348865e+03	FES = 1100	Diversity = 1.3355e+01
Generation 20:	Best = 2.316434e+03	FES = 2100	Diversity = 2.4421e+00
Generation 30:	Best = 2.310922e+03	FES = 3100	Diversity = 8.2335e-01
Generation 40:	Best = 2.306065e+03	FES = 4100	Diversity = 5.5755e-01
Generation 50:	Best = 2.303815e+03	FES = 5100	Diversity = 5.0094e-01
Generation 60:	Best = 2.302286e+03	FES = 6100	Diversity = 4.4500e-01
Generation 70:	Best = 2.302096e+03	FES = 7100	Diversity = 3.9507e-01
Generation 80:	Best = 2.301992e+03	FES = 8100	Diversity = 3.3098e-01
Generation 90:	Best = 2.301909e+03	FES = 9100	Diversity = 2.9524e-01
Generation 100:	Best = 2.301877e+03	FES = 10100	Diversity = 2.9246e-01
Generation 110:	Best = 2.301865e+03	FES = 11100	Diversity = 1.1423e-01
Generation 120:	Best = 2.301821e+03	FES = 12100	Diversity = 1.2259e-02
Generation 130:	Best = 2.301810e+03	FES = 13100	Diversity = 6.3590e-03
Generation 140:	Best = 2.301805e+03	FES = 14100	Diversity = 4.5587e-03
Generation 150:	Best = 2.301803e+03	FES = 15100	Diversity = 2.1067e-03
Generation 160:	Best = 2.301803e+03	FES = 16100	Diversity = 1.1542e-03
Generation 170:	Best = 2.301802e+03	FES = 17100	Diversity = 7.2666e-04
Generation 180:	Best = 2.301802e+03	FES = 18100	Diversity = 5.6440e-04
Generation 190:	Best = 2.301802e+03	FES = 19100	Diversity = 3.0481e-04
Generation 200:	Best = 2.301802e+03	FES = 20100	Diversity = 1.6612e-04
Generation 210:	Best = 2.301802e+03	FES = 21100	Diversity = 1.3276e-04
Generation 220:	Best = 2.301802e+03	FES = 22100	Diversity = 1.0747e-04
Generation 230:	Best = 2.301802e+03	FES = 23100	Diversity = 7.8672e-05
Generation 240:	Best = 2.301802e+03	FES = 24100	Diversity = 5.7264e-05
Generation 250:	Best = 2.301802e+03	FES = 25100	Diversity = 4.8881e-05
Generation 260:	Best = 2.301802e+03	FES = 26100	Diversity = 4.7637e-05
Generation 270:	Best = 2.301802e+03	FES = 27100	Diversity = 3.8972e-05
Generation 280:	Best = 2.301802e+03	FES = 28100	Diversity = 2.1815e-05
Generation 290:	Best = 2.301802e+03	FES = 29100	Diversity = 1.9561e-05
Generation 300:	Best = 2.301802e+03	FES = 30100	Diversity = 1.8879e-05
Generation 310:	Best = 2.301802e+03	FES = 31100	Diversity = 1.6756e-05
Generation 320:	Best = 2.301802e+03	FES = 32100	Diversity = 1.2223e-05
Generation 330:	Best = 2.301802e+03	FES = 33100	Diversity = 4.8896e-06
Generation 340:	Best = 2.301802e+03	FES = 34100	Diversity = 5.3374e-06
Generation 350:	Best = 2.301802e+03	FES = 35100	Diversity = 3.9512e-06
Generation 360:	Best = 2.301802e+03	FES = 36100	Diversity = 2.5223e-06
Generation 370:	Best = 2.301802e+03	FES = 37100	Diversity = 1.8048e-06
Generation 380:	Best = 2.301802e+03	FES = 38100	Diversity = 1.1563e-06
Generation 390:	Best = 2.301802e+03	FES = 39100	Diversity = 7.3881e-07
Generation 400:	Best = 2.301802e+03	FES = 40100	Diversity = 5.8992e-07
Generation 410:	Best = 2.301802e+03	FES = 41100	Diversity = 6.6128e-07
Generation 420:	Best = 2.301802e+03	FES = 42100	Diversity = 8.0327e-07
Generation 430:	Best = 2.301802e+03	FES = 43100	Diversity = 4.4156e-07
Generation 440:	Best = 2.301802e+03	FES = 44100	Diversity = 2.8507e-07

[illegible]

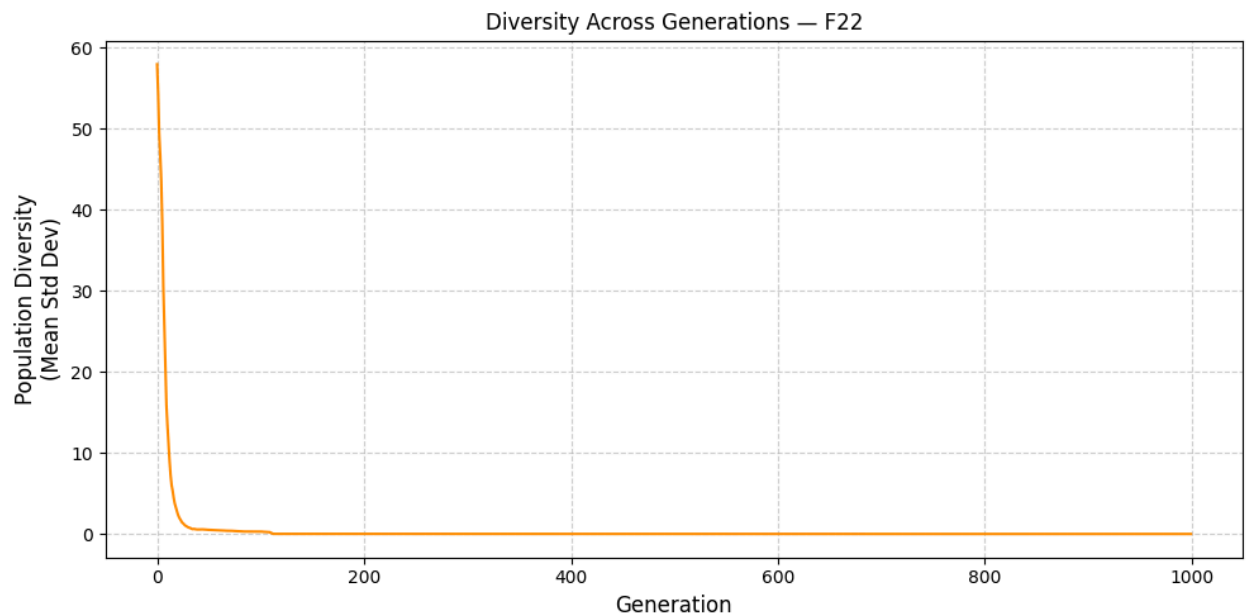
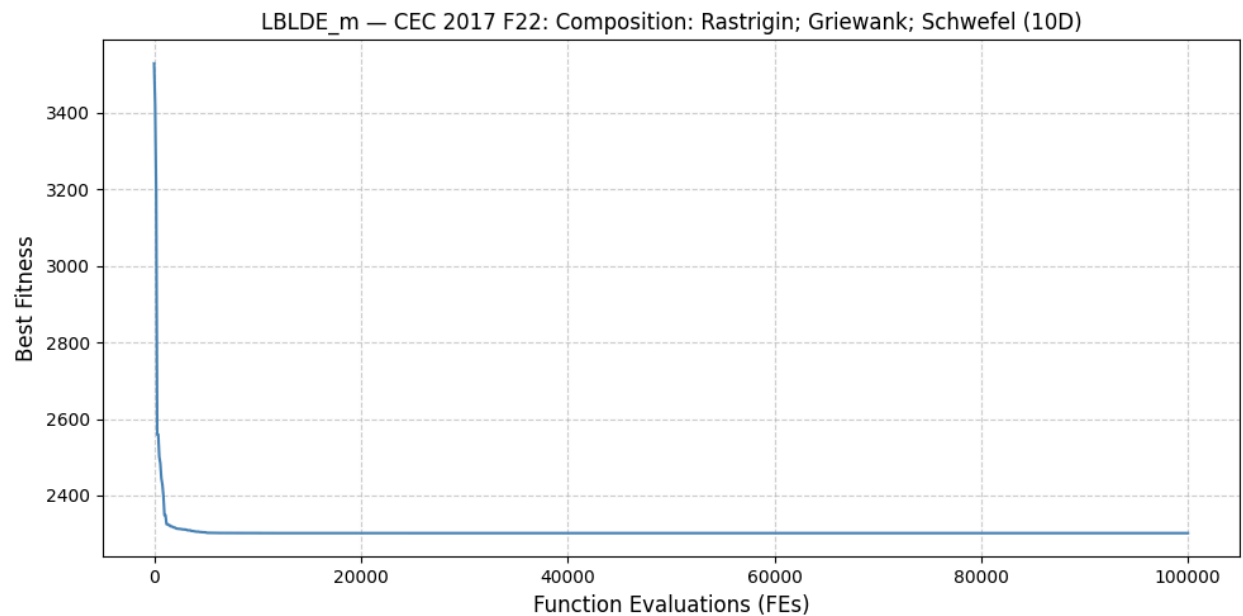
Generation 990: Best = 2.301802e+03 FES = 99100 Diversity = 3.6415e-14

Optimization completed!

Final Best Fitness = 2.301802e+03

Total Function Evaluations = 100000

```
=====
=
Results:
  Best Fitness           : 2.301802e+03
  Global Optimum         : 2.200000e+03
  Error (|best-optimum|) : 1.018022e+02
  Best Solution (5 dim)  : [-50.76016618  11.24684984  22.43533691  34.72667791
30.53370637]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F22.png

```
In [33]: # =====  
# F23 – Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)  
# =====  
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosenbrock')
```

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LBLDE_m on CEC 2017 F23 – Single Run

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Function : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (10D)
Bounds : [-100, 100]
Global optimum : 2300.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.773921e+03 | Diversity = 5.7928e+01

Generation	10:	Best = 2.681356e+03	FES = 1100	Diversity = 2.3287e+01
Generation	20:	Best = 2.649530e+03	FES = 2100	Diversity = 1.8334e+01
Generation	30:	Best = 2.646652e+03	FES = 3100	Diversity = 1.5791e+01
Generation	40:	Best = 2.645968e+03	FES = 4100	Diversity = 1.4882e+01
Generation	50:	Best = 2.623666e+03	FES = 5100	Diversity = 1.4108e+01
Generation	60:	Best = 2.614995e+03	FES = 6100	Diversity = 1.3904e+01
Generation	70:	Best = 2.614692e+03	FES = 7100	Diversity = 1.3218e+01
Generation	80:	Best = 2.614341e+03	FES = 8100	Diversity = 1.2617e+01
Generation	90:	Best = 2.613562e+03	FES = 9100	Diversity = 1.2203e+01
Generation	100:	Best = 2.613443e+03	FES = 10100	Diversity = 1.2043e+01
Generation	110:	Best = 2.613385e+03	FES = 11100	Diversity = 1.1954e+01
Generation	120:	Best = 2.613361e+03	FES = 12100	Diversity = 1.1206e+01
Generation	130:	Best = 2.613360e+03	FES = 13100	Diversity = 1.0800e+01
Generation	140:	Best = 2.613283e+03	FES = 14100	Diversity = 1.0243e+01
Generation	150:	Best = 2.613012e+03	FES = 15100	Diversity = 9.5073e+00
Generation	160:	Best = 2.612945e+03	FES = 16100	Diversity = 8.4748e+00
Generation	170:	Best = 2.611030e+03	FES = 17100	Diversity = 7.4347e+00
Generation	180:	Best = 2.610600e+03	FES = 18100	Diversity = 6.3755e+00
Generation	190:	Best = 2.610577e+03	FES = 19100	Diversity = 5.1704e+00
Generation	200:	Best = 2.609592e+03	FES = 20100	Diversity = 4.5355e+00
Generation	210:	Best = 2.609592e+03	FES = 21100	Diversity = 3.1845e+00
Generation	220:	Best = 2.609439e+03	FES = 22100	Diversity = 2.0299e+00
Generation	230:	Best = 2.609421e+03	FES = 23100	Diversity = 1.9002e+00
Generation	240:	Best = 2.609404e+03	FES = 24100	Diversity = 1.9275e+00
Generation	250:	Best = 2.609247e+03	FES = 25100	Diversity = 2.0997e+00
Generation	260:	Best = 2.609069e+03	FES = 26100	Diversity = 2.1390e+00
Generation	270:	Best = 2.609056e+03	FES = 27100	Diversity = 2.1361e+00
Generation	280:	Best = 2.608892e+03	FES = 28100	Diversity = 2.1774e+00
Generation	290:	Best = 2.608353e+03	FES = 29100	Diversity = 2.2578e+00
Generation	300:	Best = 2.608339e+03	FES = 30100	Diversity = 2.2796e+00
Generation	310:	Best = 2.608265e+03	FES = 31100	Diversity = 2.2806e+00
Generation	320:	Best = 2.608245e+03	FES = 32100	Diversity = 2.2896e+00
Generation	330:	Best = 2.608230e+03	FES = 33100	Diversity = 2.3603e+00
Generation	340:	Best = 2.608226e+03	FES = 34100	Diversity = 2.3947e+00
Generation	350:	Best = 2.608211e+03	FES = 35100	Diversity = 2.4249e+00
Generation	360:	Best = 2.608210e+03	FES = 36100	Diversity = 2.5762e+00
Generation	370:	Best = 2.608198e+03	FES = 37100	Diversity = 2.6087e+00
Generation	380:	Best = 2.608085e+03	FES = 38100	Diversity = 2.6739e+00
Generation	390:	Best = 2.607716e+03	FES = 39100	Diversity = 2.7207e+00
Generation	400:	Best = 2.607695e+03	FES = 40100	Diversity = 2.7763e+00
Generation	410:	Best = 2.607695e+03	FES = 41100	Diversity = 2.7909e+00
Generation	420:	Best = 2.607695e+03	FES = 42100	Diversity = 2.7917e+00
Generation	430:	Best = 2.607695e+03	FES = 43100	Diversity = 2.8374e+00
Generation	440:	Best = 2.607695e+03	FES = 44100	Diversity = 2.8599e+00

Generation	450:	Best = 2.607687e+03	FES = 45100	Diversity = 2.8720e+00
Generation	460:	Best = 2.607683e+03	FES = 46100	Diversity = 2.7695e+00
Generation	470:	Best = 2.607674e+03	FES = 47100	Diversity = 2.4956e+00
Generation	480:	Best = 2.607331e+03	FES = 48100	Diversity = 1.1104e+00
Generation	490:	Best = 2.607270e+03	FES = 49100	Diversity = 4.1883e-01
Generation	500:	Best = 2.607268e+03	FES = 50100	Diversity = 2.1329e-01
Generation	510:	Best = 2.607265e+03	FES = 51100	Diversity = 1.2365e-01
Generation	520:	Best = 2.607263e+03	FES = 52100	Diversity = 9.7279e-02
Generation	530:	Best = 2.607261e+03	FES = 53100	Diversity = 8.0855e-02
Generation	540:	Best = 2.607260e+03	FES = 54100	Diversity = 4.7608e-02
Generation	550:	Best = 2.607260e+03	FES = 55100	Diversity = 3.6024e-02
Generation	560:	Best = 2.607260e+03	FES = 56100	Diversity = 2.5824e-02
Generation	570:	Best = 2.607260e+03	FES = 57100	Diversity = 2.0396e-02
Generation	580:	Best = 2.607260e+03	FES = 58100	Diversity = 1.6260e-02
Generation	590:	Best = 2.607260e+03	FES = 59100	Diversity = 1.2805e-02
Generation	600:	Best = 2.607260e+03	FES = 60100	Diversity = 7.5254e-03
Generation	610:	Best = 2.607260e+03	FES = 61100	Diversity = 6.0976e-03
Generation	620:	Best = 2.607260e+03	FES = 62100	Diversity = 4.7140e-03
Generation	630:	Best = 2.607260e+03	FES = 63100	Diversity = 4.3663e-03
Generation	640:	Best = 2.607260e+03	FES = 64100	Diversity = 3.7258e-03
Generation	650:	Best = 2.607260e+03	FES = 65100	Diversity = 3.7314e-03
Generation	660:	Best = 2.607260e+03	FES = 66100	Diversity = 3.4957e-03
Generation	670:	Best = 2.607260e+03	FES = 67100	Diversity = 3.0571e-03
Generation	680:	Best = 2.607260e+03	FES = 68100	Diversity = 2.8029e-03
Generation	690:	Best = 2.607260e+03	FES = 69100	Diversity = 2.0518e-03
Generation	700:	Best = 2.607260e+03	FES = 70100	Diversity = 1.9836e-03
Generation	710:	Best = 2.607260e+03	FES = 71100	Diversity = 1.9342e-03
Generation	720:	Best = 2.607260e+03	FES = 72100	Diversity = 1.8340e-03
Generation	730:	Best = 2.607260e+03	FES = 73100	Diversity = 1.8205e-03
Generation	740:	Best = 2.607260e+03	FES = 74100	Diversity = 1.7750e-03
Generation	750:	Best = 2.607260e+03	FES = 75100	Diversity = 1.5362e-03
Generation	760:	Best = 2.607260e+03	FES = 76100	Diversity = 1.6269e-03
Generation	770:	Best = 2.607260e+03	FES = 77100	Diversity = 1.3106e-03
Generation	780:	Best = 2.607260e+03	FES = 78100	Diversity = 1.0854e-03
Generation	790:	Best = 2.607260e+03	FES = 79100	Diversity = 9.0627e-04
Generation	800:	Best = 2.607260e+03	FES = 80100	Diversity = 7.8160e-04
Generation	810:	Best = 2.607260e+03	FES = 81100	Diversity = 7.7347e-04
Generation	820:	Best = 2.607260e+03	FES = 82100	Diversity = 7.3967e-04
Generation	830:	Best = 2.607260e+03	FES = 83100	Diversity = 7.3681e-04
Generation	840:	Best = 2.607260e+03	FES = 84100	Diversity = 7.0760e-04
Generation	850:	Best = 2.607260e+03	FES = 85100	Diversity = 5.2779e-04
Generation	860:	Best = 2.607260e+03	FES = 86100	Diversity = 4.5485e-04
Generation	870:	Best = 2.607260e+03	FES = 87100	Diversity = 4.4722e-04
Generation	880:	Best = 2.607260e+03	FES = 88100	Diversity = 4.1955e-04
Generation	890:	Best = 2.607260e+03	FES = 89100	Diversity = 3.0342e-04
Generation	900:	Best = 2.607260e+03	FES = 90100	Diversity = 2.8793e-04
Generation	910:	Best = 2.607260e+03	FES = 91100	Diversity = 2.9435e-04
Generation	920:	Best = 2.607260e+03	FES = 92100	Diversity = 2.7856e-04
Generation	930:	Best = 2.607260e+03	FES = 93100	Diversity = 2.2949e-04
Generation	940:	Best = 2.607260e+03	FES = 94100	Diversity = 1.9934e-04
Generation	950:	Best = 2.607260e+03	FES = 95100	Diversity = 2.1720e-04
Generation	960:	Best = 2.607260e+03	FES = 96100	Diversity = 1.7917e-04
Generation	970:	Best = 2.607260e+03	FES = 97100	Diversity = 1.4581e-04
Generation	980:	Best = 2.607260e+03	FES = 98100	Diversity = 1.2788e-04

Generation 990: Best = 2.607260e+03 FES = 99100 Diversity = 1.1032e-04

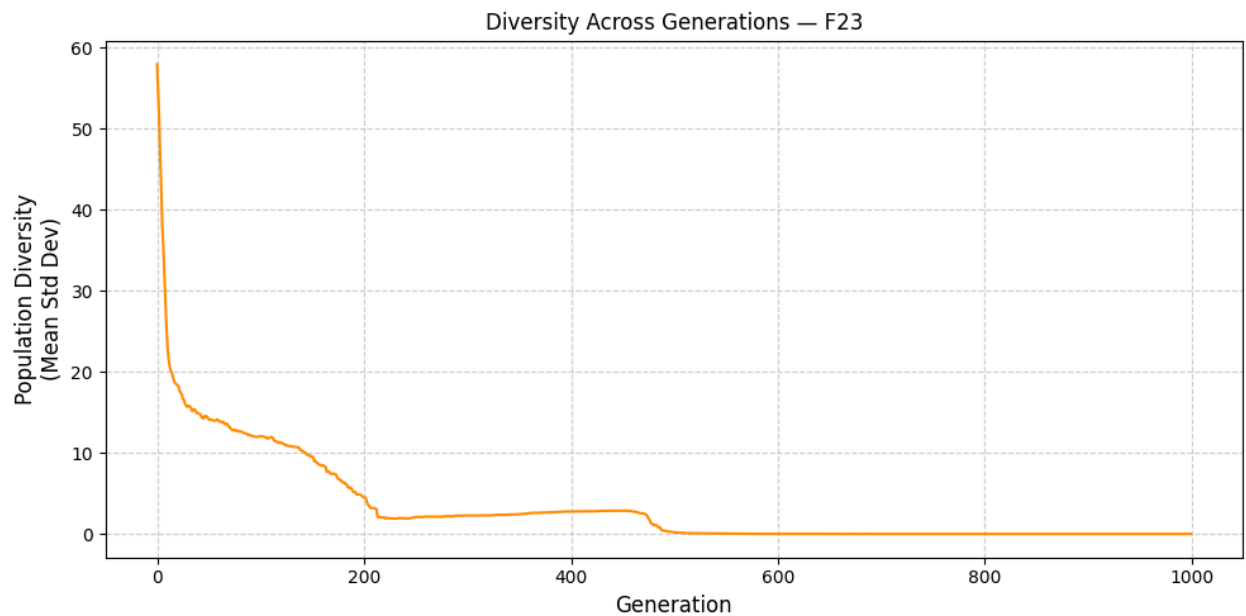
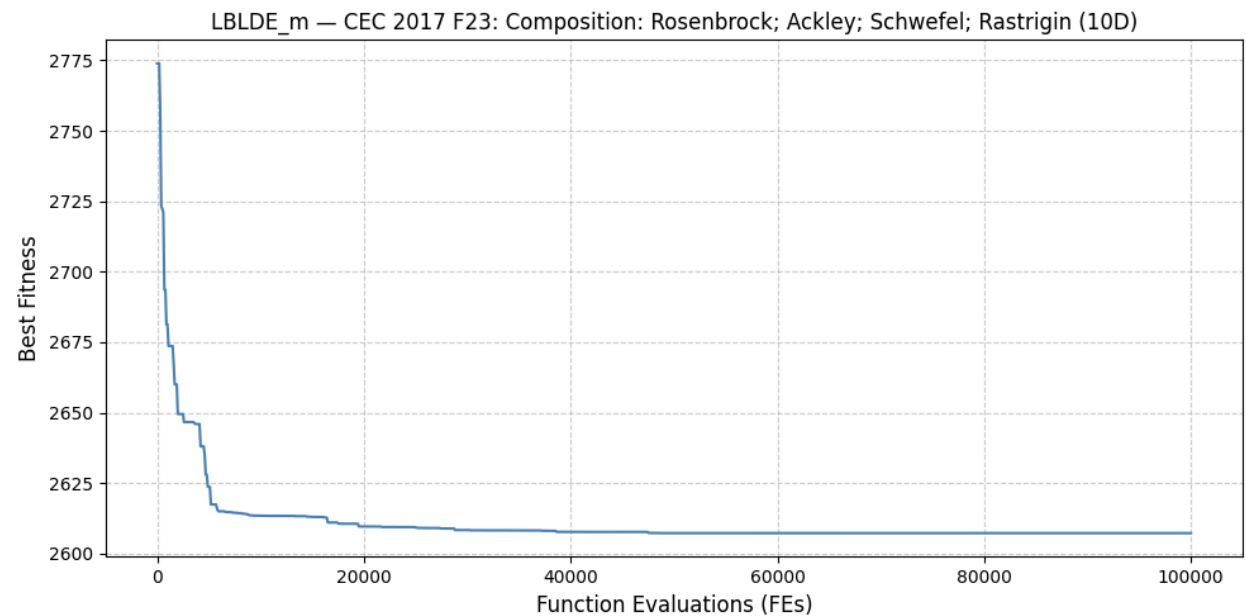
Optimization completed!

Final Best Fitness = 2.607260e+03

Total Function Evaluations = 100000

Results:

Best Fitness : 2.607260e+03
Global Optimum : 2.300000e+03
Error (|best-optimum|) : 3.072597e+02
Best Solution (5 dim) : [-78.45432684 -92.69679543 -55.35244557 51.63879894
-63.93454138]
D^(0) (init. diversity): 5.7928e+01



Graph saved as convergence_F23.png

```
In [34]: # =====  
# F24 – Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)  
# =====  
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

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LBLDE_m on CEC 2017 F24 – Single Run

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Function : Composition: Ackley; Elliptic; Griewank; Rastrigin (10D)
Bounds : [-100, 100]
Global optimum : 2400.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 2.997998e+03 | Diversity = 5.7928e+01

Generation 10:	Best = 2.795118e+03	FES = 1100	Diversity = 2.1538e+01
Generation 20:	Best = 2.785385e+03	FES = 2100	Diversity = 1.2914e+01
Generation 30:	Best = 2.766216e+03	FES = 3100	Diversity = 1.0997e+01
Generation 40:	Best = 2.758597e+03	FES = 4100	Diversity = 9.8297e+00
Generation 50:	Best = 2.757784e+03	FES = 5100	Diversity = 9.4706e+00
Generation 60:	Best = 2.745489e+03	FES = 6100	Diversity = 8.8803e+00
Generation 70:	Best = 2.743765e+03	FES = 7100	Diversity = 8.7462e+00
Generation 80:	Best = 2.743372e+03	FES = 8100	Diversity = 8.5475e+00
Generation 90:	Best = 2.740251e+03	FES = 9100	Diversity = 8.2559e+00
Generation 100:	Best = 2.734785e+03	FES = 10100	Diversity = 7.8986e+00
Generation 110:	Best = 2.728549e+03	FES = 11100	Diversity = 7.5338e+00
Generation 120:	Best = 2.727928e+03	FES = 12100	Diversity = 6.8052e+00
Generation 130:	Best = 2.726038e+03	FES = 13100	Diversity = 6.2649e+00
Generation 140:	Best = 2.722603e+03	FES = 14100	Diversity = 5.7653e+00
Generation 150:	Best = 2.721852e+03	FES = 15100	Diversity = 5.6014e+00
Generation 160:	Best = 2.721619e+03	FES = 16100	Diversity = 5.0968e+00
Generation 170:	Best = 2.721445e+03	FES = 17100	Diversity = 4.4758e+00
Generation 180:	Best = 2.721299e+03	FES = 18100	Diversity = 4.2290e+00
Generation 190:	Best = 2.721259e+03	FES = 19100	Diversity = 3.9167e+00
Generation 200:	Best = 2.721255e+03	FES = 20100	Diversity = 3.7214e+00
Generation 210:	Best = 2.721207e+03	FES = 21100	Diversity = 1.8919e+00
Generation 220:	Best = 2.721200e+03	FES = 22100	Diversity = 2.0818e-02
Generation 230:	Best = 2.721199e+03	FES = 23100	Diversity = 4.4683e-03
Generation 240:	Best = 2.721199e+03	FES = 24100	Diversity = 2.2272e-03
Generation 250:	Best = 2.721199e+03	FES = 25100	Diversity = 1.6702e-03
Generation 260:	Best = 2.721199e+03	FES = 26100	Diversity = 7.9537e-04
Generation 270:	Best = 2.721199e+03	FES = 27100	Diversity = 4.7233e-04
Generation 280:	Best = 2.721199e+03	FES = 28100	Diversity = 3.1292e-04
Generation 290:	Best = 2.721199e+03	FES = 29100	Diversity = 2.3594e-04
Generation 300:	Best = 2.721199e+03	FES = 30100	Diversity = 1.4546e-04
Generation 310:	Best = 2.721199e+03	FES = 31100	Diversity = 8.3364e-05
Generation 320:	Best = 2.721199e+03	FES = 32100	Diversity = 4.2606e-05
Generation 330:	Best = 2.721199e+03	FES = 33100	Diversity = 2.6032e-05
Generation 340:	Best = 2.721199e+03	FES = 34100	Diversity = 1.7702e-05
Generation 350:	Best = 2.721199e+03	FES = 35100	Diversity = 1.1643e-05
Generation 360:	Best = 2.721199e+03	FES = 36100	Diversity = 1.0412e-05
Generation 370:	Best = 2.721199e+03	FES = 37100	Diversity = 7.0794e-06
Generation 380:	Best = 2.721199e+03	FES = 38100	Diversity = 4.7361e-06
Generation 390:	Best = 2.721199e+03	FES = 39100	Diversity = 3.1784e-06
Generation 400:	Best = 2.721199e+03	FES = 40100	Diversity = 2.1720e-06
Generation 410:	Best = 2.721199e+03	FES = 41100	Diversity = 1.6778e-06
Generation 420:	Best = 2.721199e+03	FES = 42100	Diversity = 1.1857e-06
Generation 430:	Best = 2.721199e+03	FES = 43100	Diversity = 8.1928e-07
Generation 440:	Best = 2.721199e+03	FES = 44100	Diversity = 5.9484e-07

Generation	450:	Best = 2.721199e+03	FES = 45100	Diversity = 5.2382e-07
Generation	460:	Best = 2.721199e+03	FES = 46100	Diversity = 4.8537e-07
Generation	470:	Best = 2.721199e+03	FES = 47100	Diversity = 4.3198e-07
Generation	480:	Best = 2.721199e+03	FES = 48100	Diversity = 2.8378e-07
Generation	490:	Best = 2.721199e+03	FES = 49100	Diversity = 3.4786e-07
Generation	500:	Best = 2.721199e+03	FES = 50100	Diversity = 3.2697e-07
Generation	510:	Best = 2.721199e+03	FES = 51100	Diversity = 2.7084e-07
Generation	520:	Best = 2.721199e+03	FES = 52100	Diversity = 1.9060e-07
Generation	530:	Best = 2.721199e+03	FES = 53100	Diversity = 1.3306e-07
Generation	540:	Best = 2.721199e+03	FES = 54100	Diversity = 7.4113e-08
Generation	550:	Best = 2.721199e+03	FES = 55100	Diversity = 6.1538e-08
Generation	560:	Best = 2.721199e+03	FES = 56100	Diversity = 3.1163e-08
Generation	570:	Best = 2.721199e+03	FES = 57100	Diversity = 2.9771e-08
Generation	580:	Best = 2.721199e+03	FES = 58100	Diversity = 9.0868e-09
Generation	590:	Best = 2.721199e+03	FES = 59100	Diversity = 1.9615e-09
Generation	600:	Best = 2.721199e+03	FES = 60100	Diversity = 1.0830e-09
Generation	610:	Best = 2.721199e+03	FES = 61100	Diversity = 3.3592e-10
Generation	620:	Best = 2.721199e+03	FES = 62100	Diversity = 4.6375e-11
Generation	630:	Best = 2.721199e+03	FES = 63100	Diversity = 6.4103e-12
Generation	640:	Best = 2.721199e+03	FES = 64100	Diversity = 3.3960e-12
Generation	650:	Best = 2.721199e+03	FES = 65100	Diversity = 4.7648e-13
Generation	660:	Best = 2.721199e+03	FES = 66100	Diversity = 8.2423e-14
Generation	670:	Best = 2.721199e+03	FES = 67100	Diversity = 6.1092e-14
Generation	680:	Best = 2.721199e+03	FES = 68100	Diversity = 6.0396e-14
Generation	690:	Best = 2.721199e+03	FES = 69100	Diversity = 6.0396e-14
Generation	700:	Best = 2.721199e+03	FES = 70100	Diversity = 6.0396e-14
Generation	710:	Best = 2.721199e+03	FES = 71100	Diversity = 6.0396e-14
Generation	720:	Best = 2.721199e+03	FES = 72100	Diversity = 6.0396e-14
Generation	730:	Best = 2.721199e+03	FES = 73100	Diversity = 6.0396e-14
Generation	740:	Best = 2.721199e+03	FES = 74100	Diversity = 6.0396e-14
Generation	750:	Best = 2.721199e+03	FES = 75100	Diversity = 6.0396e-14
Generation	760:	Best = 2.721199e+03	FES = 76100	Diversity = 6.0396e-14
Generation	770:	Best = 2.721199e+03	FES = 77100	Diversity = 6.0396e-14
Generation	780:	Best = 2.721199e+03	FES = 78100	Diversity = 6.0396e-14
Generation	790:	Best = 2.721199e+03	FES = 79100	Diversity = 6.0396e-14
Generation	800:	Best = 2.721199e+03	FES = 80100	Diversity = 6.0396e-14
Generation	810:	Best = 2.721199e+03	FES = 81100	Diversity = 6.0396e-14
Generation	820:	Best = 2.721199e+03	FES = 82100	Diversity = 6.0396e-14
Generation	830:	Best = 2.721199e+03	FES = 83100	Diversity = 6.0396e-14
Generation	840:	Best = 2.721199e+03	FES = 84100	Diversity = 6.0396e-14
Generation	850:	Best = 2.721199e+03	FES = 85100	Diversity = 6.0396e-14
Generation	860:	Best = 2.721199e+03	FES = 86100	Diversity = 6.0396e-14
Generation	870:	Best = 2.721199e+03	FES = 87100	Diversity = 6.0396e-14
Generation	880:	Best = 2.721199e+03	FES = 88100	Diversity = 6.0396e-14
Generation	890:	Best = 2.721199e+03	FES = 89100	Diversity = 6.0396e-14
Generation	900:	Best = 2.721199e+03	FES = 90100	Diversity = 6.0396e-14
Generation	910:	Best = 2.721199e+03	FES = 91100	Diversity = 6.0396e-14
Generation	920:	Best = 2.721199e+03	FES = 92100	Diversity = 6.0396e-14
Generation	930:	Best = 2.721199e+03	FES = 93100	Diversity = 6.0396e-14
Generation	940:	Best = 2.721199e+03	FES = 94100	Diversity = 6.0396e-14
Generation	950:	Best = 2.721199e+03	FES = 95100	Diversity = 6.0396e-14
Generation	960:	Best = 2.721199e+03	FES = 96100	Diversity = 6.0396e-14
Generation	970:	Best = 2.721199e+03	FES = 97100	Diversity = 6.0396e-14
Generation	980:	Best = 2.721199e+03	FES = 98100	Diversity = 6.0396e-14

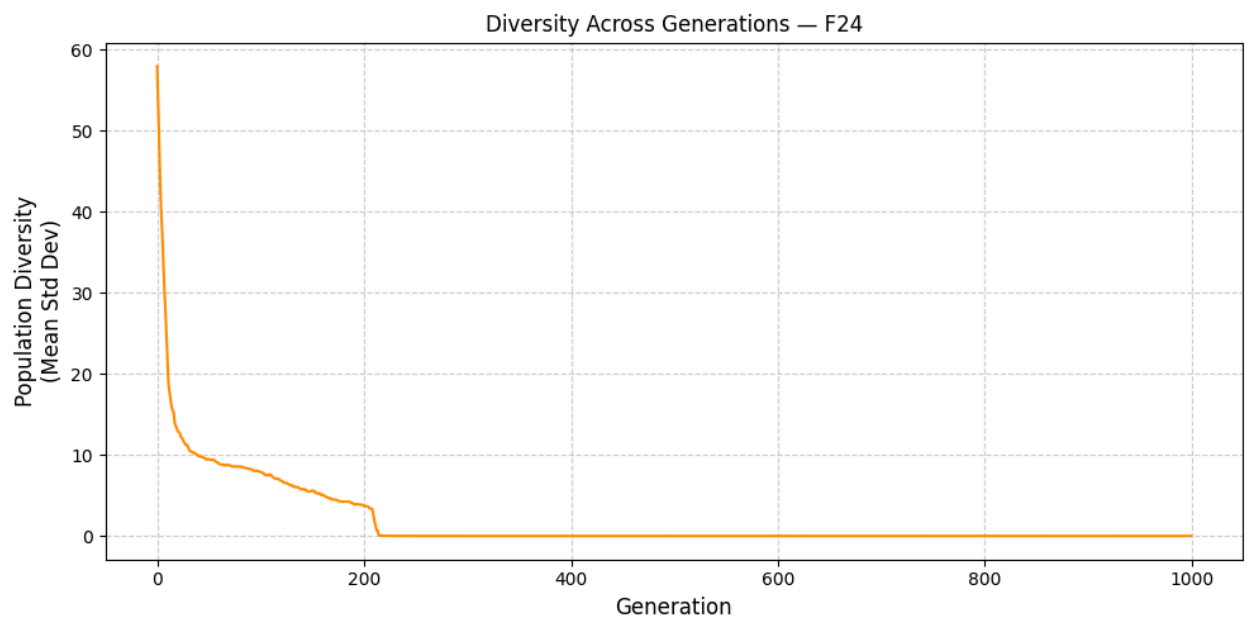
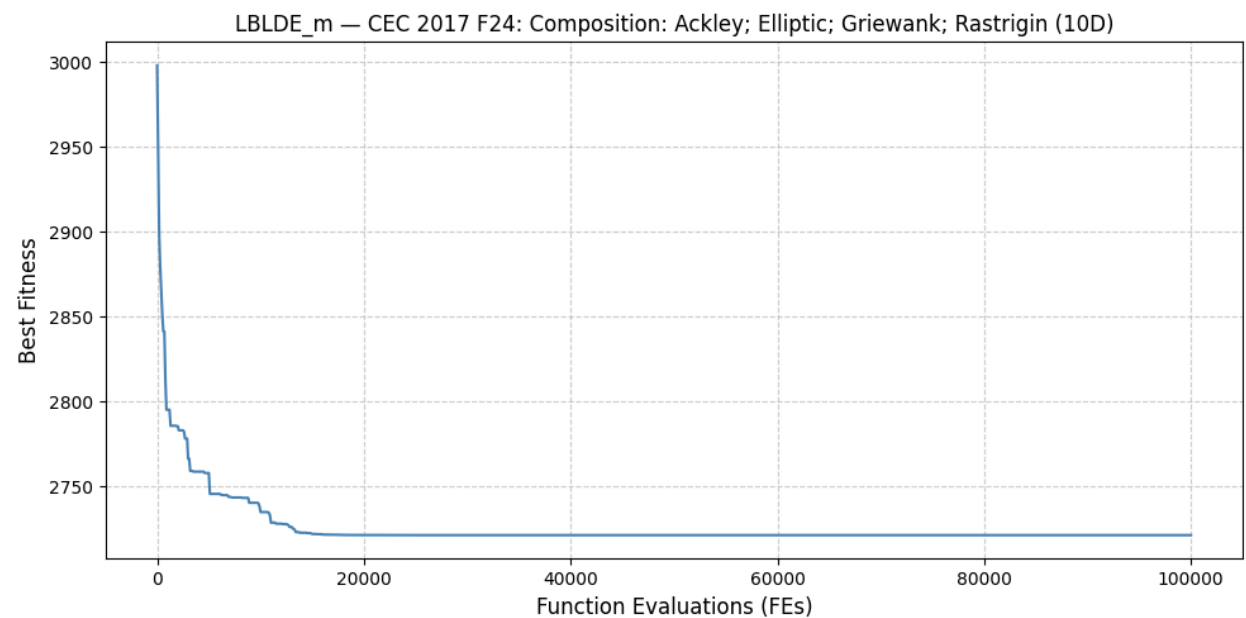
Generation 990: Best = 2.721199e+03 FES = 99100 Diversity = 6.0396e-14

Optimization completed!

Final Best Fitness = 2.721199e+03

Total Function Evaluations = 100000

```
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Results:
  Best Fitness          : 2.721199e+03
  Global Optimum        : 2.400000e+03
  Error (|best-optimum|) : 3.211989e+02
  Best Solution (5 dim)  : [ 28.43278016  53.20015073 -77.60291823 -68.82016471
53.06632468]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F24.png

```
In [35]: # =====  
# F25 – Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)  
# =====  
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

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LBLDE_m on CEC 2017 F25 – Single Run

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Function : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock (10D)

Bounds : [-100, 100]

Global optimum : 2500.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 4.081798e+03 | Diversity = 5.7928e+01

Generation 10: Best = 2.957208e+03 FES = 1100 Diversity = 1.3269e+01

Generation 20: Best = 2.949072e+03 FES = 2100 Diversity = 2.3843e+00

Generation 30: Best = 2.947054e+03 FES = 3100 Diversity = 9.8954e-01

Generation 40: Best = 2.946605e+03 FES = 4100 Diversity = 7.5196e-01

Generation 50: Best = 2.946295e+03 FES = 5100 Diversity = 6.7590e-01

Generation 60: Best = 2.946145e+03 FES = 6100 Diversity = 5.6314e-01

Generation 70: Best = 2.946127e+03 FES = 7100 Diversity = 5.3138e-01

Generation 80: Best = 2.946124e+03 FES = 8100 Diversity = 5.3118e-01

Generation 90: Best = 2.946123e+03 FES = 9100 Diversity = 5.0156e-01

Generation 100: Best = 2.946122e+03 FES = 10100 Diversity = 4.8959e-01

Generation 110: Best = 2.946115e+03 FES = 11100 Diversity = 4.3961e-01

Generation 120: Best = 2.946114e+03 FES = 12100 Diversity = 4.0732e-01

Generation 130: Best = 2.946114e+03 FES = 13100 Diversity = 3.8442e-01

Generation 140: Best = 2.946114e+03 FES = 14100 Diversity = 2.1165e-01

Generation 150: Best = 2.946110e+03 FES = 15100 Diversity = 1.0391e-02

Generation 160: Best = 2.946109e+03 FES = 16100 Diversity = 1.0156e-03

Generation 170: Best = 2.946109e+03 FES = 17100 Diversity = 4.1026e-04

Generation 180: Best = 2.946109e+03 FES = 18100 Diversity = 2.5754e-04

Generation 190: Best = 2.946109e+03 FES = 19100 Diversity = 1.4741e-04

Generation 200: Best = 2.946109e+03 FES = 20100 Diversity = 8.0166e-05

Generation 210: Best = 2.946109e+03 FES = 21100 Diversity = 4.9053e-05

Generation 220: Best = 2.946109e+03 FES = 22100 Diversity = 3.8373e-05

Generation 230: Best = 2.946109e+03 FES = 23100 Diversity = 2.4078e-05

Generation 240: Best = 2.946109e+03 FES = 24100 Diversity = 1.7726e-05

Generation 250: Best = 2.946109e+03 FES = 25100 Diversity = 1.2551e-05

Generation 260: Best = 2.946109e+03 FES = 26100 Diversity = 8.9183e-06

Generation 270: Best = 2.946109e+03 FES = 27100 Diversity = 7.6196e-06

Generation 280: Best = 2.946109e+03 FES = 28100 Diversity = 6.2613e-06

Generation 290: Best = 2.946109e+03 FES = 29100 Diversity = 4.3878e-06

Generation 300: Best = 2.946109e+03 FES = 30100 Diversity = 2.7777e-06

Generation 310: Best = 2.946109e+03 FES = 31100 Diversity = 1.9609e-06

Generation 320: Best = 2.946109e+03 FES = 32100 Diversity = 1.2587e-06

Generation 330: Best = 2.946109e+03 FES = 33100 Diversity = 1.0841e-06

Generation 340: Best = 2.946109e+03 FES = 34100 Diversity = 1.0323e-06

Generation 350: Best = 2.946109e+03 FES = 35100 Diversity = 7.6188e-07

Generation 360: Best = 2.946109e+03 FES = 36100 Diversity = 4.9225e-07

Generation 370: Best = 2.946109e+03 FES = 37100 Diversity = 4.3180e-07

Generation 380: Best = 2.946109e+03 FES = 38100 Diversity = 4.3603e-07

Generation 390: Best = 2.946109e+03 FES = 39100 Diversity = 2.6812e-07

Generation 400: Best = 2.946109e+03 FES = 40100 Diversity = 2.8505e-07

Generation 410: Best = 2.946109e+03 FES = 41100 Diversity = 1.8558e-07

Generation 420: Best = 2.946109e+03 FES = 42100 Diversity = 1.6410e-07

Generation 430: Best = 2.946109e+03 FES = 43100 Diversity = 8.3500e-08

[illegible]

Generation 980: Best = 2.946109e+03 FES = 98100 Diversity = 5.4534e-14
Generation 990: Best = 2.946109e+03 FES = 99100 Diversity = 5.4534e-14

Optimization completed!

Final Best Fitness = 2.946109e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 2.946109e+03

Global Optimum : 2.500000e+03

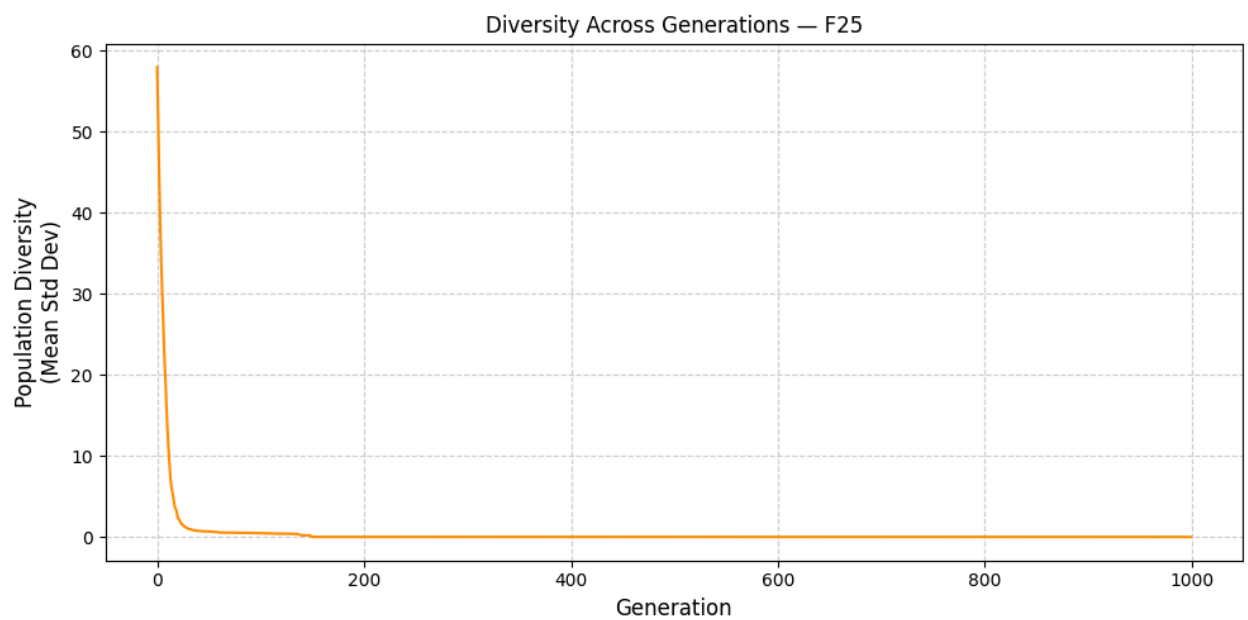
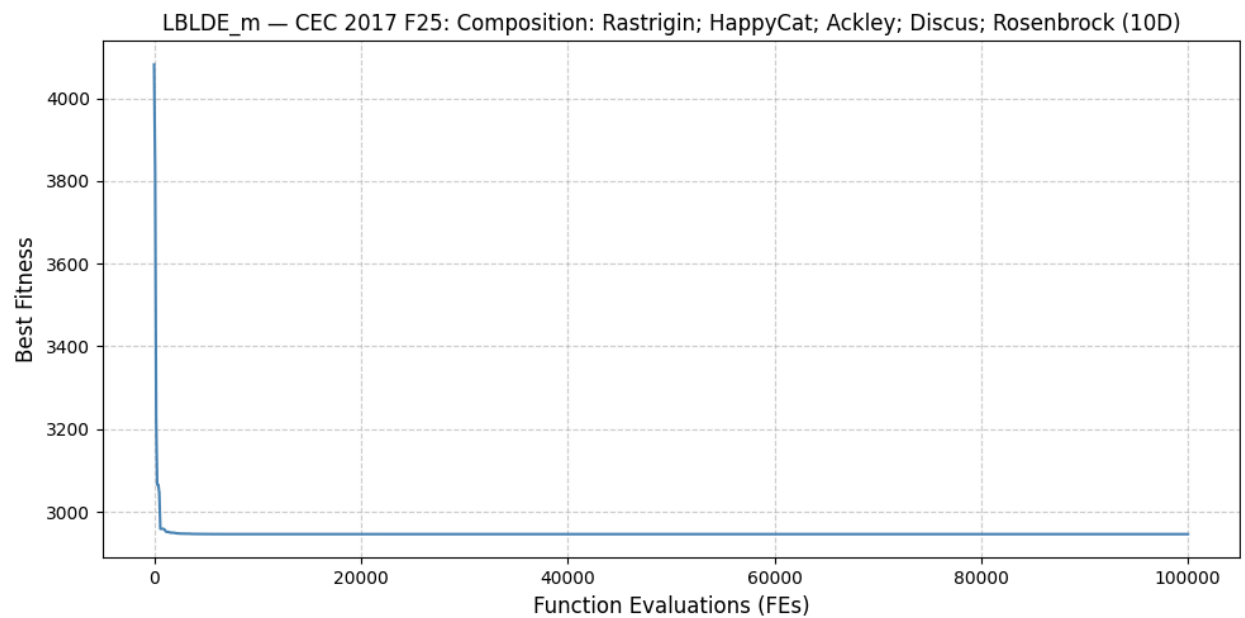
Error (|best-optimum|) : 4.461089e+02

Best Solution (5 dim) : [6.74878891 64.29867031 -64.86422351 -9.97111122
53.92206704]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F25.png

```
In [36]: # =====
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaf
```

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LBLDE_m on CEC 2017 F26 – Single Run

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Function : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (10D)

Bounds : [-100, 100]

Global optimum : 2600.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 4.061684e+03 | Diversity = 5.7928e+01

Generation 10: Best = 3.423833e+03 FES = 1100 Diversity = 2.0399e+01

Generation 20: Best = 3.114023e+03 FES = 2100 Diversity = 6.1499e+00

Generation 30: Best = 3.030493e+03 FES = 3100 Diversity = 2.7991e+00

Generation 40: Best = 3.007999e+03 FES = 4100 Diversity = 1.2726e+00

Generation 50: Best = 2.994763e+03 FES = 5100 Diversity = 8.9572e-01

Generation 60: Best = 2.982917e+03 FES = 6100 Diversity = 4.0459e-01

Generation 70: Best = 2.981077e+03 FES = 7100 Diversity = 1.7764e-01

Generation 80: Best = 2.979441e+03 FES = 8100 Diversity = 1.2808e-01

Generation 90: Best = 2.978629e+03 FES = 9100 Diversity = 9.7298e-02

Generation 100: Best = 2.977772e+03 FES = 10100 Diversity = 8.2229e-02

Generation 110: Best = 2.977310e+03 FES = 11100 Diversity = 5.8566e-02

Generation 120: Best = 2.977013e+03 FES = 12100 Diversity = 4.5650e-02

Generation 130: Best = 2.976893e+03 FES = 13100 Diversity = 2.7957e-02

Generation 140: Best = 2.976823e+03 FES = 14100 Diversity = 2.4289e-02

Generation 150: Best = 2.976767e+03 FES = 15100 Diversity = 1.8669e-02

Generation 160: Best = 2.976746e+03 FES = 16100 Diversity = 1.2254e-02

Generation 170: Best = 2.976725e+03 FES = 17100 Diversity = 1.0427e-02

Generation 180: Best = 2.976706e+03 FES = 18100 Diversity = 9.5853e-03

Generation 190: Best = 2.976698e+03 FES = 19100 Diversity = 5.7846e-03

Generation 200: Best = 2.976691e+03 FES = 20100 Diversity = 4.6463e-03

Generation 210: Best = 2.976688e+03 FES = 21100 Diversity = 3.6131e-03

Generation 220: Best = 2.976685e+03 FES = 22100 Diversity = 3.8786e-03

Generation 230: Best = 2.976683e+03 FES = 23100 Diversity = 2.7098e-03

Generation 240: Best = 2.976682e+03 FES = 24100 Diversity = 2.7332e-03

Generation 250: Best = 2.976681e+03 FES = 25100 Diversity = 2.3086e-03

Generation 260: Best = 2.976680e+03 FES = 26100 Diversity = 1.7976e-03

Generation 270: Best = 2.976680e+03 FES = 27100 Diversity = 1.4589e-03

Generation 280: Best = 2.976679e+03 FES = 28100 Diversity = 1.2720e-03

Generation 290: Best = 2.976679e+03 FES = 29100 Diversity = 1.1255e-03

Generation 300: Best = 2.976679e+03 FES = 30100 Diversity = 5.6005e-04

Generation 310: Best = 2.976679e+03 FES = 31100 Diversity = 4.4009e-04

Generation 320: Best = 2.976679e+03 FES = 32100 Diversity = 3.7966e-04

Generation 330: Best = 2.976679e+03 FES = 33100 Diversity = 2.0432e-04

Generation 340: Best = 2.976679e+03 FES = 34100 Diversity = 1.5777e-04

Generation 350: Best = 2.976679e+03 FES = 35100 Diversity = 1.5674e-04

Generation 360: Best = 2.976679e+03 FES = 36100 Diversity = 1.0940e-04

Generation 370: Best = 2.976679e+03 FES = 37100 Diversity = 1.0714e-04

Generation 380: Best = 2.976679e+03 FES = 38100 Diversity = 6.0687e-05

Generation 390: Best = 2.976679e+03 FES = 39100 Diversity = 3.6165e-05

Generation 400: Best = 2.976679e+03 FES = 40100 Diversity = 3.6290e-05

Generation 410: Best = 2.976679e+03 FES = 41100 Diversity = 3.2935e-05

Generation 420: Best = 2.976679e+03 FES = 42100 Diversity = 2.3539e-05

Generation 430: Best = 2.976679e+03 FES = 43100 Diversity = 2.0190e-05

Generation	440:	Best = 2.976679e+03	FES = 44100	Diversity = 1.2530e-05
Generation	450:	Best = 2.976679e+03	FES = 45100	Diversity = 1.0367e-05
Generation	460:	Best = 2.976679e+03	FES = 46100	Diversity = 8.8502e-06
Generation	470:	Best = 2.976679e+03	FES = 47100	Diversity = 5.4502e-06
Generation	480:	Best = 2.976679e+03	FES = 48100	Diversity = 4.9962e-06
Generation	490:	Best = 2.976679e+03	FES = 49100	Diversity = 5.6539e-06
Generation	500:	Best = 2.976679e+03	FES = 50100	Diversity = 3.0176e-06
Generation	510:	Best = 2.976679e+03	FES = 51100	Diversity = 2.5639e-06
Generation	520:	Best = 2.976679e+03	FES = 52100	Diversity = 1.8070e-06
Generation	530:	Best = 2.976679e+03	FES = 53100	Diversity = 1.3725e-06
Generation	540:	Best = 2.976679e+03	FES = 54100	Diversity = 1.2915e-06
Generation	550:	Best = 2.976679e+03	FES = 55100	Diversity = 1.0811e-06
Generation	560:	Best = 2.976679e+03	FES = 56100	Diversity = 7.3840e-07
Generation	570:	Best = 2.976679e+03	FES = 57100	Diversity = 4.3756e-07
Generation	580:	Best = 2.976679e+03	FES = 58100	Diversity = 3.1308e-07
Generation	590:	Best = 2.976679e+03	FES = 59100	Diversity = 2.6512e-07
Generation	600:	Best = 2.976679e+03	FES = 60100	Diversity = 2.1370e-07
Generation	610:	Best = 2.976679e+03	FES = 61100	Diversity = 2.1021e-07
Generation	620:	Best = 2.976679e+03	FES = 62100	Diversity = 7.6770e-08
Generation	630:	Best = 2.976679e+03	FES = 63100	Diversity = 6.7571e-08
Generation	640:	Best = 2.976679e+03	FES = 64100	Diversity = 4.3717e-08
Generation	650:	Best = 2.976679e+03	FES = 65100	Diversity = 3.5920e-08
Generation	660:	Best = 2.976679e+03	FES = 66100	Diversity = 1.9536e-08
Generation	670:	Best = 2.976679e+03	FES = 67100	Diversity = 3.8038e-09
Generation	680:	Best = 2.976679e+03	FES = 68100	Diversity = 5.4381e-10
Generation	690:	Best = 2.976679e+03	FES = 69100	Diversity = 6.1708e-11
Generation	700:	Best = 2.976679e+03	FES = 70100	Diversity = 4.8734e-12
Generation	710:	Best = 2.976679e+03	FES = 71100	Diversity = 1.0963e-13
Generation	720:	Best = 2.976679e+03	FES = 72100	Diversity = 7.2227e-14
Generation	730:	Best = 2.976679e+03	FES = 73100	Diversity = 6.8850e-14
Generation	740:	Best = 2.976679e+03	FES = 74100	Diversity = 6.8567e-14
Generation	750:	Best = 2.976679e+03	FES = 75100	Diversity = 6.8567e-14
Generation	760:	Best = 2.976679e+03	FES = 76100	Diversity = 6.8567e-14
Generation	770:	Best = 2.976679e+03	FES = 77100	Diversity = 6.8567e-14
Generation	780:	Best = 2.976679e+03	FES = 78100	Diversity = 6.8567e-14
Generation	790:	Best = 2.976679e+03	FES = 79100	Diversity = 6.8567e-14
Generation	800:	Best = 2.976679e+03	FES = 80100	Diversity = 6.8567e-14
Generation	810:	Best = 2.976679e+03	FES = 81100	Diversity = 6.8567e-14
Generation	820:	Best = 2.976679e+03	FES = 82100	Diversity = 6.8567e-14
Generation	830:	Best = 2.976679e+03	FES = 83100	Diversity = 6.8567e-14
Generation	840:	Best = 2.976679e+03	FES = 84100	Diversity = 6.8567e-14
Generation	850:	Best = 2.976679e+03	FES = 85100	Diversity = 6.8567e-14
Generation	860:	Best = 2.976679e+03	FES = 86100	Diversity = 6.8567e-14
Generation	870:	Best = 2.976679e+03	FES = 87100	Diversity = 6.8567e-14
Generation	880:	Best = 2.976679e+03	FES = 88100	Diversity = 6.8567e-14
Generation	890:	Best = 2.976679e+03	FES = 89100	Diversity = 6.8567e-14
Generation	900:	Best = 2.976679e+03	FES = 90100	Diversity = 6.8567e-14
Generation	910:	Best = 2.976679e+03	FES = 91100	Diversity = 6.8567e-14
Generation	920:	Best = 2.976679e+03	FES = 92100	Diversity = 6.8567e-14
Generation	930:	Best = 2.976679e+03	FES = 93100	Diversity = 6.8567e-14
Generation	940:	Best = 2.976679e+03	FES = 94100	Diversity = 6.8567e-14
Generation	950:	Best = 2.976679e+03	FES = 95100	Diversity = 6.8567e-14
Generation	960:	Best = 2.976679e+03	FES = 96100	Diversity = 6.8567e-14
Generation	970:	Best = 2.976679e+03	FES = 97100	Diversity = 6.8567e-14

Generation 980: Best = 2.976679e+03 FES = 98100 Diversity = 6.8567e-14
Generation 990: Best = 2.976679e+03 FES = 99100 Diversity = 6.8567e-14

Optimization completed!

Final Best Fitness = 2.976679e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 2.976679e+03

Global Optimum : 2.600000e+03

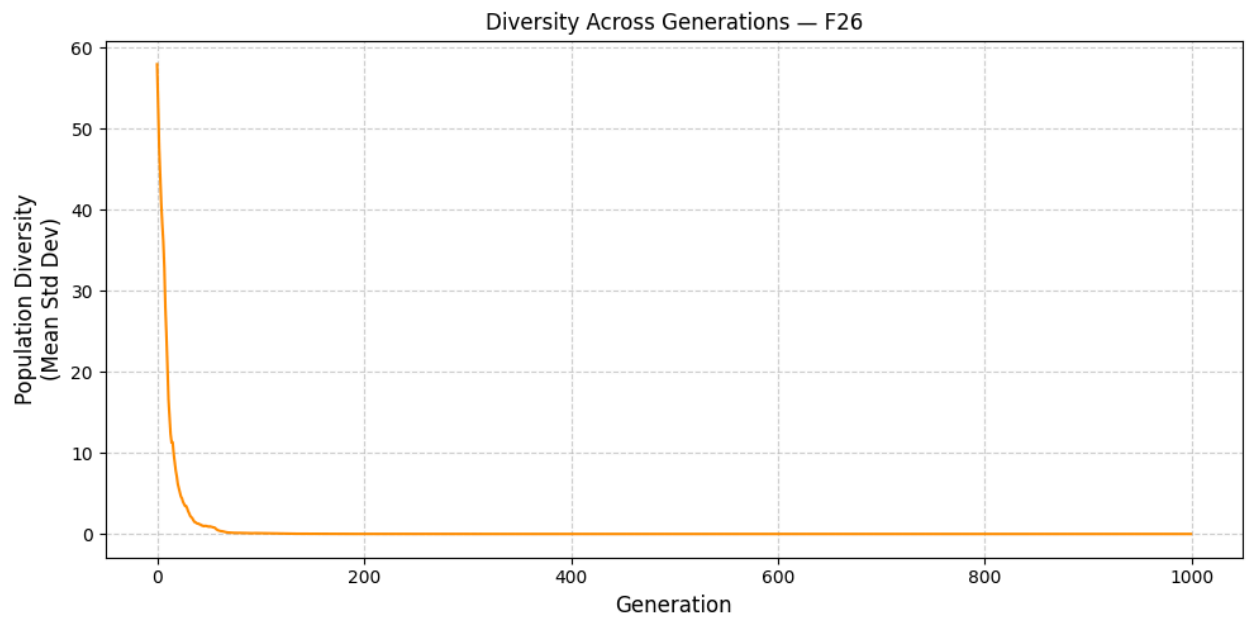
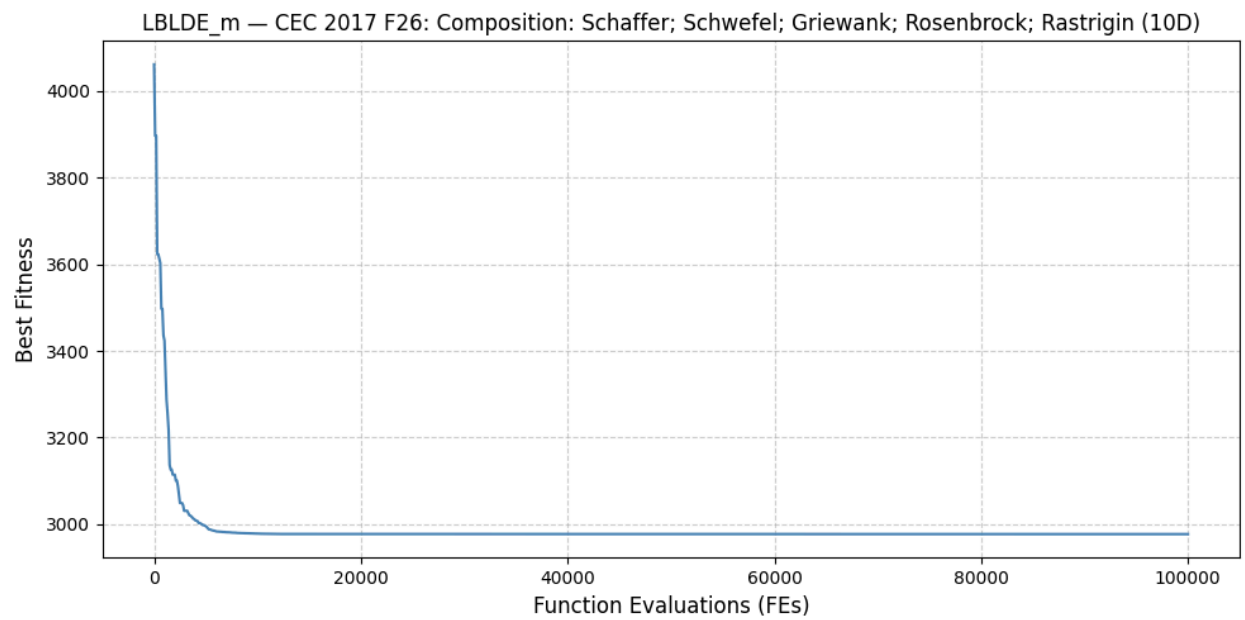
Error (|best-optimum|) : 3.766788e+02

Best Solution (5 dim) : [23.18736642 -9.3453962 74.17672188 -47.40449342
37.0772174]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F26.png

```
In [37]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic)
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat')
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LBLDE_m on CEC 2017 F27 – Single Run

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Function : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (10D)

Bounds : [-100, 100]

Global optimum : 2700.0

Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 3.284356e+03 | Diversity = 5.7928e+01

Generation 10: Best = 3.143600e+03 FES = 1100 Diversity = 3.2813e+01

Generation 20: Best = 3.106315e+03 FES = 2100 Diversity = 1.0257e+01

Generation 30: Best = 3.096229e+03 FES = 3100 Diversity = 4.6477e+00

Generation 40: Best = 3.094197e+03 FES = 4100 Diversity = 3.9345e+00

Generation 50: Best = 3.093777e+03 FES = 5100 Diversity = 3.5598e+00

Generation 60: Best = 3.093718e+03 FES = 6100 Diversity = 3.2215e+00

Generation 70: Best = 3.093689e+03 FES = 7100 Diversity = 2.6162e+00

Generation 80: Best = 3.093680e+03 FES = 8100 Diversity = 2.4932e+00

Generation 90: Best = 3.093677e+03 FES = 9100 Diversity = 2.2406e+00

Generation 100: Best = 3.093627e+03 FES = 10100 Diversity = 2.1694e+00

Generation 110: Best = 3.093597e+03 FES = 11100 Diversity = 2.1698e+00

Generation 120: Best = 3.093581e+03 FES = 12100 Diversity = 2.1070e+00

Generation 130: Best = 3.093570e+03 FES = 13100 Diversity = 2.1072e+00

Generation 140: Best = 3.093566e+03 FES = 14100 Diversity = 2.1108e+00

Generation 150: Best = 3.093564e+03 FES = 15100 Diversity = 1.9290e+00

Generation 160: Best = 3.093564e+03 FES = 16100 Diversity = 1.9292e+00

Generation 170: Best = 3.093563e+03 FES = 17100 Diversity = 1.8862e+00

Generation 180: Best = 3.093563e+03 FES = 18100 Diversity = 1.8850e+00

Generation 190: Best = 3.093563e+03 FES = 19100 Diversity = 1.8818e+00

Generation 200: Best = 3.093563e+03 FES = 20100 Diversity = 1.8819e+00

Generation 210: Best = 3.093563e+03 FES = 21100 Diversity = 1.8726e+00

Generation 220: Best = 3.093563e+03 FES = 22100 Diversity = 1.8726e+00

Generation 230: Best = 3.093563e+03 FES = 23100 Diversity = 1.8726e+00

Generation 240: Best = 3.093563e+03 FES = 24100 Diversity = 1.8733e+00

Generation 250: Best = 3.093563e+03 FES = 25100 Diversity = 1.8746e+00

Generation 260: Best = 3.093563e+03 FES = 26100 Diversity = 1.8746e+00

Generation 270: Best = 3.093563e+03 FES = 27100 Diversity = 1.8746e+00

Generation 280: Best = 3.093563e+03 FES = 28100 Diversity = 1.8746e+00

Generation 290: Best = 3.093563e+03 FES = 29100 Diversity = 1.8746e+00

Generation 300: Best = 3.093563e+03 FES = 30100 Diversity = 1.8746e+00

Generation 310: Best = 3.093563e+03 FES = 31100 Diversity = 1.8746e+00

Generation 320: Best = 3.093563e+03 FES = 32100 Diversity = 1.8736e+00

Generation 330: Best = 3.093563e+03 FES = 33100 Diversity = 1.8736e+00

Generation 340: Best = 3.093563e+03 FES = 34100 Diversity = 1.8443e+00

Generation 350: Best = 3.093563e+03 FES = 35100 Diversity = 1.6027e+00

Generation 360: Best = 3.093563e+03 FES = 36100 Diversity = 1.6027e+00

Generation 370: Best = 3.093563e+03 FES = 37100 Diversity = 1.6027e+00

Generation 380: Best = 3.093563e+03 FES = 38100 Diversity = 1.6027e+00

Generation 390: Best = 3.093563e+03 FES = 39100 Diversity = 1.6014e+00

Generation 400: Best = 3.093563e+03 FES = 40100 Diversity = 1.6014e+00

Generation 410: Best = 3.093563e+03 FES = 41100 Diversity = 1.6009e+00

Generation 420: Best = 3.093563e+03 FES = 42100 Diversity = 1.6009e+00

Generation 430: Best = 3.093563e+03 FES = 43100 Diversity = 1.6009e+00

Generation	440:	Best = 3.093563e+03	FES = 44100	Diversity = 1.6009e+00
Generation	450:	Best = 3.093563e+03	FES = 45100	Diversity = 1.5887e+00
Generation	460:	Best = 3.093563e+03	FES = 46100	Diversity = 1.5836e+00
Generation	470:	Best = 3.093563e+03	FES = 47100	Diversity = 1.5836e+00
Generation	480:	Best = 3.093563e+03	FES = 48100	Diversity = 1.5836e+00
Generation	490:	Best = 3.093563e+03	FES = 49100	Diversity = 1.5836e+00
Generation	500:	Best = 3.093563e+03	FES = 50100	Diversity = 1.5836e+00
Generation	510:	Best = 3.093563e+03	FES = 51100	Diversity = 1.5836e+00
Generation	520:	Best = 3.093563e+03	FES = 52100	Diversity = 1.5776e+00
Generation	530:	Best = 3.093563e+03	FES = 53100	Diversity = 1.5776e+00
Generation	540:	Best = 3.093563e+03	FES = 54100	Diversity = 1.5776e+00
Generation	550:	Best = 3.093563e+03	FES = 55100	Diversity = 1.5776e+00
Generation	560:	Best = 3.093563e+03	FES = 56100	Diversity = 1.5776e+00
Generation	570:	Best = 3.093563e+03	FES = 57100	Diversity = 1.5776e+00
Generation	580:	Best = 3.093563e+03	FES = 58100	Diversity = 1.5776e+00
Generation	590:	Best = 3.093563e+03	FES = 59100	Diversity = 1.5776e+00
Generation	600:	Best = 3.093563e+03	FES = 60100	Diversity = 1.5776e+00
Generation	610:	Best = 3.093563e+03	FES = 61100	Diversity = 1.5776e+00
Generation	620:	Best = 3.093563e+03	FES = 62100	Diversity = 1.5776e+00
Generation	630:	Best = 3.093545e+03	FES = 63100	Diversity = 4.0566e-02
Generation	640:	Best = 3.093539e+03	FES = 64100	Diversity = 1.5485e-02
Generation	650:	Best = 3.093538e+03	FES = 65100	Diversity = 7.2511e-03
Generation	660:	Best = 3.093537e+03	FES = 66100	Diversity = 3.9154e-03
Generation	670:	Best = 3.093537e+03	FES = 67100	Diversity = 3.2719e-03
Generation	680:	Best = 3.093536e+03	FES = 68100	Diversity = 2.9123e-03
Generation	690:	Best = 3.093536e+03	FES = 69100	Diversity = 2.2468e-03
Generation	700:	Best = 3.093536e+03	FES = 70100	Diversity = 1.3151e-03
Generation	710:	Best = 3.093536e+03	FES = 71100	Diversity = 9.3797e-04
Generation	720:	Best = 3.093536e+03	FES = 72100	Diversity = 6.8721e-04
Generation	730:	Best = 3.093536e+03	FES = 73100	Diversity = 3.2083e-04
Generation	740:	Best = 3.093536e+03	FES = 74100	Diversity = 2.3893e-04
Generation	750:	Best = 3.093536e+03	FES = 75100	Diversity = 1.4017e-04
Generation	760:	Best = 3.093536e+03	FES = 76100	Diversity = 1.0514e-04
Generation	770:	Best = 3.093536e+03	FES = 77100	Diversity = 4.2114e-05
Generation	780:	Best = 3.093536e+03	FES = 78100	Diversity = 2.8384e-05
Generation	790:	Best = 3.093536e+03	FES = 79100	Diversity = 1.7145e-05
Generation	800:	Best = 3.093536e+03	FES = 80100	Diversity = 1.4360e-05
Generation	810:	Best = 3.093536e+03	FES = 81100	Diversity = 1.0410e-05
Generation	820:	Best = 3.093536e+03	FES = 82100	Diversity = 8.0243e-06
Generation	830:	Best = 3.093536e+03	FES = 83100	Diversity = 4.2789e-06
Generation	840:	Best = 3.093536e+03	FES = 84100	Diversity = 2.3188e-06
Generation	850:	Best = 3.093536e+03	FES = 85100	Diversity = 2.4769e-06
Generation	860:	Best = 3.093536e+03	FES = 86100	Diversity = 2.1403e-06
Generation	870:	Best = 3.093536e+03	FES = 87100	Diversity = 2.0309e-06
Generation	880:	Best = 3.093536e+03	FES = 88100	Diversity = 1.4553e-06
Generation	890:	Best = 3.093536e+03	FES = 89100	Diversity = 1.0888e-06
Generation	900:	Best = 3.093536e+03	FES = 90100	Diversity = 1.2546e-06
Generation	910:	Best = 3.093536e+03	FES = 91100	Diversity = 1.0655e-06
Generation	920:	Best = 3.093536e+03	FES = 92100	Diversity = 8.2816e-07
Generation	930:	Best = 3.093536e+03	FES = 93100	Diversity = 3.8115e-07
Generation	940:	Best = 3.093536e+03	FES = 94100	Diversity = 2.5406e-07
Generation	950:	Best = 3.093536e+03	FES = 95100	Diversity = 1.6612e-07
Generation	960:	Best = 3.093536e+03	FES = 96100	Diversity = 8.9415e-08
Generation	970:	Best = 3.093536e+03	FES = 97100	Diversity = 1.1409e-07

Generation 980: Best = 3.093536e+03 FES = 98100 Diversity = 1.5577e-07
Generation 990: Best = 3.093536e+03 FES = 99100 Diversity = 4.5772e-08

Optimization completed!

Final Best Fitness = 3.093536e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 3.093536e+03

Global Optimum : 2.700000e+03

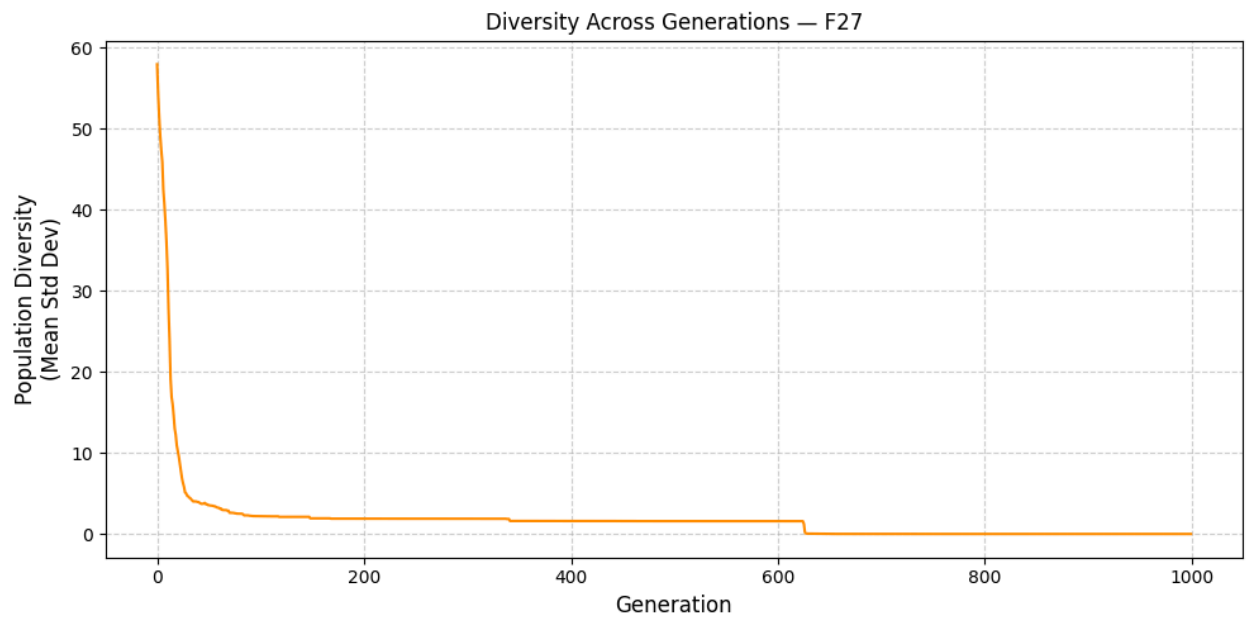
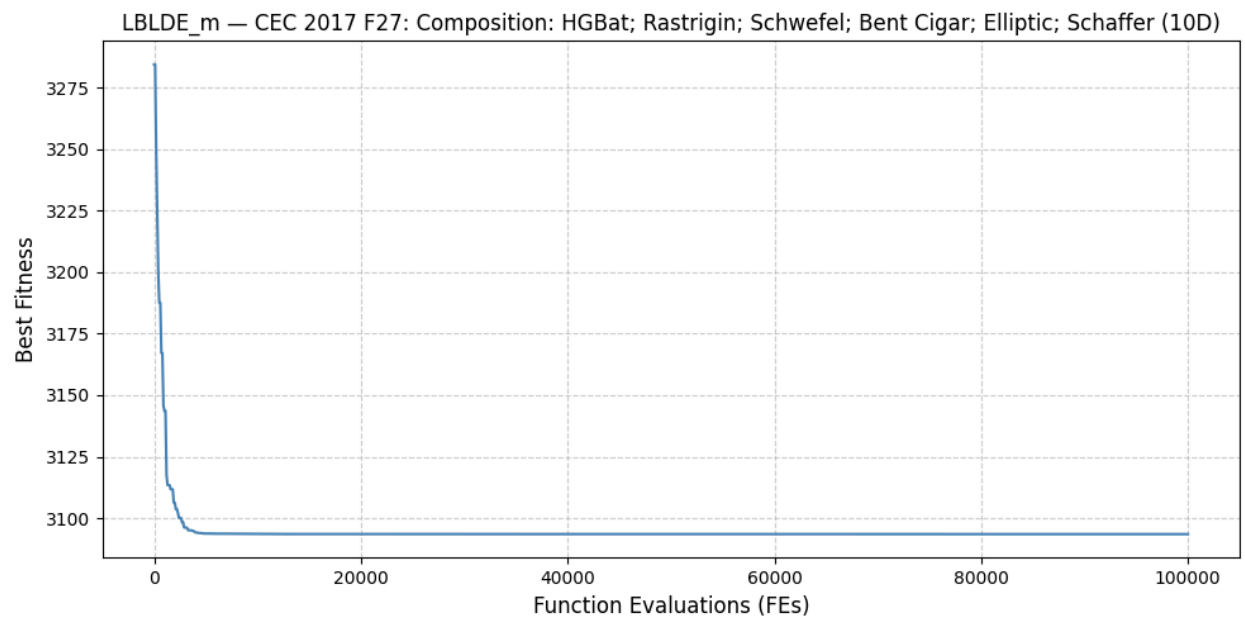
Error (|best-optimum|) : 3.935359e+02

Best Solution (5 dim) : [-26.61655096 -69.41877715 -95.53738557 -50.94905352
94.52821141]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F27.png

```
In [38]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

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LBLDE_m on CEC 2017 F28 – Single Run

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Function : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (10D)

Bounds : [-100, 100]

Global optimum : 2800.0

Initial $D^*(0) = 5.7928e+01$ | Diversity threshold = 5.7928e+00

Generation 0: Best Fitness = 4.188047e+03 | Diversity = 5.7928e+01

Generation 10: Best = 3.270183e+03 FES = 1100 Diversity = 3.3193e+01

Generation 20: Best = 3.181375e+03 FES = 2100 Diversity = 1.2744e+01

Generation 30: Best = 3.175718e+03 FES = 3100 Diversity = 1.9757e+00

Generation 40: Best = 3.173296e+03 FES = 4100 Diversity = 1.5241e+00

Generation 50: Best = 3.170347e+03 FES = 5100 Diversity = 1.4447e+00

Generation 60: Best = 3.164278e+03 FES = 6100 Diversity = 1.4411e+00

Generation 70: Best = 3.155417e+03 FES = 7100 Diversity = 1.2133e+00

Generation 80: Best = 3.148275e+03 FES = 8100 Diversity = 9.4956e-01

Generation 90: Best = 3.144672e+03 FES = 9100 Diversity = 6.4447e-01

Generation 100: Best = 3.142000e+03 FES = 10100 Diversity = 4.6596e-01

Generation 110: Best = 3.140762e+03 FES = 11100 Diversity = 2.8669e-01

Generation 120: Best = 3.140216e+03 FES = 12100 Diversity = 1.8543e-01

Generation 130: Best = 3.139912e+03 FES = 13100 Diversity = 1.0592e-01

Generation 140: Best = 3.139746e+03 FES = 14100 Diversity = 8.2123e-02

Generation 150: Best = 3.139672e+03 FES = 15100 Diversity = 5.1804e-02

Generation 160: Best = 3.139624e+03 FES = 16100 Diversity = 3.7293e-02

Generation 170: Best = 3.139599e+03 FES = 17100 Diversity = 2.3809e-02

Generation 180: Best = 3.139583e+03 FES = 18100 Diversity = 2.1151e-02

Generation 190: Best = 3.139568e+03 FES = 19100 Diversity = 1.7124e-02

Generation 200: Best = 3.139550e+03 FES = 20100 Diversity = 1.8294e-02

Generation 210: Best = 3.139531e+03 FES = 21100 Diversity = 1.8326e-02

Generation 220: Best = 3.139521e+03 FES = 22100 Diversity = 1.5674e-02

Generation 230: Best = 3.139513e+03 FES = 23100 Diversity = 1.0800e-02

Generation 240: Best = 3.139506e+03 FES = 24100 Diversity = 1.4753e-02

Generation 250: Best = 3.139503e+03 FES = 25100 Diversity = 1.1535e-02

Generation 260: Best = 3.139502e+03 FES = 26100 Diversity = 5.3257e-03

Generation 270: Best = 3.139501e+03 FES = 27100 Diversity = 3.9203e-03

Generation 280: Best = 3.139500e+03 FES = 28100 Diversity = 4.5523e-03

Generation 290: Best = 3.139500e+03 FES = 29100 Diversity = 3.4213e-03

Generation 300: Best = 3.139499e+03 FES = 30100 Diversity = 2.4706e-03

Generation 310: Best = 3.139499e+03 FES = 31100 Diversity = 1.6734e-03

Generation 320: Best = 3.139499e+03 FES = 32100 Diversity = 1.6183e-03

Generation 330: Best = 3.139499e+03 FES = 33100 Diversity = 1.3513e-03

Generation 340: Best = 3.139499e+03 FES = 34100 Diversity = 1.2887e-03

Generation 350: Best = 3.139499e+03 FES = 35100 Diversity = 9.0941e-04

Generation 360: Best = 3.139499e+03 FES = 36100 Diversity = 5.7347e-04

Generation 370: Best = 3.139499e+03 FES = 37100 Diversity = 3.0583e-04

Generation 380: Best = 3.139499e+03 FES = 38100 Diversity = 3.2994e-04

Generation 390: Best = 3.139499e+03 FES = 39100 Diversity = 2.3538e-04

Generation 400: Best = 3.139499e+03 FES = 40100 Diversity = 2.1318e-04

Generation 410: Best = 3.139499e+03 FES = 41100 Diversity = 1.4691e-04

Generation 420: Best = 3.139499e+03 FES = 42100 Diversity = 1.6194e-04

Generation 430: Best = 3.139499e+03 FES = 43100 Diversity = 1.7652e-04

Generation	440:	Best = 3.139499e+03	FES = 44100	Diversity = 1.3919e-04
Generation	450:	Best = 3.139499e+03	FES = 45100	Diversity = 7.9104e-05
Generation	460:	Best = 3.139499e+03	FES = 46100	Diversity = 6.0269e-05
Generation	470:	Best = 3.139499e+03	FES = 47100	Diversity = 3.2213e-05
Generation	480:	Best = 3.139499e+03	FES = 48100	Diversity = 2.9997e-05
Generation	490:	Best = 3.139499e+03	FES = 49100	Diversity = 2.8151e-05
Generation	500:	Best = 3.139499e+03	FES = 50100	Diversity = 2.1055e-05
Generation	510:	Best = 3.139499e+03	FES = 51100	Diversity = 1.6920e-05
Generation	520:	Best = 3.139499e+03	FES = 52100	Diversity = 1.4339e-05
Generation	530:	Best = 3.139499e+03	FES = 53100	Diversity = 5.8458e-06
Generation	540:	Best = 3.139499e+03	FES = 54100	Diversity = 4.9145e-06
Generation	550:	Best = 3.139499e+03	FES = 55100	Diversity = 4.1702e-06
Generation	560:	Best = 3.139499e+03	FES = 56100	Diversity = 2.3609e-06
Generation	570:	Best = 3.139499e+03	FES = 57100	Diversity = 1.9730e-06
Generation	580:	Best = 3.139499e+03	FES = 58100	Diversity = 1.3887e-06
Generation	590:	Best = 3.139499e+03	FES = 59100	Diversity = 1.7430e-06
Generation	600:	Best = 3.139499e+03	FES = 60100	Diversity = 1.0936e-06
Generation	610:	Best = 3.139499e+03	FES = 61100	Diversity = 1.3082e-06
Generation	620:	Best = 3.139499e+03	FES = 62100	Diversity = 7.6562e-07
Generation	630:	Best = 3.139499e+03	FES = 63100	Diversity = 1.2538e-06
Generation	640:	Best = 3.139499e+03	FES = 64100	Diversity = 8.0152e-07
Generation	650:	Best = 3.139499e+03	FES = 65100	Diversity = 3.6494e-07
Generation	660:	Best = 3.139499e+03	FES = 66100	Diversity = 2.4384e-07
Generation	670:	Best = 3.139499e+03	FES = 67100	Diversity = 1.6295e-07
Generation	680:	Best = 3.139499e+03	FES = 68100	Diversity = 9.7154e-08
Generation	690:	Best = 3.139499e+03	FES = 69100	Diversity = 5.0379e-08
Generation	700:	Best = 3.139499e+03	FES = 70100	Diversity = 1.0498e-09
Generation	710:	Best = 3.139499e+03	FES = 71100	Diversity = 2.8353e-11
Generation	720:	Best = 3.139499e+03	FES = 72100	Diversity = 3.7288e-12
Generation	730:	Best = 3.139499e+03	FES = 73100	Diversity = 5.2888e-13
Generation	740:	Best = 3.139499e+03	FES = 74100	Diversity = 5.6660e-14
Generation	750:	Best = 3.139499e+03	FES = 75100	Diversity = 5.6843e-14
Generation	760:	Best = 3.139499e+03	FES = 76100	Diversity = 5.6843e-14
Generation	770:	Best = 3.139499e+03	FES = 77100	Diversity = 5.6843e-14
Generation	780:	Best = 3.139499e+03	FES = 78100	Diversity = 5.6843e-14
Generation	790:	Best = 3.139499e+03	FES = 79100	Diversity = 5.6843e-14
Generation	800:	Best = 3.139499e+03	FES = 80100	Diversity = 5.6843e-14
Generation	810:	Best = 3.139499e+03	FES = 81100	Diversity = 5.6843e-14
Generation	820:	Best = 3.139499e+03	FES = 82100	Diversity = 5.6843e-14
Generation	830:	Best = 3.139499e+03	FES = 83100	Diversity = 5.6843e-14
Generation	840:	Best = 3.139499e+03	FES = 84100	Diversity = 5.6843e-14
Generation	850:	Best = 3.139499e+03	FES = 85100	Diversity = 5.6843e-14
Generation	860:	Best = 3.139499e+03	FES = 86100	Diversity = 5.6843e-14
Generation	870:	Best = 3.139499e+03	FES = 87100	Diversity = 5.6843e-14
Generation	880:	Best = 3.139499e+03	FES = 88100	Diversity = 5.6843e-14
Generation	890:	Best = 3.139499e+03	FES = 89100	Diversity = 5.6843e-14
Generation	900:	Best = 3.139499e+03	FES = 90100	Diversity = 5.6843e-14
Generation	910:	Best = 3.139499e+03	FES = 91100	Diversity = 5.6843e-14
Generation	920:	Best = 3.139499e+03	FES = 92100	Diversity = 5.6843e-14
Generation	930:	Best = 3.139499e+03	FES = 93100	Diversity = 5.6843e-14
Generation	940:	Best = 3.139499e+03	FES = 94100	Diversity = 5.6843e-14
Generation	950:	Best = 3.139499e+03	FES = 95100	Diversity = 5.6843e-14
Generation	960:	Best = 3.139499e+03	FES = 96100	Diversity = 5.6843e-14
Generation	970:	Best = 3.139499e+03	FES = 97100	Diversity = 5.6843e-14

Generation 980: Best = 3.139499e+03 FES = 98100 Diversity = 5.6843e-14
Generation 990: Best = 3.139499e+03 FES = 99100 Diversity = 5.6843e-14

Optimization completed!

Final Best Fitness = 3.139499e+03

Total Function Evaluations = 100000

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Results:

Best Fitness : 3.139499e+03

Global Optimum : 2.800000e+03

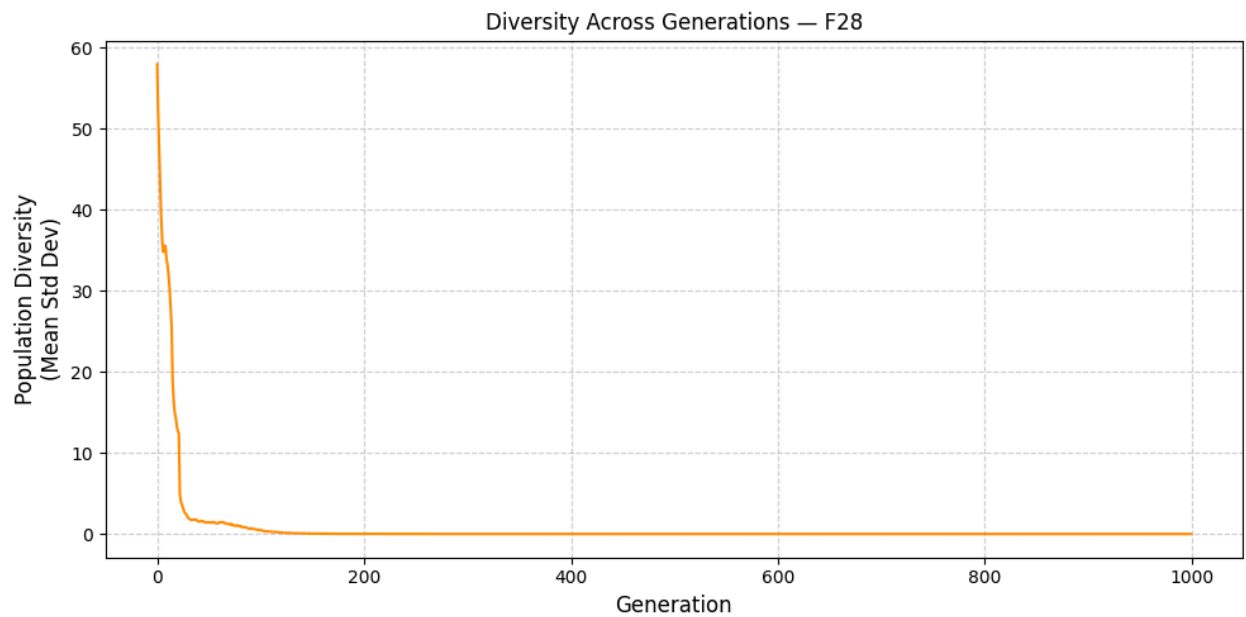
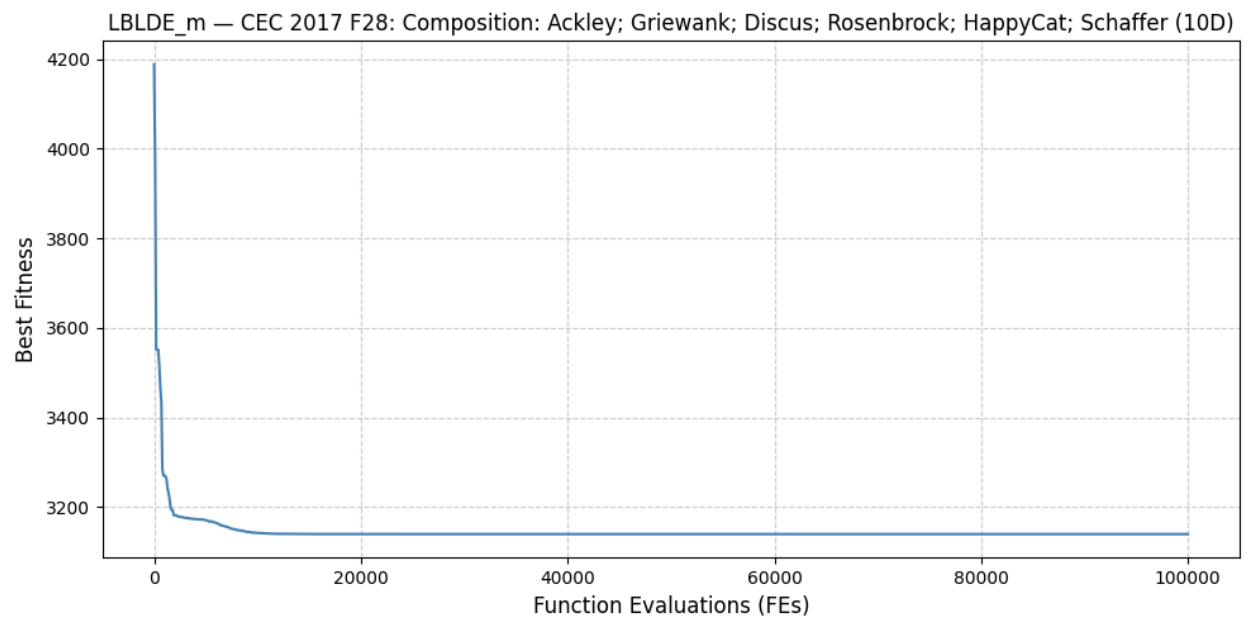
Error (|best-optimum|) : 3.394990e+02

Best Solution (5 dim) : [-34.32282187 -40.01258951 -51.36365732 48.25146467
-66.28982121]

D^(0) (init. diversity): 5.7928e+01

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Graph saved as convergence_F28.png

```
In [39]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

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LBLDE_m on CEC 2017 F29 – Single Run

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Function : Composition: F15; F16; F17 (10D)
Bounds : [-100, 100]
Global optimum : 2900.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 3.805371e+03 | Diversity = 5.7928e+01

Generation 10:	Best = 3.450402e+03	FES = 1100	Diversity = 4.3503e+01
Generation 20:	Best = 3.254916e+03	FES = 2100	Diversity = 3.7397e+01
Generation 30:	Best = 3.224870e+03	FES = 3100	Diversity = 3.2564e+01
Generation 40:	Best = 3.222870e+03	FES = 4100	Diversity = 2.9490e+01
Generation 50:	Best = 3.222870e+03	FES = 5100	Diversity = 2.6756e+01
Generation 60:	Best = 3.210205e+03	FES = 6100	Diversity = 2.4008e+01
Generation 70:	Best = 3.203366e+03	FES = 7100	Diversity = 2.3078e+01
Generation 80:	Best = 3.203366e+03	FES = 8100	Diversity = 2.0863e+01
Generation 90:	Best = 3.203366e+03	FES = 9100	Diversity = 2.0947e+01
Generation 100:	Best = 3.203366e+03	FES = 10100	Diversity = 2.0823e+01
Generation 110:	Best = 3.201757e+03	FES = 11100	Diversity = 2.0377e+01
Generation 120:	Best = 3.186669e+03	FES = 12100	Diversity = 1.9728e+01
Generation 130:	Best = 3.178637e+03	FES = 13100	Diversity = 1.9615e+01
Generation 140:	Best = 3.178637e+03	FES = 14100	Diversity = 1.9342e+01
Generation 150:	Best = 3.170702e+03	FES = 15100	Diversity = 1.9599e+01
Generation 160:	Best = 3.170702e+03	FES = 16100	Diversity = 1.9579e+01
Generation 170:	Best = 3.170702e+03	FES = 17100	Diversity = 1.9122e+01
Generation 180:	Best = 3.170702e+03	FES = 18100	Diversity = 1.9395e+01
Generation 190:	Best = 3.167036e+03	FES = 19100	Diversity = 1.8930e+01
Generation 200:	Best = 3.167036e+03	FES = 20100	Diversity = 1.8835e+01
Generation 210:	Best = 3.167036e+03	FES = 21100	Diversity = 1.8247e+01
Generation 220:	Best = 3.165765e+03	FES = 22100	Diversity = 1.8217e+01
Generation 230:	Best = 3.165765e+03	FES = 23100	Diversity = 1.8179e+01
Generation 240:	Best = 3.165608e+03	FES = 24100	Diversity = 1.7926e+01
Generation 250:	Best = 3.165608e+03	FES = 25100	Diversity = 1.8103e+01
Generation 260:	Best = 3.165608e+03	FES = 26100	Diversity = 1.8444e+01
Generation 270:	Best = 3.165608e+03	FES = 27100	Diversity = 1.8336e+01
Generation 280:	Best = 3.165608e+03	FES = 28100	Diversity = 1.8309e+01
Generation 290:	Best = 3.162272e+03	FES = 29100	Diversity = 1.8927e+01
Generation 300:	Best = 3.162272e+03	FES = 30100	Diversity = 1.8914e+01
Generation 310:	Best = 3.162272e+03	FES = 31100	Diversity = 1.8956e+01
Generation 320:	Best = 3.160853e+03	FES = 32100	Diversity = 1.8698e+01
Generation 330:	Best = 3.160853e+03	FES = 33100	Diversity = 1.8262e+01
Generation 340:	Best = 3.160853e+03	FES = 34100	Diversity = 1.8224e+01
Generation 350:	Best = 3.159374e+03	FES = 35100	Diversity = 1.8485e+01
Generation 360:	Best = 3.159374e+03	FES = 36100	Diversity = 1.8485e+01
Generation 370:	Best = 3.159374e+03	FES = 37100	Diversity = 1.8071e+01
Generation 380:	Best = 3.159374e+03	FES = 38100	Diversity = 1.8097e+01
Generation 390:	Best = 3.159374e+03	FES = 39100	Diversity = 1.8242e+01
Generation 400:	Best = 3.159374e+03	FES = 40100	Diversity = 1.8411e+01
Generation 410:	Best = 3.159374e+03	FES = 41100	Diversity = 1.8406e+01
Generation 420:	Best = 3.159374e+03	FES = 42100	Diversity = 1.8289e+01
Generation 430:	Best = 3.159374e+03	FES = 43100	Diversity = 1.8456e+01
Generation 440:	Best = 3.158353e+03	FES = 44100	Diversity = 1.8313e+01

Generation	450:	Best = 3.158353e+03	FES = 45100	Diversity = 1.8339e+01
Generation	460:	Best = 3.158353e+03	FES = 46100	Diversity = 1.8033e+01
Generation	470:	Best = 3.158353e+03	FES = 47100	Diversity = 1.7990e+01
Generation	480:	Best = 3.158353e+03	FES = 48100	Diversity = 1.8041e+01
Generation	490:	Best = 3.158353e+03	FES = 49100	Diversity = 1.7824e+01
Generation	500:	Best = 3.158353e+03	FES = 50100	Diversity = 1.7828e+01
Generation	510:	Best = 3.158353e+03	FES = 51100	Diversity = 1.7837e+01
Generation	520:	Best = 3.158353e+03	FES = 52100	Diversity = 1.7649e+01
Generation	530:	Best = 3.158100e+03	FES = 53100	Diversity = 1.7661e+01
Generation	540:	Best = 3.158100e+03	FES = 54100	Diversity = 1.7758e+01
Generation	550:	Best = 3.157493e+03	FES = 55100	Diversity = 1.7608e+01
Generation	560:	Best = 3.157493e+03	FES = 56100	Diversity = 1.7587e+01
Generation	570:	Best = 3.156685e+03	FES = 57100	Diversity = 1.7215e+01
Generation	580:	Best = 3.156685e+03	FES = 58100	Diversity = 1.7080e+01
Generation	590:	Best = 3.154129e+03	FES = 59100	Diversity = 1.7019e+01
Generation	600:	Best = 3.152922e+03	FES = 60100	Diversity = 1.7024e+01
Generation	610:	Best = 3.152922e+03	FES = 61100	Diversity = 1.7207e+01
Generation	620:	Best = 3.152922e+03	FES = 62100	Diversity = 1.7347e+01
Generation	630:	Best = 3.152886e+03	FES = 63100	Diversity = 1.7323e+01
Generation	640:	Best = 3.152886e+03	FES = 64100	Diversity = 1.7426e+01
Generation	650:	Best = 3.150065e+03	FES = 65100	Diversity = 1.7379e+01
Generation	660:	Best = 3.149619e+03	FES = 66100	Diversity = 1.7333e+01
Generation	670:	Best = 3.149619e+03	FES = 67100	Diversity = 1.7209e+01
Generation	680:	Best = 3.149619e+03	FES = 68100	Diversity = 1.7174e+01
Generation	690:	Best = 3.149619e+03	FES = 69100	Diversity = 1.7069e+01
Generation	700:	Best = 3.149619e+03	FES = 70100	Diversity = 1.7134e+01
Generation	710:	Best = 3.149619e+03	FES = 71100	Diversity = 1.7140e+01
Generation	720:	Best = 3.149619e+03	FES = 72100	Diversity = 1.6866e+01
Generation	730:	Best = 3.149619e+03	FES = 73100	Diversity = 1.6759e+01
Generation	740:	Best = 3.149619e+03	FES = 74100	Diversity = 1.6796e+01
Generation	750:	Best = 3.149619e+03	FES = 75100	Diversity = 1.6670e+01
Generation	760:	Best = 3.149619e+03	FES = 76100	Diversity = 1.6628e+01
Generation	770:	Best = 3.149619e+03	FES = 77100	Diversity = 1.6771e+01
Generation	780:	Best = 3.149619e+03	FES = 78100	Diversity = 1.6765e+01
Generation	790:	Best = 3.149619e+03	FES = 79100	Diversity = 1.6698e+01
Generation	800:	Best = 3.149619e+03	FES = 80100	Diversity = 1.6830e+01
Generation	810:	Best = 3.145399e+03	FES = 81100	Diversity = 1.7042e+01
Generation	820:	Best = 3.145399e+03	FES = 82100	Diversity = 1.7057e+01
Generation	830:	Best = 3.145399e+03	FES = 83100	Diversity = 1.7058e+01
Generation	840:	Best = 3.145399e+03	FES = 84100	Diversity = 1.7058e+01
Generation	850:	Best = 3.145399e+03	FES = 85100	Diversity = 1.7008e+01
Generation	860:	Best = 3.145399e+03	FES = 86100	Diversity = 1.7154e+01
Generation	870:	Best = 3.145399e+03	FES = 87100	Diversity = 1.7169e+01
Generation	880:	Best = 3.145399e+03	FES = 88100	Diversity = 1.7193e+01
Generation	890:	Best = 3.145399e+03	FES = 89100	Diversity = 1.6604e+01
Generation	900:	Best = 3.144532e+03	FES = 90100	Diversity = 1.6604e+01
Generation	910:	Best = 3.144532e+03	FES = 91100	Diversity = 1.6675e+01
Generation	920:	Best = 3.144210e+03	FES = 92100	Diversity = 1.6668e+01
Generation	930:	Best = 3.141694e+03	FES = 93100	Diversity = 1.6670e+01
Generation	940:	Best = 3.141694e+03	FES = 94100	Diversity = 1.6693e+01
Generation	950:	Best = 3.141485e+03	FES = 95100	Diversity = 1.6698e+01
Generation	960:	Best = 3.141485e+03	FES = 96100	Diversity = 1.6668e+01
Generation	970:	Best = 3.141485e+03	FES = 97100	Diversity = 1.6743e+01
Generation	980:	Best = 3.141368e+03	FES = 98100	Diversity = 1.6846e+01

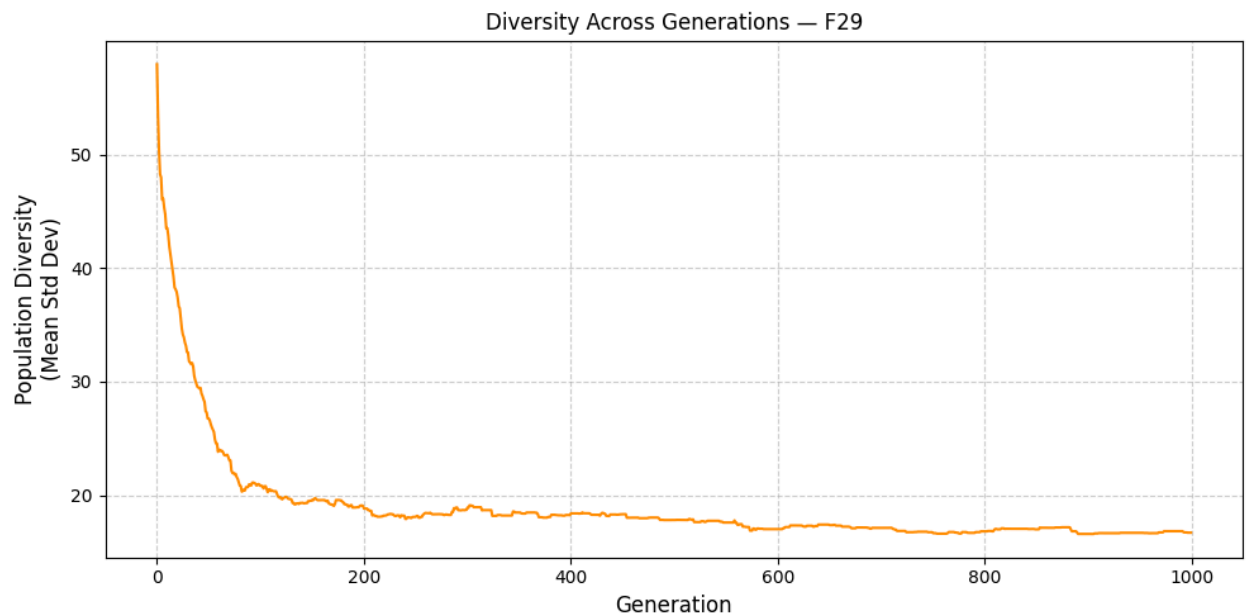
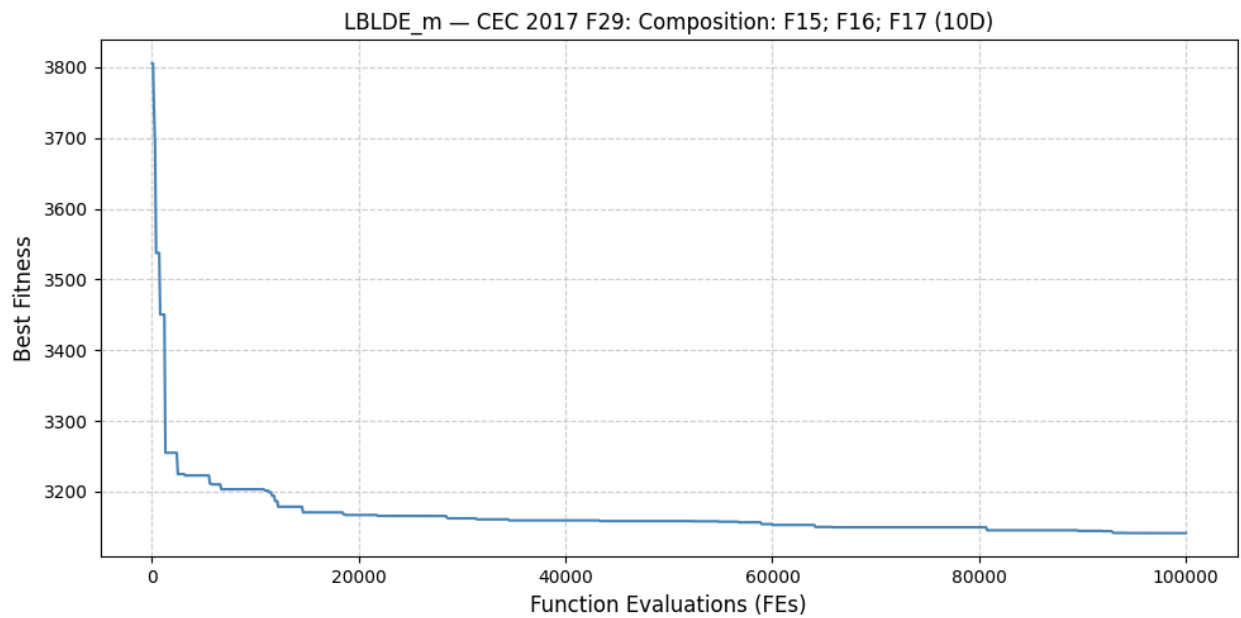
Generation 990: Best = 3.141368e+03 FES = 99100 Diversity = 1.6844e+01

Optimization completed!

Final Best Fitness = 3.141368e+03

Total Function Evaluations = 100000

```
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Results:
  Best Fitness           : 3.141368e+03
  Global Optimum         : 2.900000e+03
  Error (|best-optimum|) : 2.413678e+02
  Best Solution (5 dim)  : [ 23.11501226 -61.76788984 -56.5113145  57.57066678
-62.90439179]
  D^(0) (init. diversity): 5.7928e+01
=====
=
```



Graph saved as convergence_F29.png

```
In [40]: # =====  
# F30 – Composition Function 10 (F15; F18; F19)  
# =====  
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

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LBLDE_m on CEC 2017 F30 – Single Run

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Function : Composition: F15; F18; F19 (10D)
Bounds : [-100, 100]
Global optimum : 3000.0
Initial $D^{(0)} = 5.7928e+01$ | Diversity threshold = 5.7928e+00
Generation 0: Best Fitness = 1.165390e+08 | Diversity = 5.7928e+01

Generation	10:	Best = 1.124585e+07	FES = 1100	Diversity = 4.3531e+01
Generation	20:	Best = 8.137738e+06	FES = 2100	Diversity = 3.5997e+01
Generation	30:	Best = 3.908748e+06	FES = 3100	Diversity = 3.3433e+01
Generation	40:	Best = 7.519525e+05	FES = 4100	Diversity = 3.0909e+01
Generation	50:	Best = 2.968729e+05	FES = 5100	Diversity = 2.9195e+01
Generation	60:	Best = 2.731725e+05	FES = 6100	Diversity = 2.3833e+01
Generation	70:	Best = 2.214610e+05	FES = 7100	Diversity = 1.9753e+01
Generation	80:	Best = 6.094907e+04	FES = 8100	Diversity = 2.0132e+01
Generation	90:	Best = 4.669540e+04	FES = 9100	Diversity = 1.8017e+01
Generation	100:	Best = 2.895193e+04	FES = 10100	Diversity = 1.5857e+01
Generation	110:	Best = 1.315498e+04	FES = 11100	Diversity = 1.6937e+01
Generation	120:	Best = 1.096787e+04	FES = 12100	Diversity = 2.1103e+01
Generation	130:	Best = 9.017102e+03	FES = 13100	Diversity = 2.0564e+01
Generation	140:	Best = 7.773096e+03	FES = 14100	Diversity = 1.6063e+01
Generation	150:	Best = 7.254754e+03	FES = 15100	Diversity = 1.7738e+01
Generation	160:	Best = 6.862485e+03	FES = 16100	Diversity = 1.5506e+01
Generation	170:	Best = 6.223484e+03	FES = 17100	Diversity = 1.3551e+01
Generation	180:	Best = 6.212967e+03	FES = 18100	Diversity = 1.0320e+01
Generation	190:	Best = 6.072884e+03	FES = 19100	Diversity = 8.6756e+00
Generation	200:	Best = 5.902084e+03	FES = 20100	Diversity = 6.7927e+00
Generation	210:	Best = 5.850870e+03	FES = 21100	Diversity = 6.4509e+00
Generation	220:	Best = 5.806145e+03	FES = 22100	Diversity = 5.6754e+00
Generation	230:	Best = 5.685405e+03	FES = 23100	Diversity = 5.8315e+00
Generation	240:	Best = 5.664969e+03	FES = 24100	Diversity = 5.6106e+00
Generation	250:	Best = 5.601391e+03	FES = 25100	Diversity = 5.4234e+00
Generation	260:	Best = 5.536256e+03	FES = 26100	Diversity = 5.4001e+00
Generation	270:	Best = 5.509893e+03	FES = 27100	Diversity = 6.0590e+00
Generation	280:	Best = 5.489241e+03	FES = 28100	Diversity = 5.3925e+00
Generation	290:	Best = 5.464134e+03	FES = 29100	Diversity = 4.8223e+00
Generation	300:	Best = 5.454774e+03	FES = 30100	Diversity = 5.2051e+00
Generation	310:	Best = 5.439597e+03	FES = 31100	Diversity = 6.0793e+00
Generation	320:	Best = 5.394982e+03	FES = 32100	Diversity = 6.7674e+00
Generation	330:	Best = 5.341071e+03	FES = 33100	Diversity = 6.0047e+00
Generation	340:	Best = 5.294139e+03	FES = 34100	Diversity = 4.7630e+00
Generation	350:	Best = 5.252473e+03	FES = 35100	Diversity = 4.1281e+00
Generation	360:	Best = 5.222123e+03	FES = 36100	Diversity = 3.3160e+00
Generation	370:	Best = 5.204144e+03	FES = 37100	Diversity = 2.9641e+00
Generation	380:	Best = 5.193722e+03	FES = 38100	Diversity = 2.6058e+00
Generation	390:	Best = 5.184259e+03	FES = 39100	Diversity = 2.0043e+00
Generation	400:	Best = 5.178986e+03	FES = 40100	Diversity = 1.8257e+00
Generation	410:	Best = 5.177054e+03	FES = 41100	Diversity = 1.7683e+00
Generation	420:	Best = 5.173636e+03	FES = 42100	Diversity = 1.6019e+00
Generation	430:	Best = 5.162891e+03	FES = 43100	Diversity = 1.8164e+00
Generation	440:	Best = 5.159902e+03	FES = 44100	Diversity = 1.6814e+00

Generation	450:	Best = 5.144764e+03	FES = 45100	Diversity = 1.8920e+00
Generation	460:	Best = 5.133763e+03	FES = 46100	Diversity = 2.5735e+00
Generation	470:	Best = 5.111975e+03	FES = 47100	Diversity = 2.9245e+00
Generation	480:	Best = 5.067285e+03	FES = 48100	Diversity = 3.9222e+00
Generation	490:	Best = 4.952461e+03	FES = 49100	Diversity = 3.2214e+00
Generation	500:	Best = 4.912931e+03	FES = 50100	Diversity = 2.2878e+00
Generation	510:	Best = 4.894797e+03	FES = 51100	Diversity = 2.0054e+00
Generation	520:	Best = 4.863433e+03	FES = 52100	Diversity = 2.3924e+00
Generation	530:	Best = 4.848729e+03	FES = 53100	Diversity = 3.2275e+00
Generation	540:	Best = 4.848729e+03	FES = 54100	Diversity = 3.1526e+00
Generation	550:	Best = 4.844001e+03	FES = 55100	Diversity = 2.3815e+00
Generation	560:	Best = 4.834001e+03	FES = 56100	Diversity = 2.1682e+00
Generation	570:	Best = 4.824084e+03	FES = 57100	Diversity = 2.4392e+00
Generation	580:	Best = 4.811241e+03	FES = 58100	Diversity = 2.1848e+00
Generation	590:	Best = 4.808597e+03	FES = 59100	Diversity = 2.1269e+00
Generation	600:	Best = 4.801028e+03	FES = 60100	Diversity = 2.5244e+00
Generation	610:	Best = 4.795629e+03	FES = 61100	Diversity = 1.9887e+00
Generation	620:	Best = 4.784053e+03	FES = 62100	Diversity = 2.0346e+00
Generation	630:	Best = 4.784053e+03	FES = 63100	Diversity = 2.2572e+00
Generation	640:	Best = 4.783234e+03	FES = 64100	Diversity = 2.0853e+00
Generation	650:	Best = 4.782344e+03	FES = 65100	Diversity = 2.1648e+00
Generation	660:	Best = 4.774850e+03	FES = 66100	Diversity = 2.5087e+00
Generation	670:	Best = 4.771366e+03	FES = 67100	Diversity = 2.7827e+00
Generation	680:	Best = 4.762057e+03	FES = 68100	Diversity = 2.1028e+00
Generation	690:	Best = 4.758274e+03	FES = 69100	Diversity = 1.5994e+00
Generation	700:	Best = 4.748239e+03	FES = 70100	Diversity = 1.7734e+00
Generation	710:	Best = 4.742375e+03	FES = 71100	Diversity = 1.7501e+00
Generation	720:	Best = 4.735887e+03	FES = 72100	Diversity = 2.1395e+00
Generation	730:	Best = 4.726777e+03	FES = 73100	Diversity = 2.9189e+00
Generation	740:	Best = 4.692053e+03	FES = 74100	Diversity = 2.4262e+00
Generation	750:	Best = 4.648370e+03	FES = 75100	Diversity = 2.8962e+00
Generation	760:	Best = 4.627981e+03	FES = 76100	Diversity = 2.2630e+00
Generation	770:	Best = 4.614422e+03	FES = 77100	Diversity = 1.5414e+00
Generation	780:	Best = 4.612082e+03	FES = 78100	Diversity = 1.4688e+00
Generation	790:	Best = 4.603679e+03	FES = 79100	Diversity = 1.2653e+00
Generation	800:	Best = 4.590954e+03	FES = 80100	Diversity = 1.1425e+00
Generation	810:	Best = 4.588418e+03	FES = 81100	Diversity = 1.0674e+00
Generation	820:	Best = 4.586755e+03	FES = 82100	Diversity = 1.0836e+00
Generation	830:	Best = 4.583559e+03	FES = 83100	Diversity = 8.2855e-01
Generation	840:	Best = 4.583559e+03	FES = 84100	Diversity = 6.8927e-01
Generation	850:	Best = 4.580223e+03	FES = 85100	Diversity = 6.0566e-01
Generation	860:	Best = 4.580223e+03	FES = 86100	Diversity = 5.7527e-01
Generation	870:	Best = 4.580223e+03	FES = 87100	Diversity = 5.4836e-01
Generation	880:	Best = 4.580223e+03	FES = 88100	Diversity = 5.7183e-01
Generation	890:	Best = 4.580223e+03	FES = 89100	Diversity = 6.1628e-01
Generation	900:	Best = 4.577738e+03	FES = 90100	Diversity = 7.3949e-01
Generation	910:	Best = 4.577231e+03	FES = 91100	Diversity = 7.7545e-01
Generation	920:	Best = 4.575689e+03	FES = 92100	Diversity = 6.4555e-01
Generation	930:	Best = 4.575689e+03	FES = 93100	Diversity = 6.1989e-01
Generation	940:	Best = 4.575689e+03	FES = 94100	Diversity = 6.0615e-01
Generation	950:	Best = 4.574520e+03	FES = 95100	Diversity = 6.0744e-01
Generation	960:	Best = 4.573107e+03	FES = 96100	Diversity = 5.6454e-01
Generation	970:	Best = 4.573107e+03	FES = 97100	Diversity = 4.7245e-01
Generation	980:	Best = 4.573080e+03	FES = 98100	Diversity = 4.8180e-01

Generation 990: Best = 4.573080e+03 FES = 99100 Diversity = 4.6646e-01

Optimization completed!

Final Best Fitness = 4.573080e+03

Total Function Evaluations = 100000

Results:

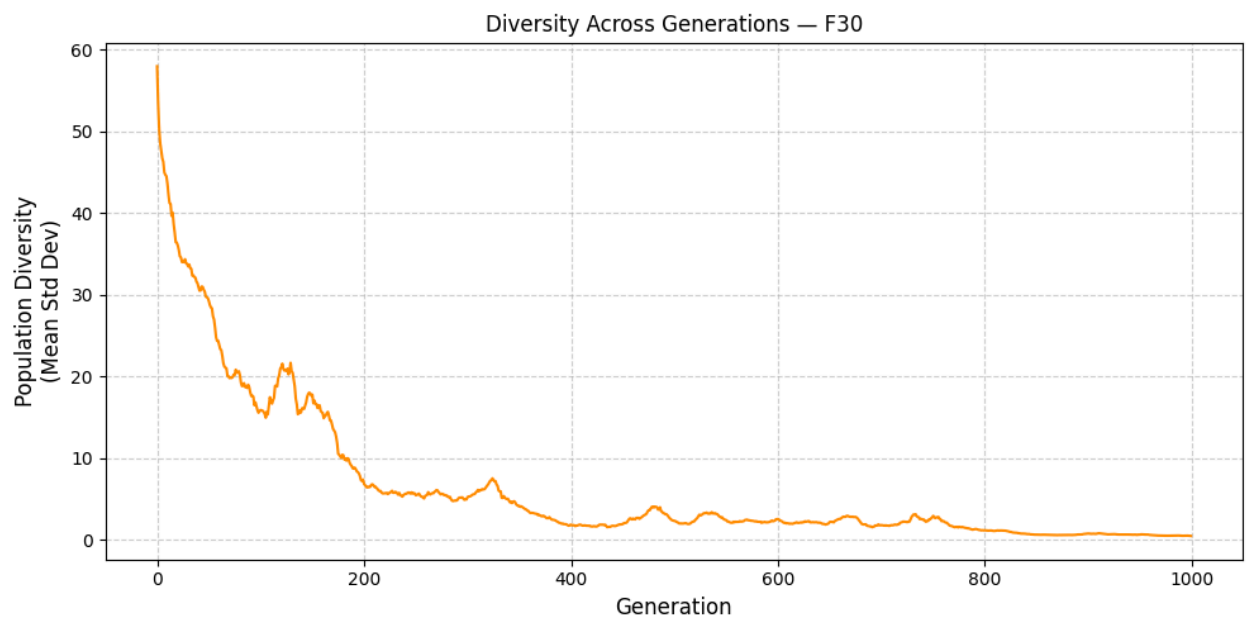
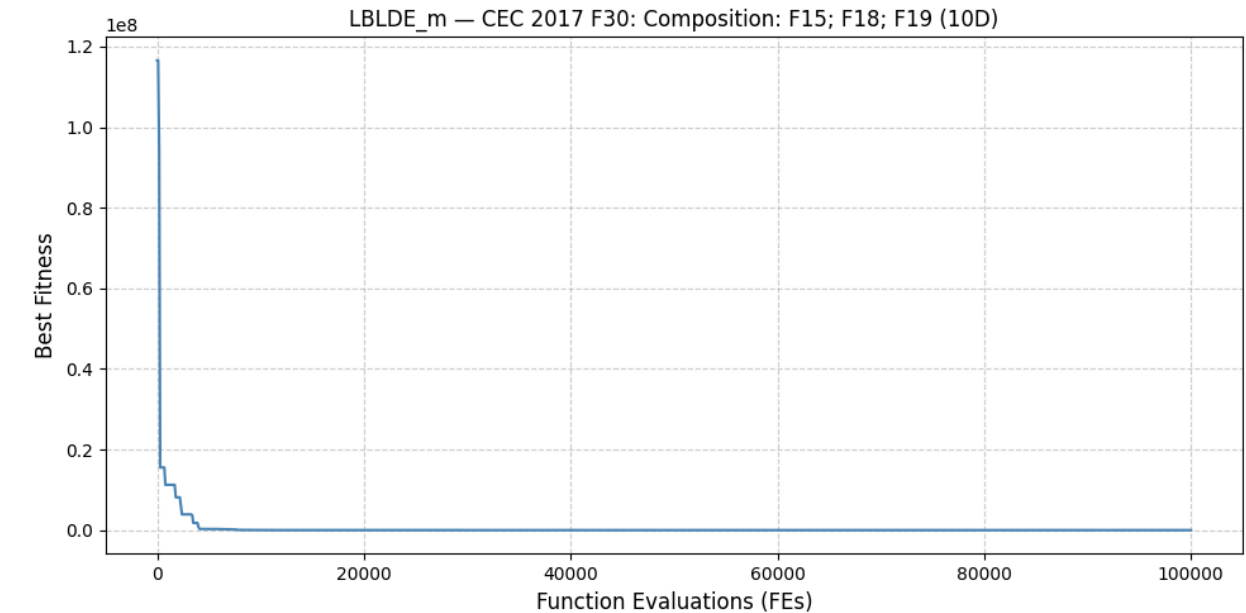
Best Fitness : 4.573080e+03

Global Optimum : 3.000000e+03

Error (|best-optimum|) : 1.573080e+03

Best Solution (5 dim) : [-34.92792129 92.8558777 -98.79618875 38.8772645
-99.62182156]

D^(0) (init. diversity): 5.7928e+01



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [42]: print('=' * 80)
print('SUMMARY – LBLDE_m on CEC 2017 (10D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["err
print('=' * 80)
```

=====			
=			
SUMMARY – LBLDE_m on CEC 2017 (10D), Single Run, Seed=42			
=====			
=			
Function	Best Fitness	Global Optimum	Error best-opt

F1	1.516031e+07	100.00	1.516021e+07
F2	2.372558e+03	200.00	2.172558e+03
F3	4.002712e+02	300.00	1.002712e+02
F4	4.039156e+02	400.00	3.915600e+00
F5	5.019918e+02	500.00	1.991785e+00
F6	6.000578e+02	600.00	5.775676e-02
F7	7.124535e+02	700.00	1.245351e+01
F8	8.039825e+02	800.00	3.982451e+00
F9	9.001974e+02	900.00	1.974188e-01
F10	1.003487e+03	1000.00	3.487353e+00
F11	1.110019e+03	1100.00	1.001882e+01
F12	1.318652e+03	1200.00	1.186518e+02
F13	1.300995e+03	1300.00	9.949601e-01
F14	1.401990e+03	1400.00	1.989918e+00
F15	1.504733e+03	1500.00	4.732737e+00
F16	1.602058e+03	1600.00	2.057710e+00
F17	1.702757e+03	1700.00	2.756999e+00
F18	1.801077e+03	1800.00	1.077054e+00
F19	1.901991e+03	1900.00	1.991105e+00
F20	2.005599e+03	2000.00	5.599194e+00
F21	2.200928e+03	2100.00	1.009283e+02
F22	2.301802e+03	2200.00	1.018022e+02
F23	2.607260e+03	2300.00	3.072597e+02
F24	2.721199e+03	2400.00	3.211989e+02
F25	2.946109e+03	2500.00	4.461089e+02
F26	2.976679e+03	2600.00	3.766788e+02
F27	3.093536e+03	2700.00	3.935359e+02
F28	3.139499e+03	2800.00	3.394990e+02
F29	3.141368e+03	2900.00	2.413678e+02
F30	4.573080e+03	3000.00	1.573080e+03
=====			
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